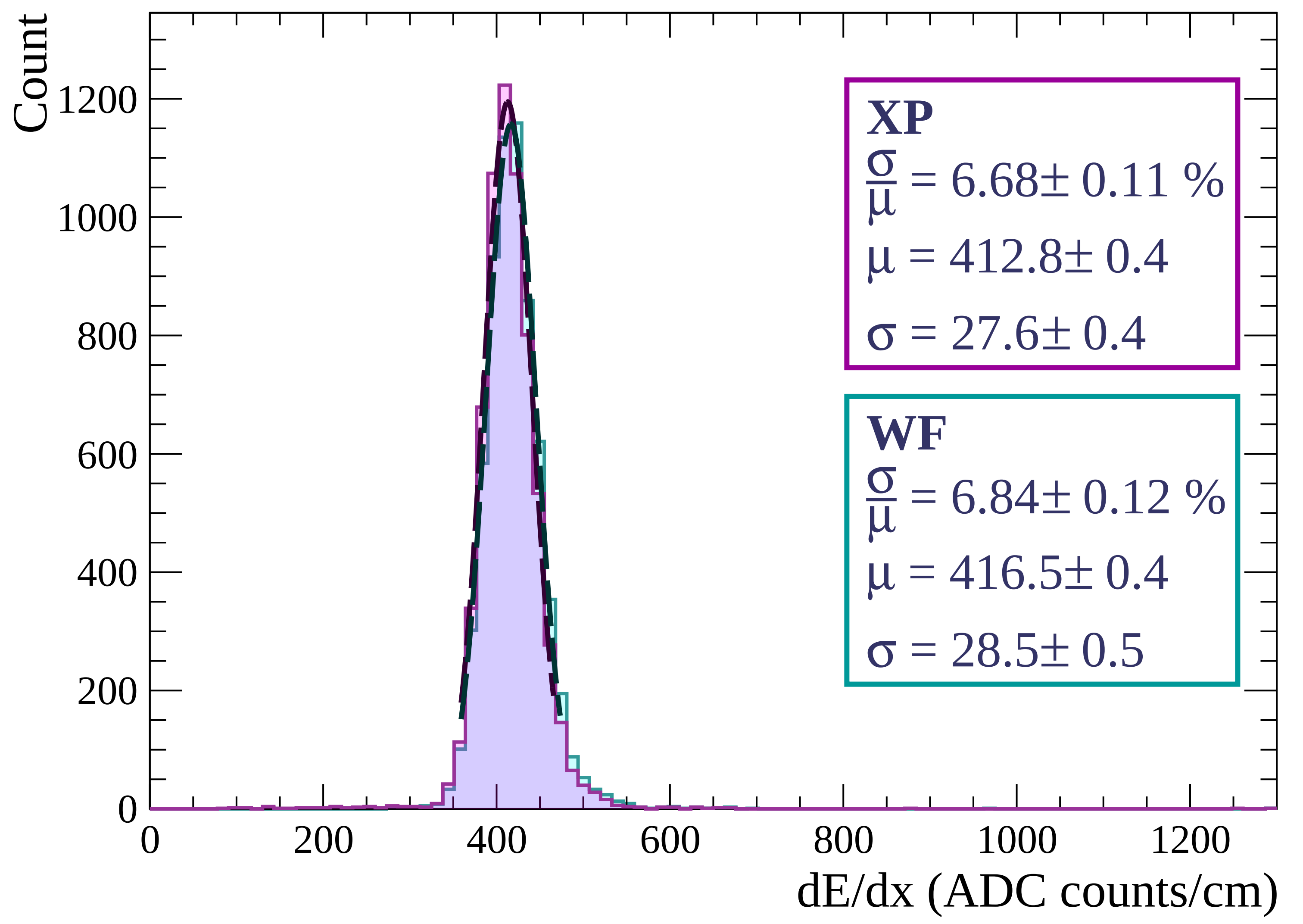
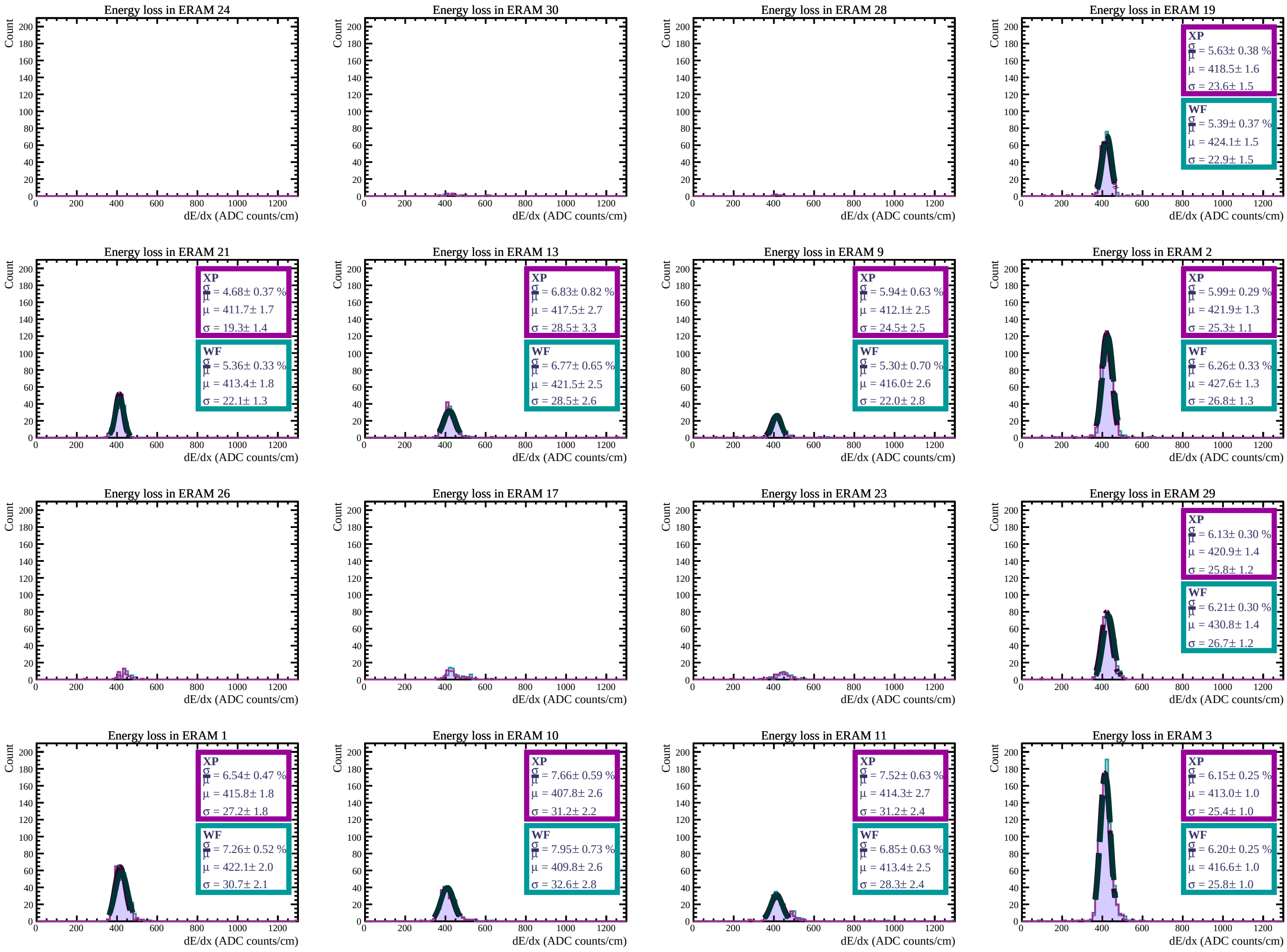
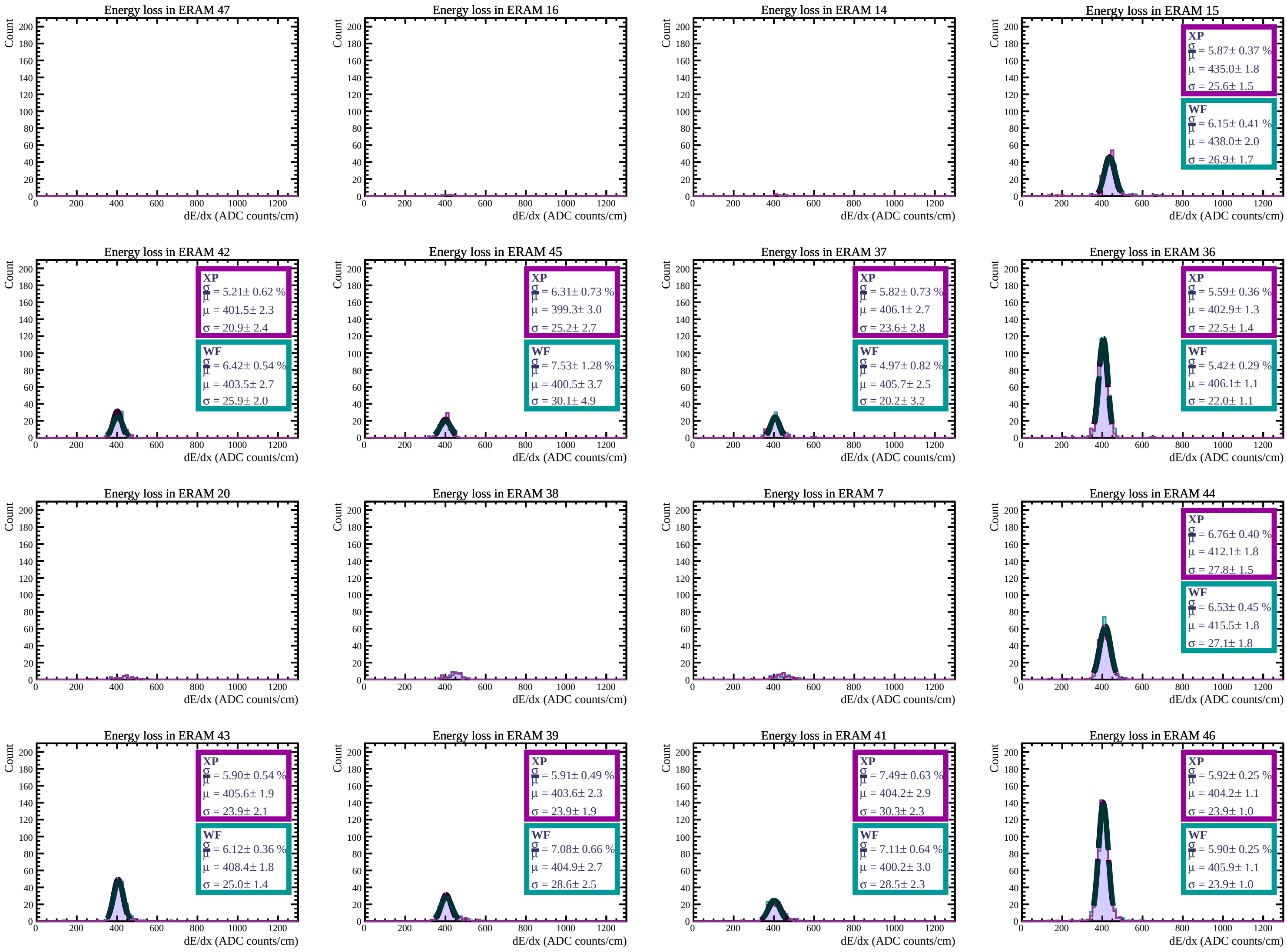
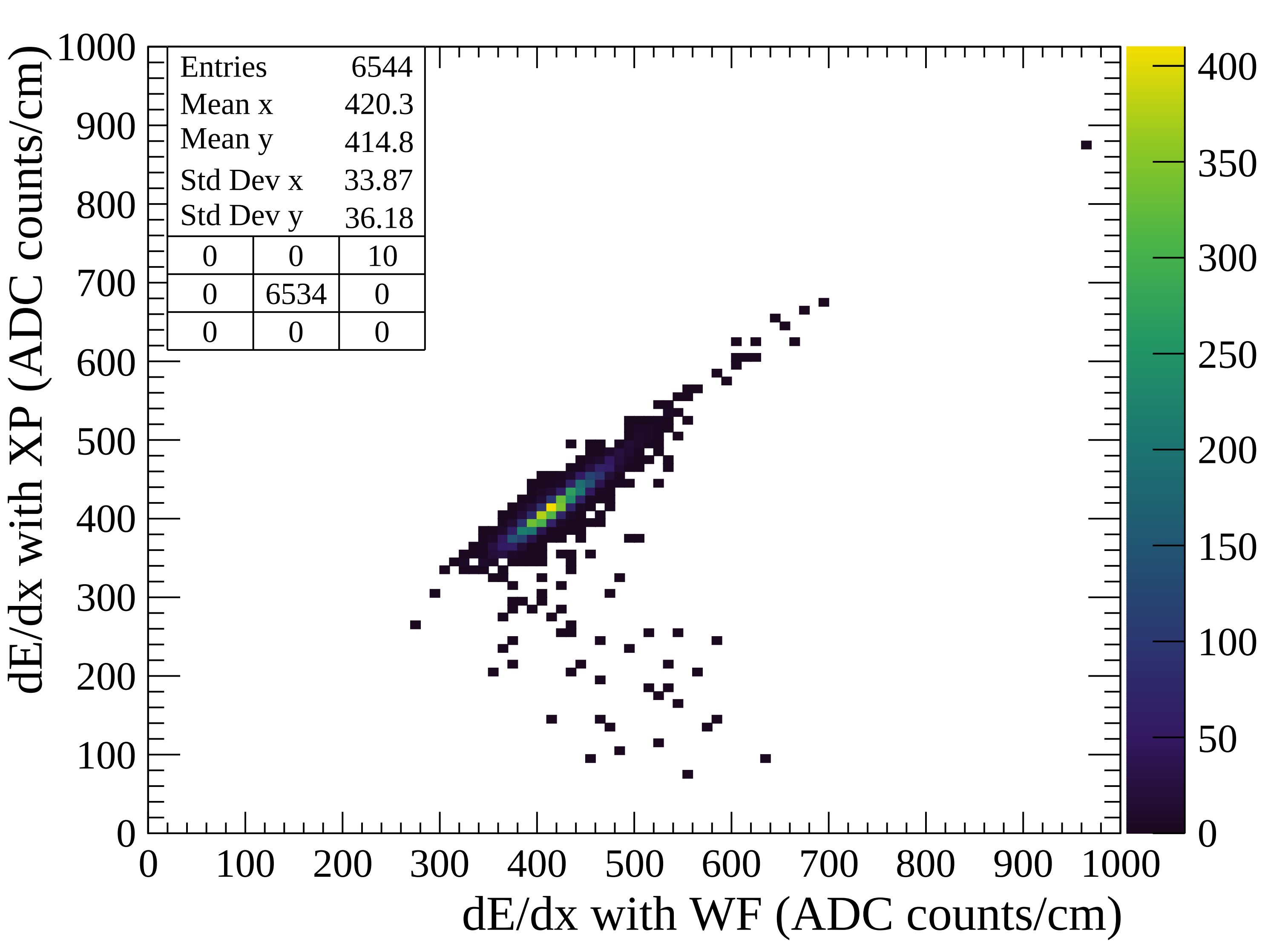


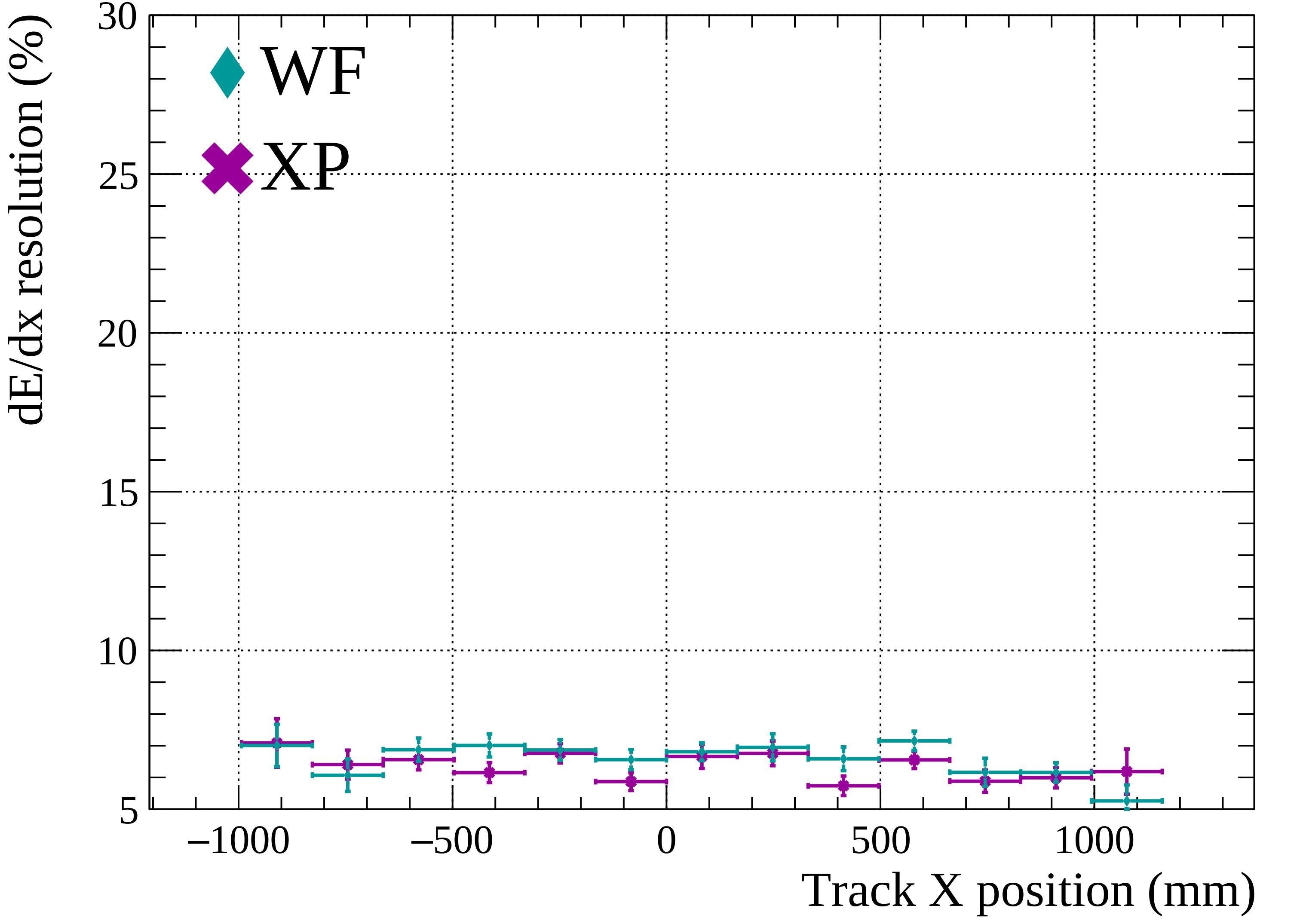
Count
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.68± 0.11 %
μ = 412.8± 0.4
σ = 27.6± 0.4
WF
σ/μ = 6.84± 0.12 %
μ = 416.5± 0.4
σ = 28.5± 0.5

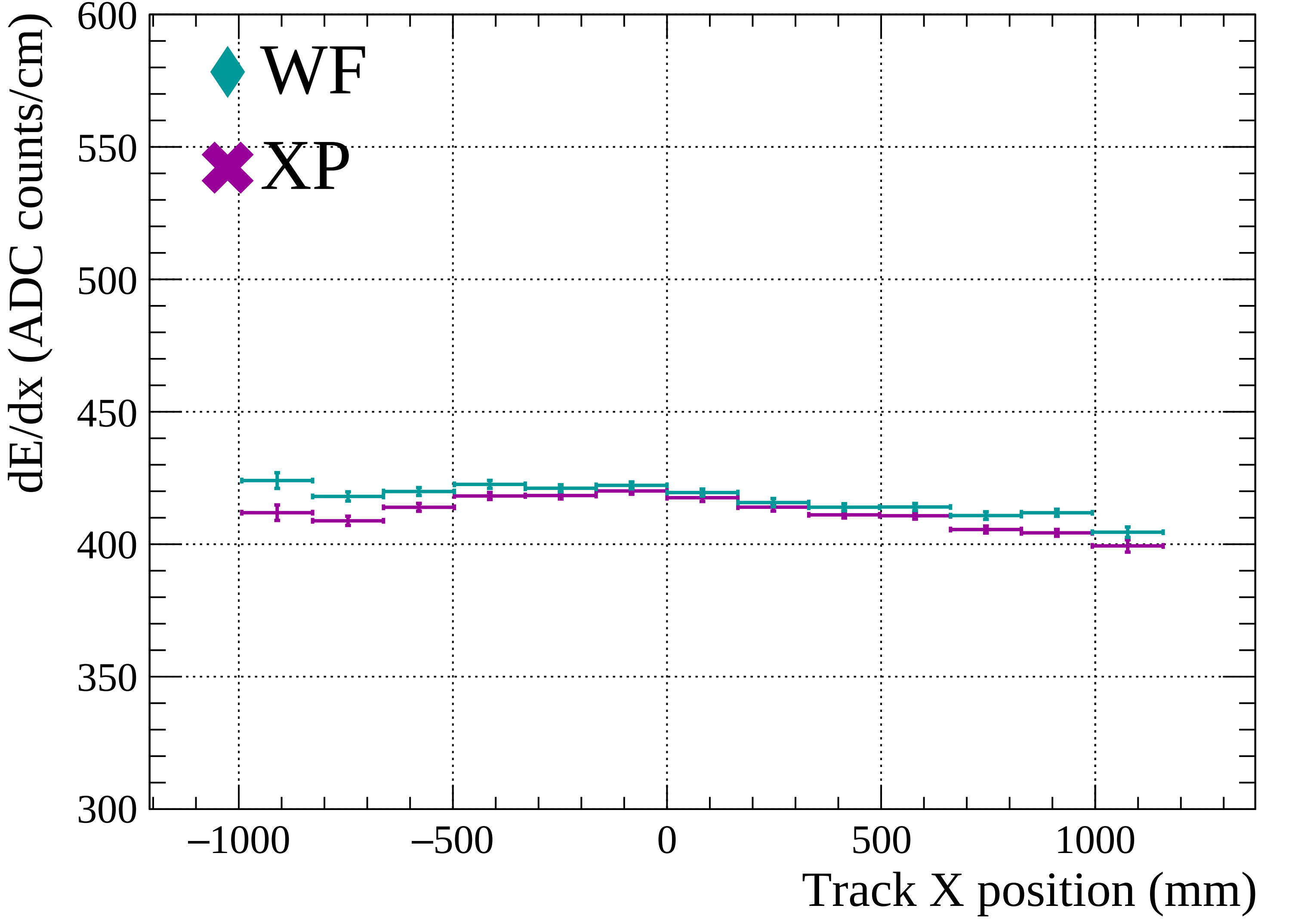


Energy loss in ERAM 24
Energy loss in ERAM 30
Energy loss in ERAM 28
Energy loss in ERAM 19
XP
σ/μ = 5.63± 0.38 %
μ = 418.5± 1.6
σ = 23.6± 1.5
WF
σ/μ = 5.39± 0.37 %
μ = 424.1± 1.5
σ = 22.9± 1.5
Energy loss in ERAM 21
XP
σ/μ = 4.68± 0.37 %
μ = 411.7± 1.7
σ = 19.3± 1.4
WF
σ/μ = 5.36± 0.33 %
μ = 413.4± 1.8
σ = 22.1± 1.3
Energy loss in ERAM 13
XP
σ/μ = 6.83± 0.82 %
μ = 417.5± 2.7
σ = 28.5± 3.3
WF
σ/μ = 6.77± 0.65 %
μ = 421.5± 2.5
σ = 28.5± 2.6
Energy loss in ERAM 9
XP
σ/μ = 5.94± 0.63 %
μ = 412.1± 2.5
σ = 24.5± 2.5
WF
σ/μ = 5.30± 0.70 %
μ = 416.0± 2.6
σ = 22.0± 2.8
Energy loss in ERAM 2
XP
σ/μ = 5.99± 0.29 %
μ = 421.9± 1.3
σ = 25.3± 1.1
WF
σ/μ = 6.26± 0.33 %
μ = 427.6± 1.3
σ = 26.8± 1.3
Energy loss in ERAM 26
Energy loss in ERAM 17
Energy loss in ERAM 23
Energy loss in ERAM 29
XP
σ/μ = 6.13± 0.30 %
μ = 420.9± 1.4
σ = 25.8± 1.2
WF
σ/μ = 6.21± 0.30 %
μ = 430.8± 1.4
σ = 26.7± 1.2
Energy loss in ERAM 1
XP
σ/μ = 6.54± 0.47 %
μ = 415.8± 1.8
σ = 27.2± 1.8
WF
σ/μ = 7.26± 0.52 %
μ = 422.1± 2.0
σ = 30.7± 2.1
Energy loss in ERAM 10
XP
σ/μ = 7.66± 0.59 %
μ = 407.8± 2.6
σ = 31.2± 2.2
WF
σ/μ = 7.95± 0.73 %
μ = 409.8± 2.6
σ = 32.6± 2.8
Energy loss in ERAM 11
XP
σ/μ = 7.52± 0.63 %
μ = 414.3± 2.7
σ = 31.2± 2.4
WF
σ/μ = 6.85± 0.63 %
μ = 413.4± 2.5
σ = 28.3± 2.4
Energy loss in ERAM 3
XP
σ/μ = 6.15± 0.25 %
μ = 413.0± 1.0
σ = 25.4± 1.0
WF
σ/μ = 6.20± 0.25 %
μ = 416.6± 1.0
σ = 25.8± 1.0
Count
dE/dx (ADC counts/cm)

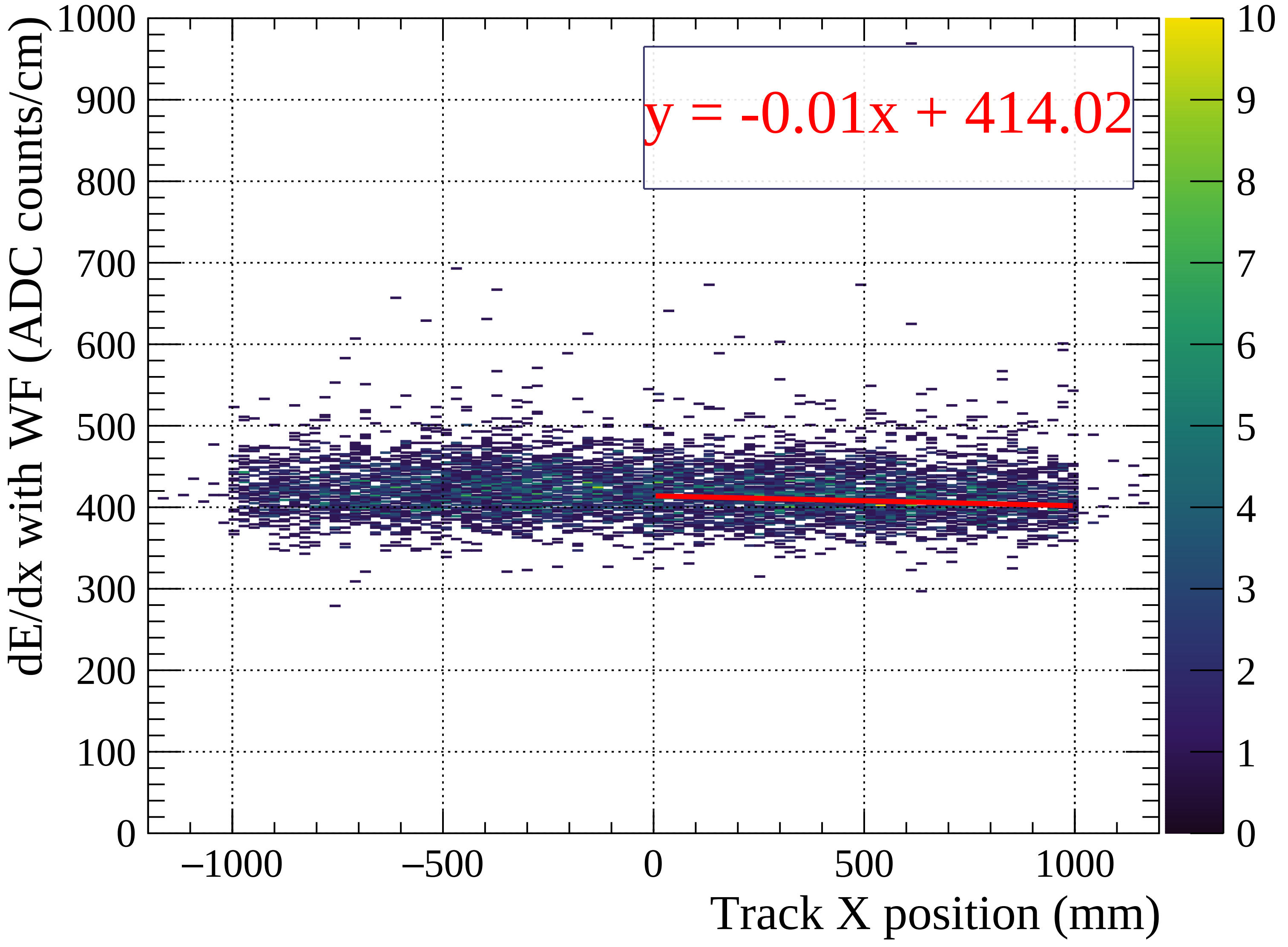

Energy loss in ERAM 47
Energy loss in ERAM 16
Energy loss in ERAM 14
Energy loss in ERAM 15
XP
σ/μ = 5.87± 0.37 %
μ = 435.0± 1.8
σ = 25.6± 1.5
WF
σ/μ = 6.15± 0.41 %
μ = 438.0± 2.0
σ = 26.9± 1.7
Energy loss in ERAM 42
XP
σ/μ = 5.21± 0.62 %
μ = 401.5± 2.3
σ = 20.9± 2.4
WF
σ/μ = 6.42± 0.54 %
μ = 403.5± 2.7
σ = 25.9± 2.0
Energy loss in ERAM 45
XP
σ/μ = 6.31± 0.73 %
μ = 399.3± 3.0
σ = 25.2± 2.7
WF
σ/μ = 7.53± 1.28 %
μ = 400.5± 3.7
σ = 30.1± 4.9
Energy loss in ERAM 37
XP
σ/μ = 5.82± 0.73 %
μ = 406.1± 2.7
σ = 23.6± 2.8
WF
σ/μ = 4.97± 0.82 %
μ = 405.7± 2.5
σ = 20.2± 3.2
Energy loss in ERAM 36
XP
σ/μ = 5.59± 0.36 %
μ = 402.9± 1.3
σ = 22.5± 1.4
WF
σ/μ = 5.42± 0.29 %
μ = 406.1± 1.1
σ = 22.0± 1.1
Energy loss in ERAM 20
Energy loss in ERAM 38
Energy loss in ERAM 7
Energy loss in ERAM 44
XP
σ/μ = 6.76± 0.40 %
μ = 412.1± 1.8
σ = 27.8± 1.5
WF
σ/μ = 6.53± 0.45 %
μ = 415.5± 1.8
σ = 27.1± 1.8
Energy loss in ERAM 43
XP
σ/μ = 5.90± 0.54 %
μ = 405.6± 1.9
σ = 23.9± 2.1
WF
σ/μ = 6.12± 0.36 %
μ = 408.4± 1.8
σ = 25.0± 1.4
Energy loss in ERAM 39
XP
σ/μ = 5.91± 0.49 %
μ = 403.6± 2.3
σ = 23.9± 1.9
WF
σ/μ = 7.08± 0.66 %
μ = 404.9± 2.7
σ = 28.6± 2.5
Energy loss in ERAM 41
XP
σ/μ = 7.49± 0.63 %
μ = 404.2± 2.9
σ = 30.3± 2.3
WF
σ/μ = 7.11± 0.64 %
μ = 400.2± 3.0
σ = 28.5± 2.3
Energy loss in ERAM 46
XP
σ/μ = 5.92± 0.25 %
μ = 404.2± 1.1
σ = 23.9± 1.0
WF
σ/μ = 5.90± 0.25 %
μ = 405.9± 1.1
σ = 23.9± 1.0
Count
dE/dx (ADC counts/cm)

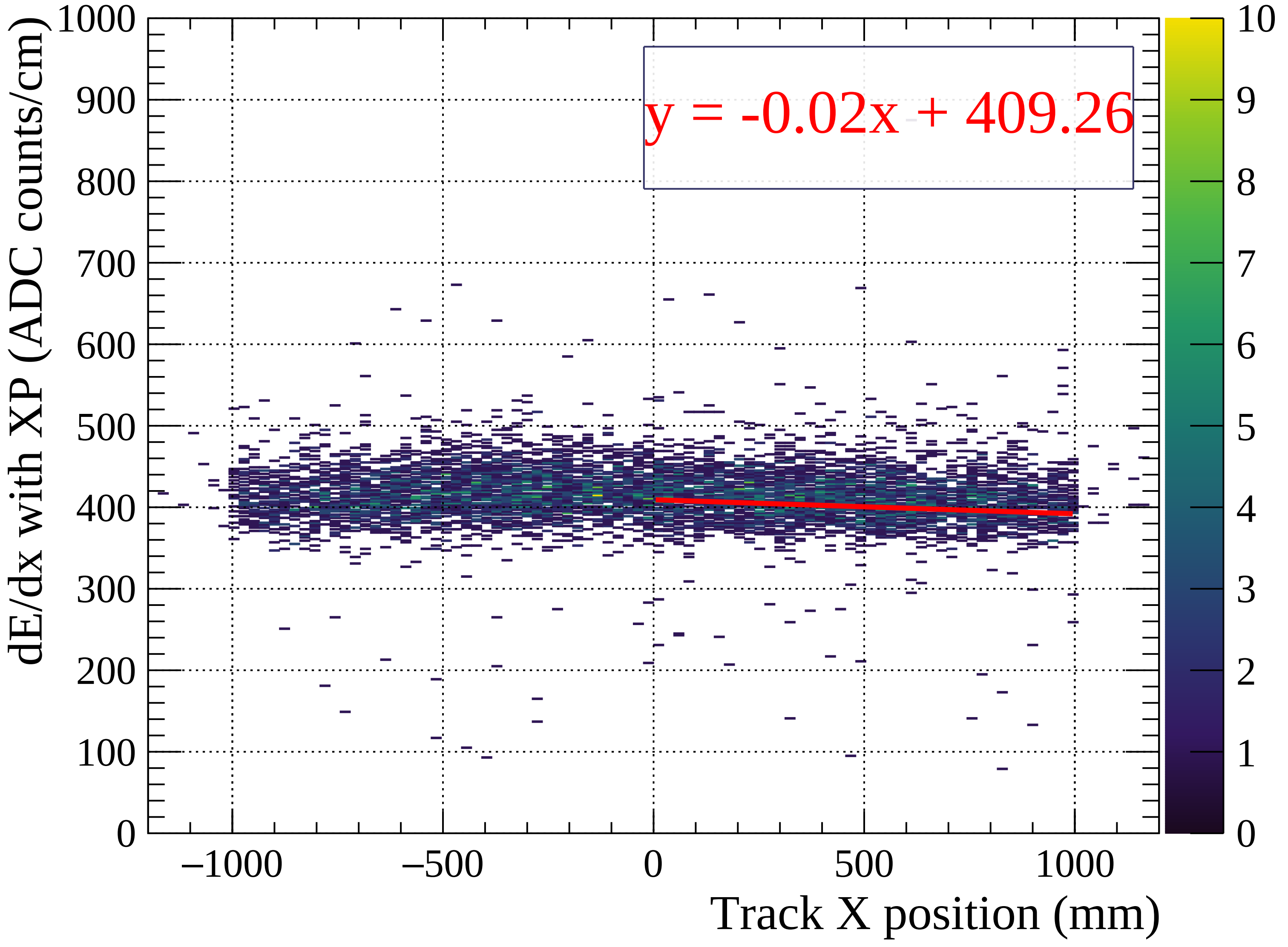

Entries 6544
Mean x 420.3
Mean y 414.8
Std Dev x 33.87
Std Dev y 36.18
0 0 10
0 6534 0
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
0
50
100
150
200
250
300
350
400

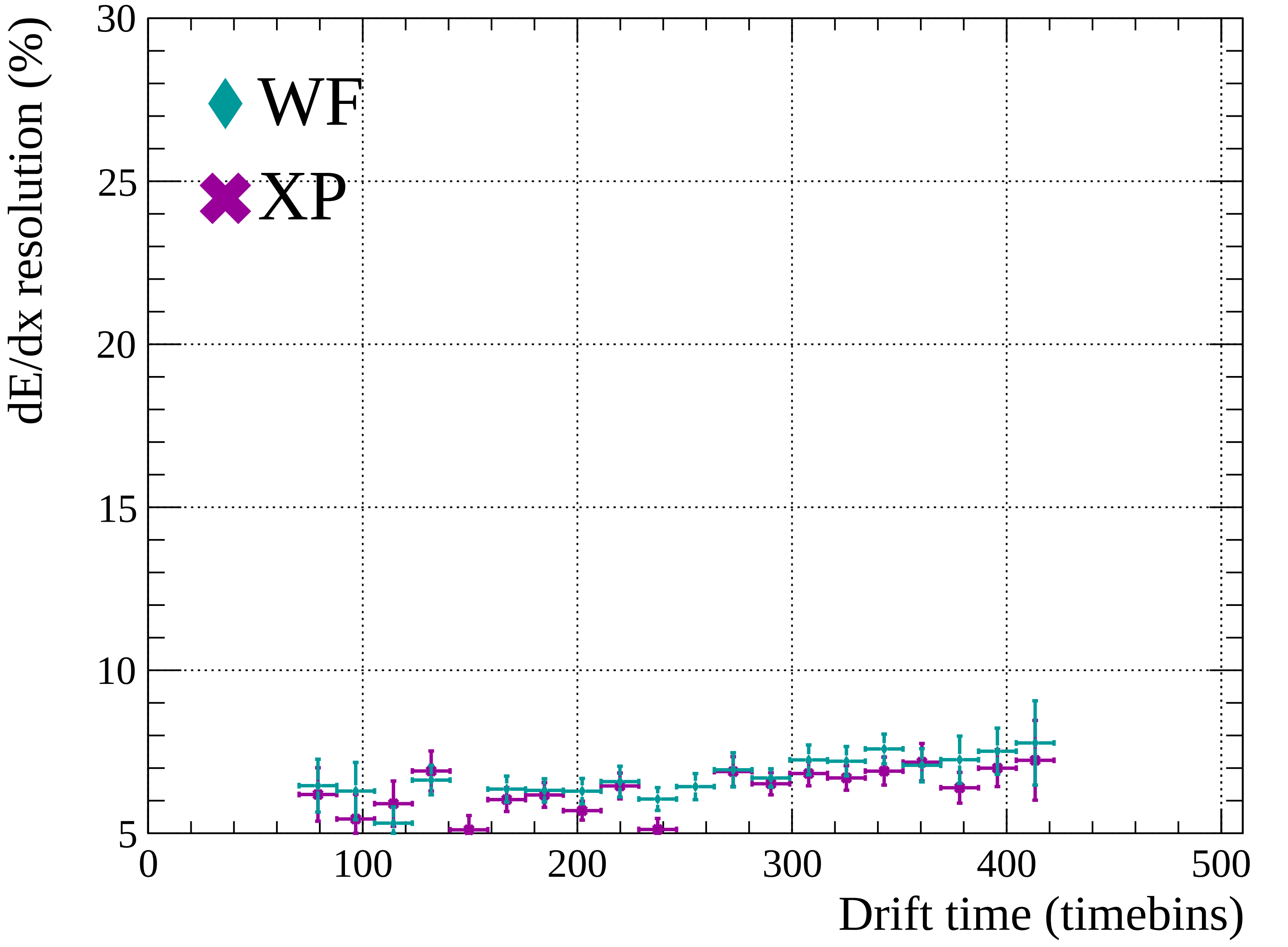

WF
XP
dE/dx resolution (%)
Track X position (mm)
30
25
20
15
10
5
−1000
−500
0
500
1000



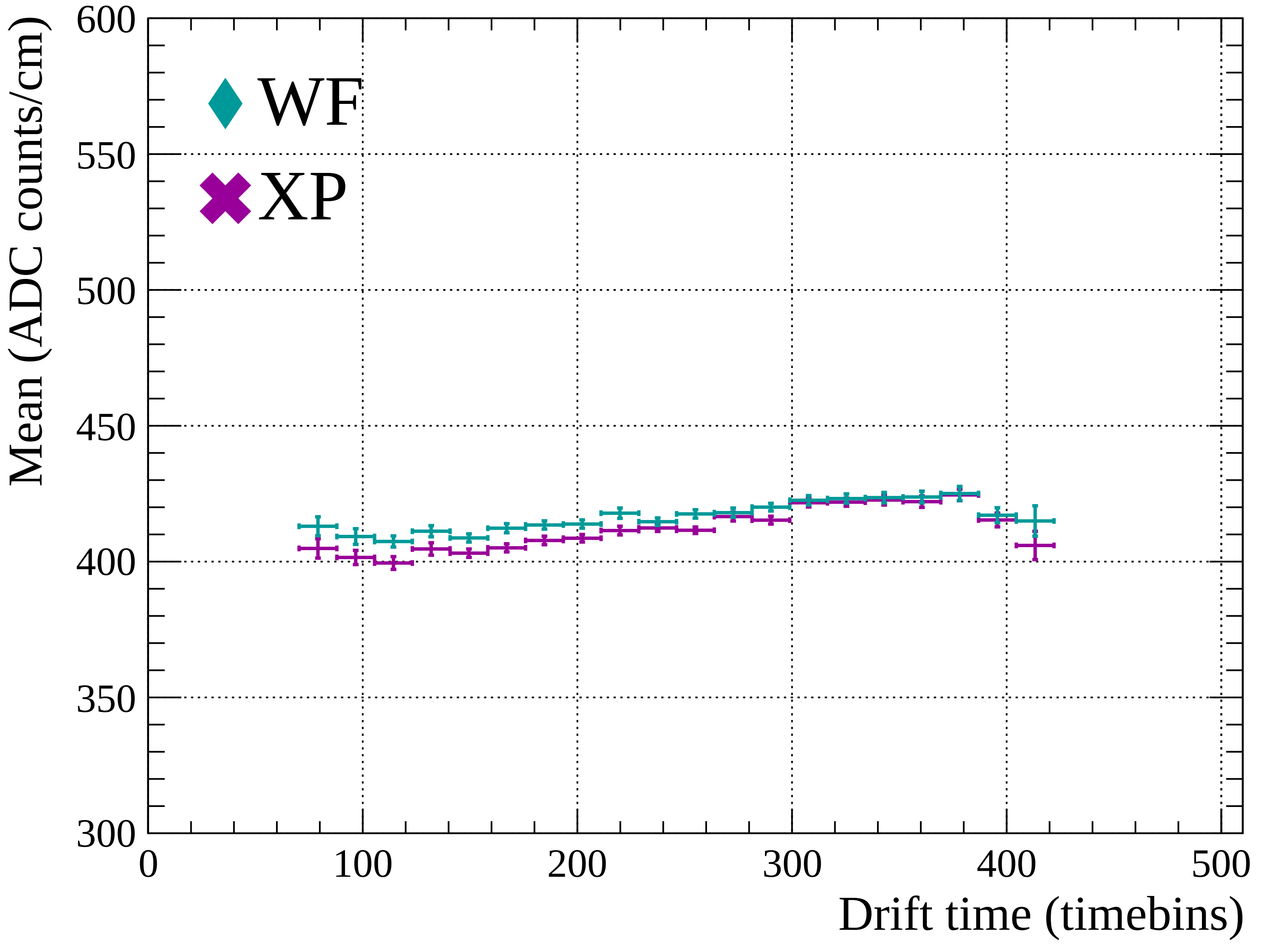
WF
XP
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
−1000
−500
0
500
1000
Track X position (mm)

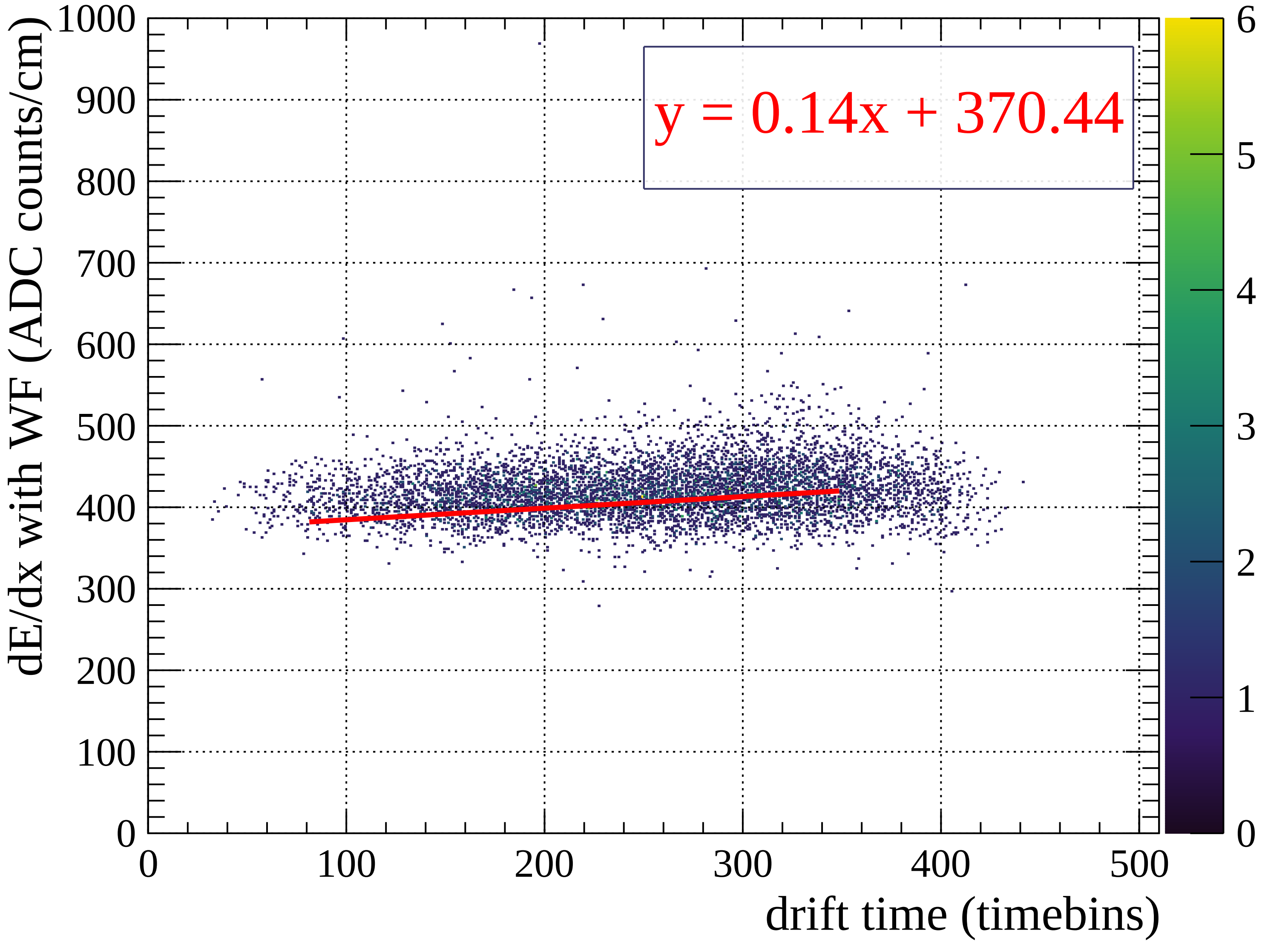

y = -0.01x + 414.02
dE/dx with WF (ADC counts/cm)
Track X position (mm)
1000
900
800
700
600
500
400
300
200
100
0
−1000
−500
0
500
1000
10
9
8
7
6
5
4
3
2
1
0


y = -0.02x + 409.26
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
1
2
3
4
5
6
7
8
9
10

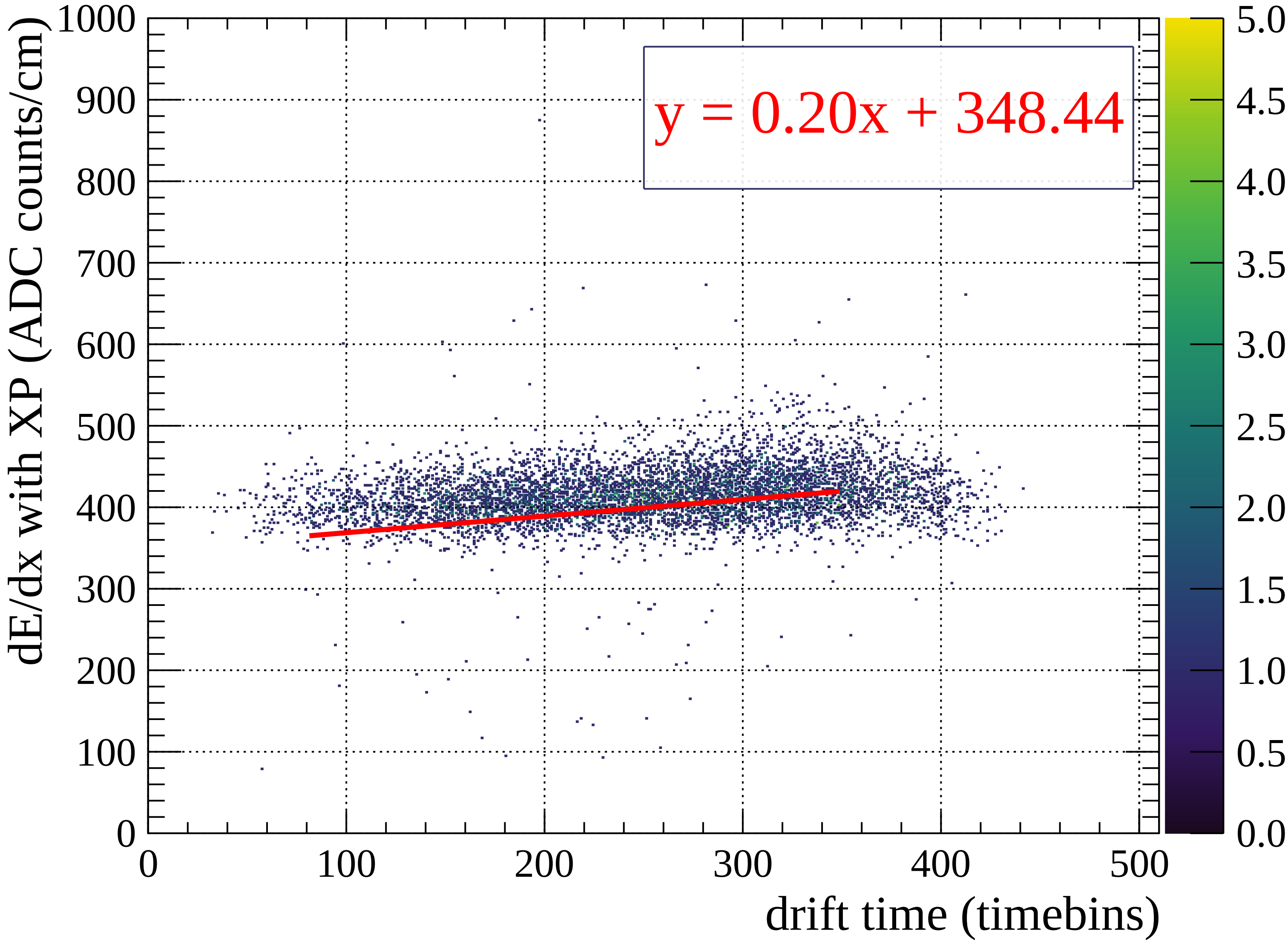


WF
XP
dE/dx resolution (%)
Drift time (timebins)
30
25
20
15
10
5
0
100
200
300
400
500

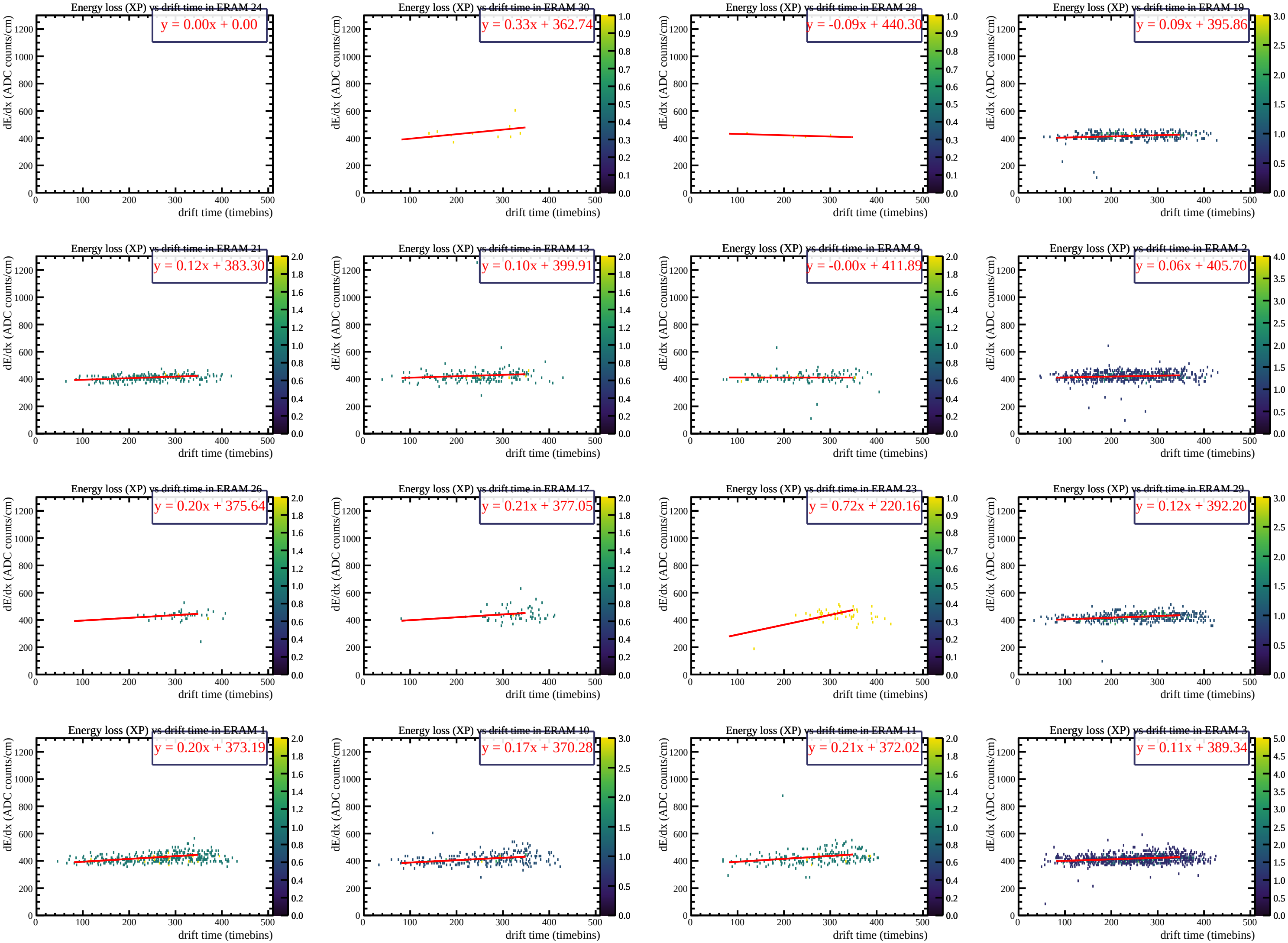

WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)



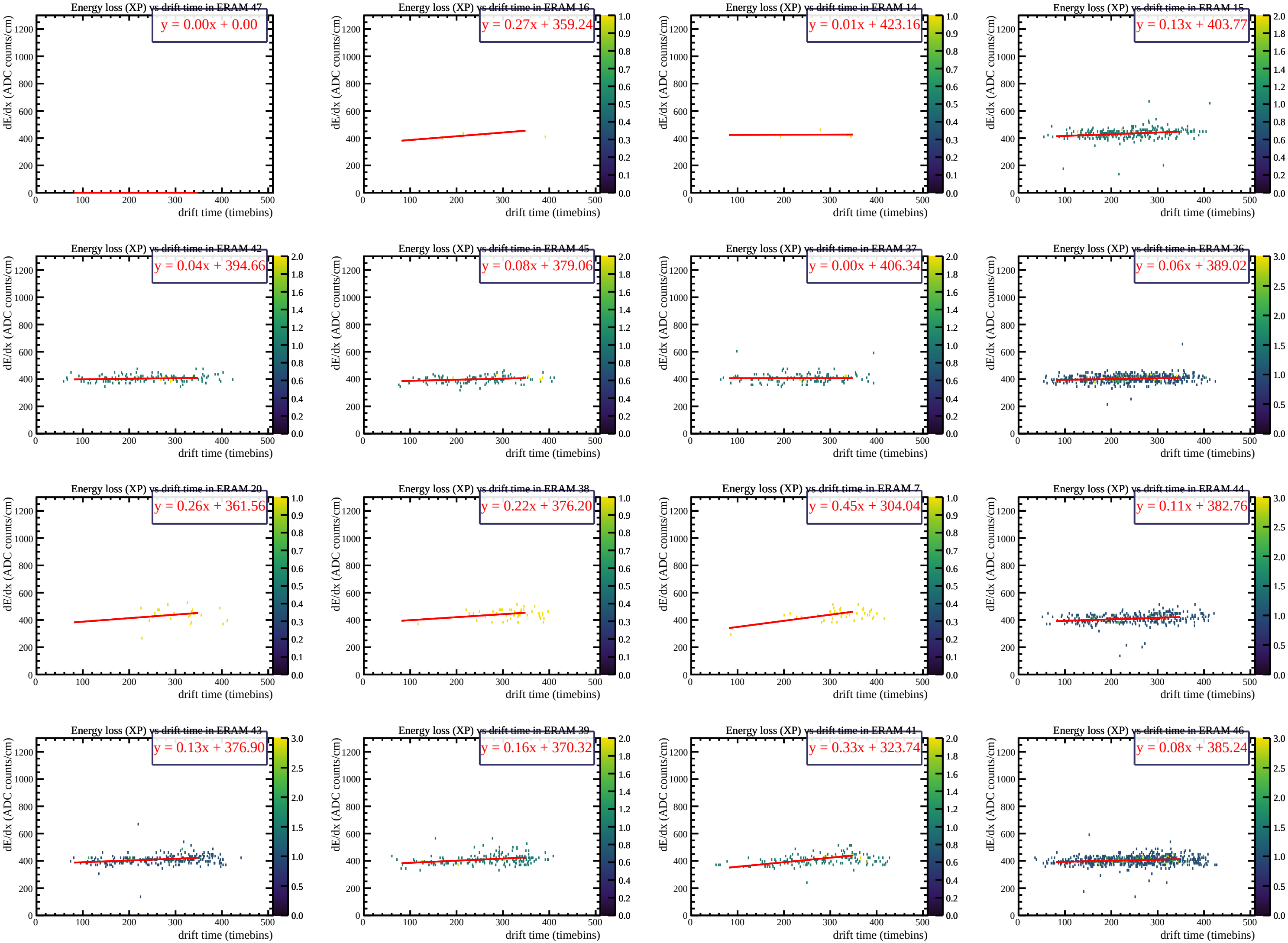
y = 0.14x + 370.44
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
1
2
3
4
5
6

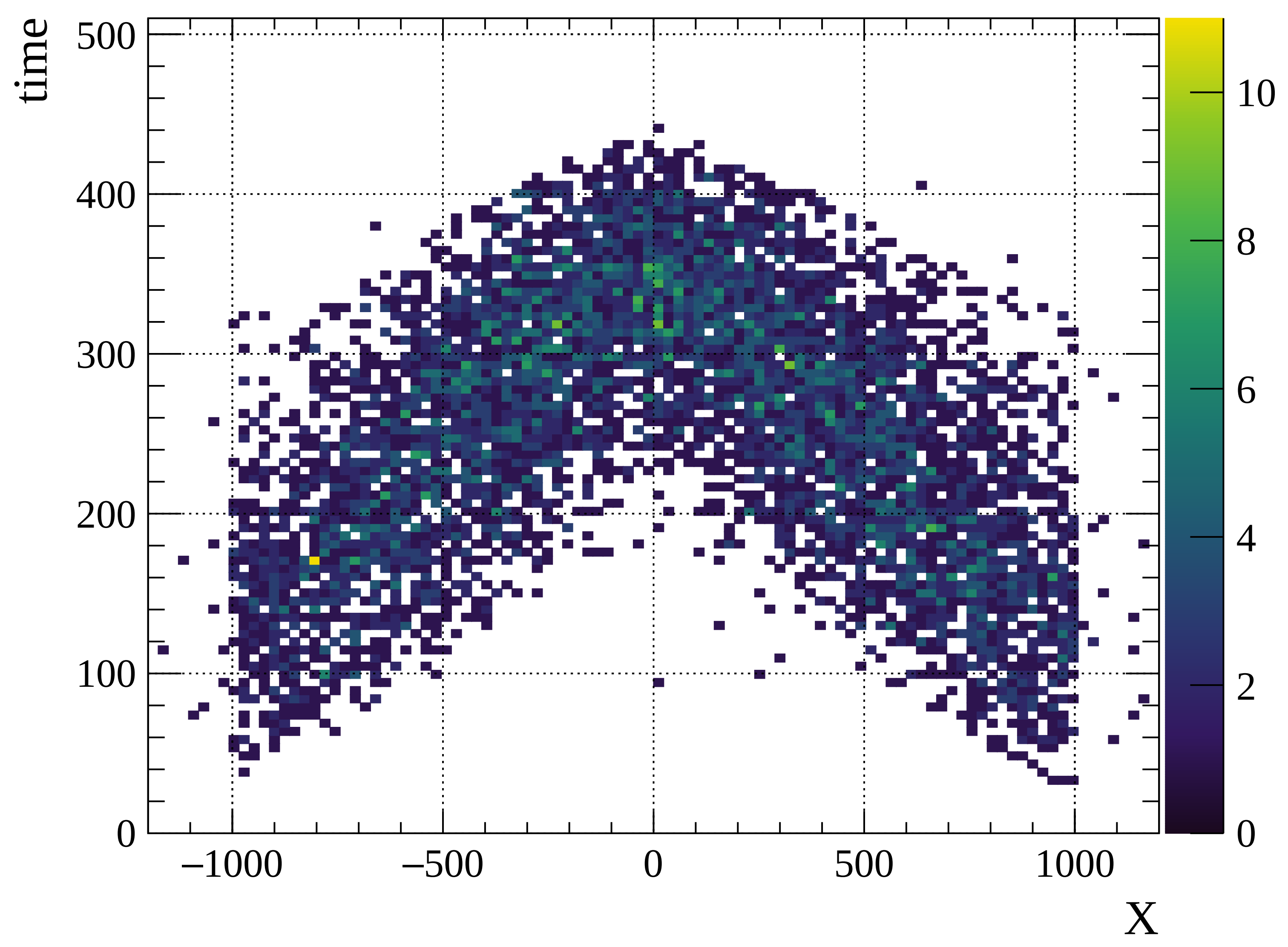


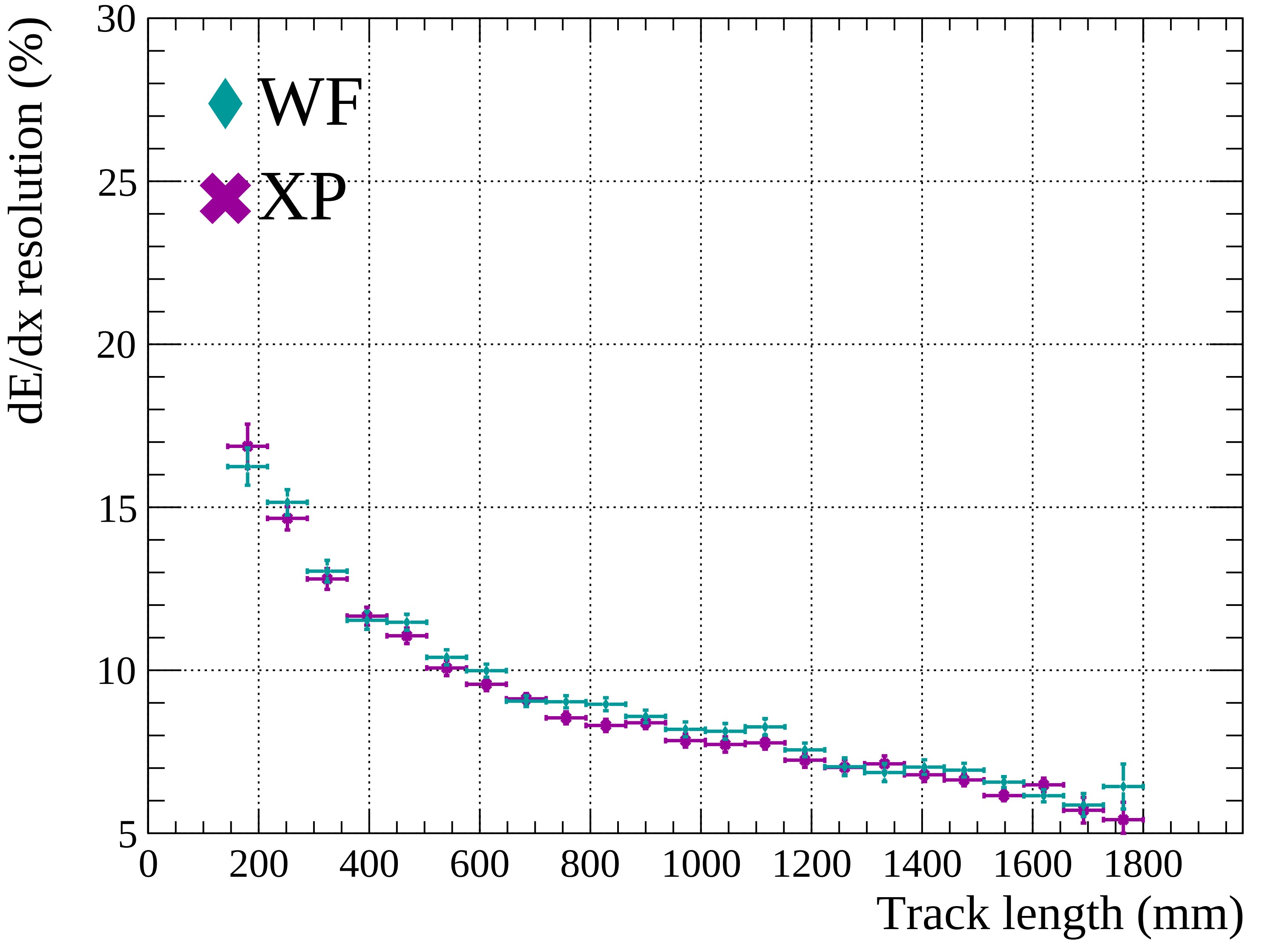
y = 0.20x + 348.44
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0



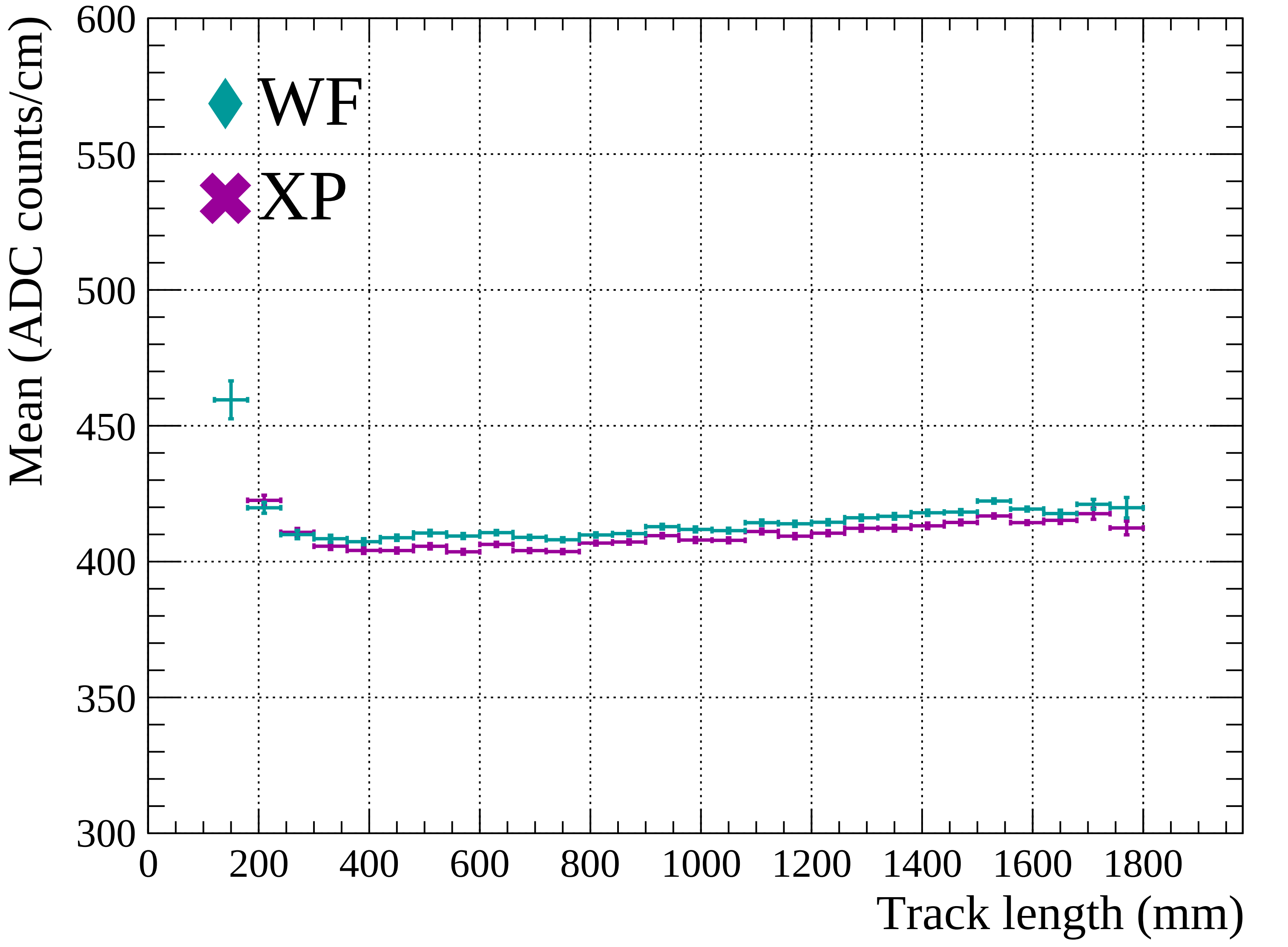
Energy loss (XP) vs drift time in ERAM 24
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 30
y = 0.33x + 362.74
Energy loss (XP) vs drift time in ERAM 28
y = -0.09x + 440.30
Energy loss (XP) vs drift time in ERAM 19
y = 0.09x + 395.86
Energy loss (XP) vs drift time in ERAM 21
y = 0.12x + 383.30
Energy loss (XP) vs drift time in ERAM 13
y = 0.10x + 399.91
Energy loss (XP) vs drift time in ERAM 9
y = -0.00x + 411.89
Energy loss (XP) vs drift time in ERAM 2
y = 0.06x + 405.70
Energy loss (XP) vs drift time in ERAM 26
y = 0.20x + 375.64
Energy loss (XP) vs drift time in ERAM 17
y = 0.21x + 377.05
Energy loss (XP) vs drift time in ERAM 23
y = 0.72x + 220.16
Energy loss (XP) vs drift time in ERAM 29
y = 0.12x + 392.20
Energy loss (XP) vs drift time in ERAM 1
y = 0.20x + 373.19
Energy loss (XP) vs drift time in ERAM 10
y = 0.17x + 370.28
Energy loss (XP) vs drift time in ERAM 11
y = 0.21x + 372.02
Energy loss (XP) vs drift time in ERAM 3
y = 0.11x + 389.34
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (XP) vs drift time in ERAM 47
y = 0.00x + 0.00
Energy loss (XP) vs drift time in ERAM 16
y = 0.27x + 359.24
Energy loss (XP) vs drift time in ERAM 14
y = 0.01x + 423.16
Energy loss (XP) vs drift time in ERAM 15
y = 0.13x + 403.77
Energy loss (XP) vs drift time in ERAM 42
y = 0.04x + 394.66
Energy loss (XP) vs drift time in ERAM 45
y = 0.08x + 379.06
Energy loss (XP) vs drift time in ERAM 37
y = 0.00x + 406.34
Energy loss (XP) vs drift time in ERAM 36
y = 0.06x + 389.02
Energy loss (XP) vs drift time in ERAM 20
y = 0.26x + 361.56
Energy loss (XP) vs drift time in ERAM 38
y = 0.22x + 376.20
Energy loss (XP) vs drift time in ERAM 7
y = 0.45x + 304.04
Energy loss (XP) vs drift time in ERAM 44
y = 0.11x + 382.76
Energy loss (XP) vs drift time in ERAM 43
y = 0.13x + 376.90
Energy loss (XP) vs drift time in ERAM 39
y = 0.16x + 370.32
Energy loss (XP) vs drift time in ERAM 41
y = 0.33x + 323.74
Energy loss (XP) vs drift time in ERAM 46
y = 0.08x + 385.24
dE/dx (ADC counts/cm)
drift time (timebins)

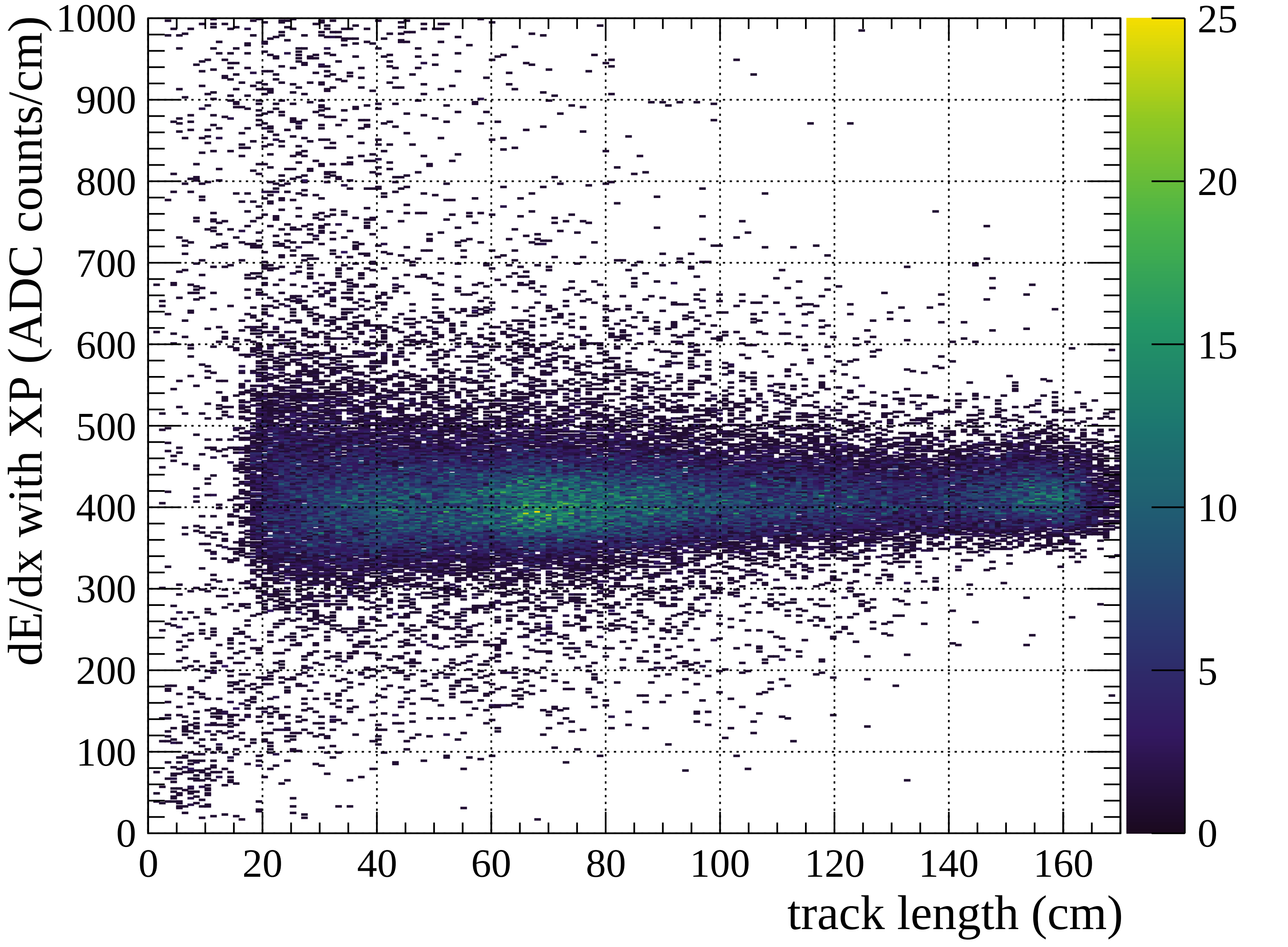

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
10
8
6
4
2
0



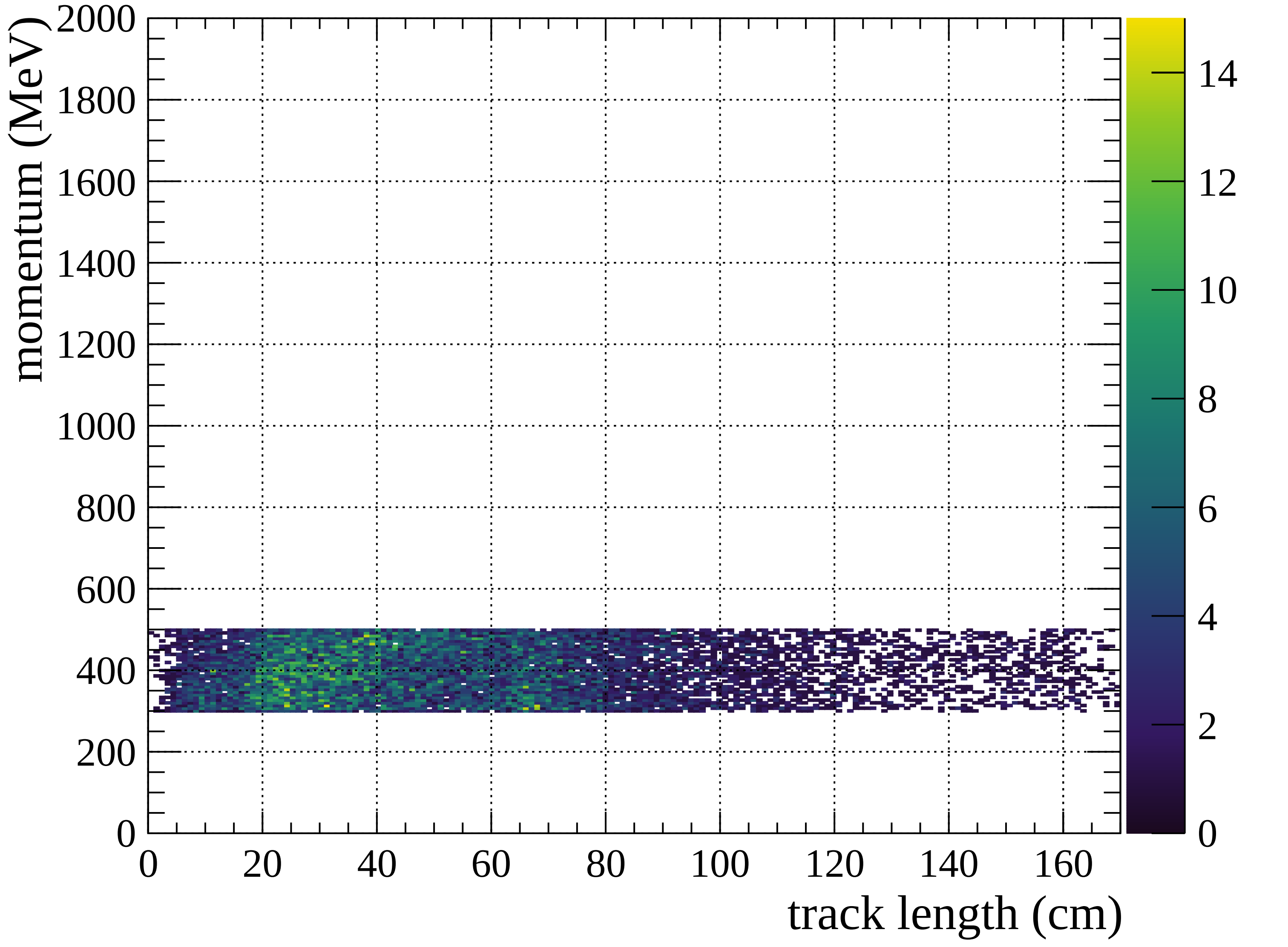
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
200
400
600
800
1000
1200
1400
1600
1800
Track length (mm)



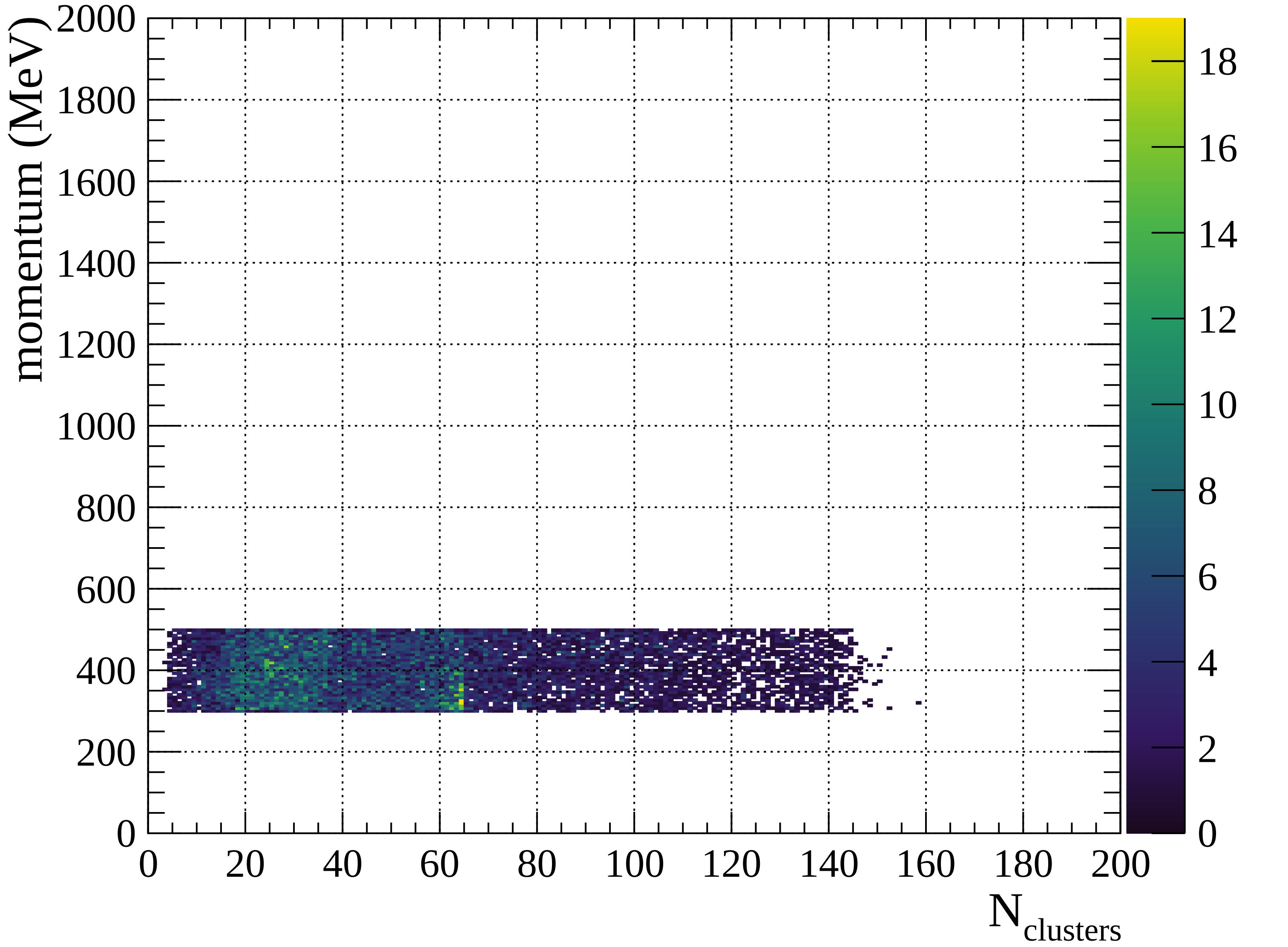
WF
XP
Mean (ADC counts/cm)
Track length (mm)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800



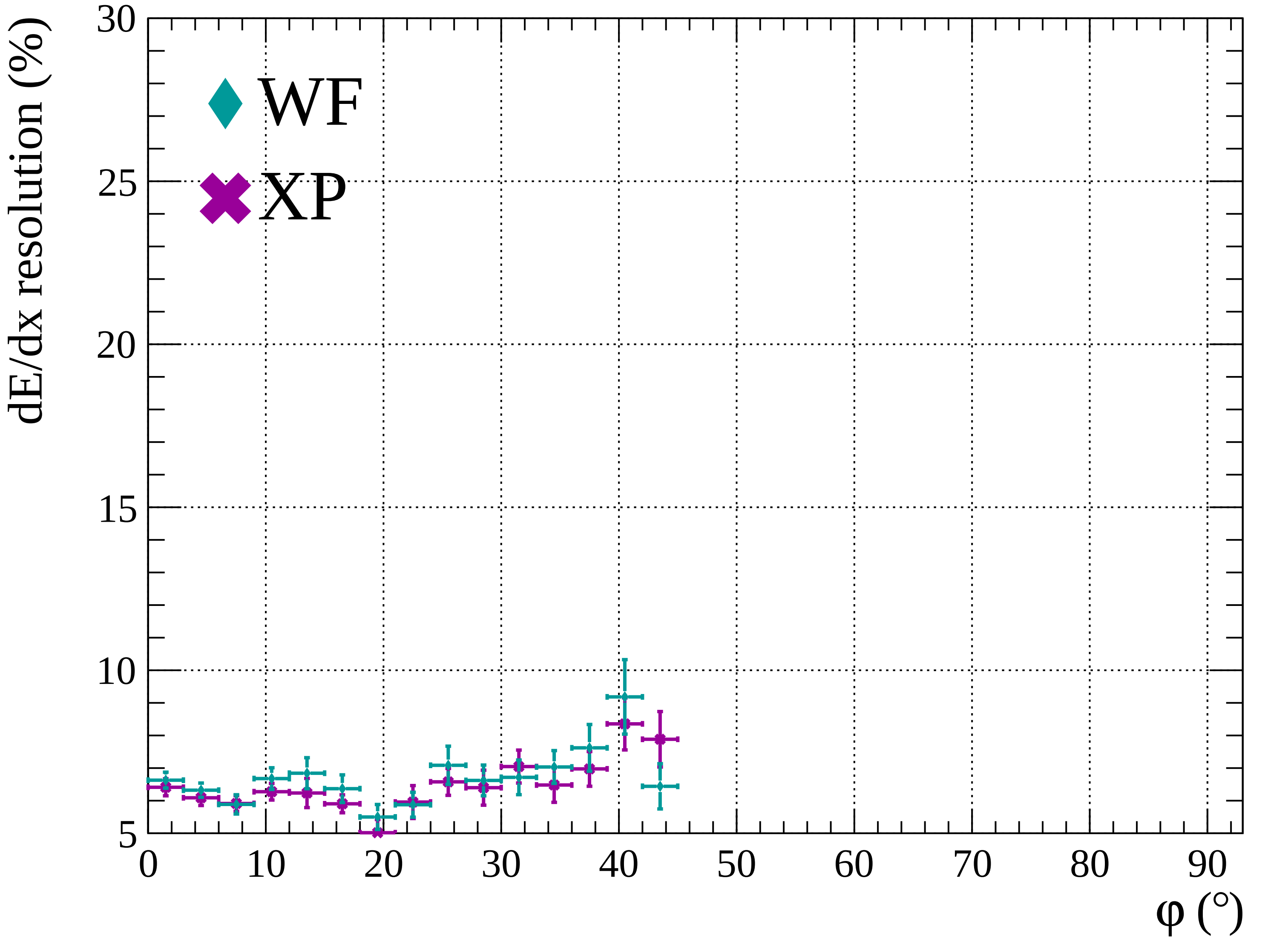
dE/dx with XP (ADC counts/cm)
track length (cm)
0
20
40
60
80
100
120
140
160
0
100
200
300
400
500
600
700
800
900
1000
0
5
10
15
20
25



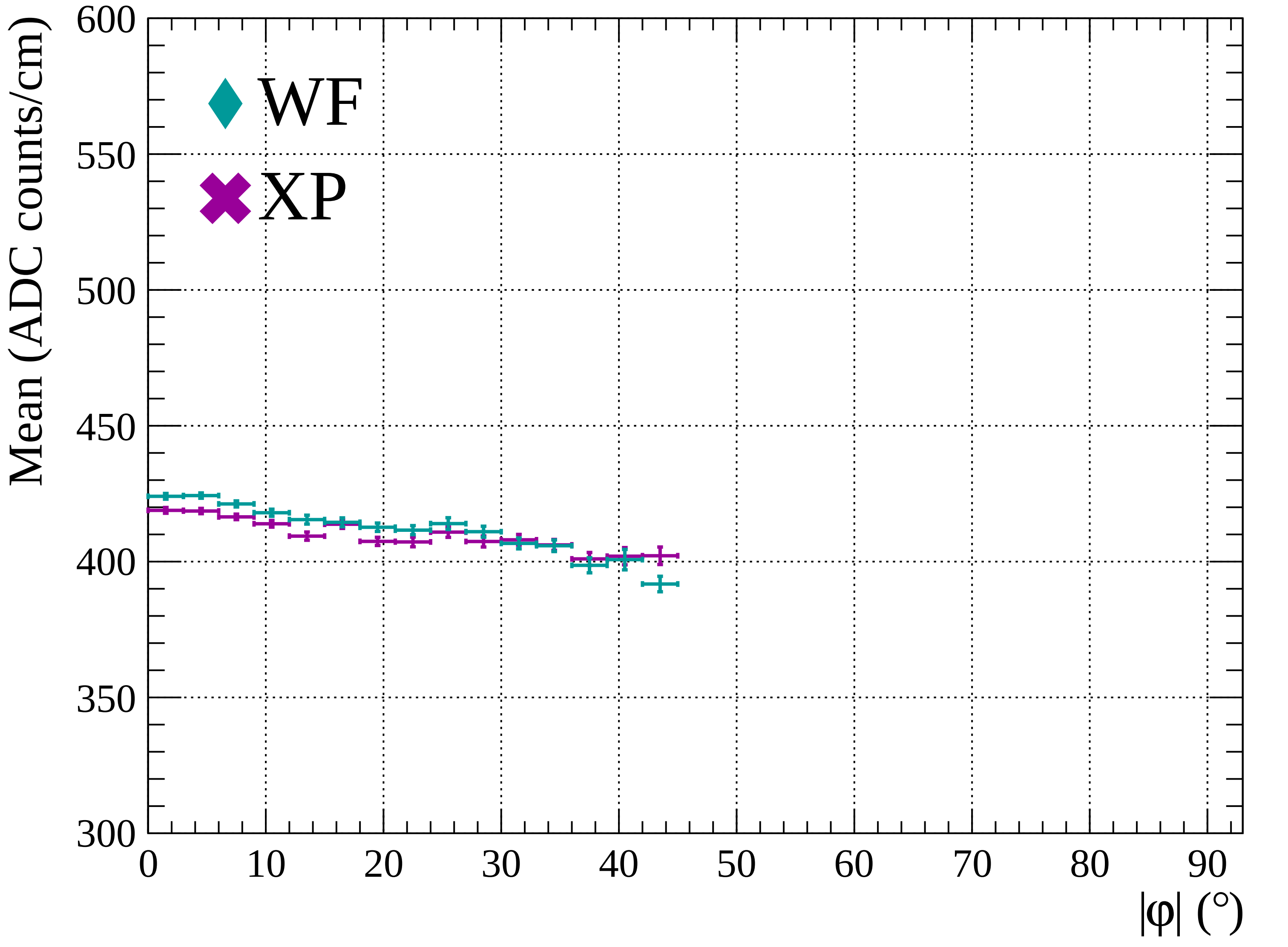
momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
0
20
40
60
80
100
120
140
160
track length (cm)
14
12
10
8
6
4
2
0



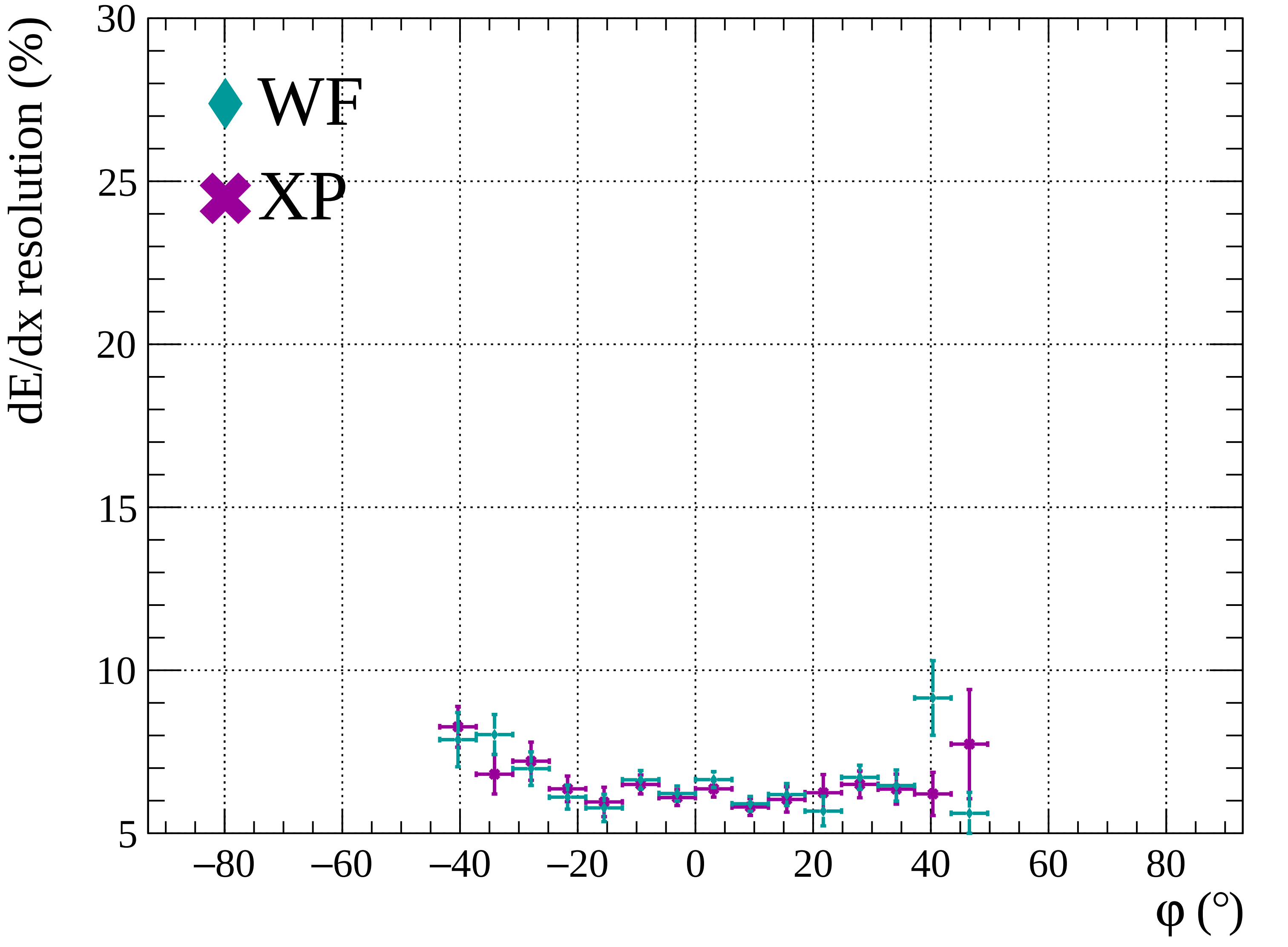
momentum (MeV)
2000
1800
1600
1400
1200
1000
800
600
400
200
0
20
40
60
80
100
120
140
160
180
200
$N_{clusters}$
18
16
14
12
10
8
6
4
2



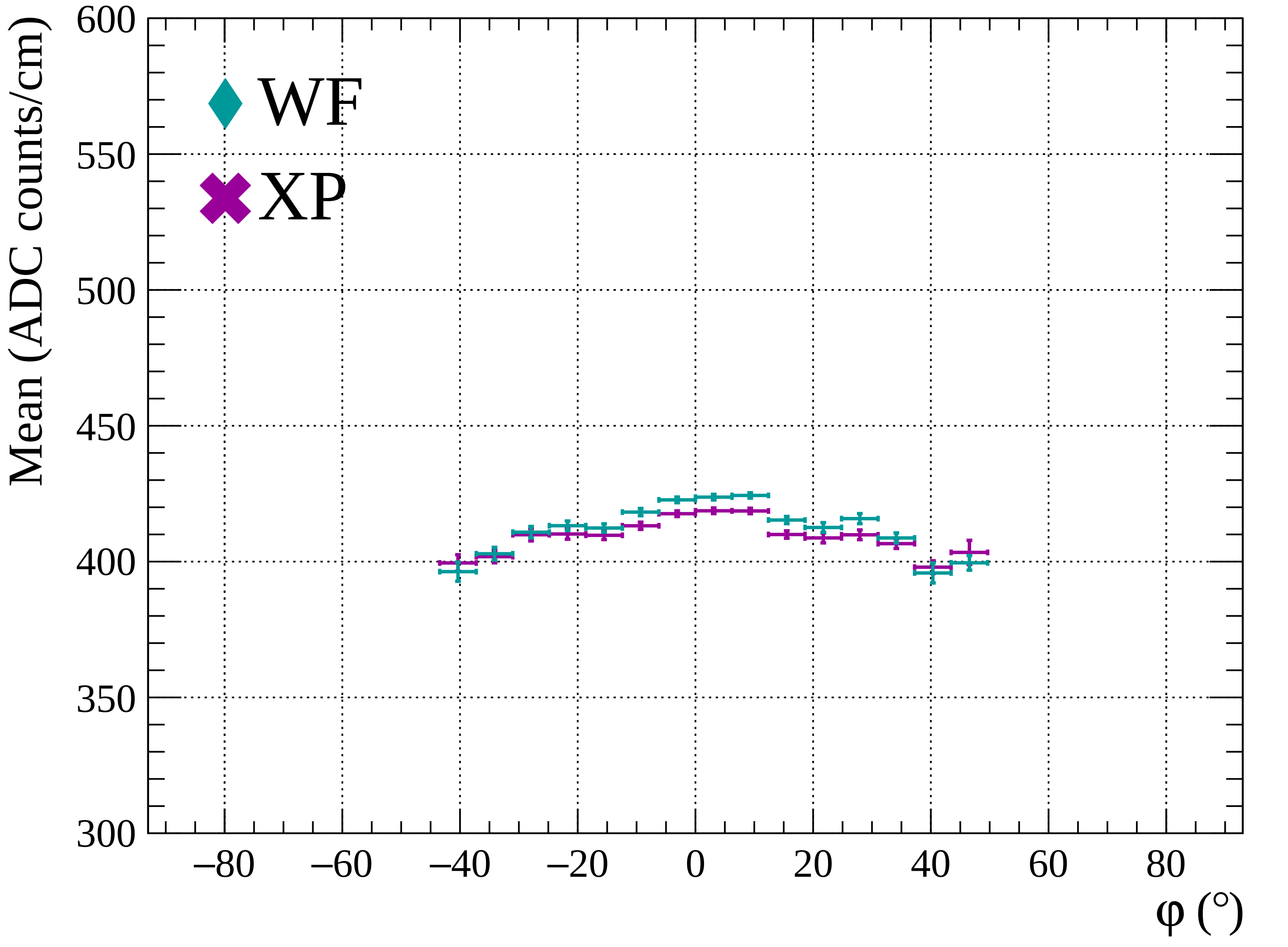
dE/dx resolution (%)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
φ (°)
WF
XP



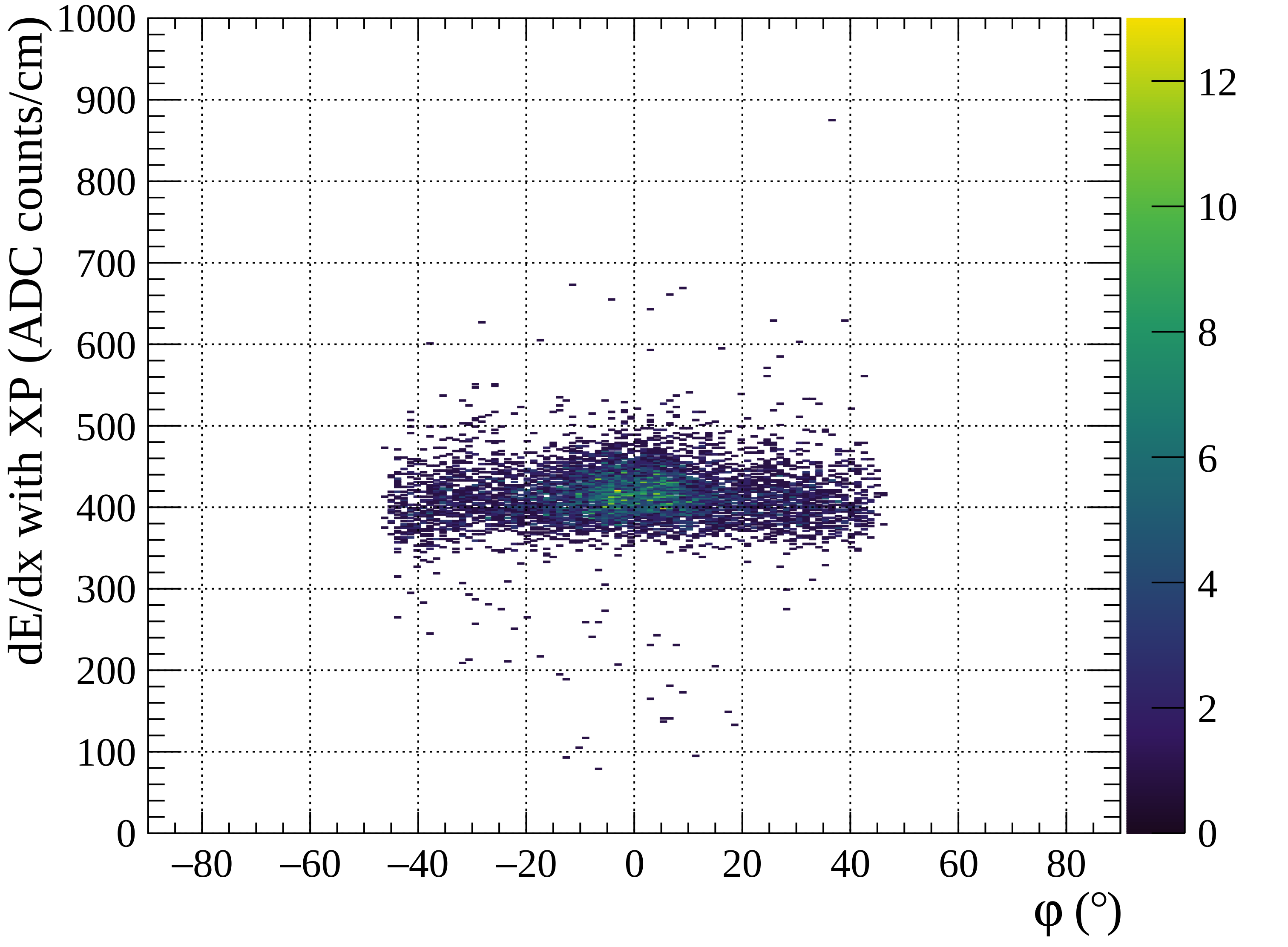
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
|φ| (°)
WF
XP



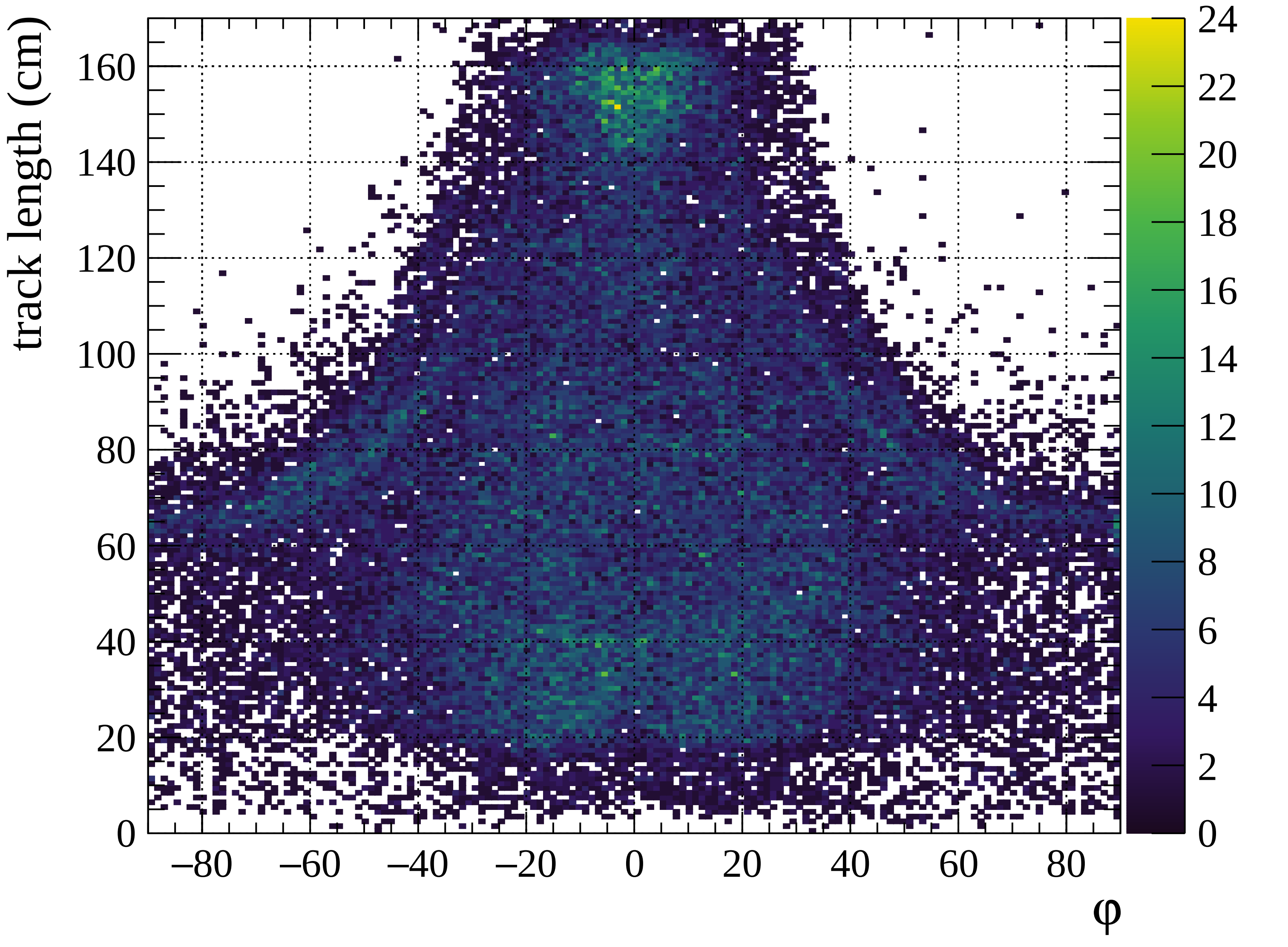
WF
XP
dE/dx resolution (%)
φ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80

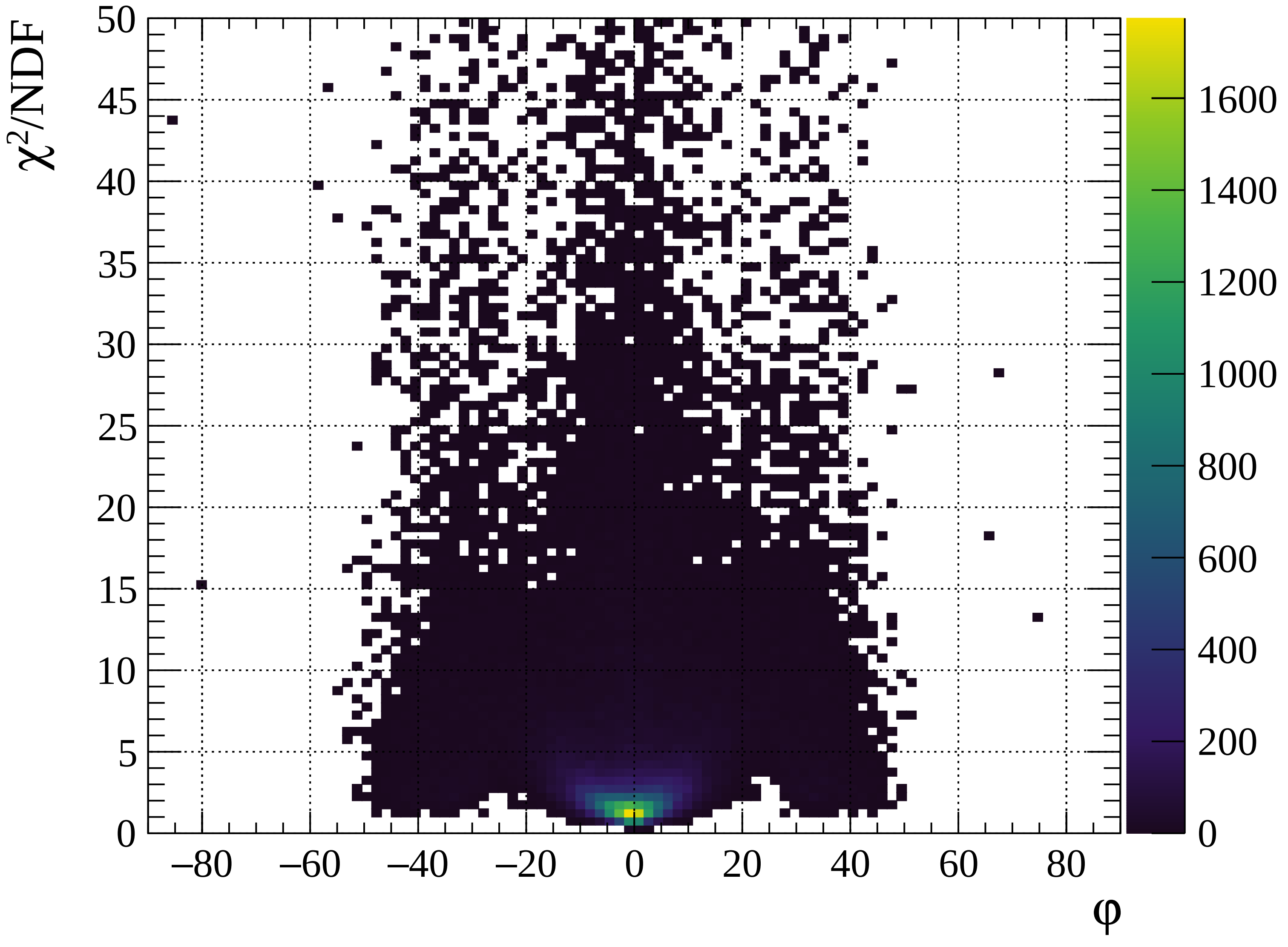


Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)

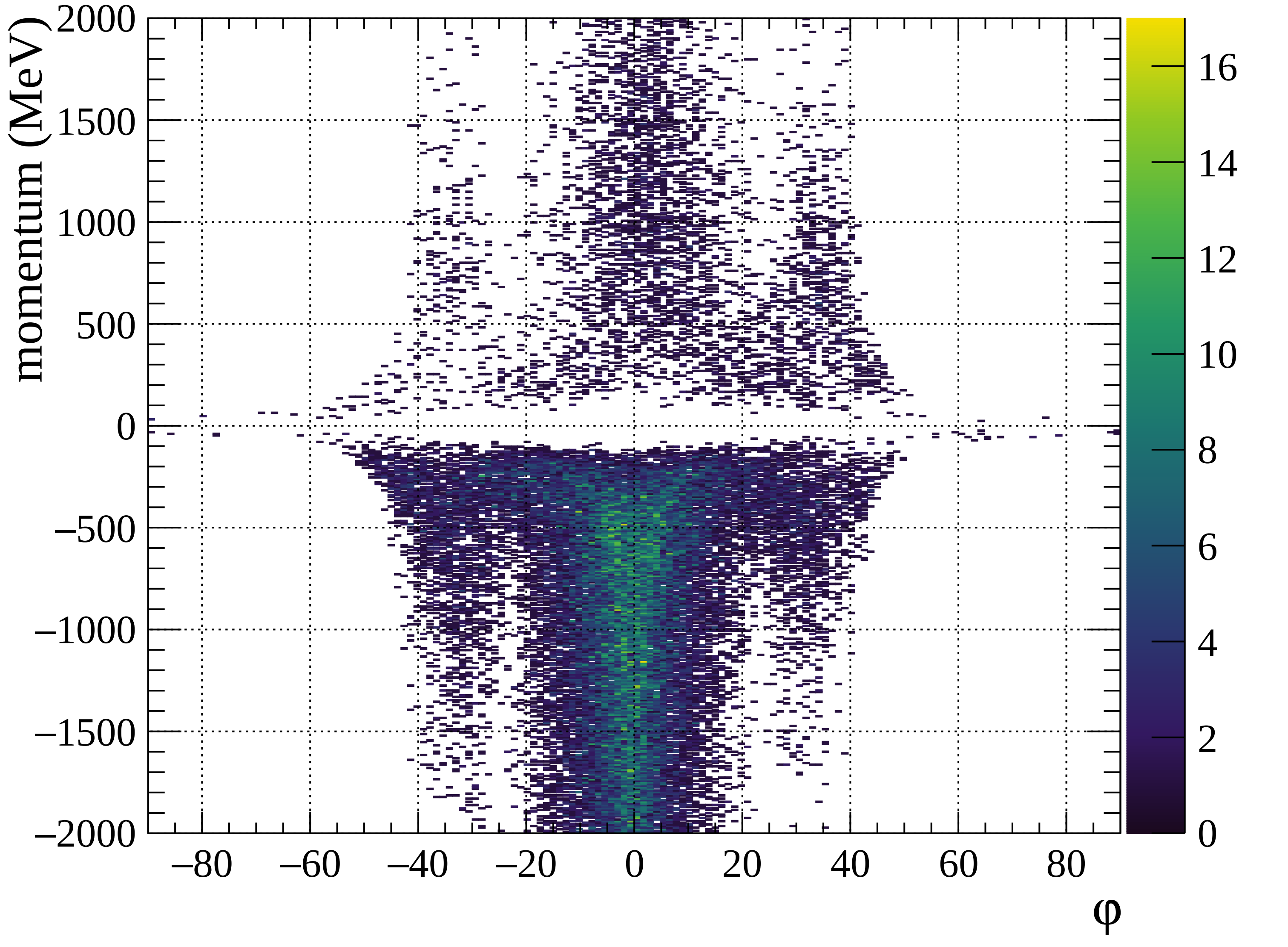


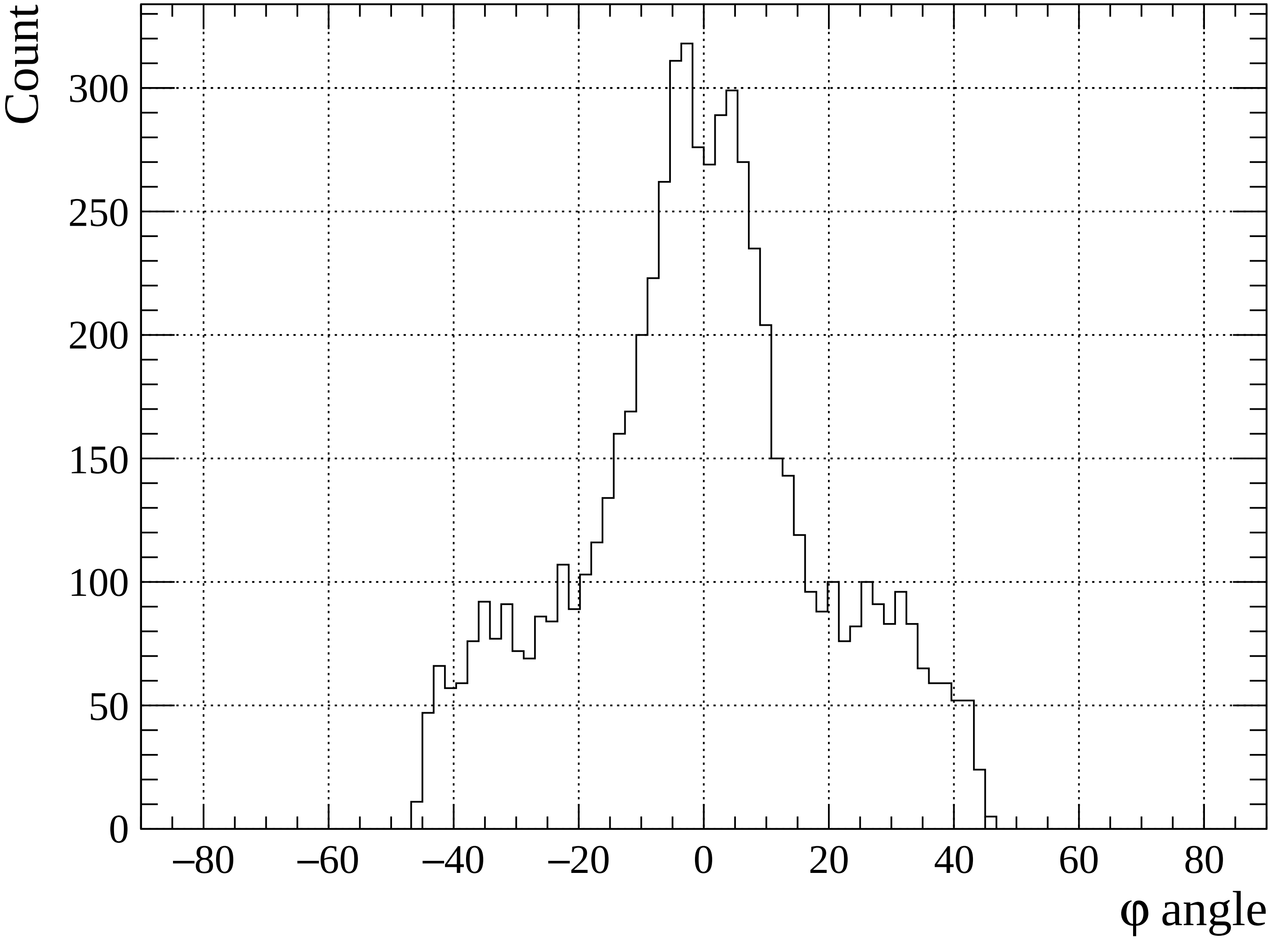
dE/dx with XP (ADC counts/cm)
φ (°)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
12
10
8
6
4
2
0

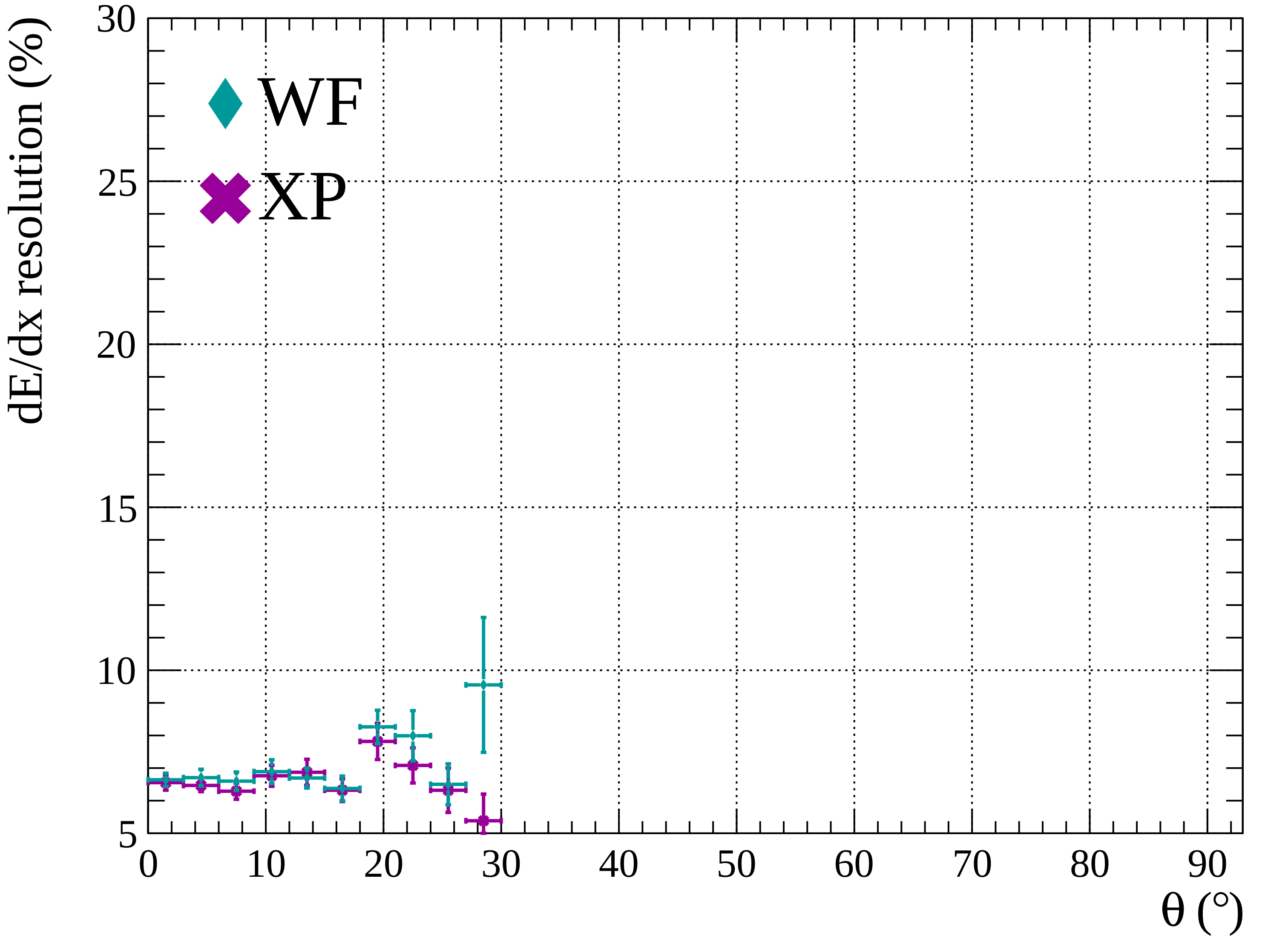

track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
φ
0
2
4
6
8
10
12
14
16
18
20
22
24



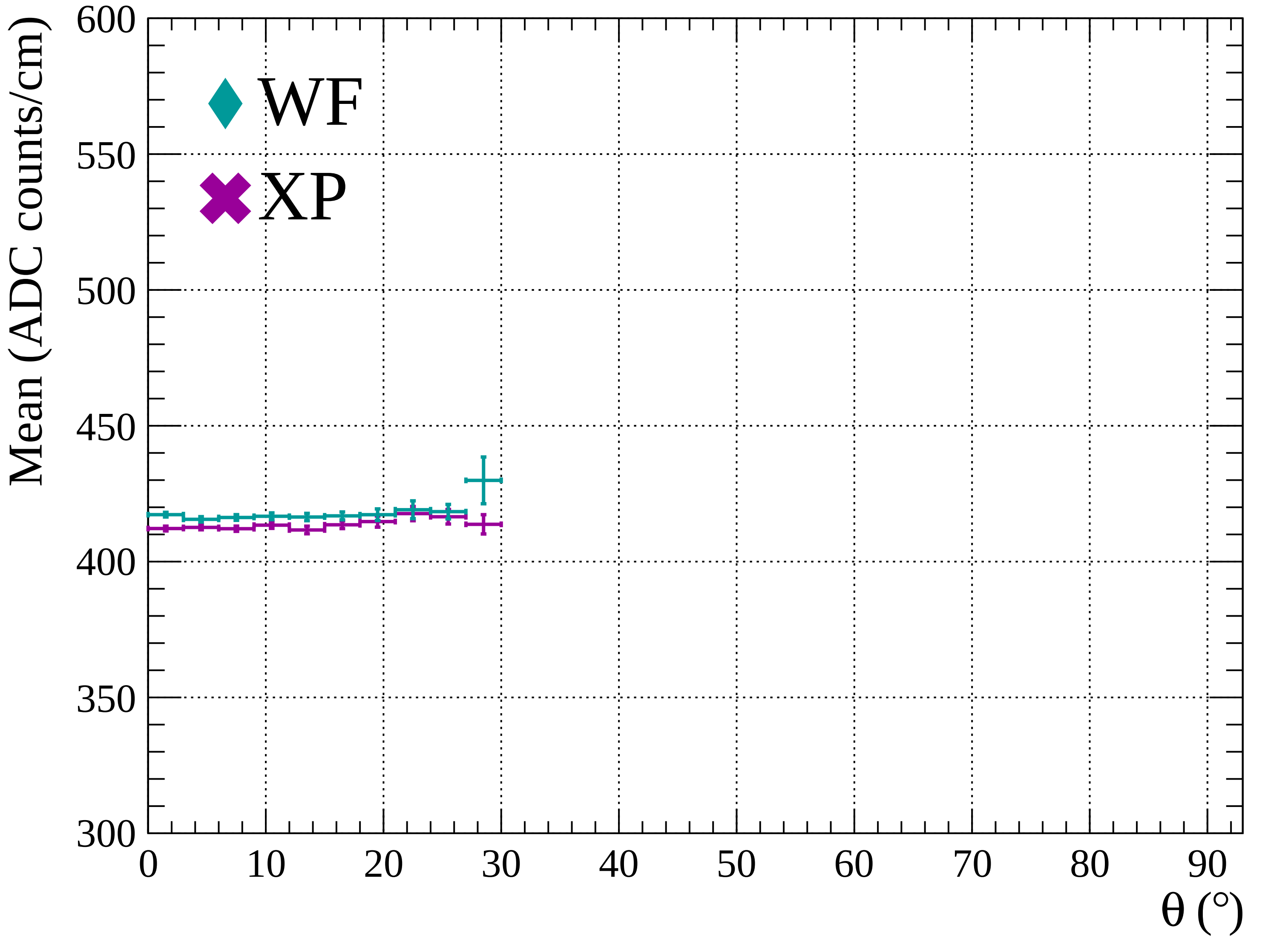
χ^2/NDF
φ
0
5
10
15
20
25
30
35
40
45
50
−80
−60
−40
−20
0
20
40
60
80
0
200
400
600
800
1000
1200
1400
1600

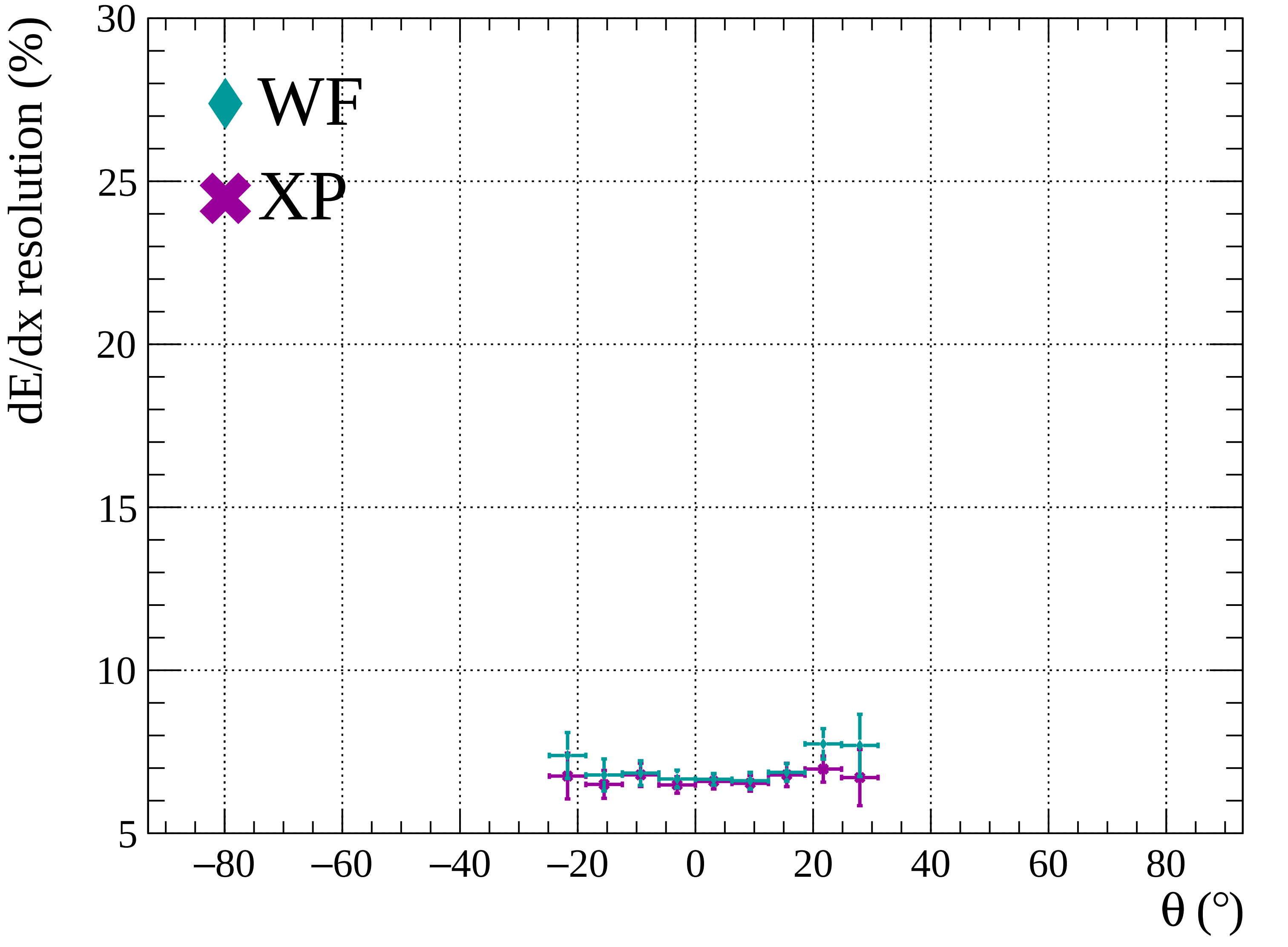

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
16
14
12
10
8
6
4
2
0


Count
300
250
200
150
100
50
0
−80
−60
−40
−20
0
20
40
60
80
φ angle

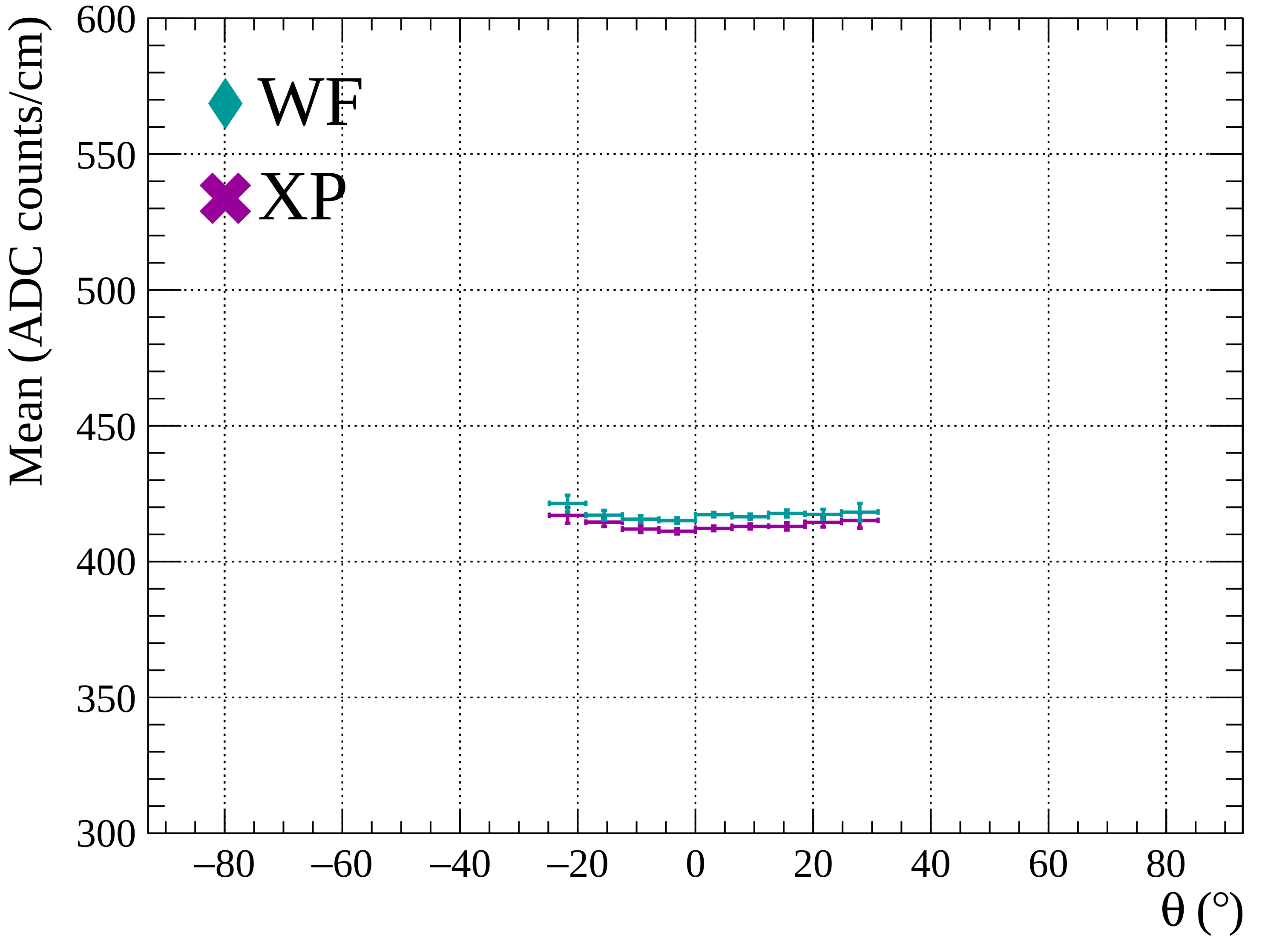


dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
0
10
20
30
40
50
60
70
80
90
θ (°)

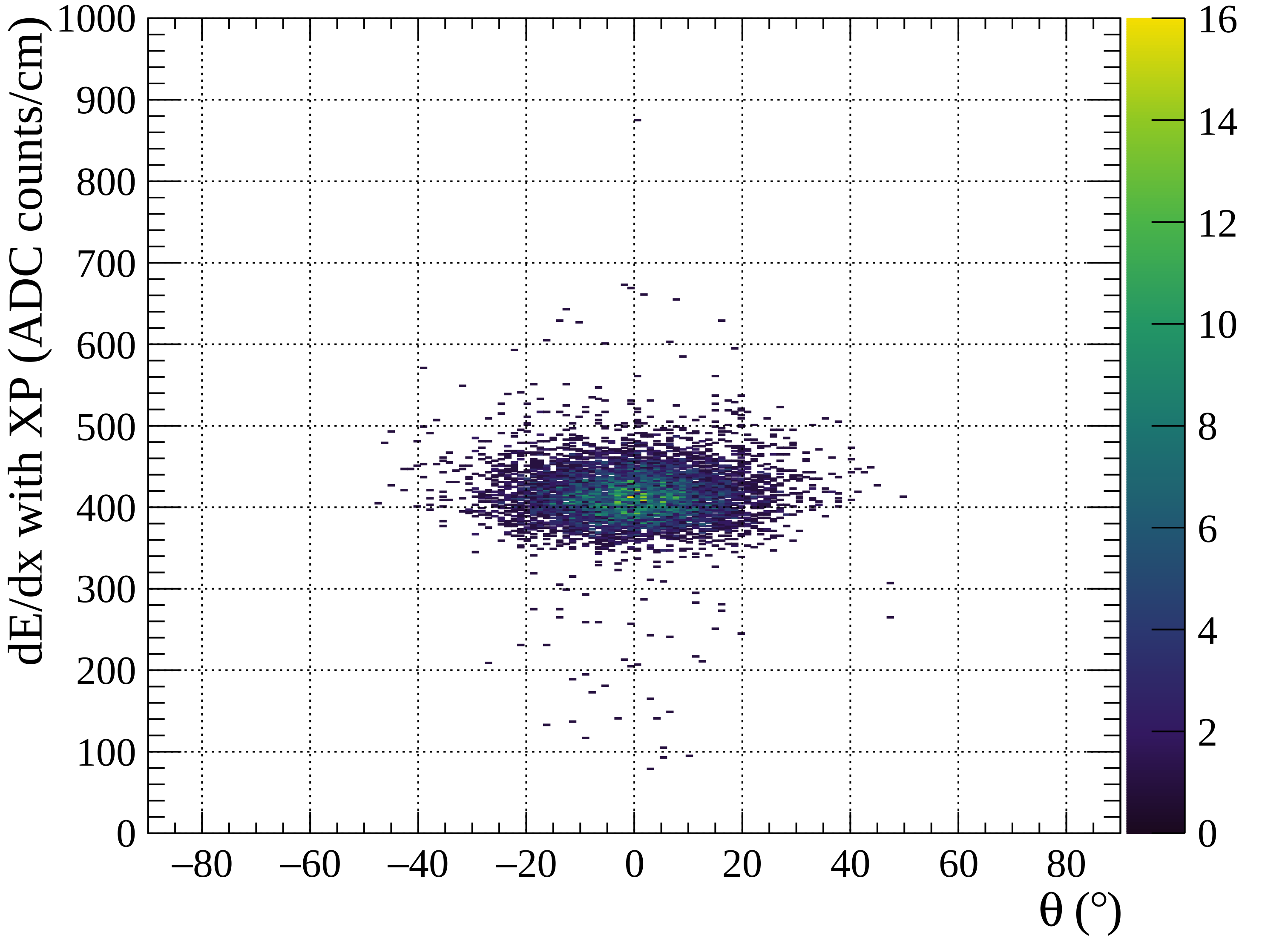

WF
XP
Mean (ADC counts/cm)
θ (°)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90



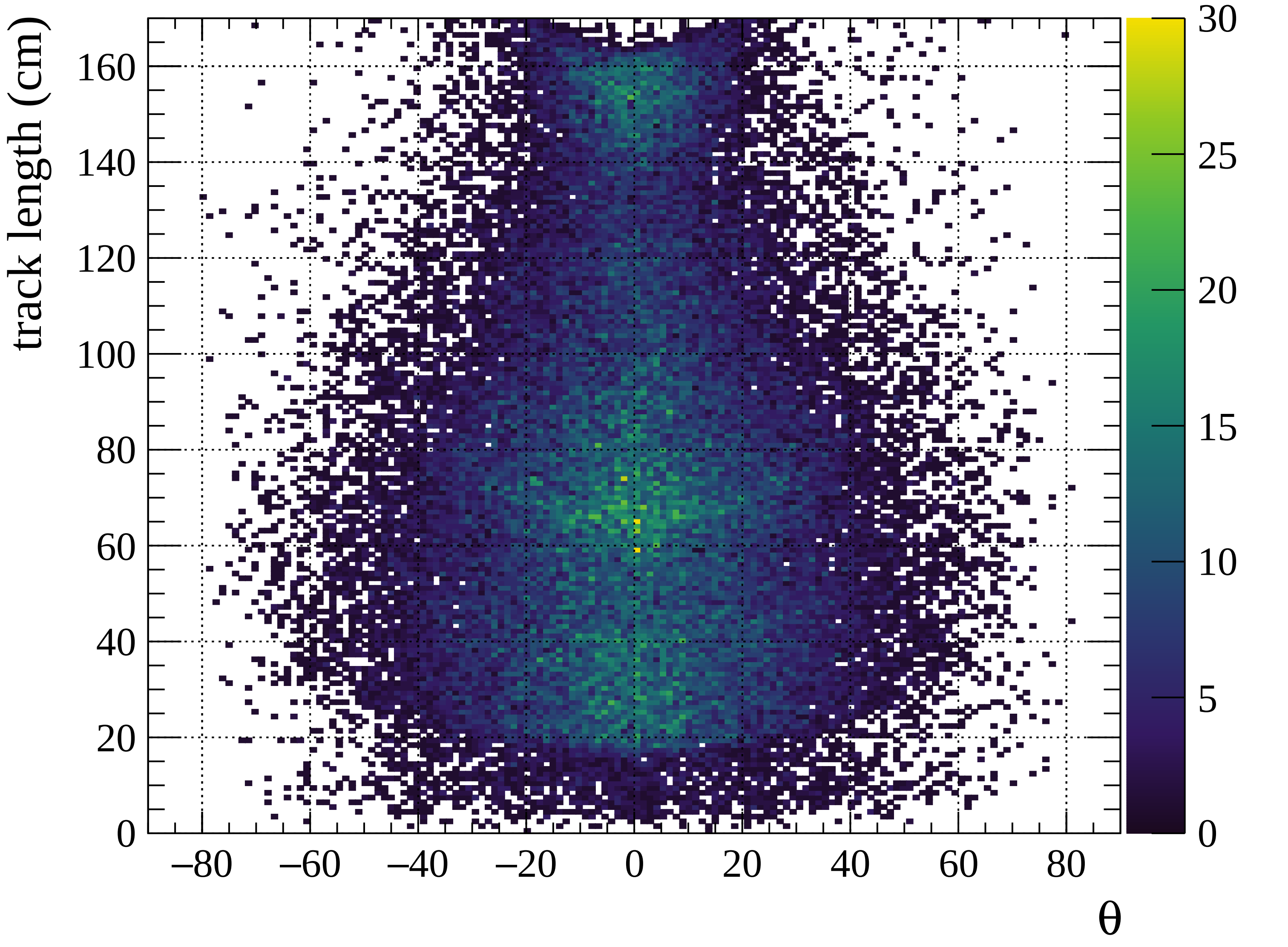
WF
XP
dE/dx resolution (%)
θ (°)
30
25
20
15
10
5
−80
−60
−40
−20
0
20
40
60
80



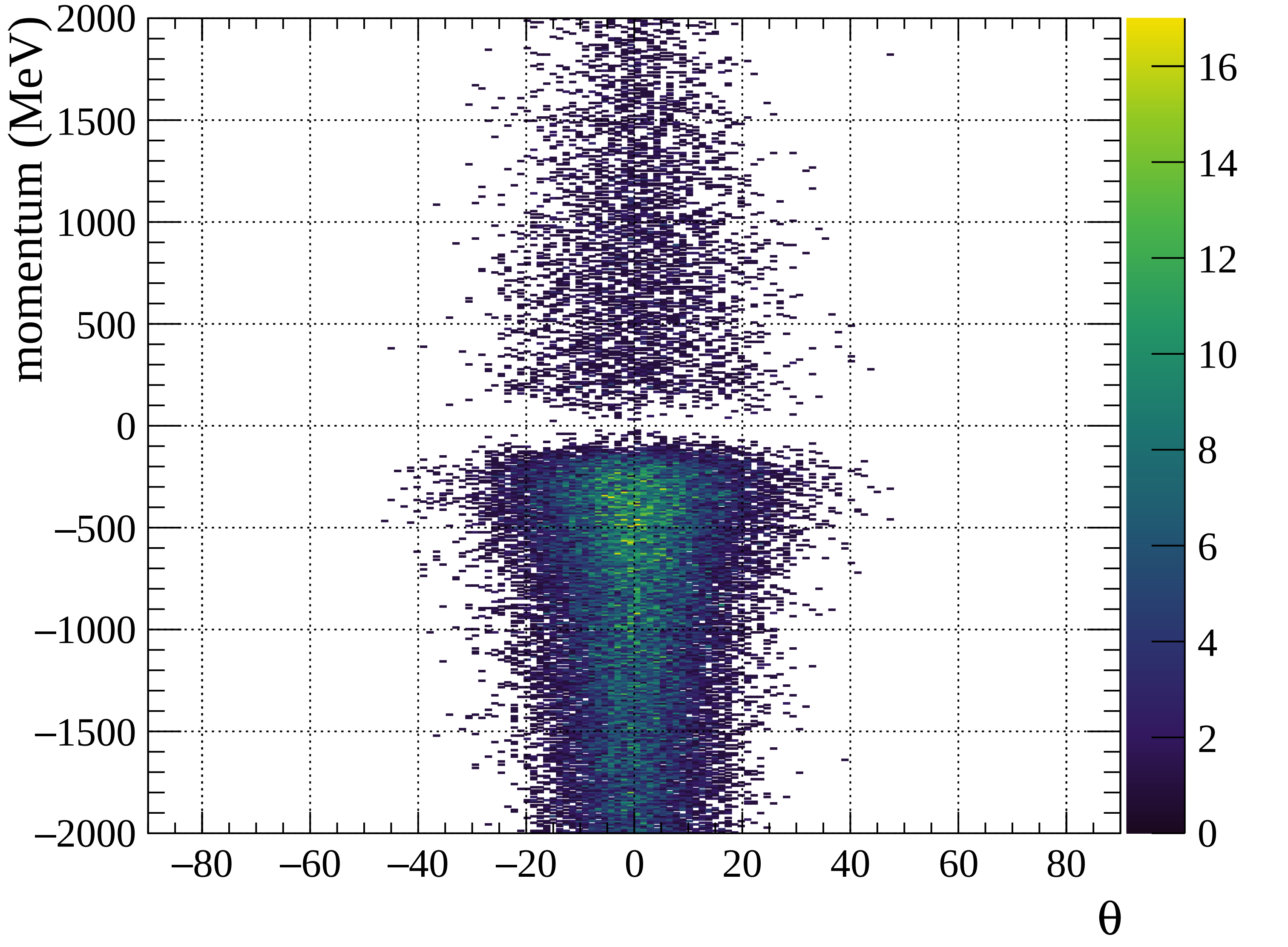
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



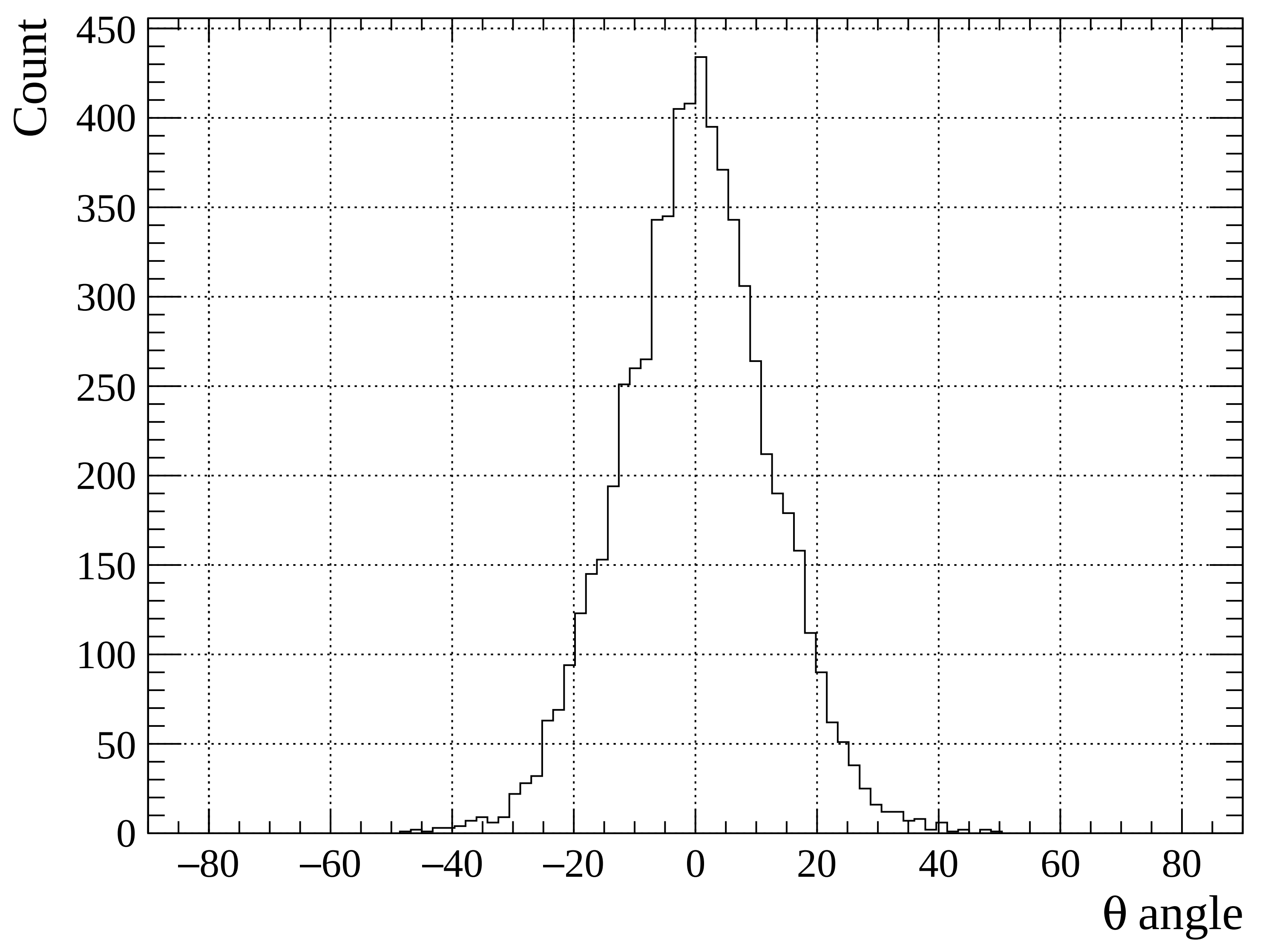
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
16
14
12
10
8
6
4
2
0



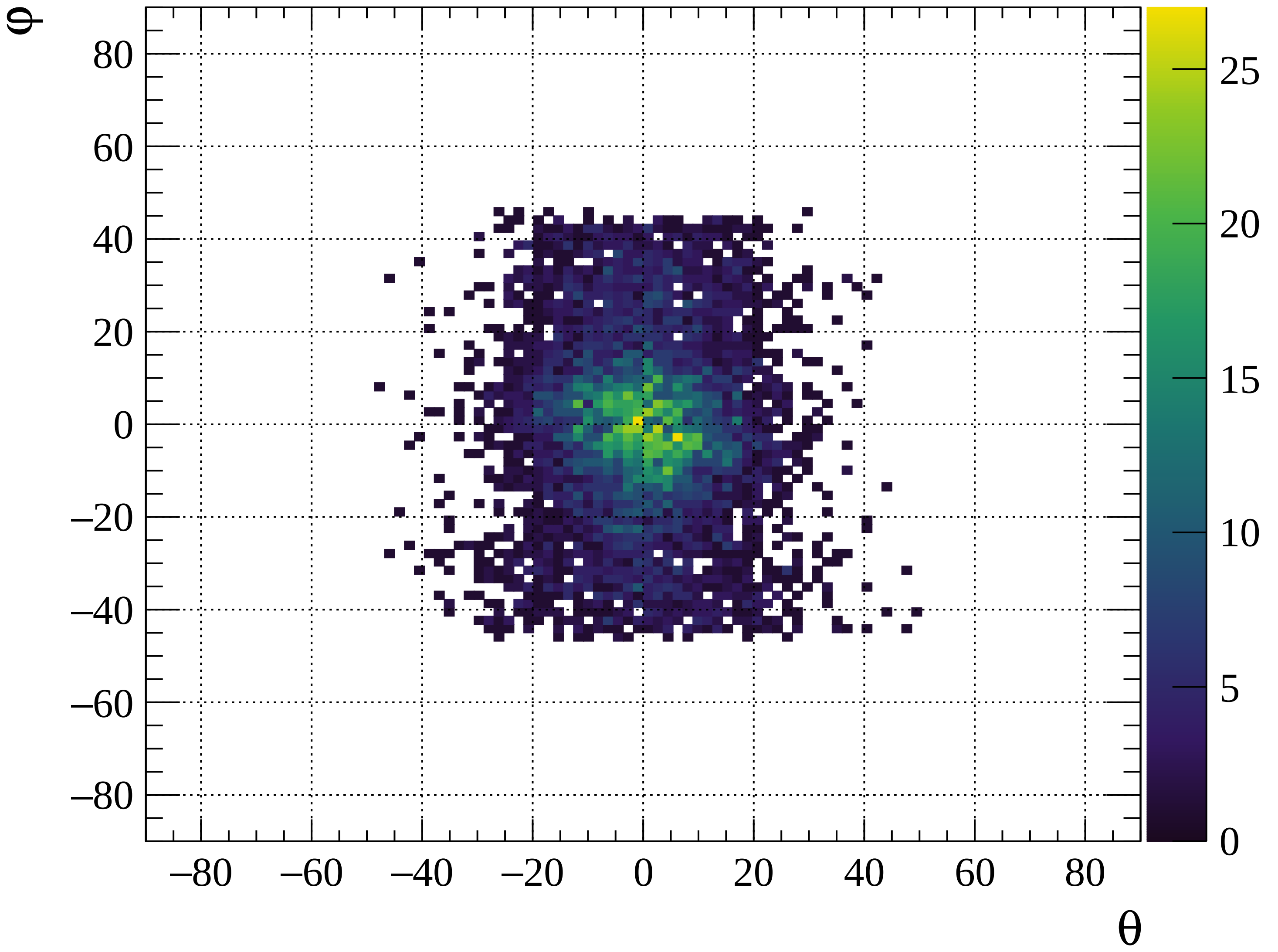
track length (cm)
0
20
40
60
80
100
120
140
160
−80
−60
−40
−20
0
20
40
60
80
θ
0
5
10
15
20
25
30

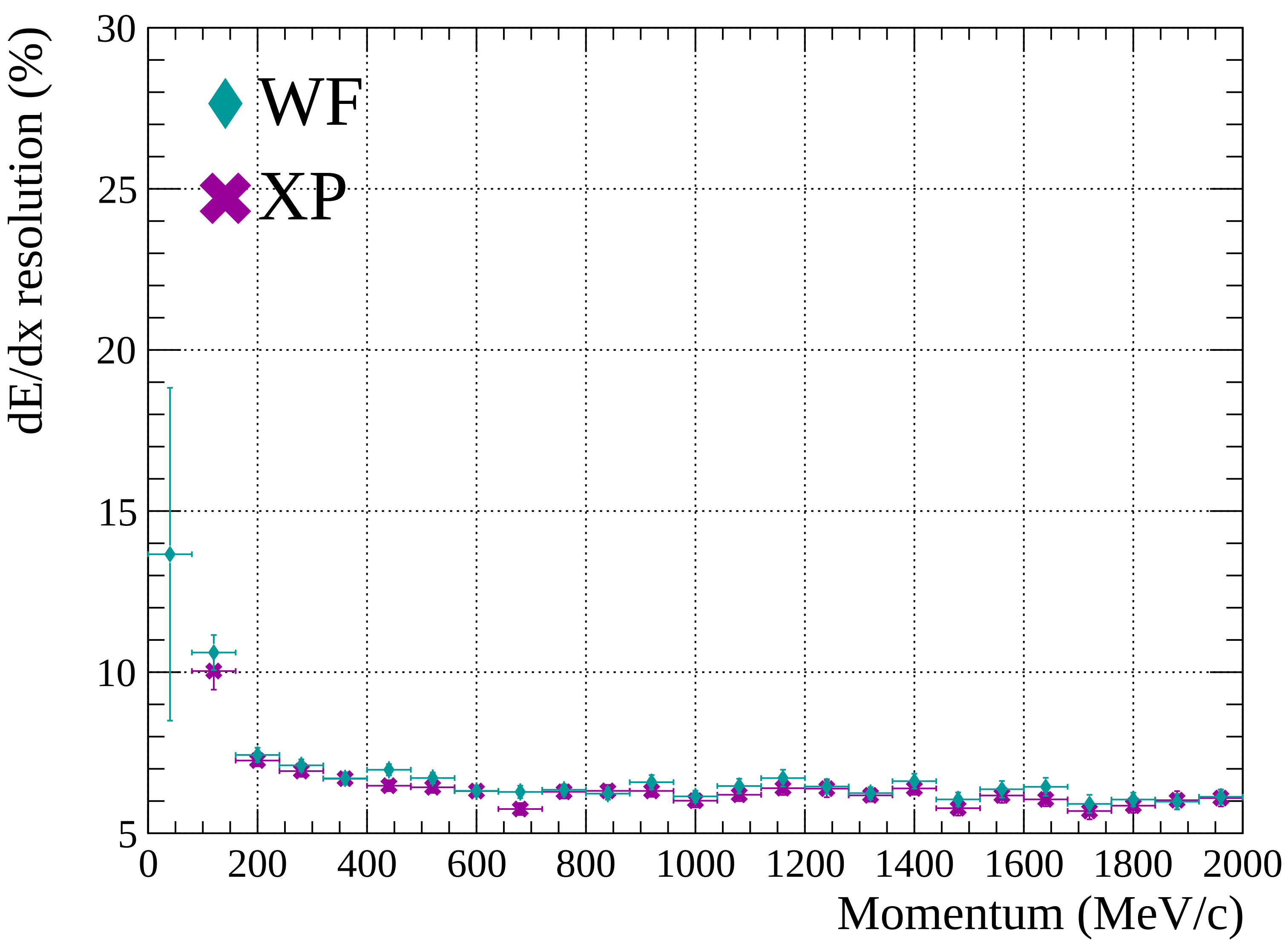


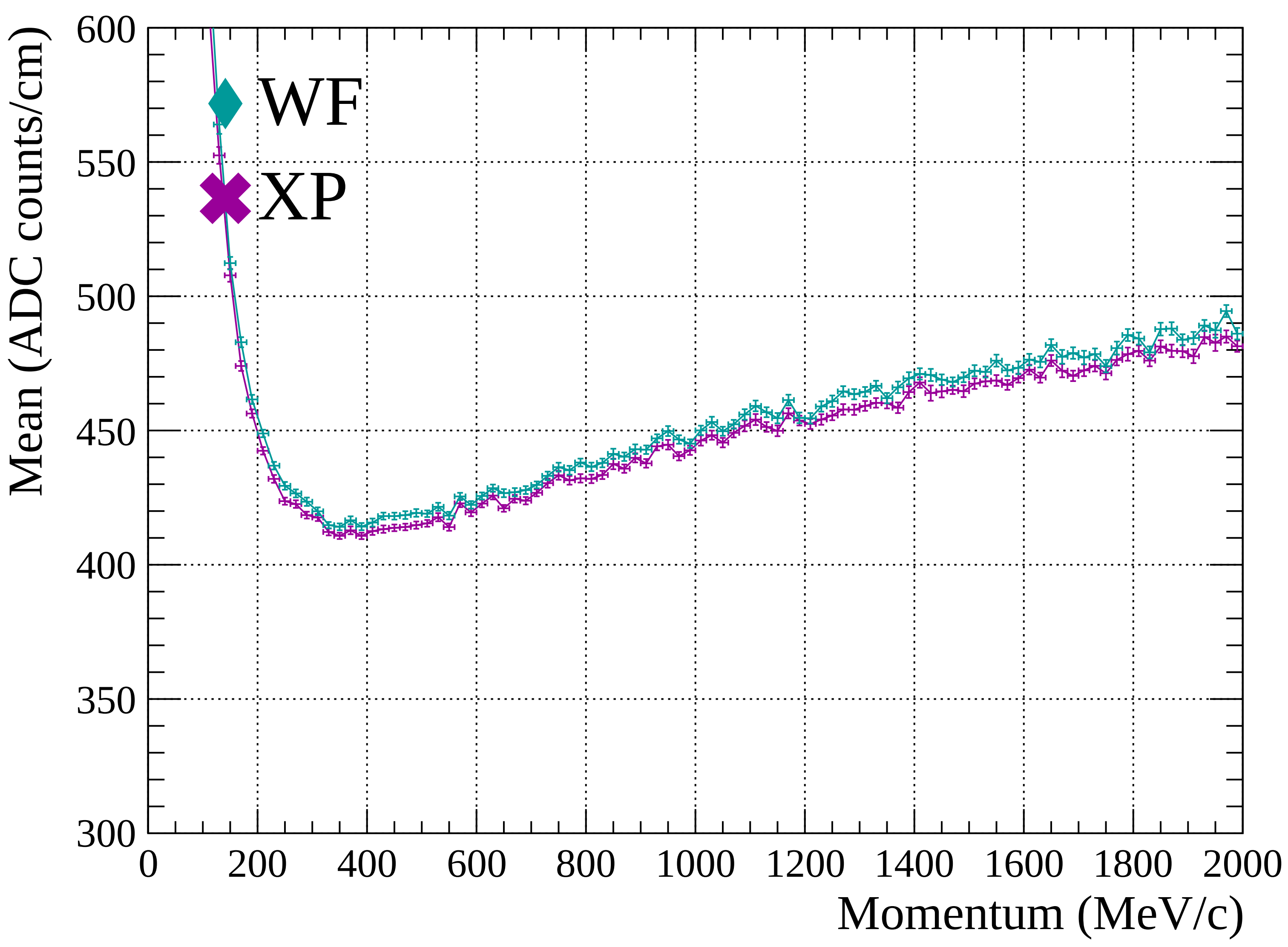
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
16
14
12
10
8
6
4
2
0



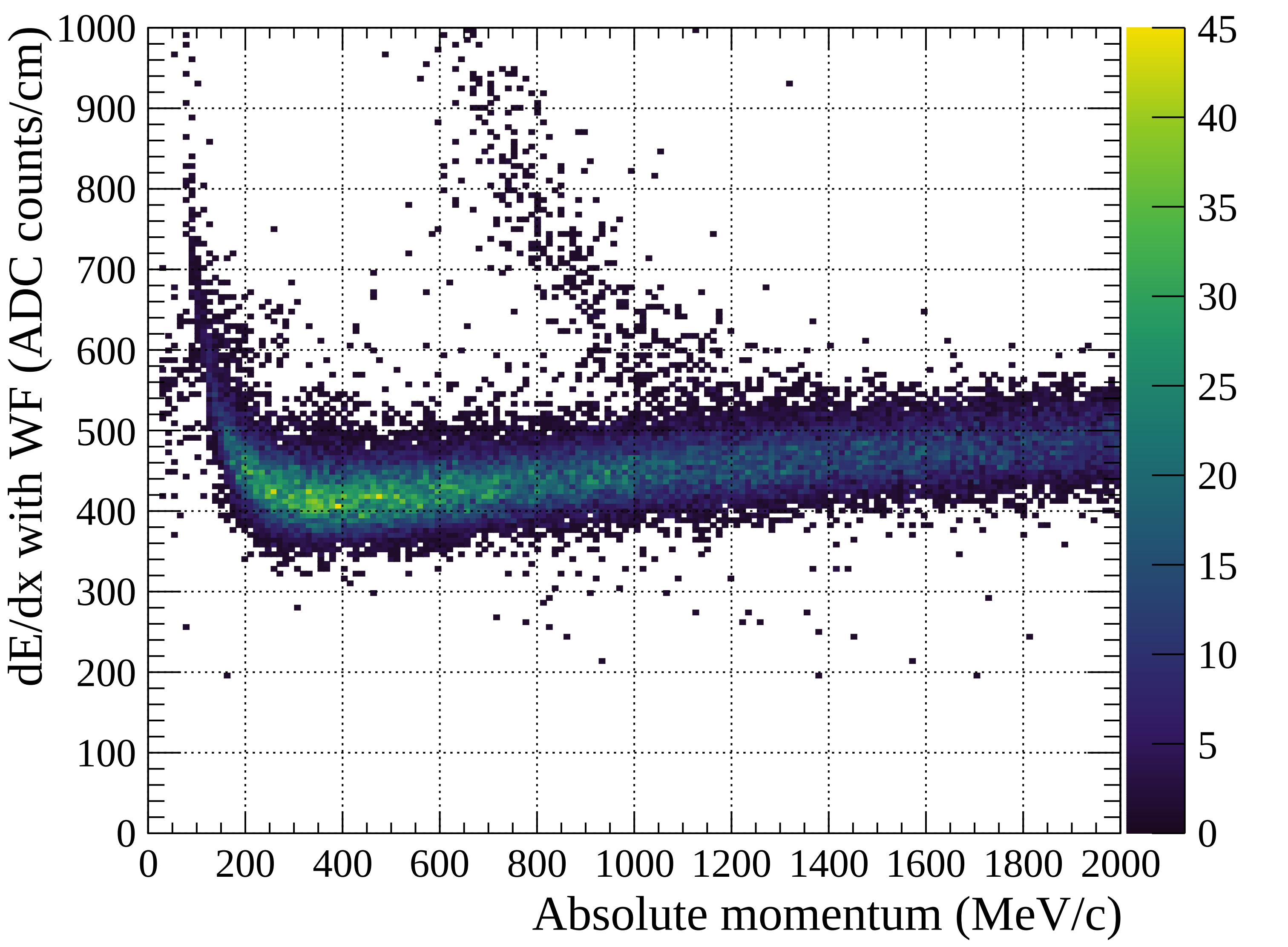
Count
450
400
350
300
250
200
150
100
50
0
−80
−60
−40
−20
0
20
40
60
80
θ angle


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
25
20
15
10
5
0

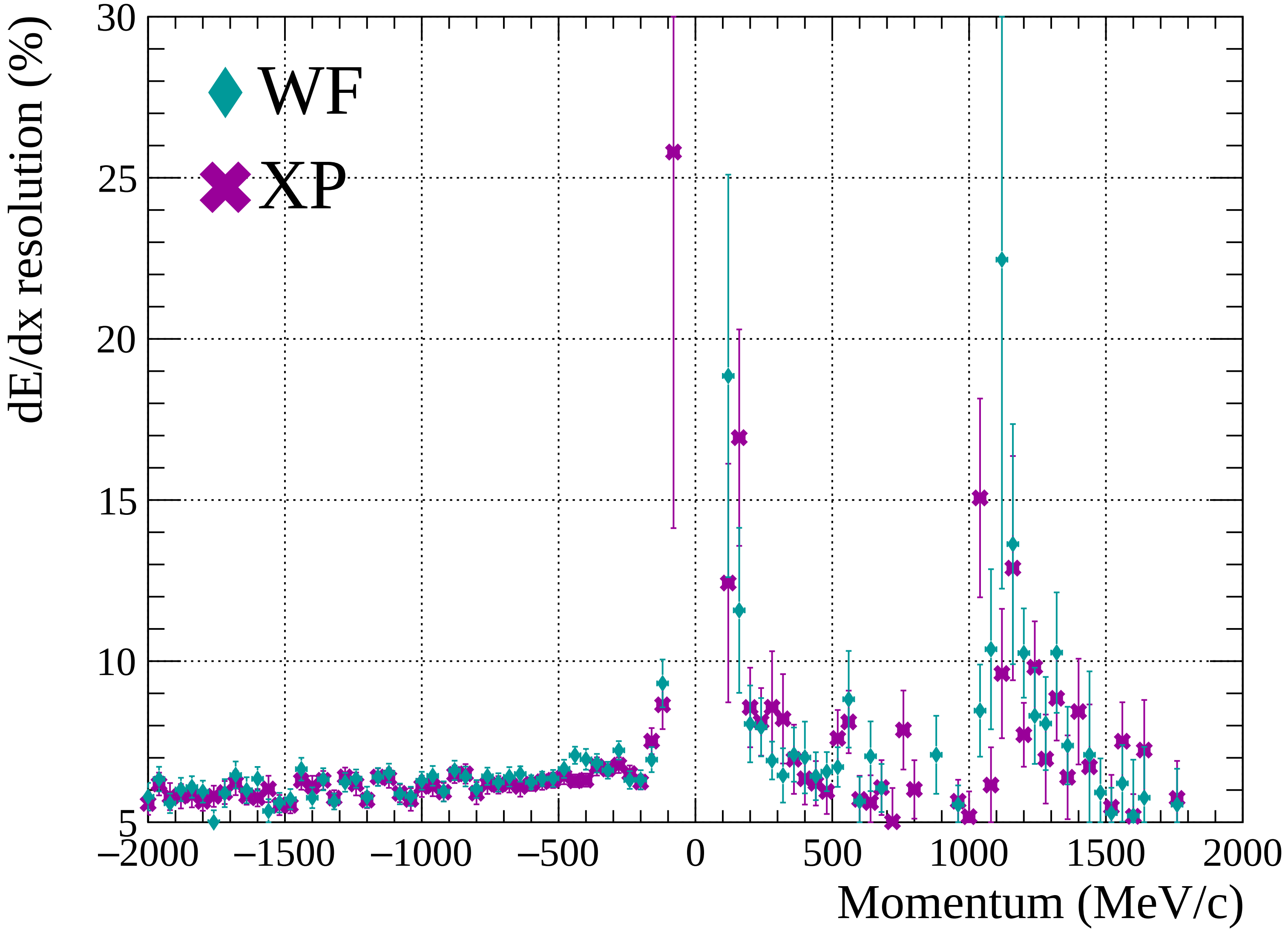

dE/dx resolution (%)
WF
XP
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000
Momentum (MeV/c)



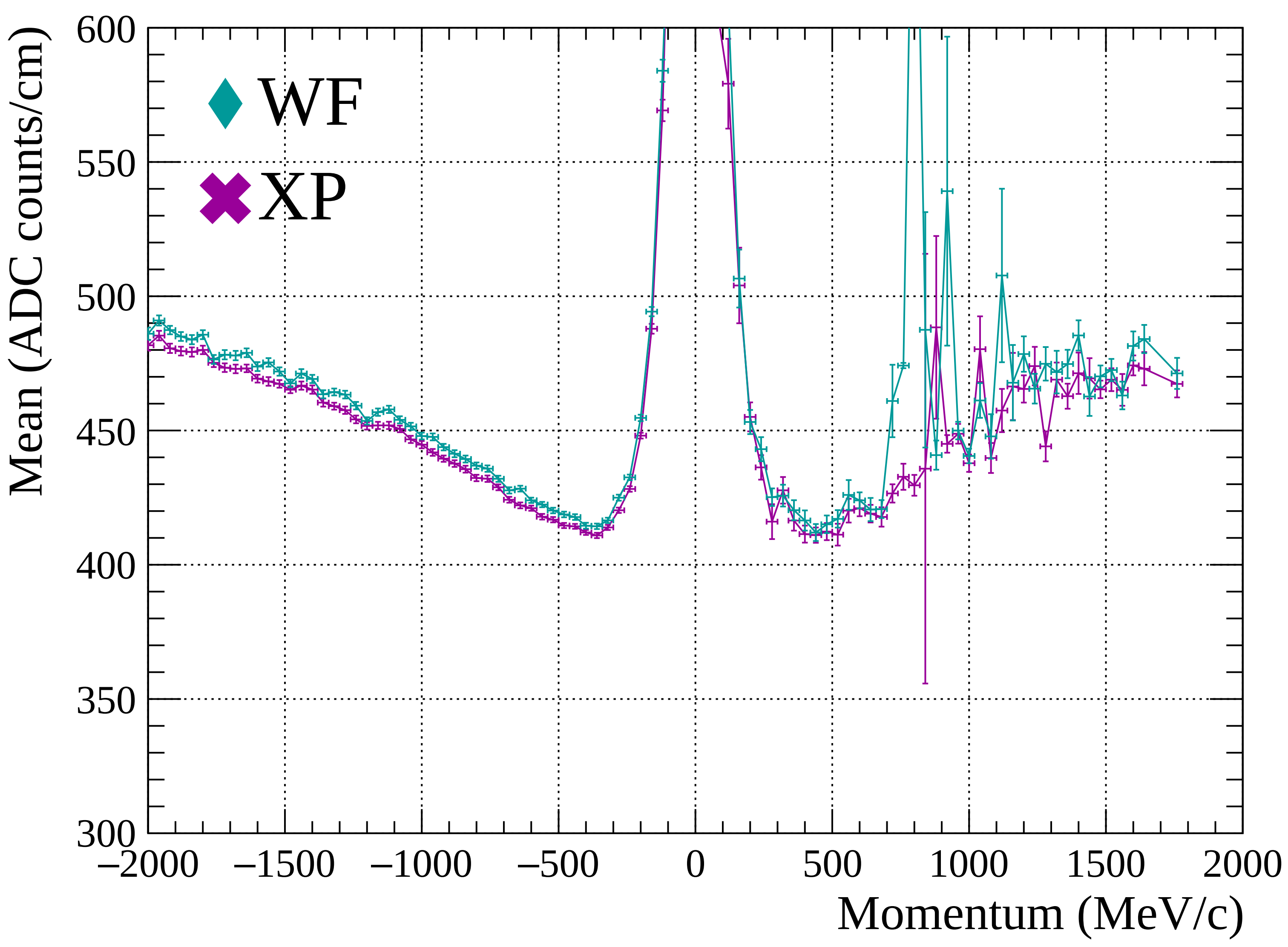
WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000



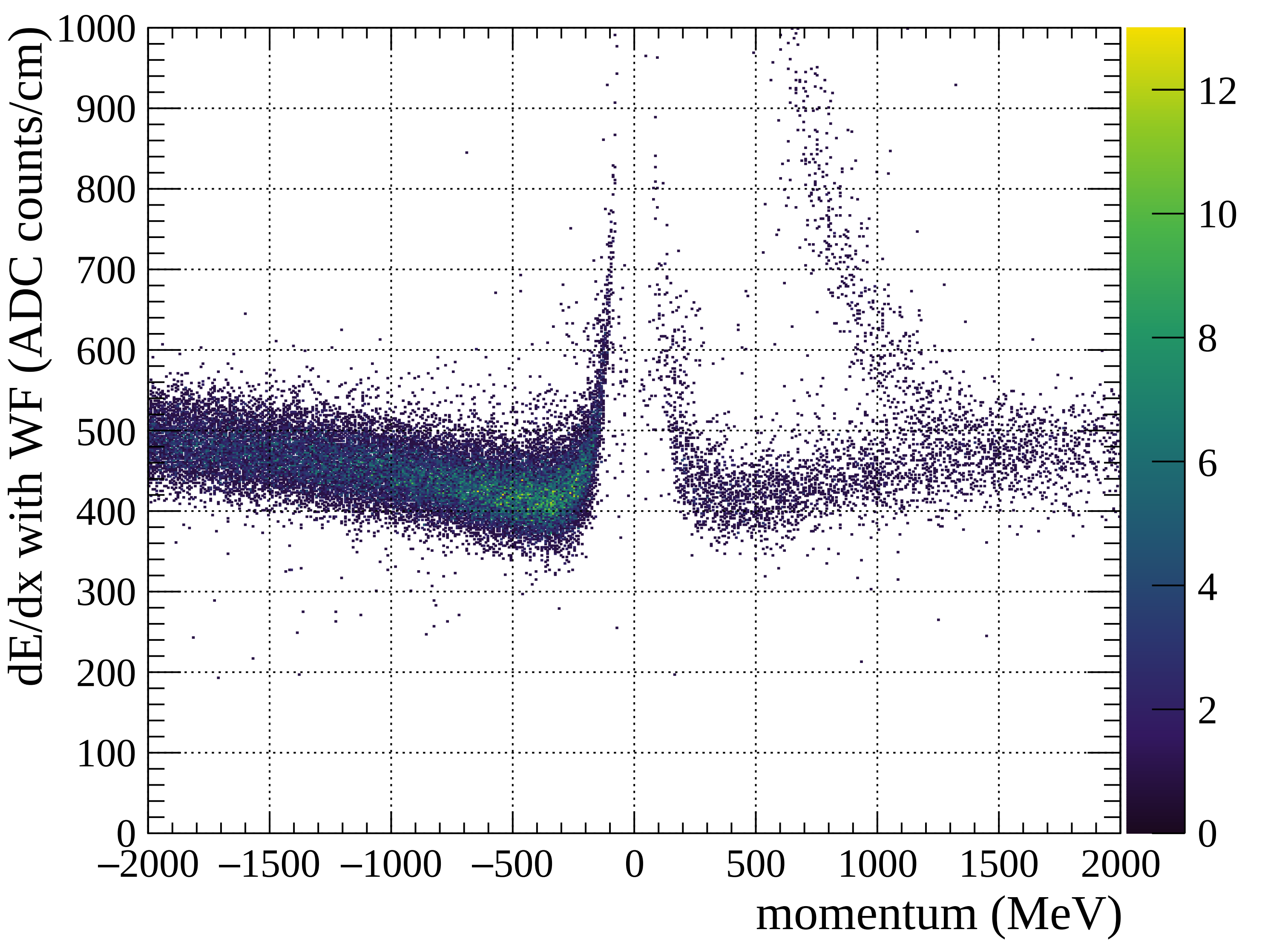
dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
5
10
15
20
25
30
35
40
45



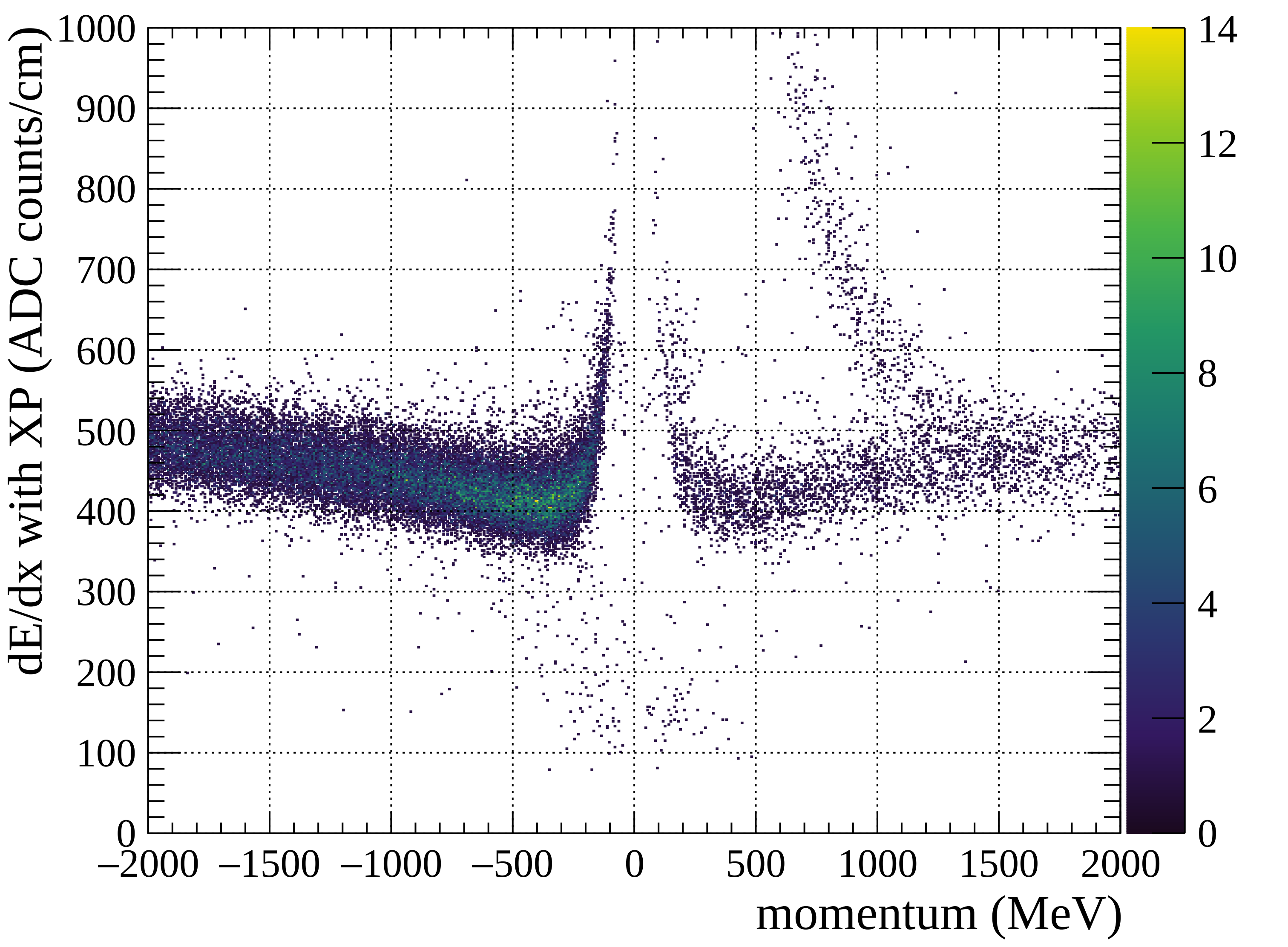
dE/dx resolution (%)
Momentum (MeV/c)
WF
XP
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

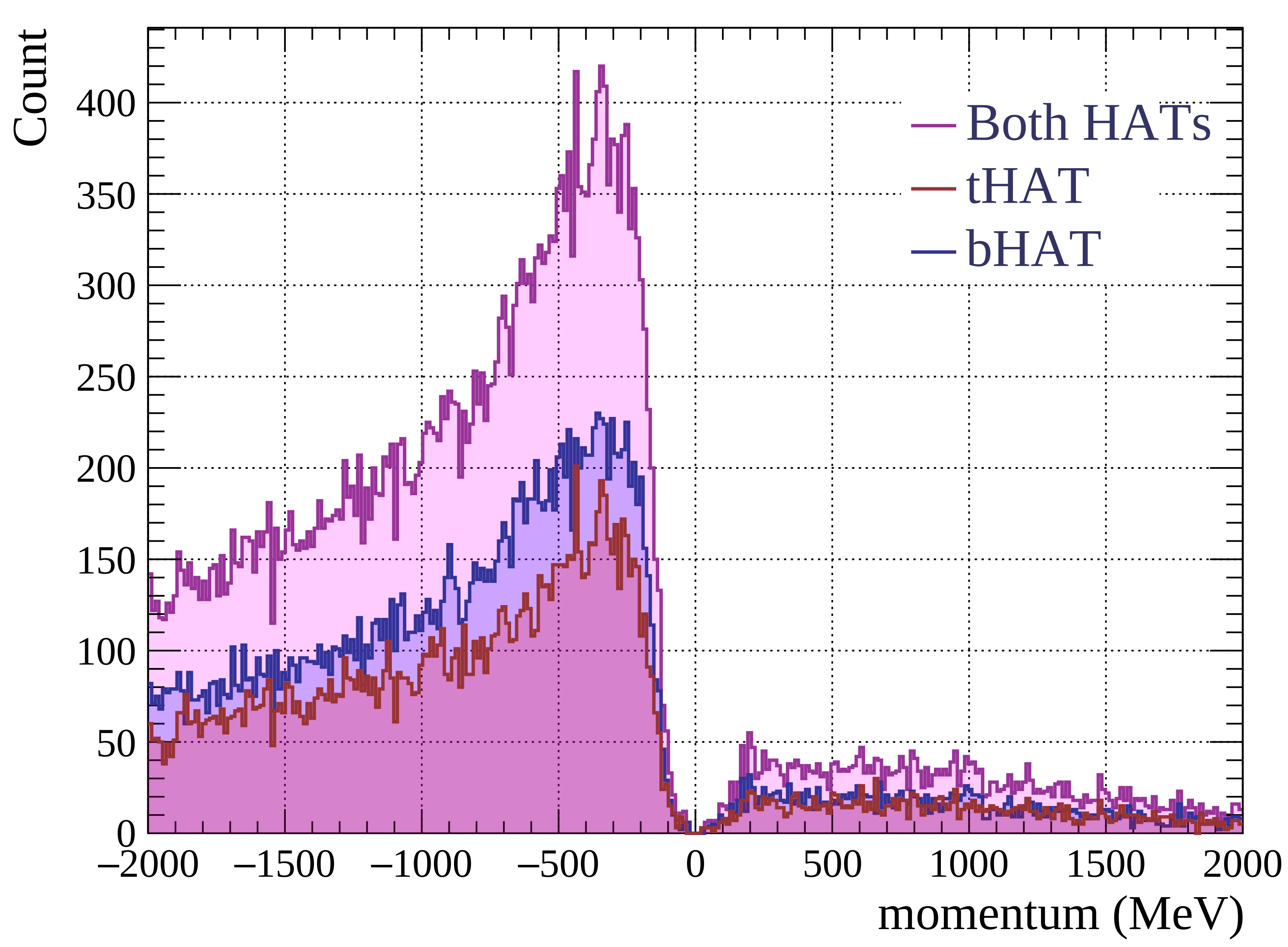


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

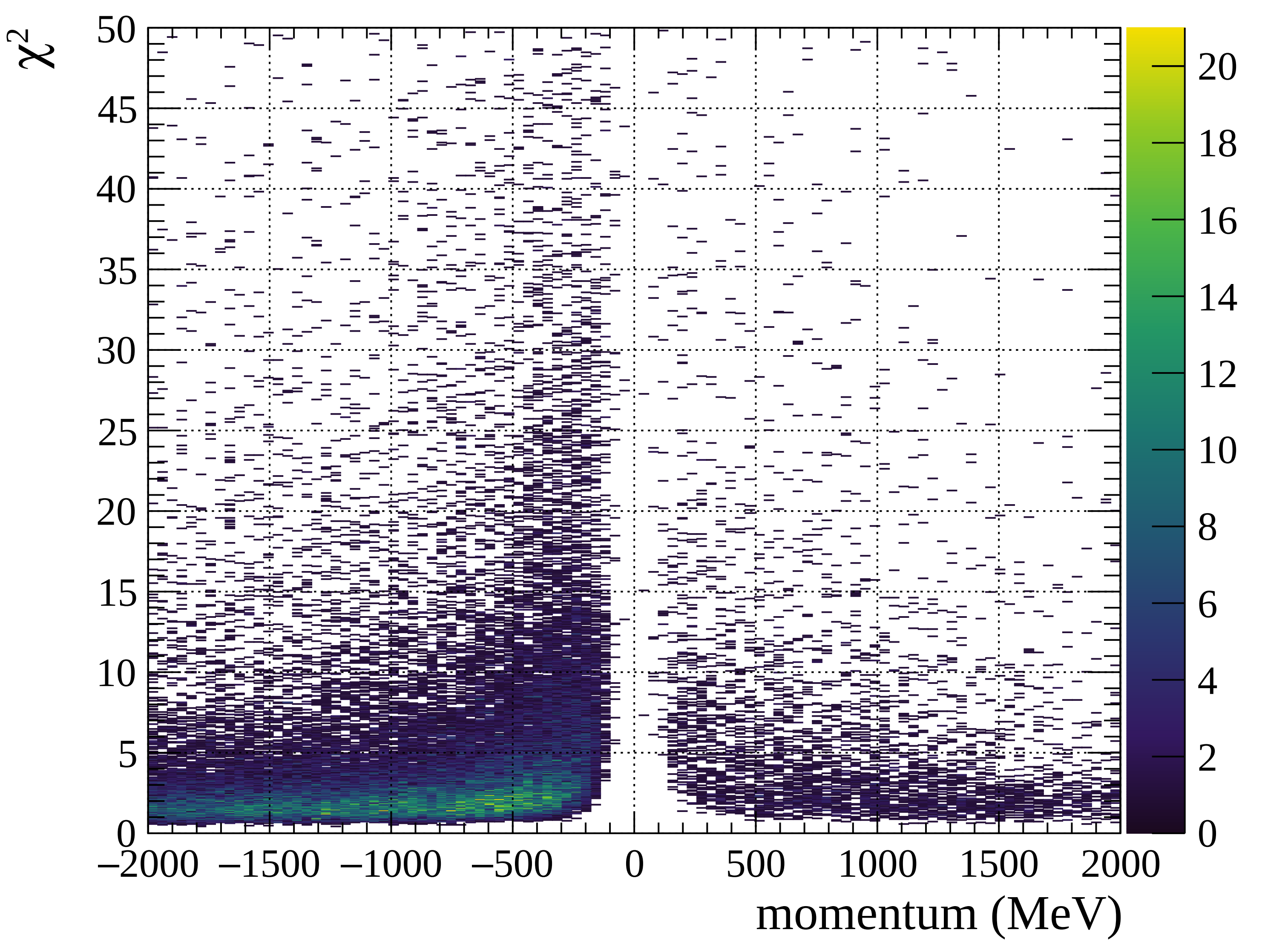


dE/dx with WF (ADC counts/cm)
momentum (MeV)

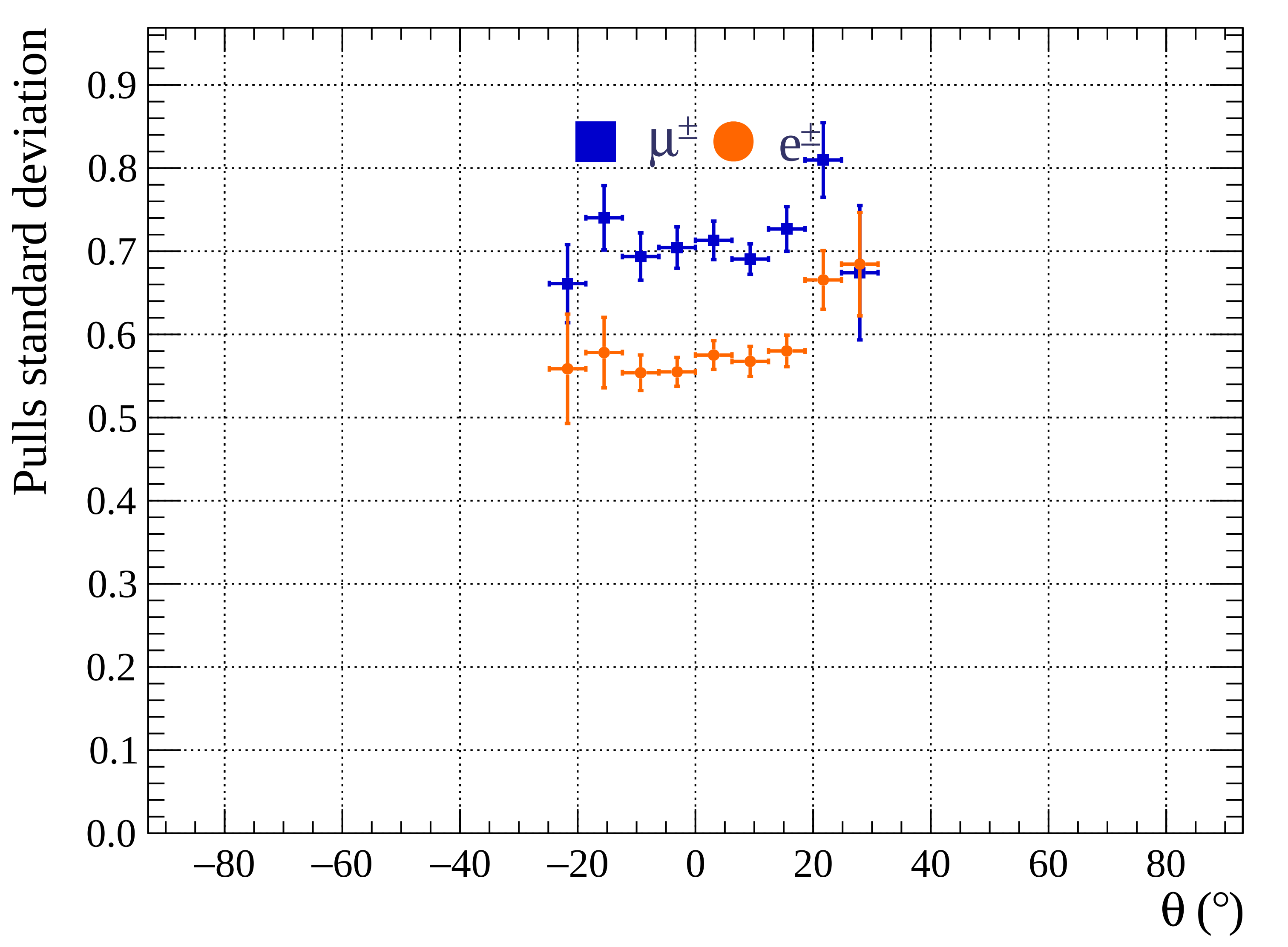

dE/dx with XP (ADC counts/cm)
momentum (MeV)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
0
2
4
6
8
10
12
14



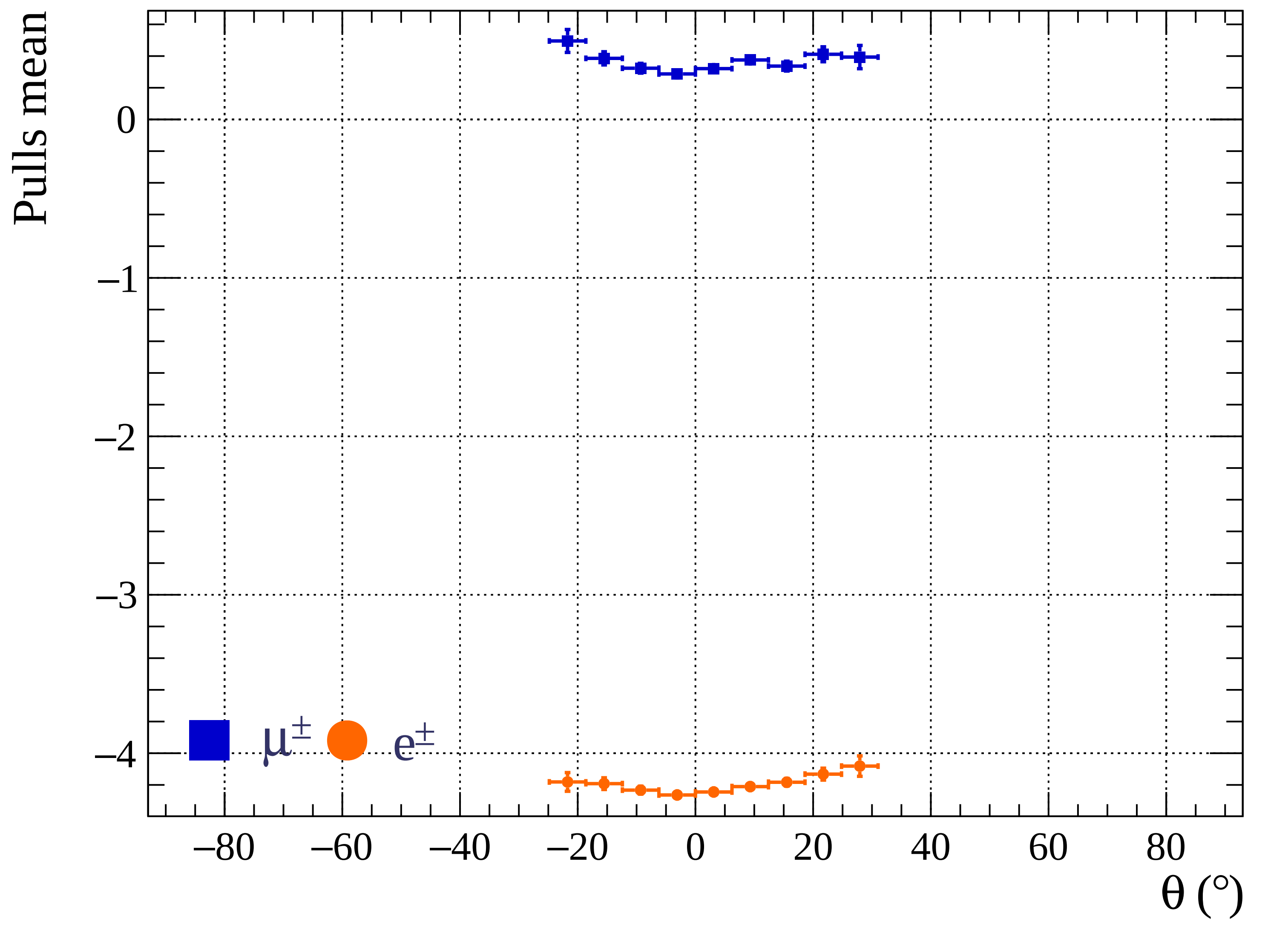
Count
400
350
300
250
200
150
100
50
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



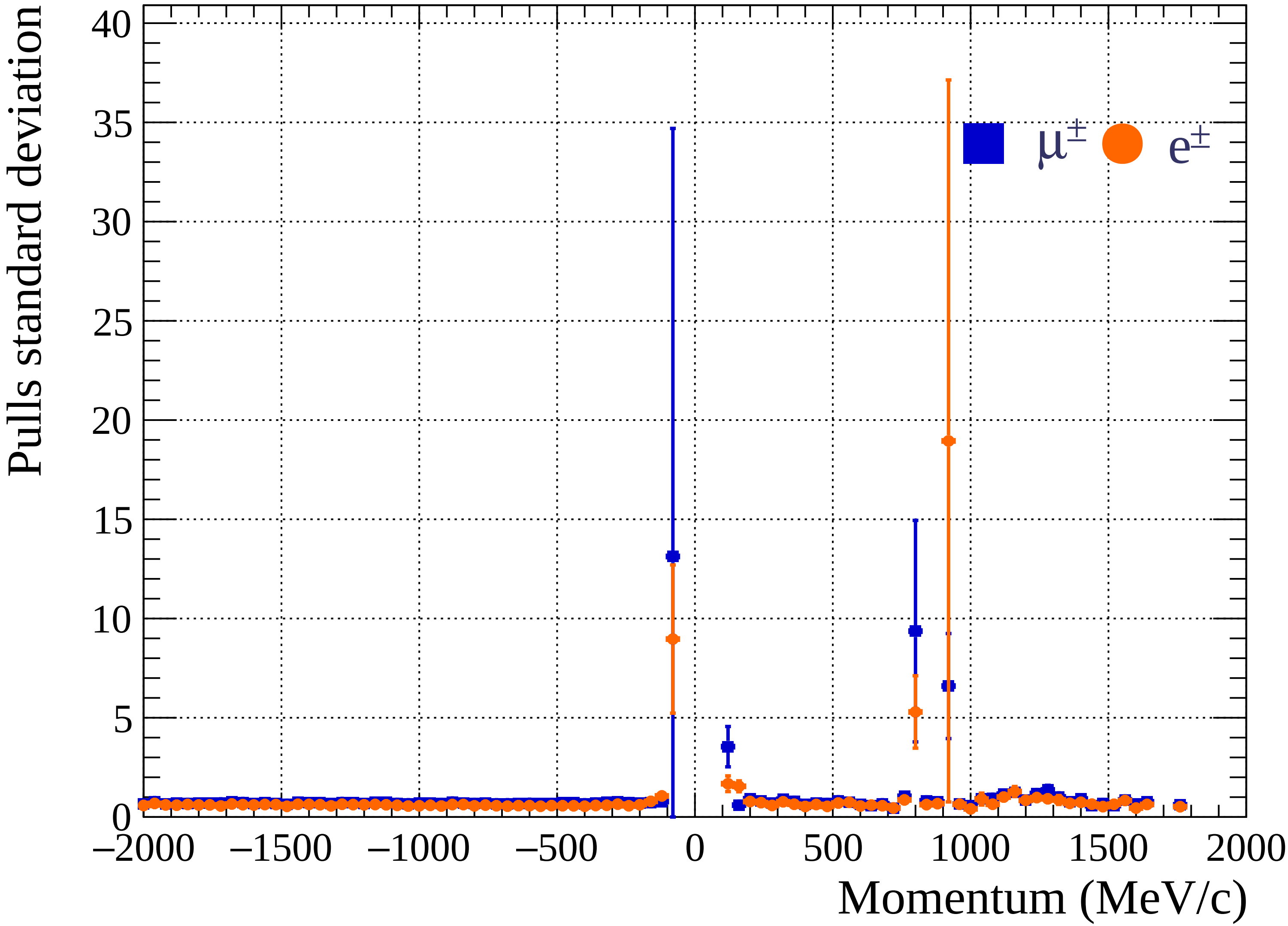
χ^2
momentum (MeV)



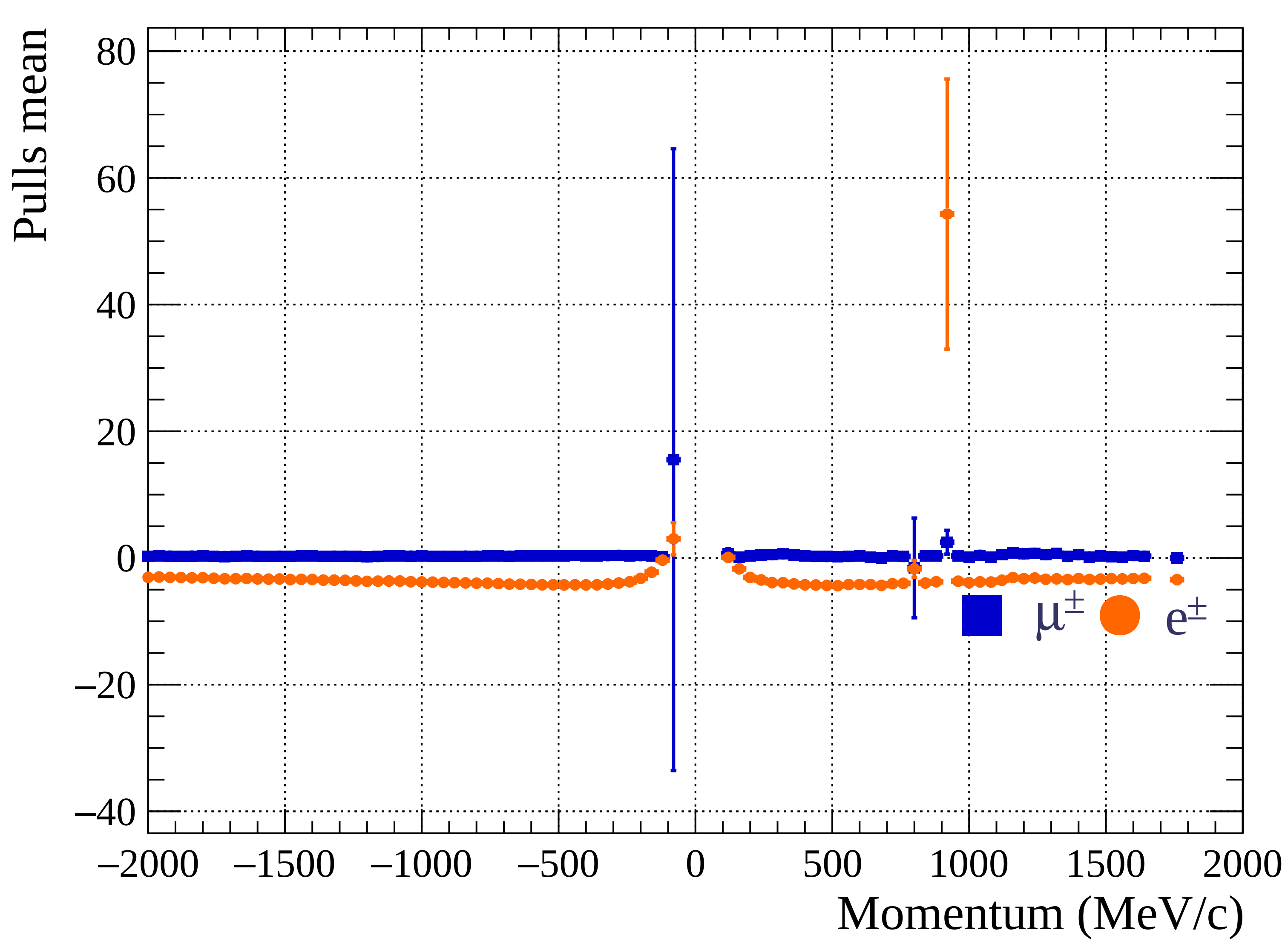
Pulls standard deviation
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



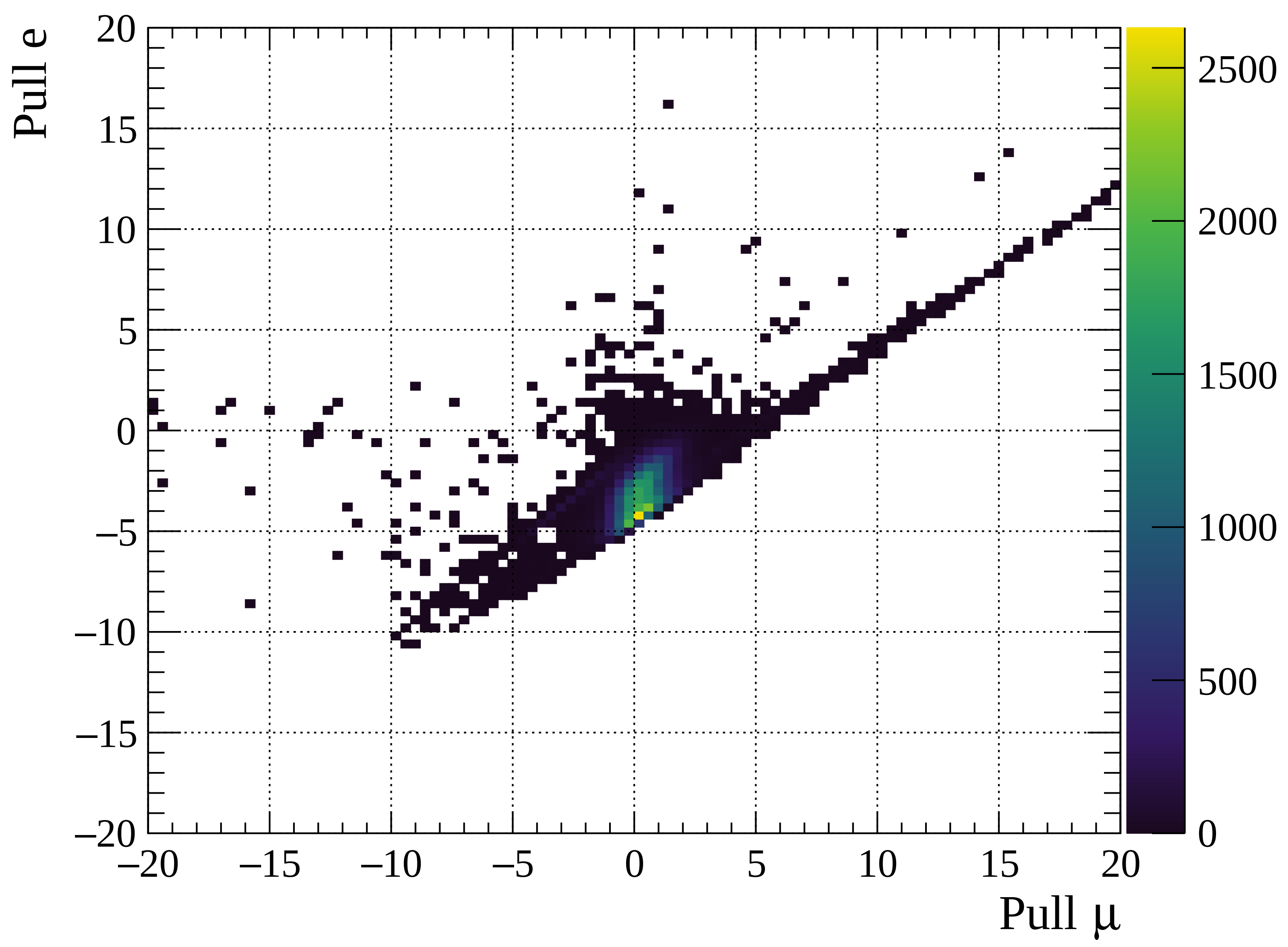
Pulls mean
0
−1
−2
−3
−4
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±

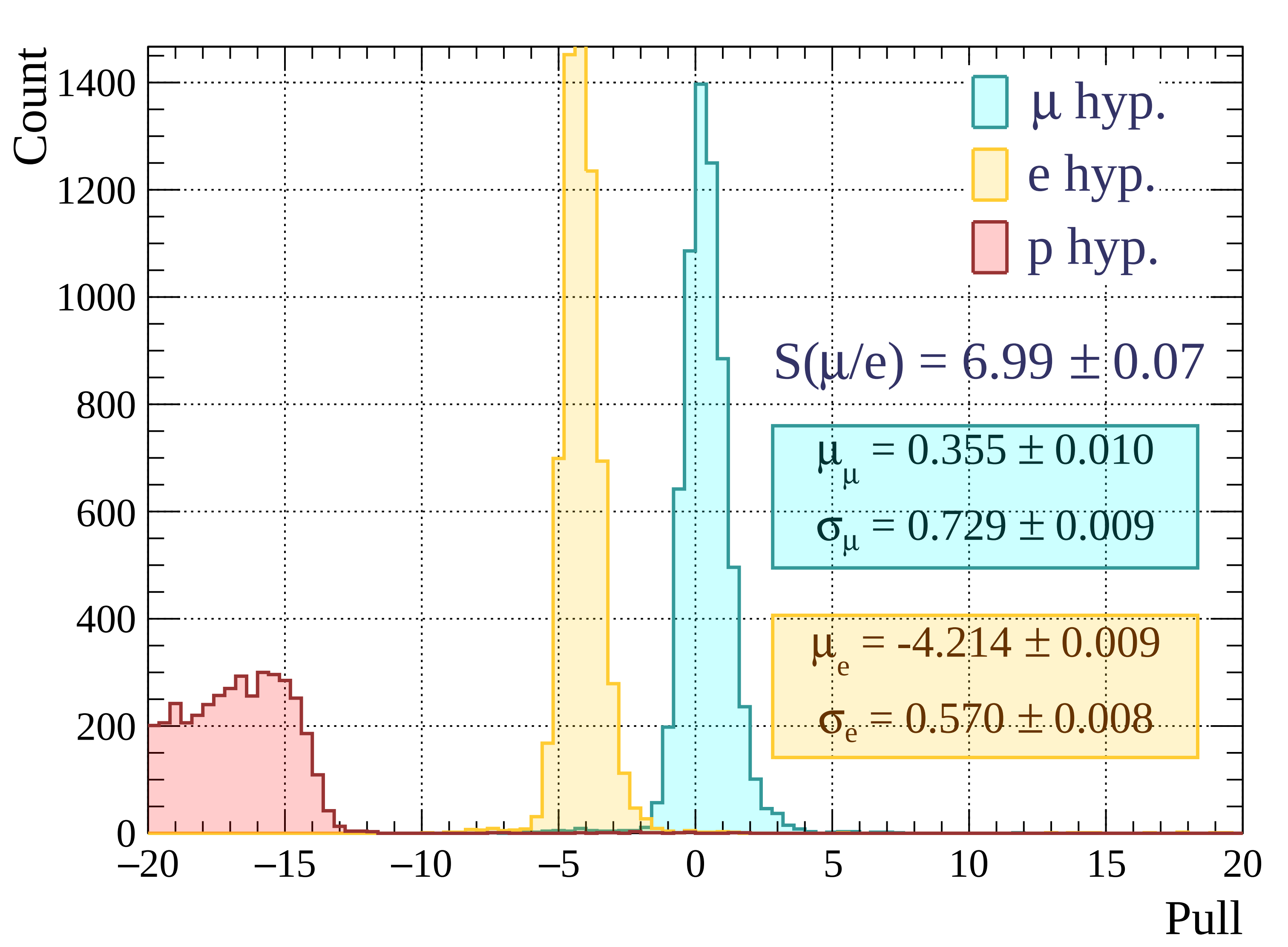


Pulls standard deviation
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

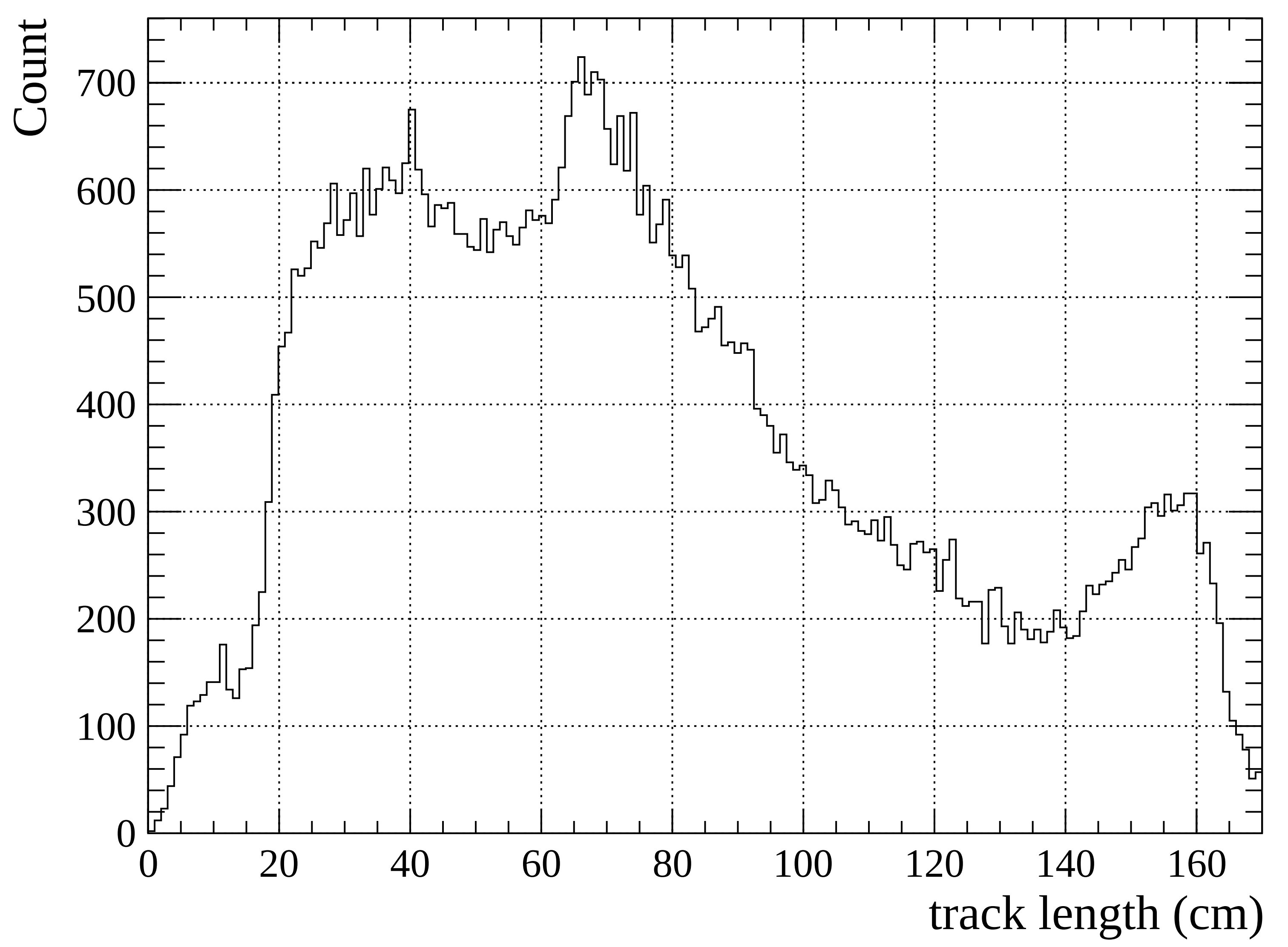


Pulls mean
80
60
40
20
0
−20
−40
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
$\mu^{\pm}$
$e^{\pm}$

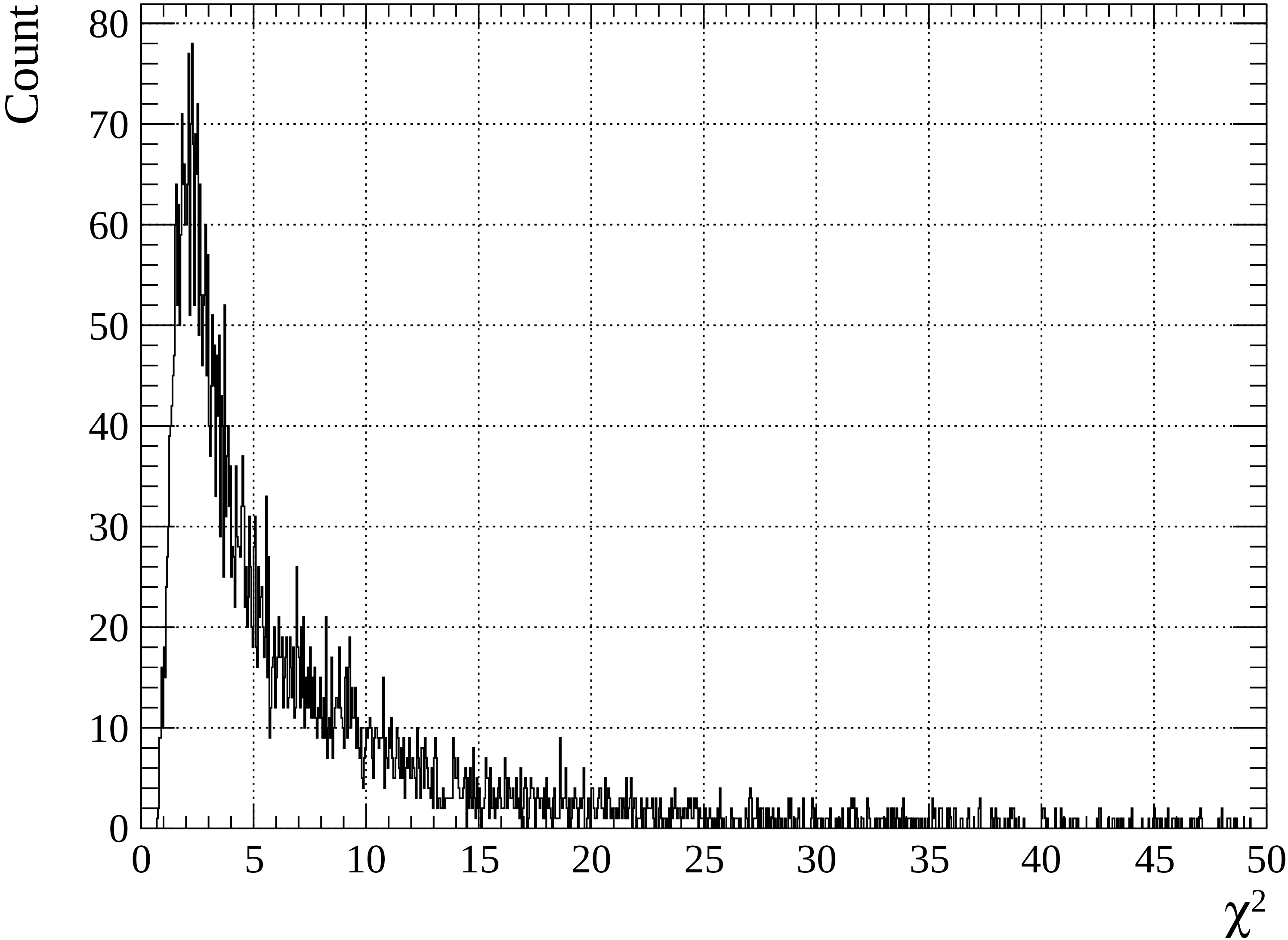

Pull e
Pull μ
20
15
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
15
20
2500
2000
1500
1000
500
0



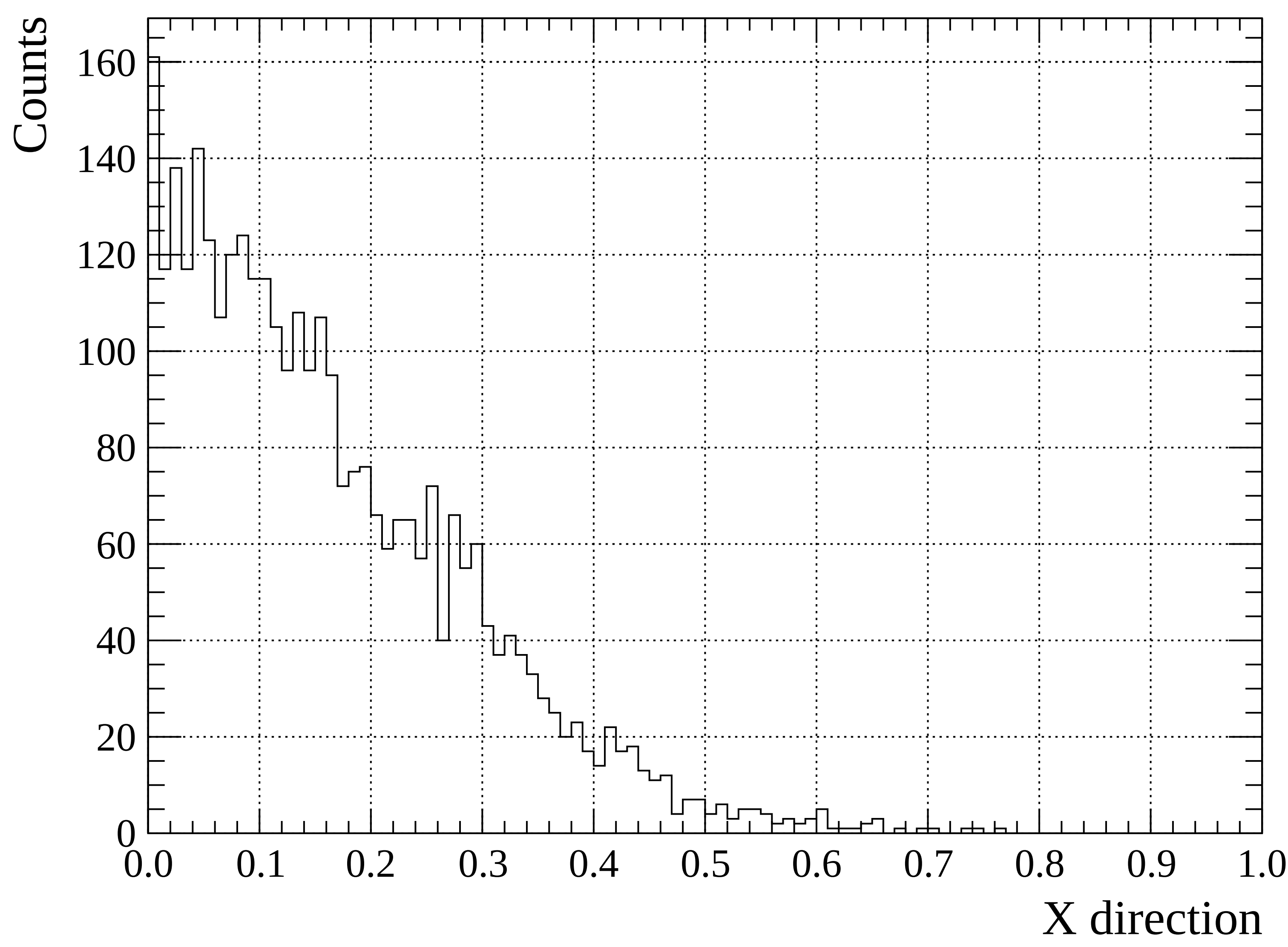
Count
1400
1200
1000
800
600
400
200
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 6.99 ± 0.07
μ_μ = 0.355 ± 0.010
σ_μ = 0.729 ± 0.009
μ_e = -4.214 ± 0.009
σ_e = 0.570 ± 0.008



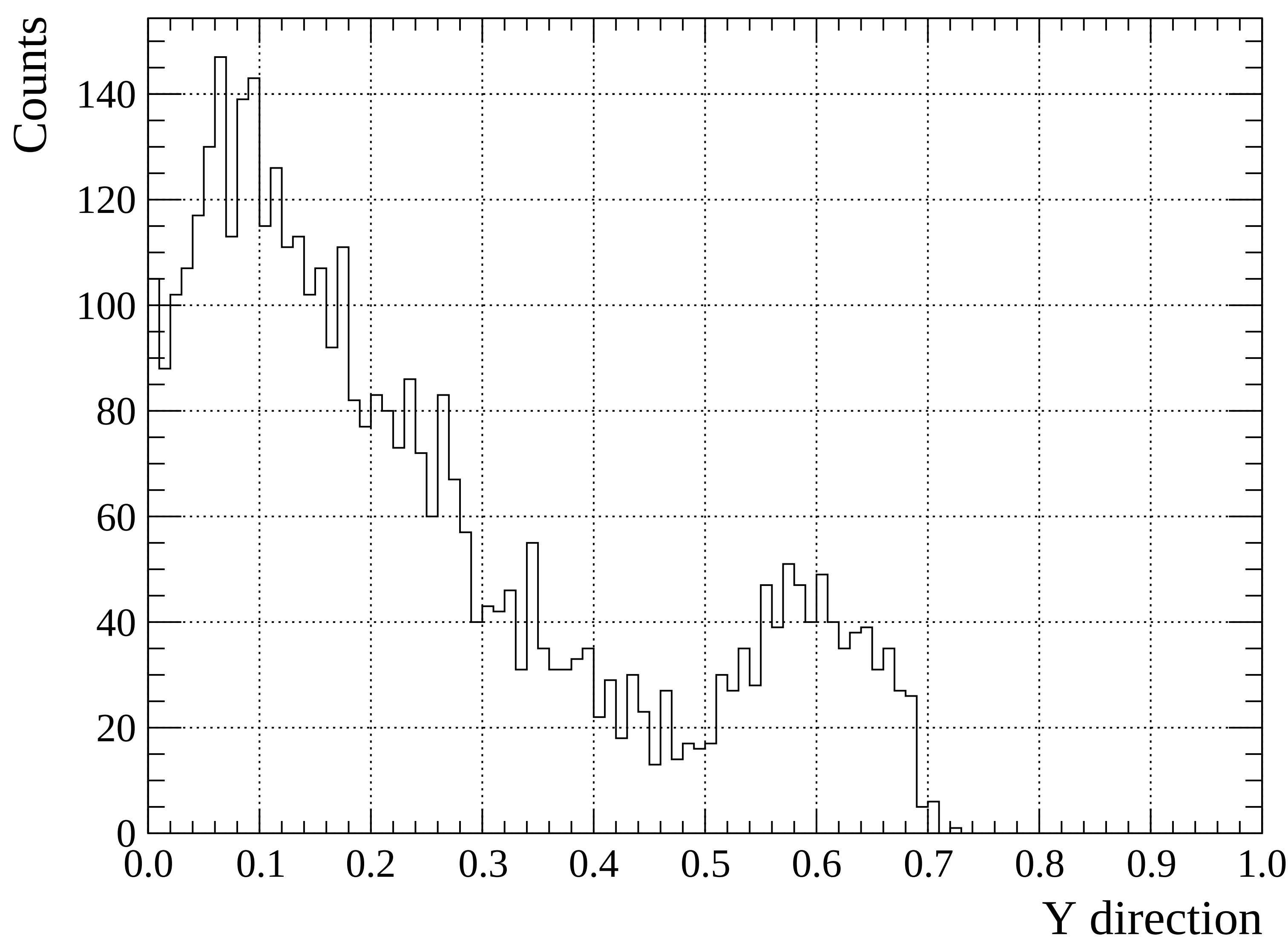
Count
0
100
200
300
400
500
600
700
0
20
40
60
80
100
120
140
160
track length (cm)

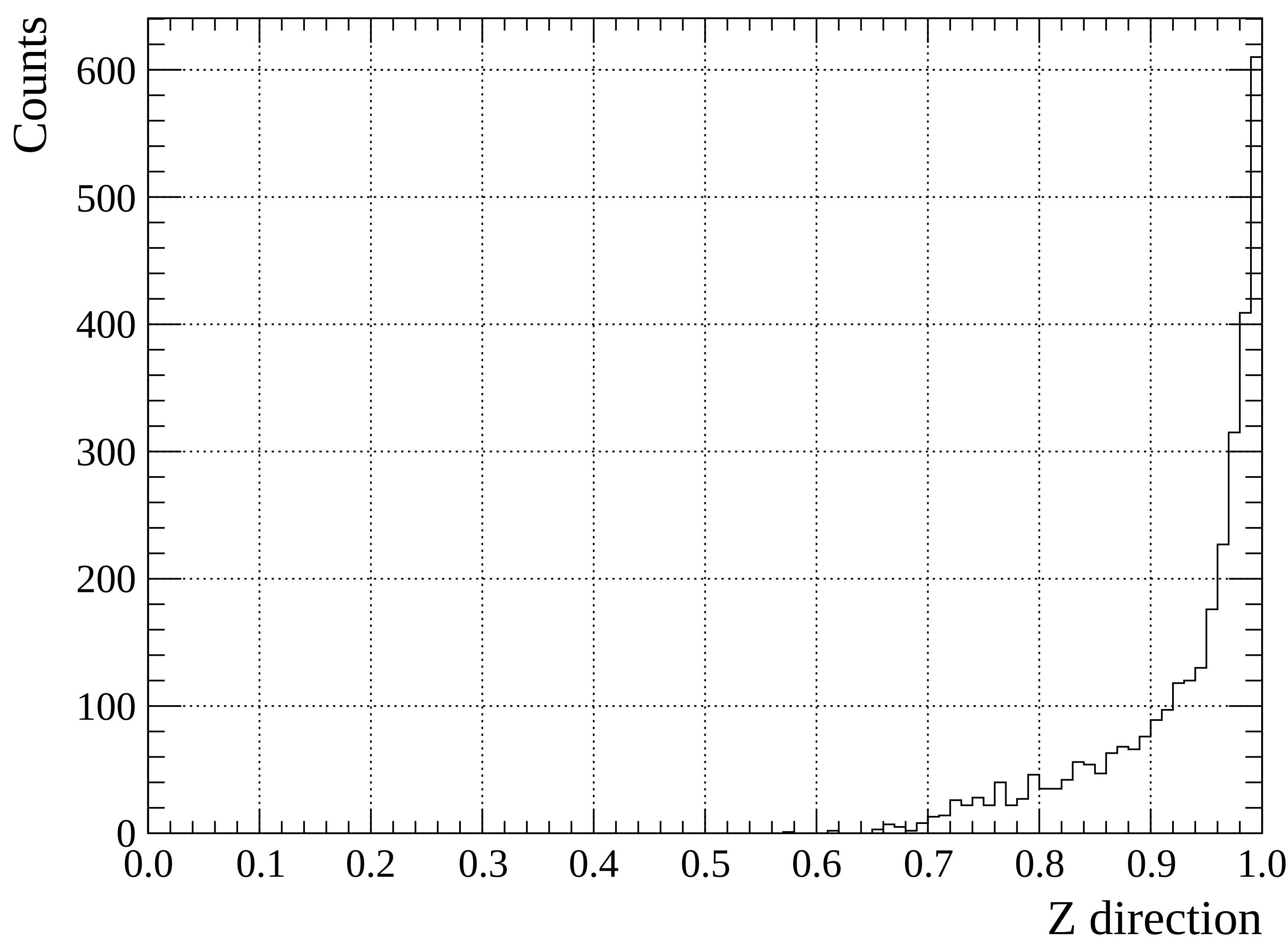


Count
0
10
20
30
40
50
60
70
80
0
5
10
15
20
25
30
35
40
45
50
χ^2

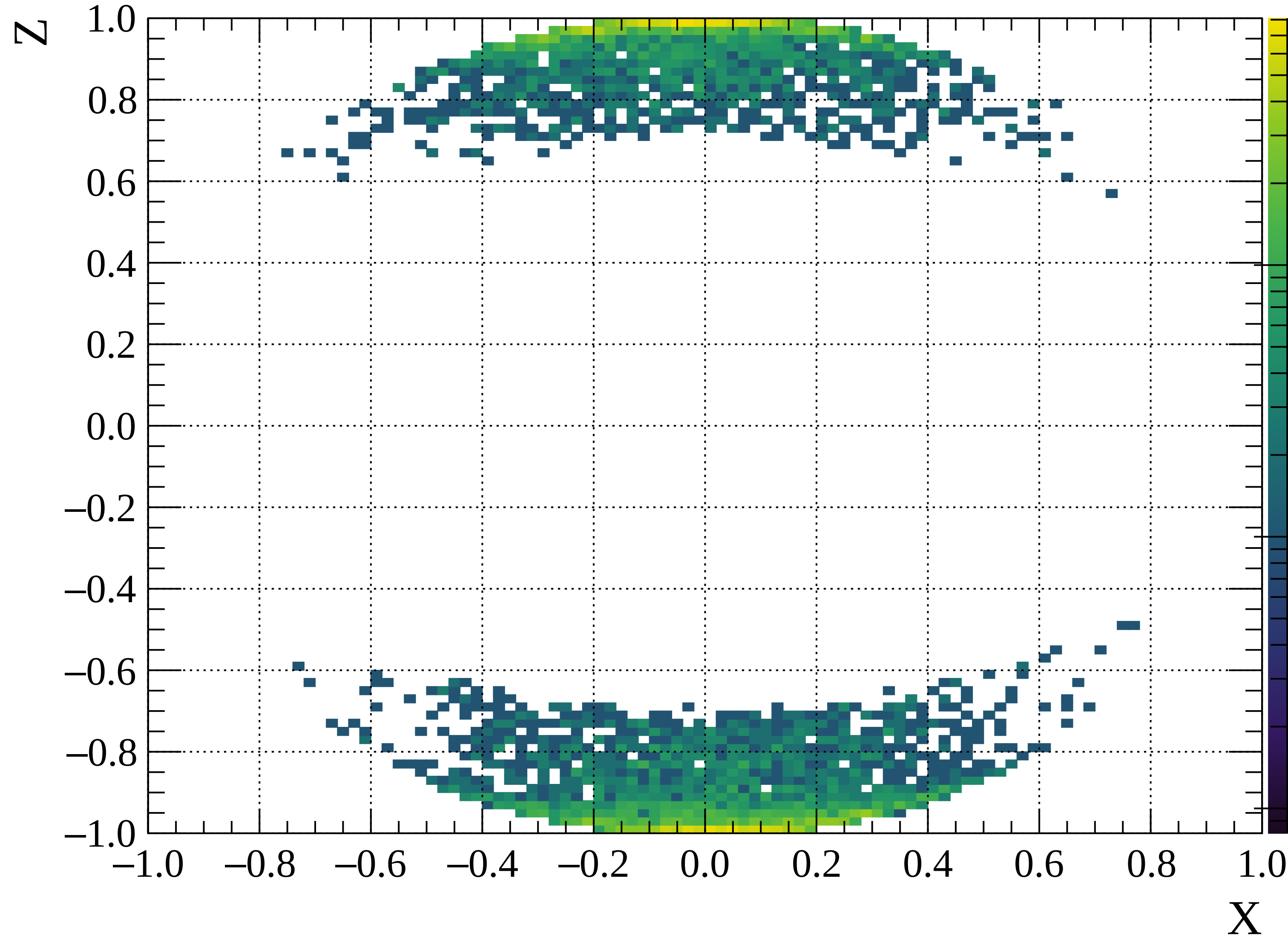


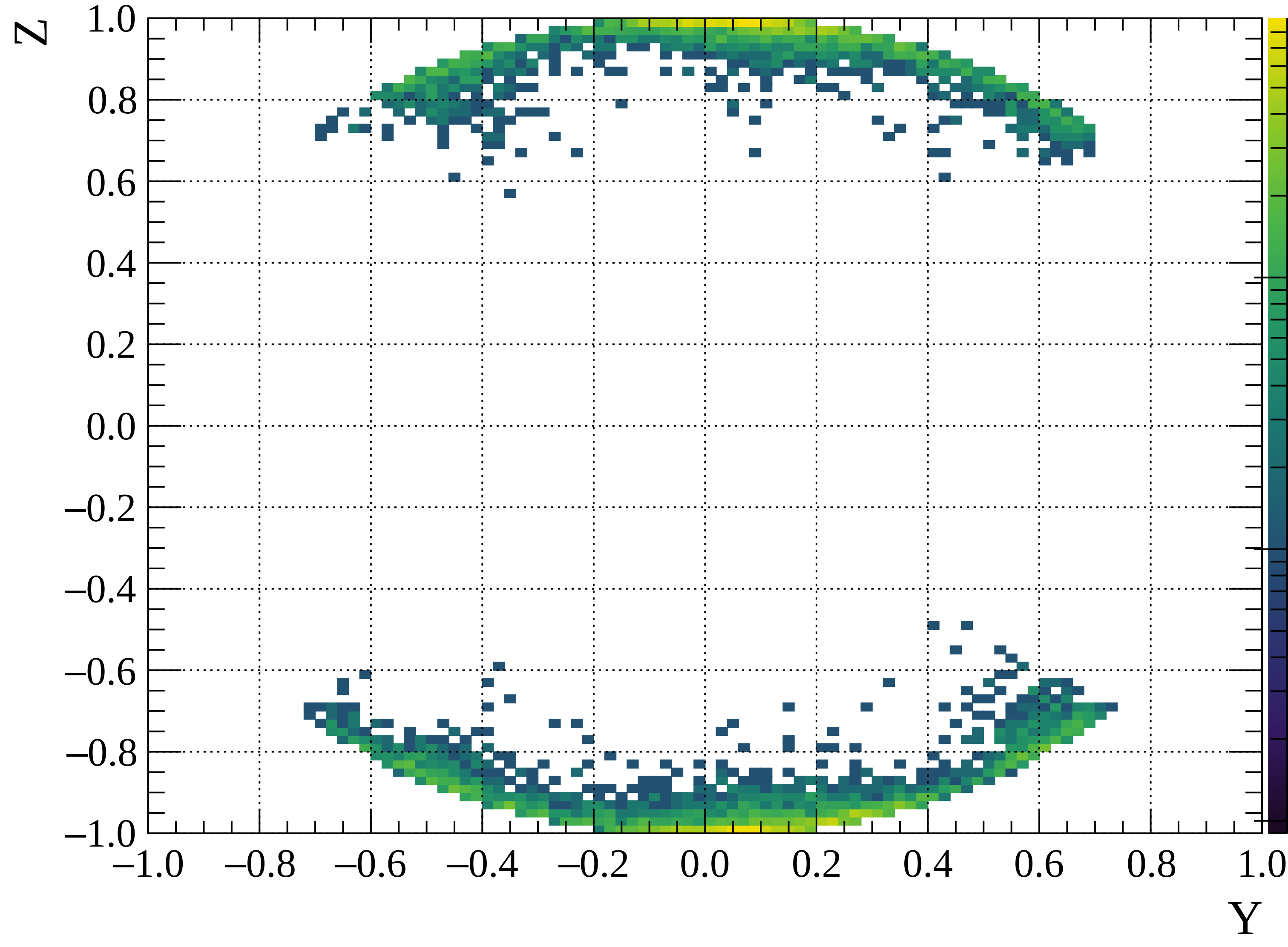
Counts
160
140
120
100
80
60
40
20
0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
X direction


Counts
0
20
40
60
80
100
120
140
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Y direction

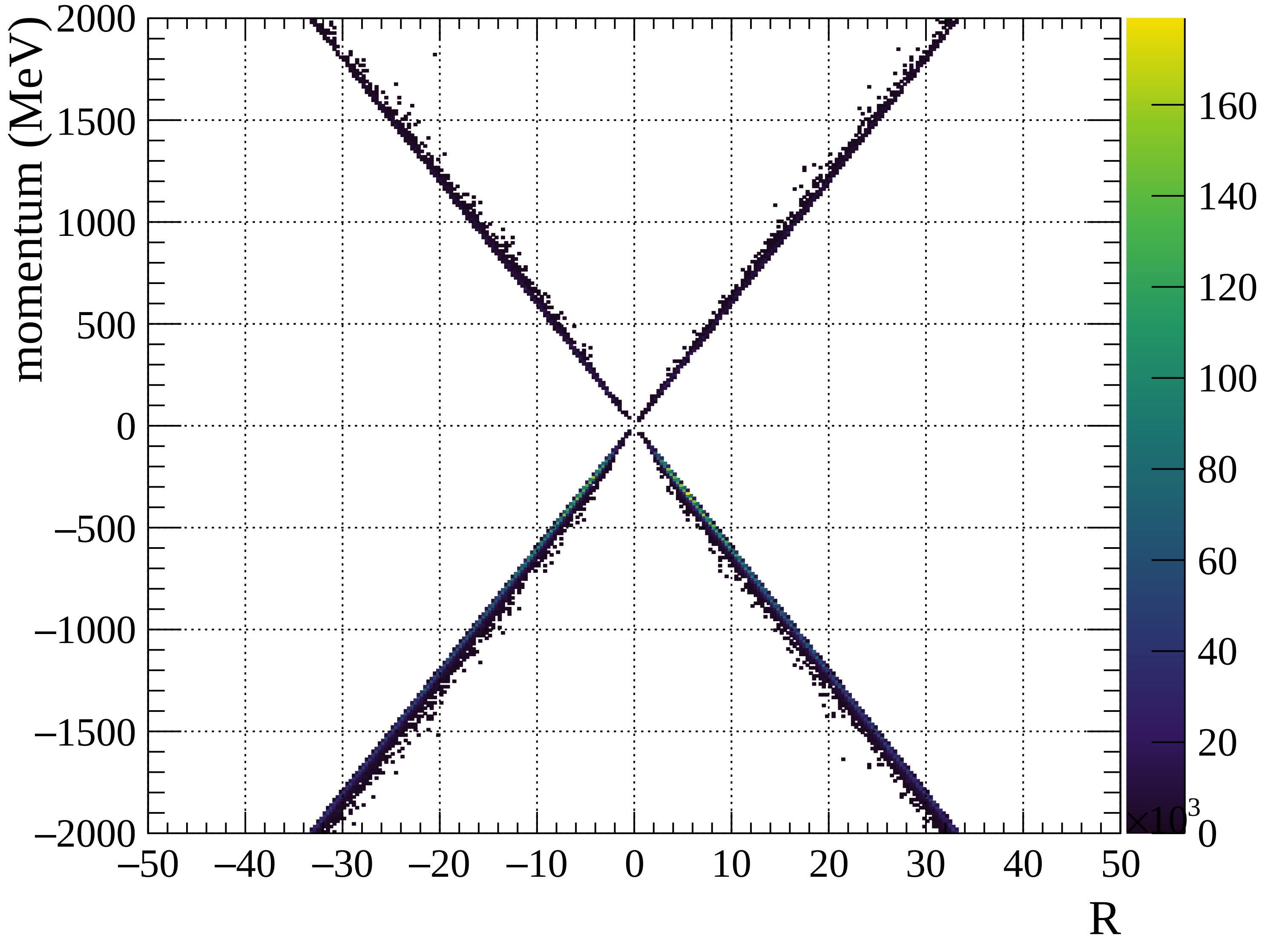


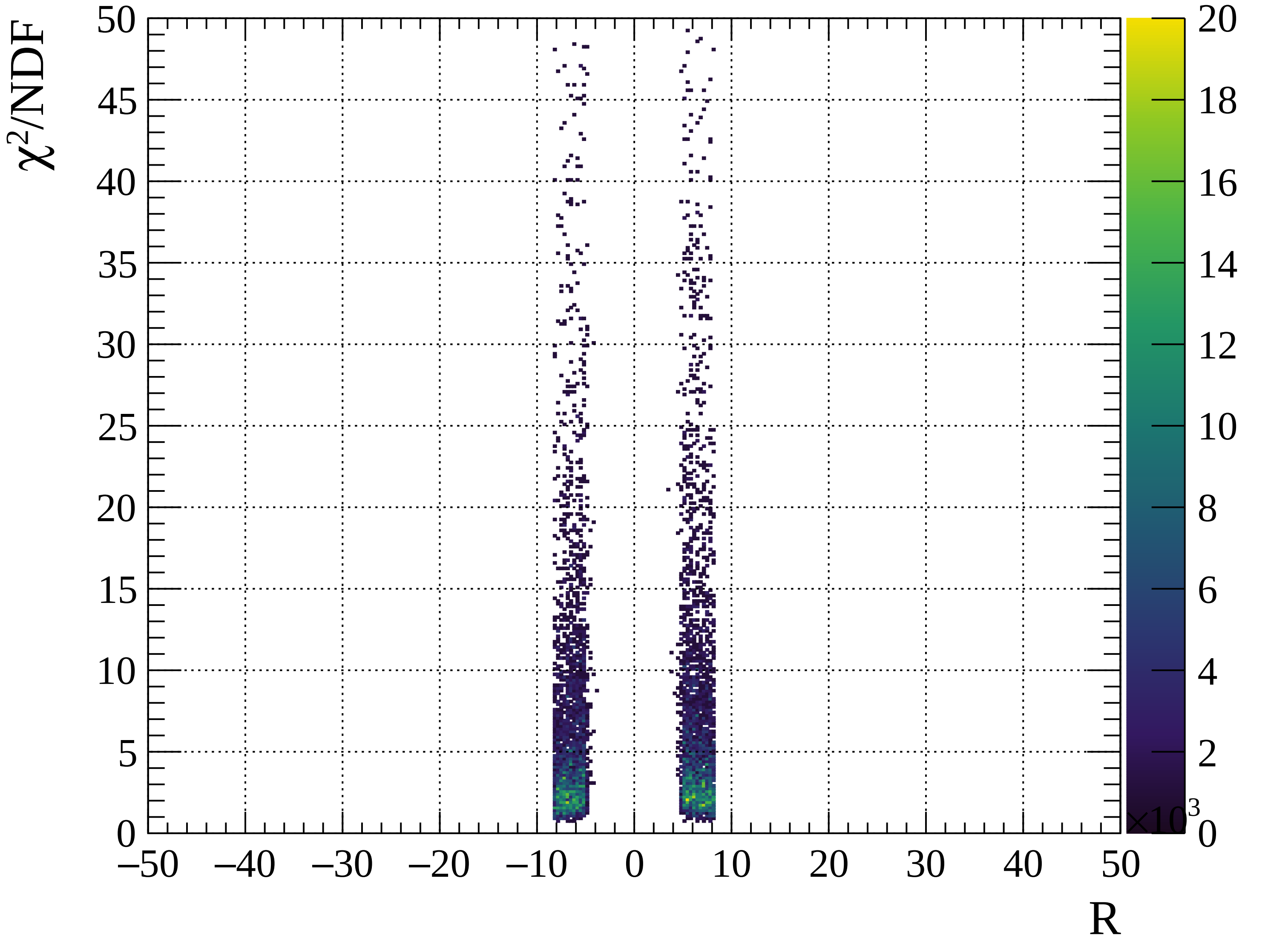
Counts
0
100
200
300
400
500
600
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Z direction


Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X

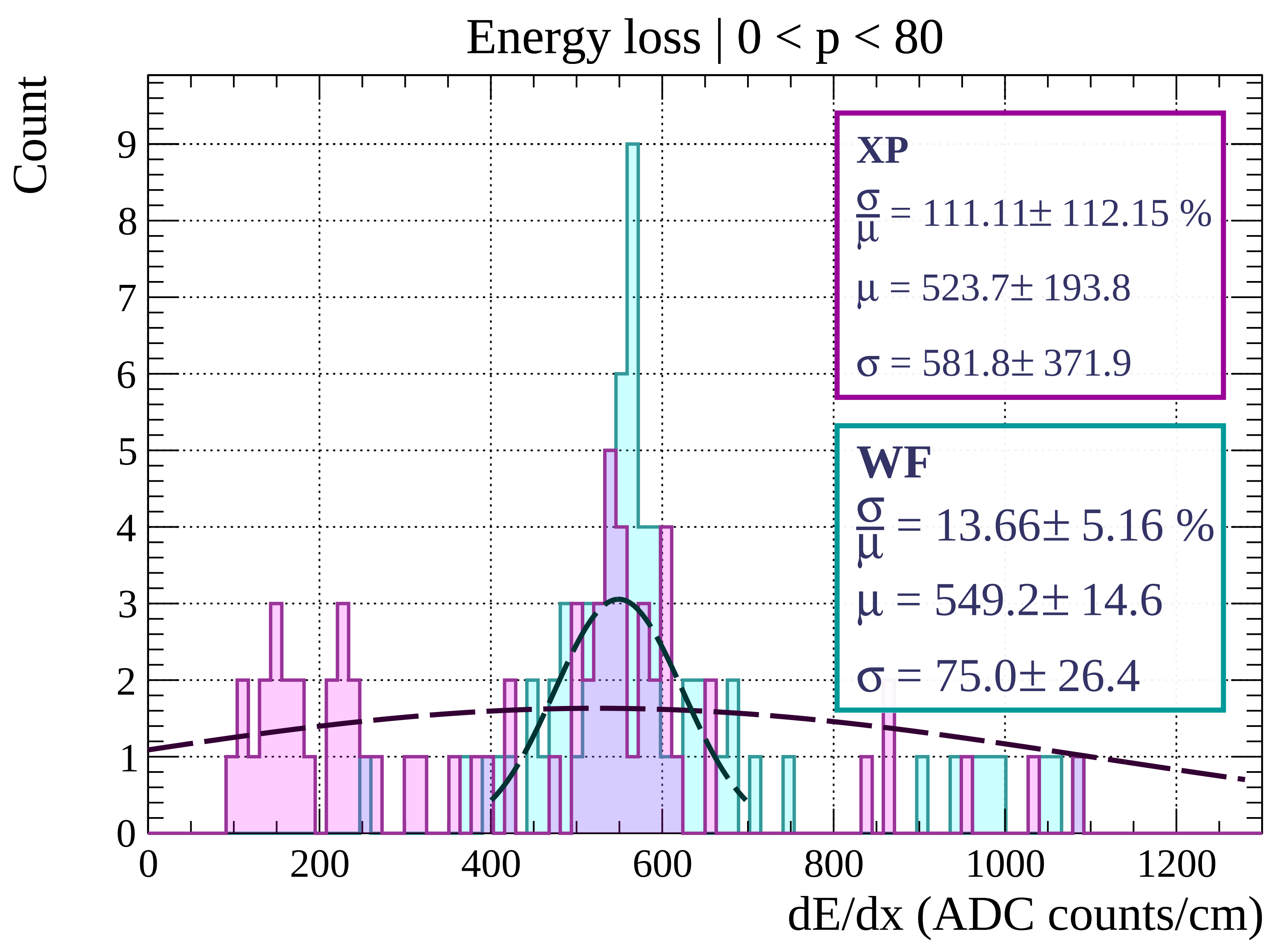


Z
Y
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0

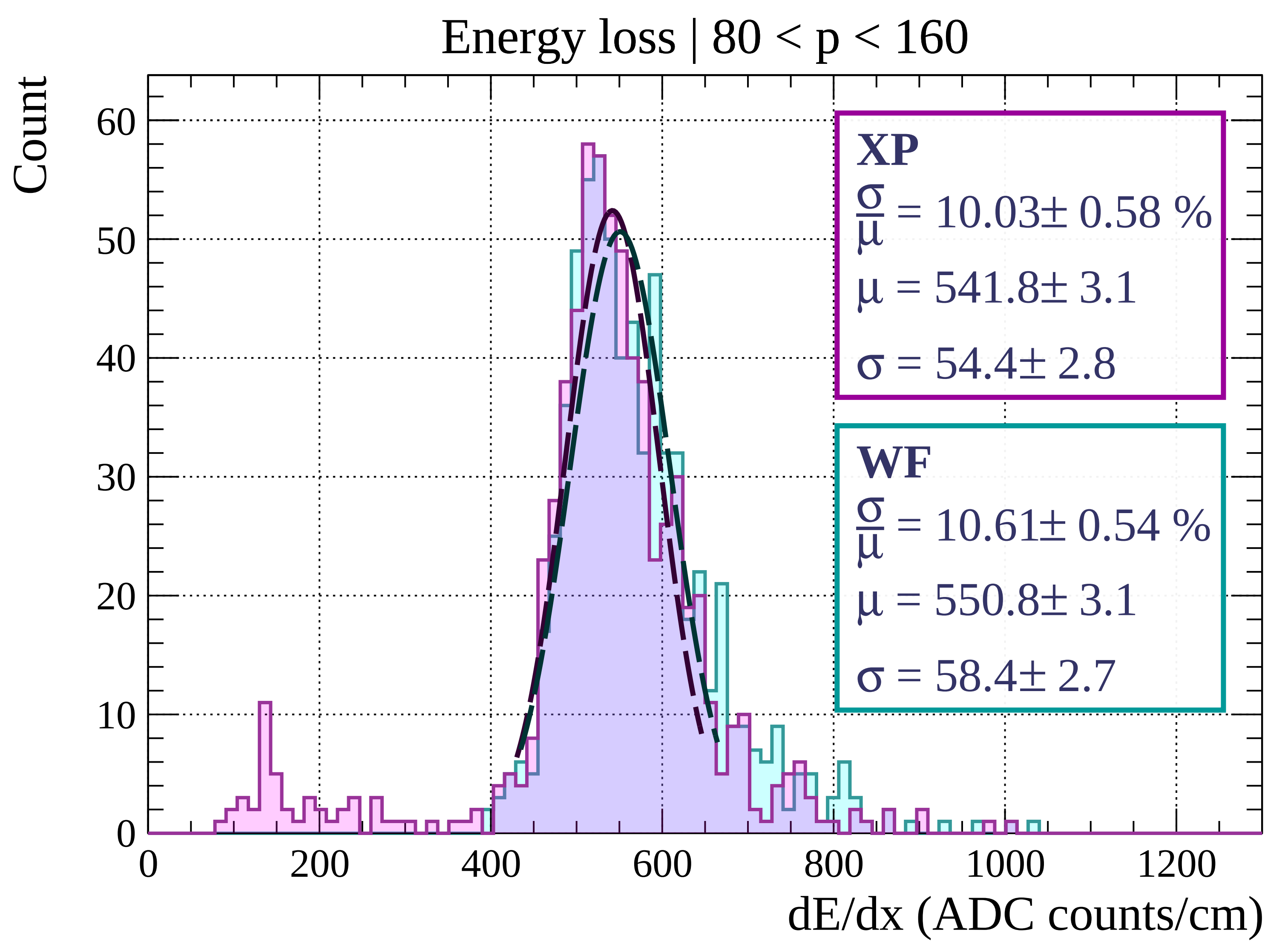

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
160
140
120
100
80
60
40
20
0
$\times 10^3$



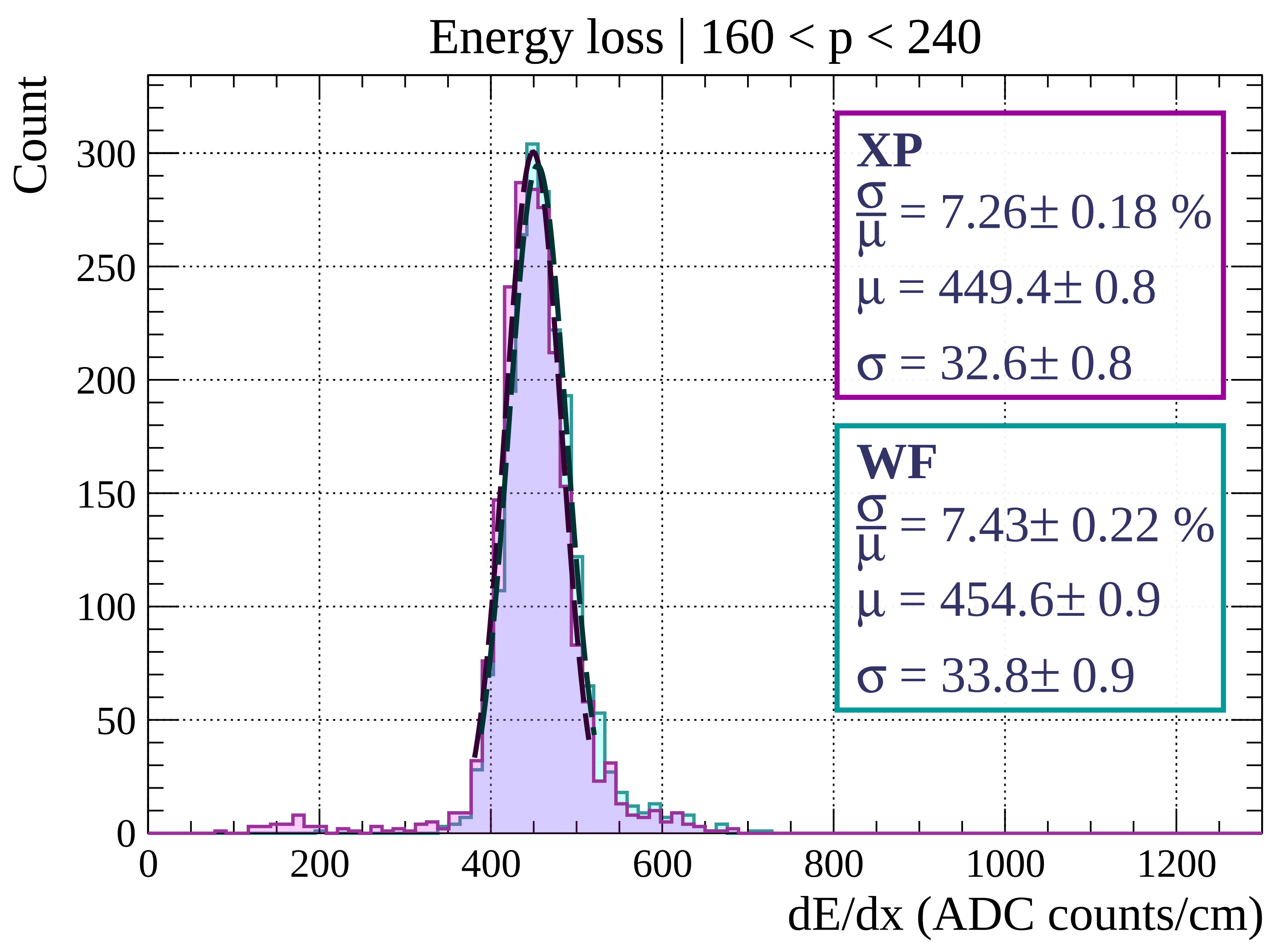
χ^2/NDF
R
$\times 10^3$

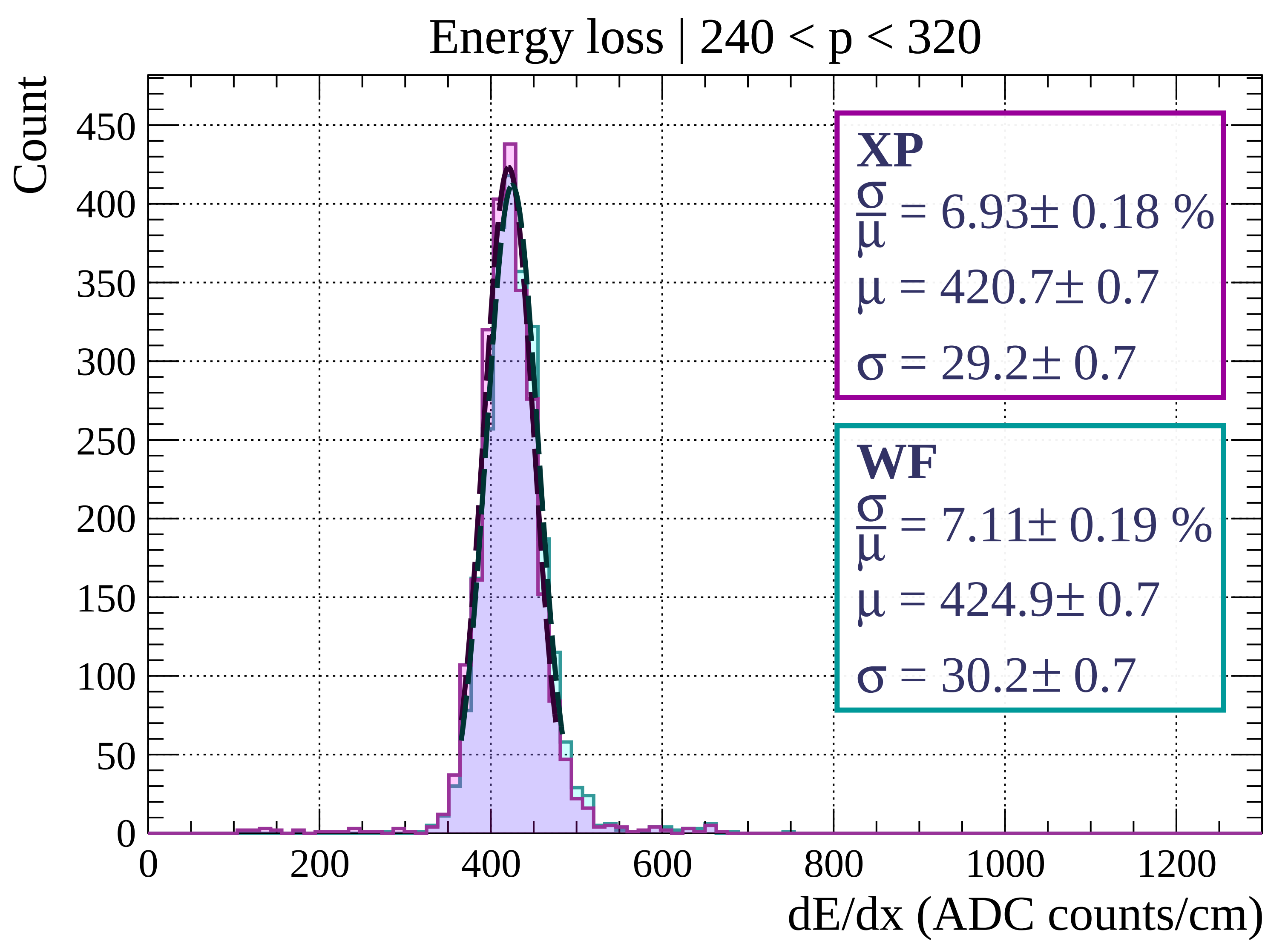


Energy loss | 0 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
9
XP
σ/μ = 111.11± 112.15 %
μ = 523.7± 193.8
σ = 581.8± 371.9
WF
σ/μ = 13.66± 5.16 %
μ = 549.2± 14.6
σ = 75.0± 26.4

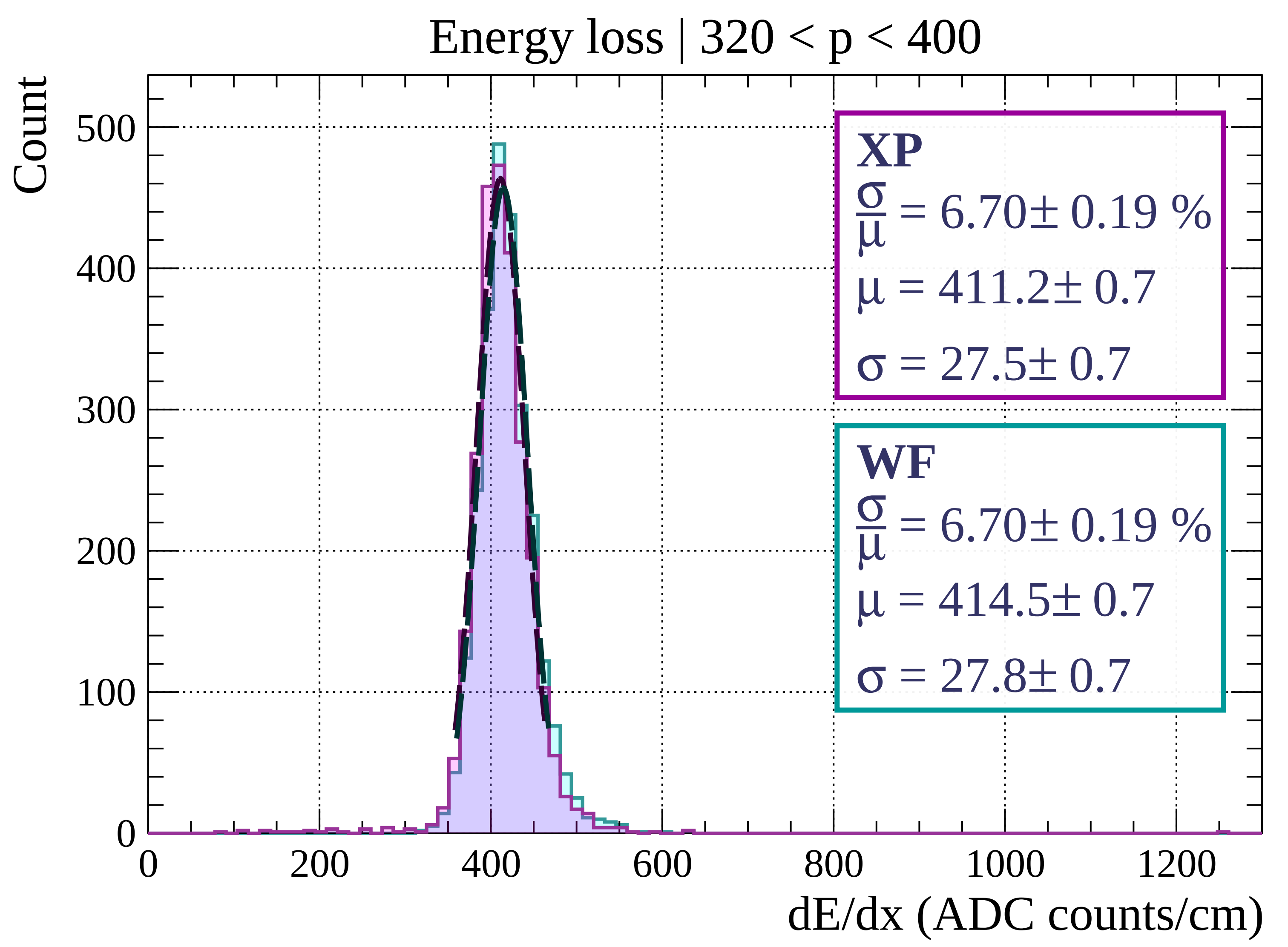


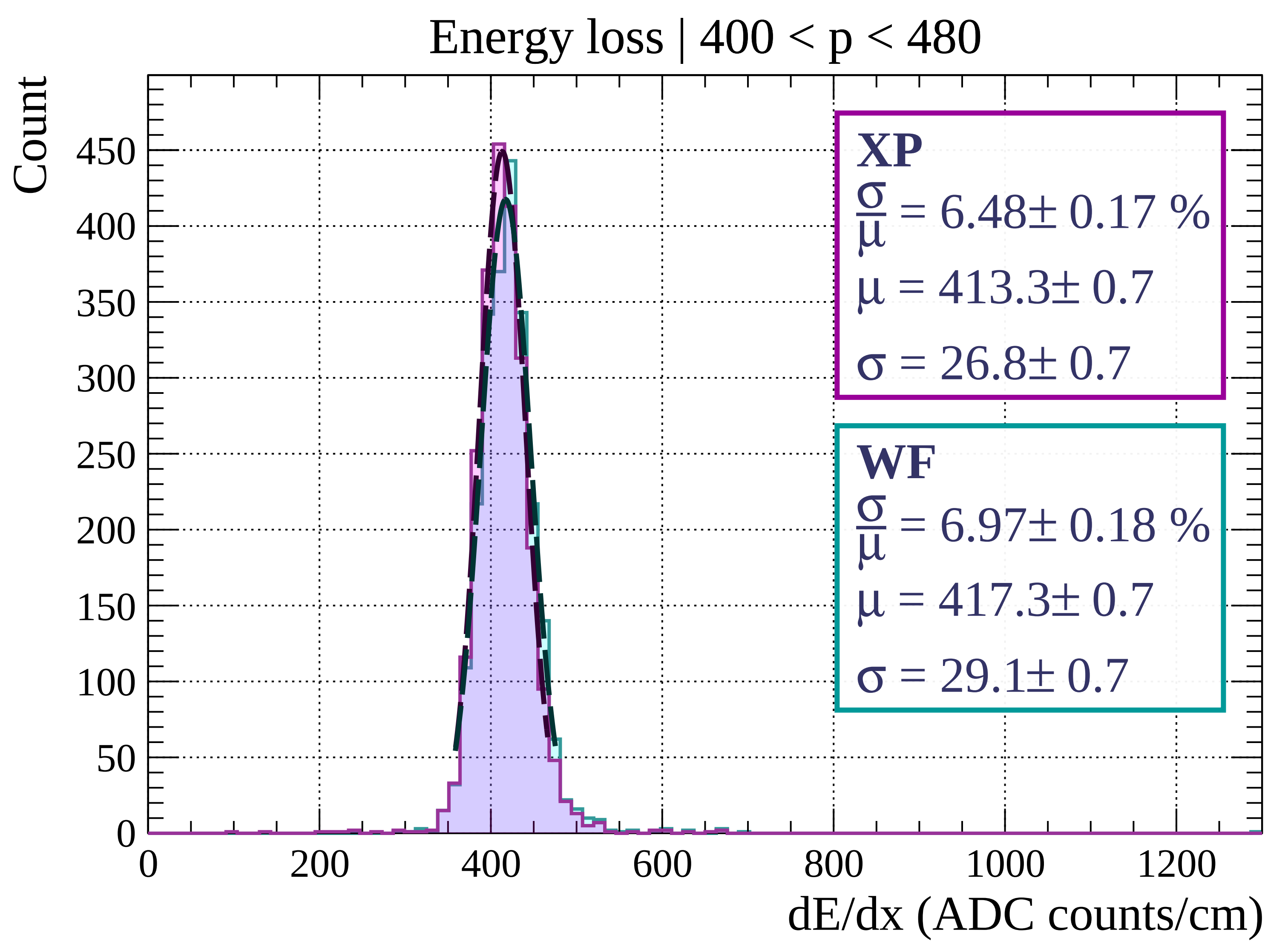
Energy loss | 80 < p < 160
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
60
XP
σ/μ = 10.03± 0.58 %
μ = 541.8± 3.1
σ = 54.4± 2.8
WF
σ/μ = 10.61± 0.54 %
μ = 550.8± 3.1
σ = 58.4± 2.7


Energy loss | 160 < p < 240
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.26± 0.18 %
μ = 449.4± 0.8
σ = 32.6± 0.8
WF
σ/μ = 7.43± 0.22 %
μ = 454.6± 0.9
σ = 33.8± 0.9

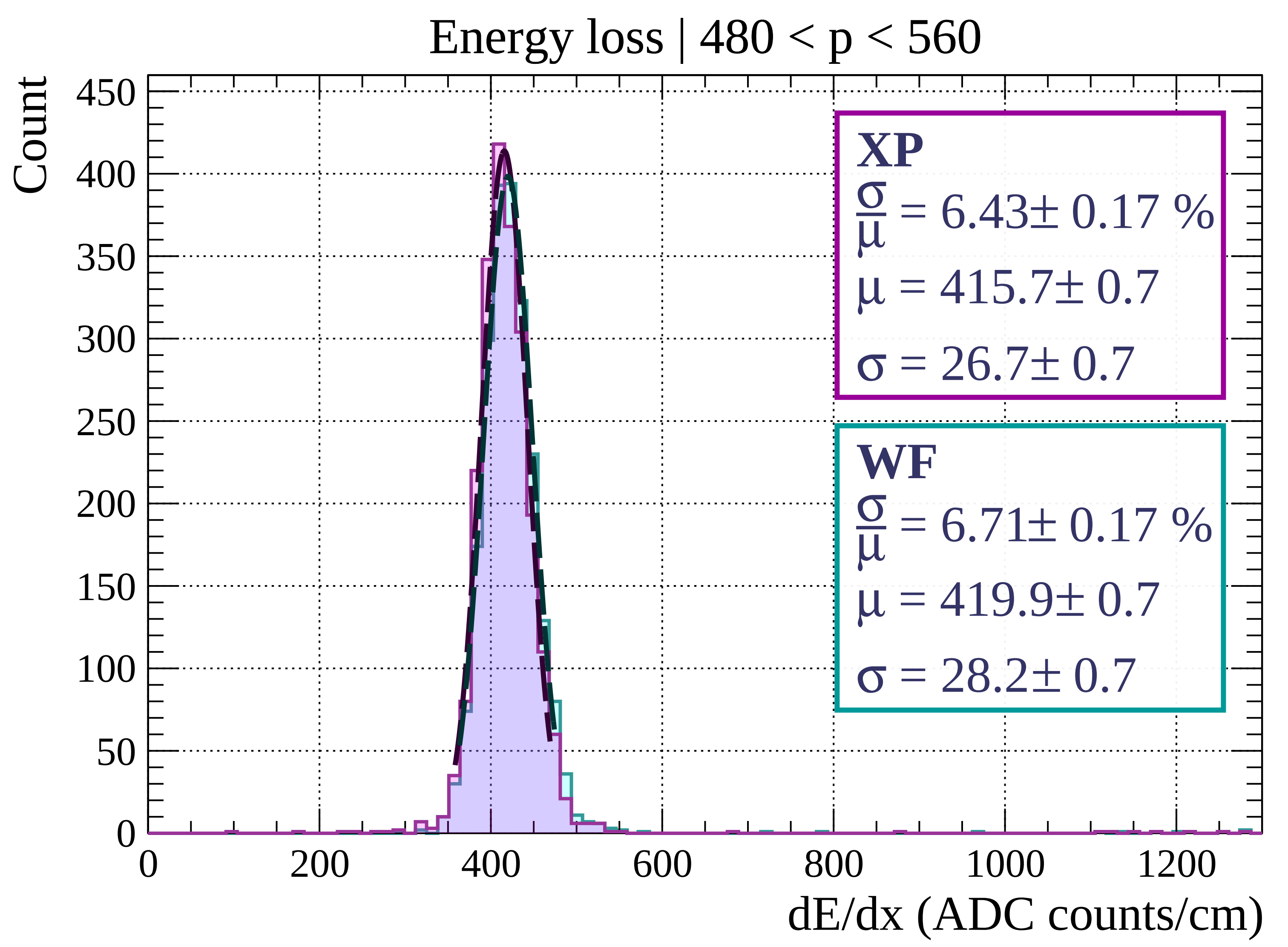


Energy loss | 240 < p < 320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.93± 0.18 %
μ = 420.7± 0.7
σ = 29.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.11± 0.19 %
μ = 424.9± 0.7
σ = 30.2± 0.7

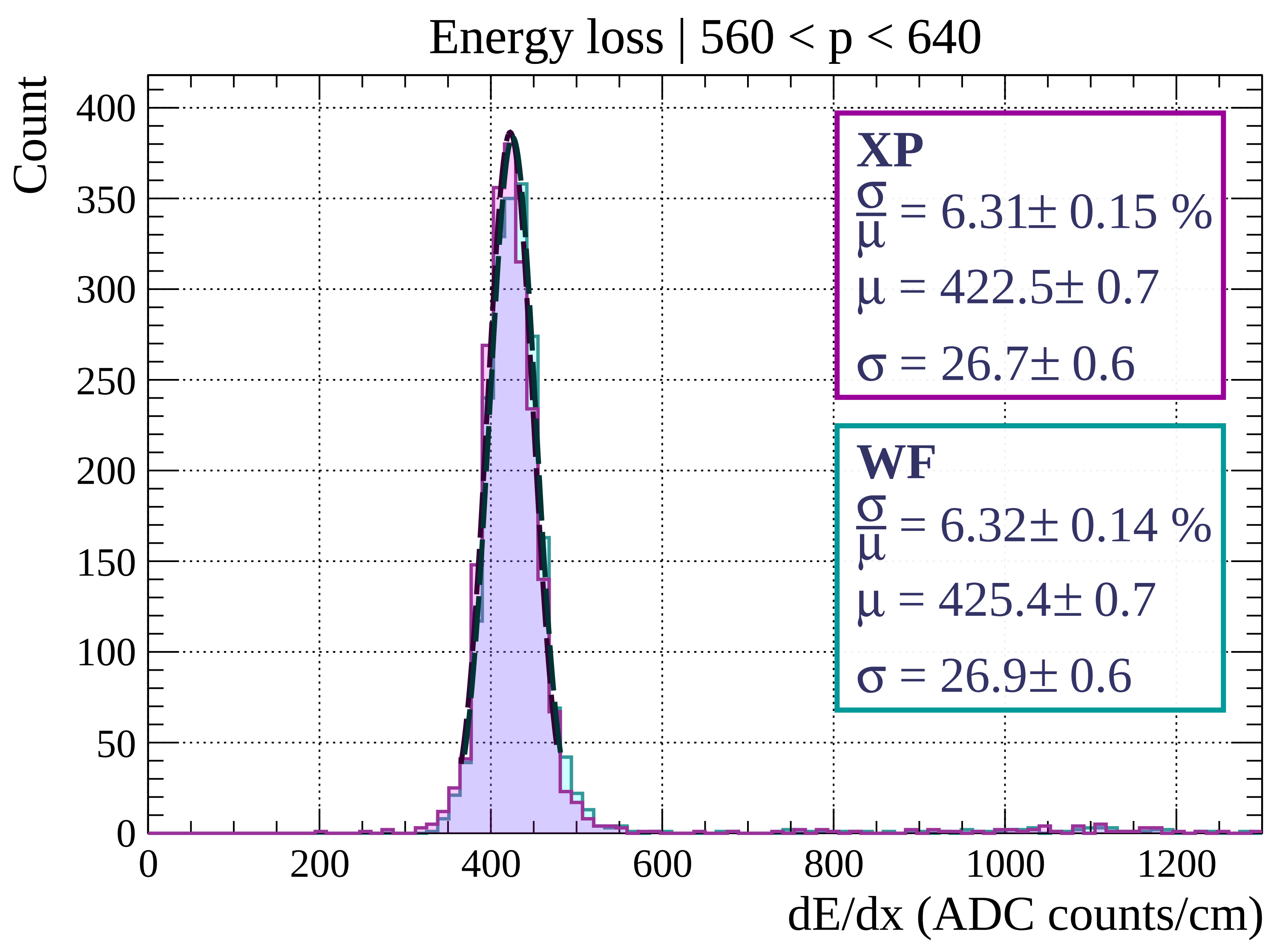

Energy loss | 320 < p < 400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 6.70± 0.19 %
μ = 411.2± 0.7
σ = 27.5± 0.7
WF
σ/μ = 6.70± 0.19 %
μ = 414.5± 0.7
σ = 27.8± 0.7



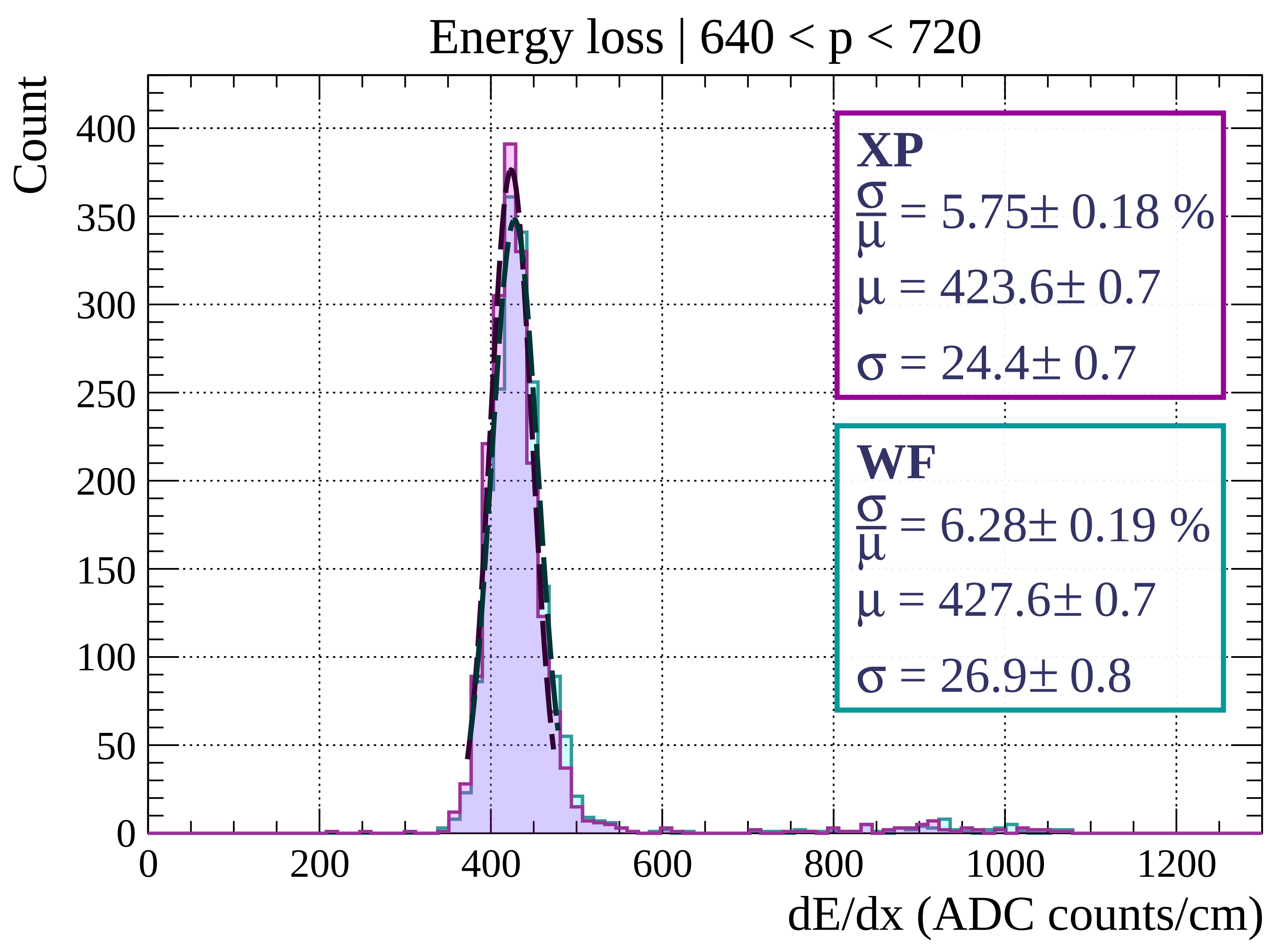
Energy loss | 400 < p < 480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.48± 0.17 %
μ = 413.3± 0.7
σ = 26.8± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.97± 0.18 %
μ = 417.3± 0.7
σ = 29.1± 0.7

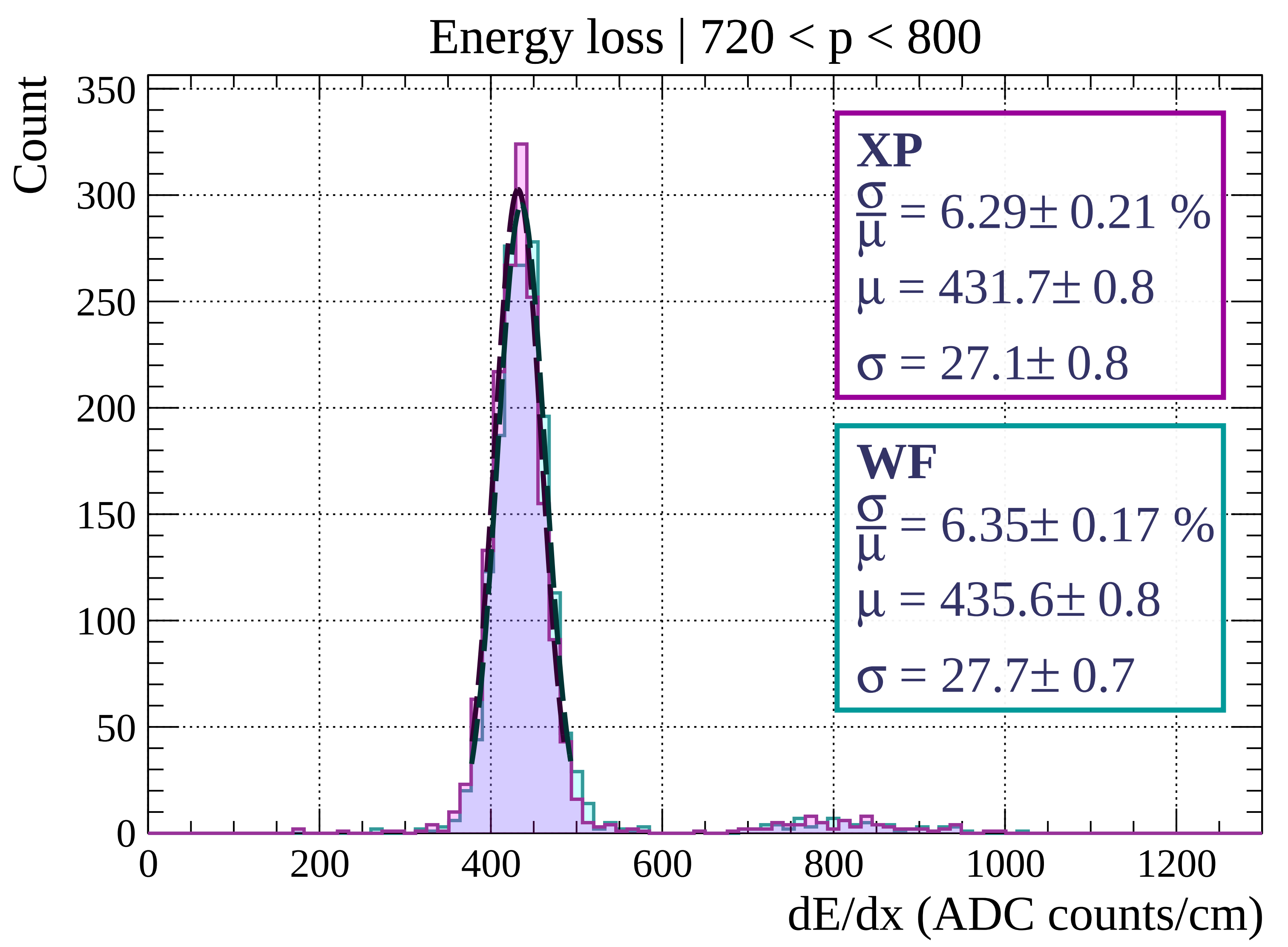


Energy loss | 480 < p < 560
Count
XP
$\frac{\sigma}{\mu}$ = 6.43± 0.17 %
μ = 415.7± 0.7
σ = 26.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 6.71± 0.17 %
μ = 419.9± 0.7
σ = 28.2± 0.7
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200

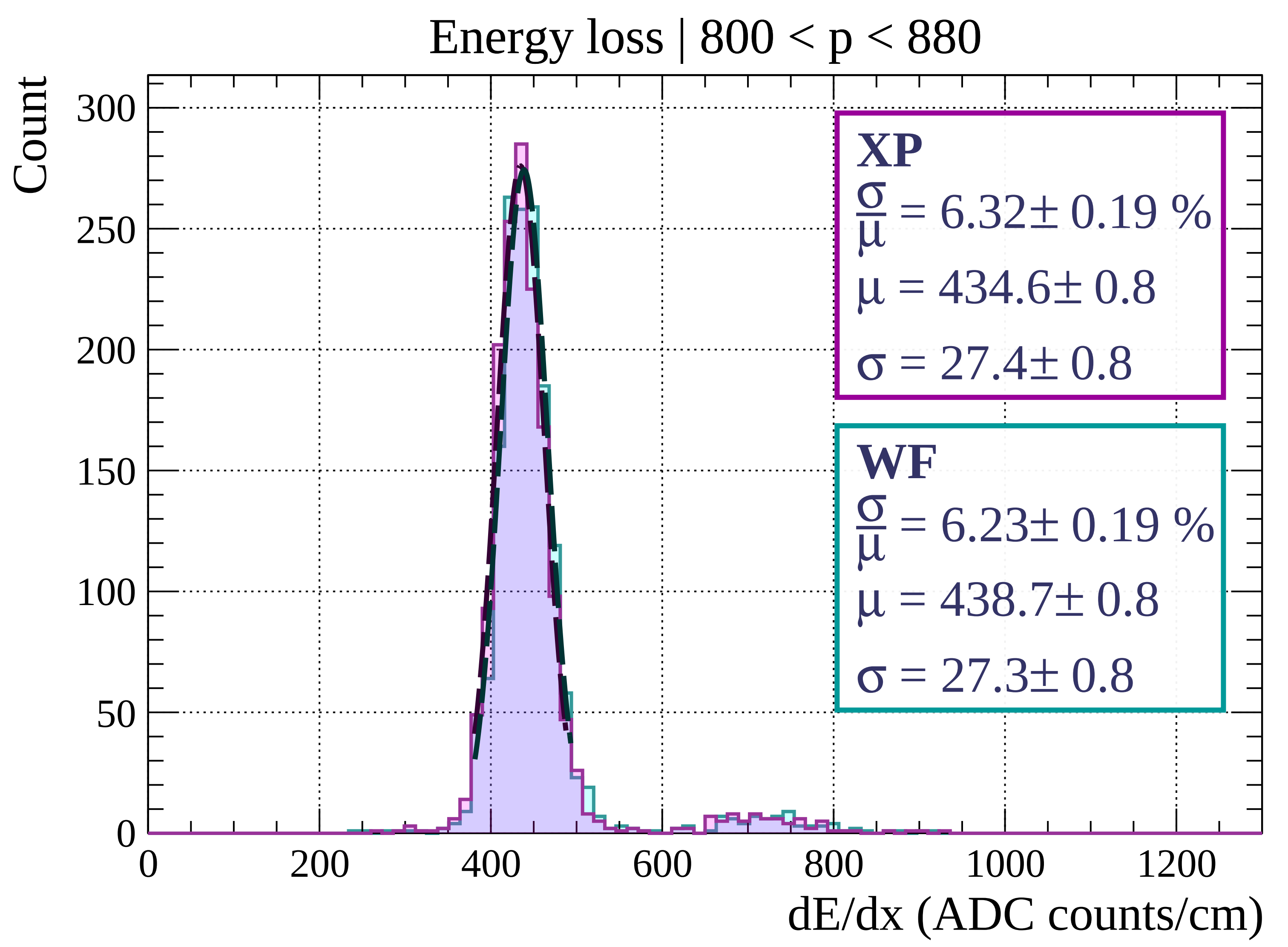


Energy loss | 560 < p < 640
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.31± 0.15 %
μ = 422.5± 0.7
σ = 26.7± 0.6
WF
$\frac{\sigma}{\mu}$ = 6.32± 0.14 %
μ = 425.4± 0.7
σ = 26.9± 0.6
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

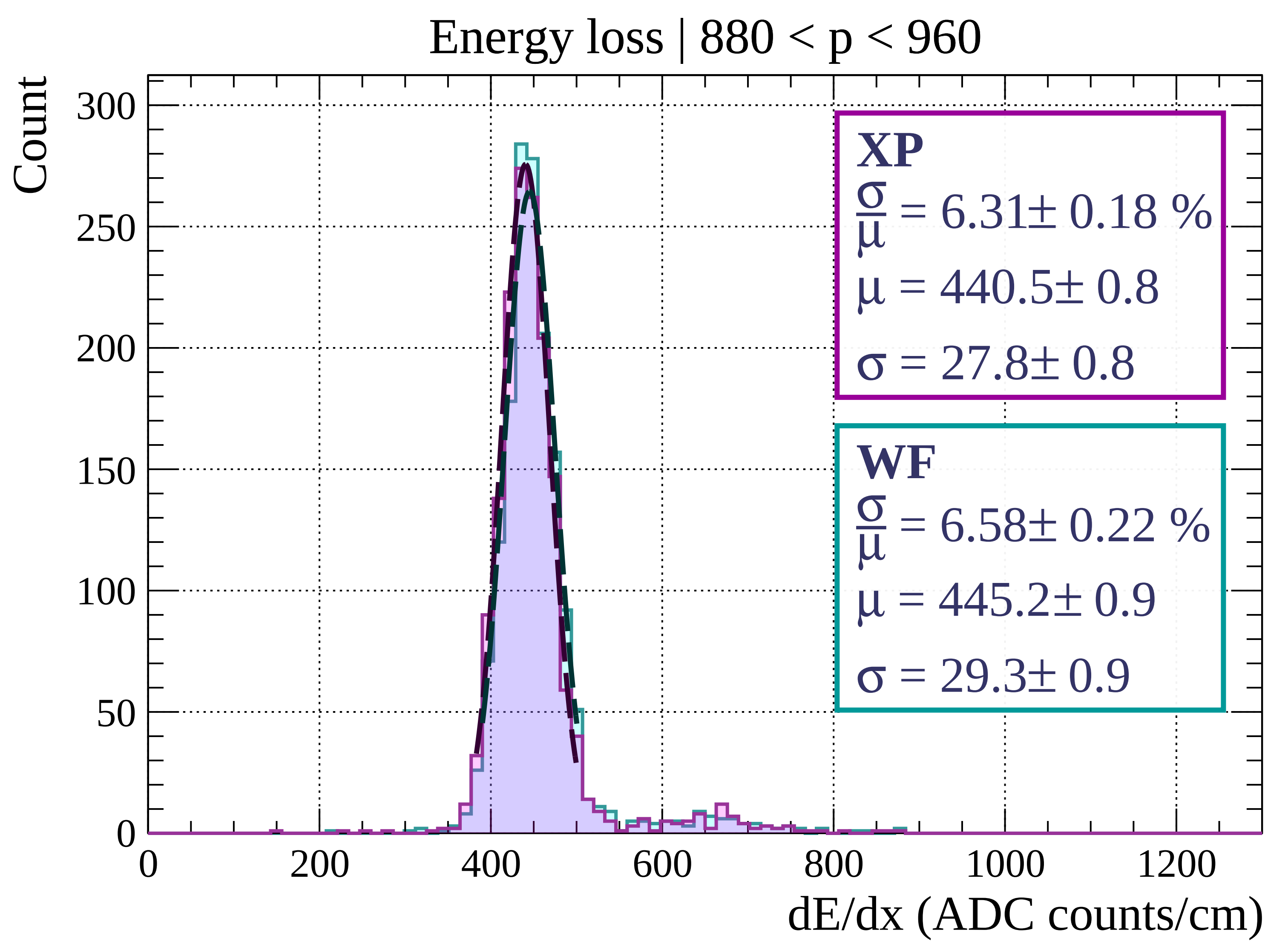

Energy loss | 640 < p < 720
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 5.75± 0.18 %
μ = 423.6± 0.7
σ = 24.4± 0.7
WF
σ/μ = 6.28± 0.19 %
μ = 427.6± 0.7
σ = 26.9± 0.8

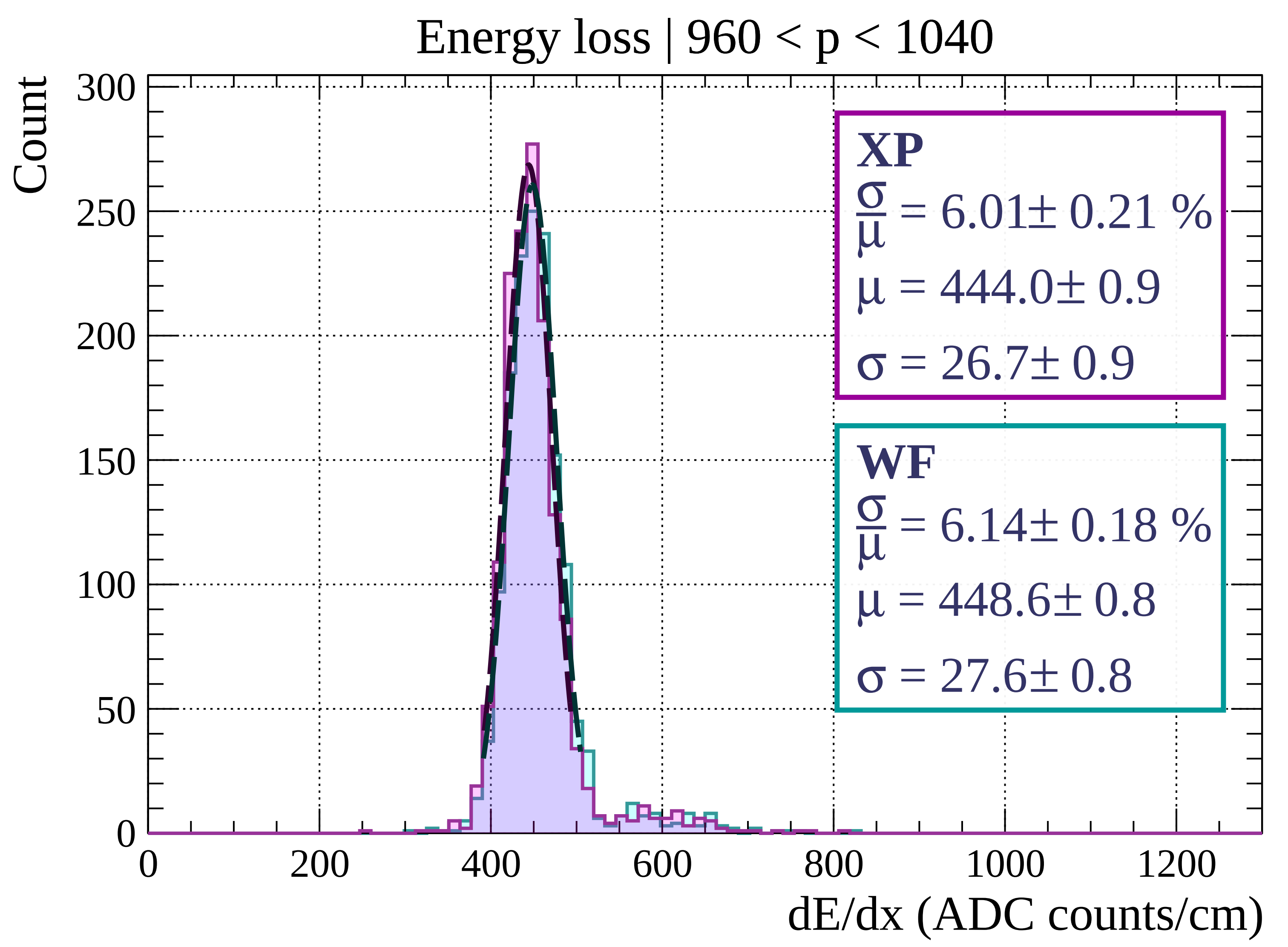


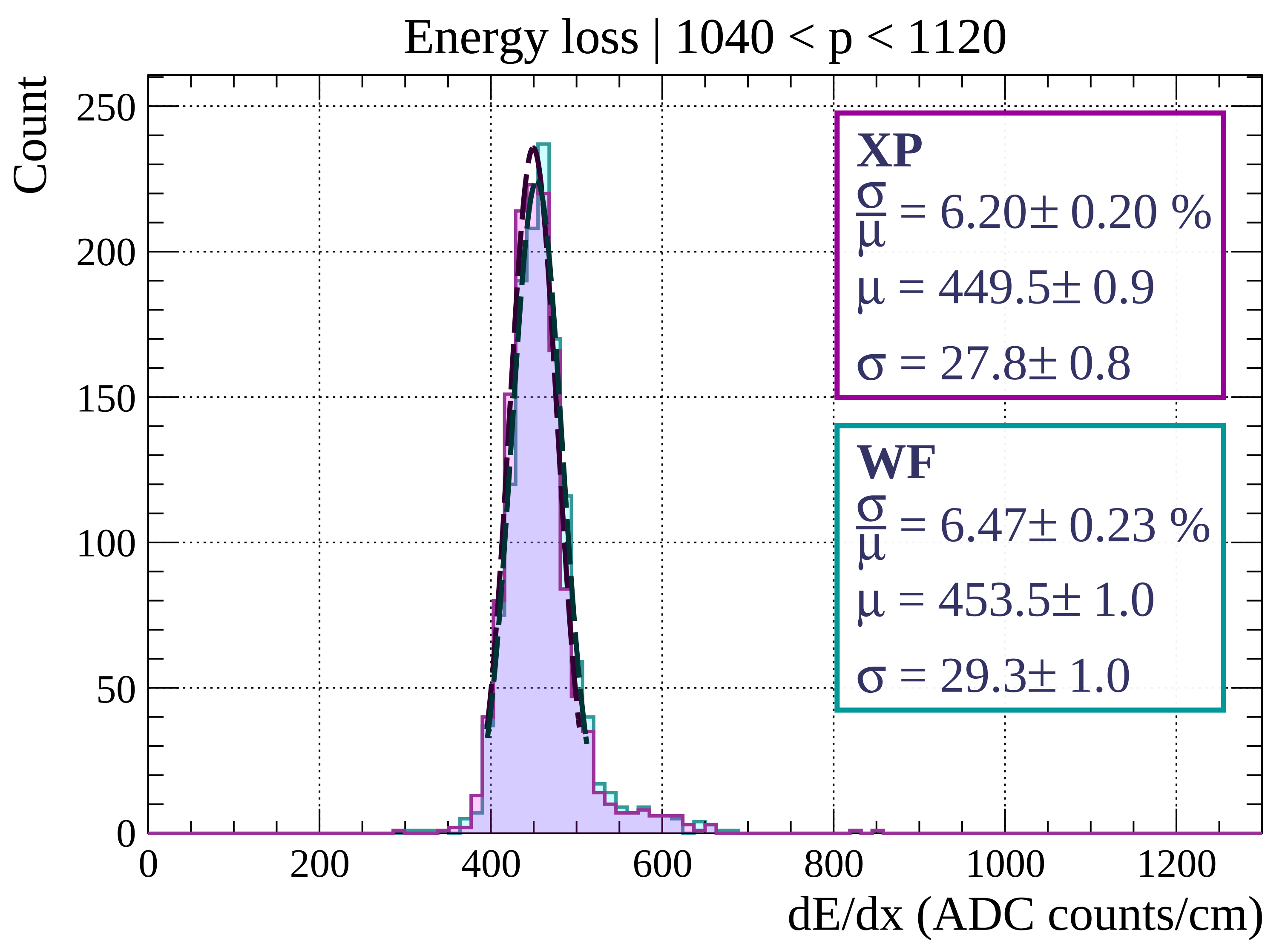
Energy loss | 720 < p < 800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
$\frac{\sigma}{\mu}$ = 6.29± 0.21 %
μ = 431.7± 0.8
σ = 27.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.35± 0.17 %
μ = 435.6± 0.8
σ = 27.7± 0.7



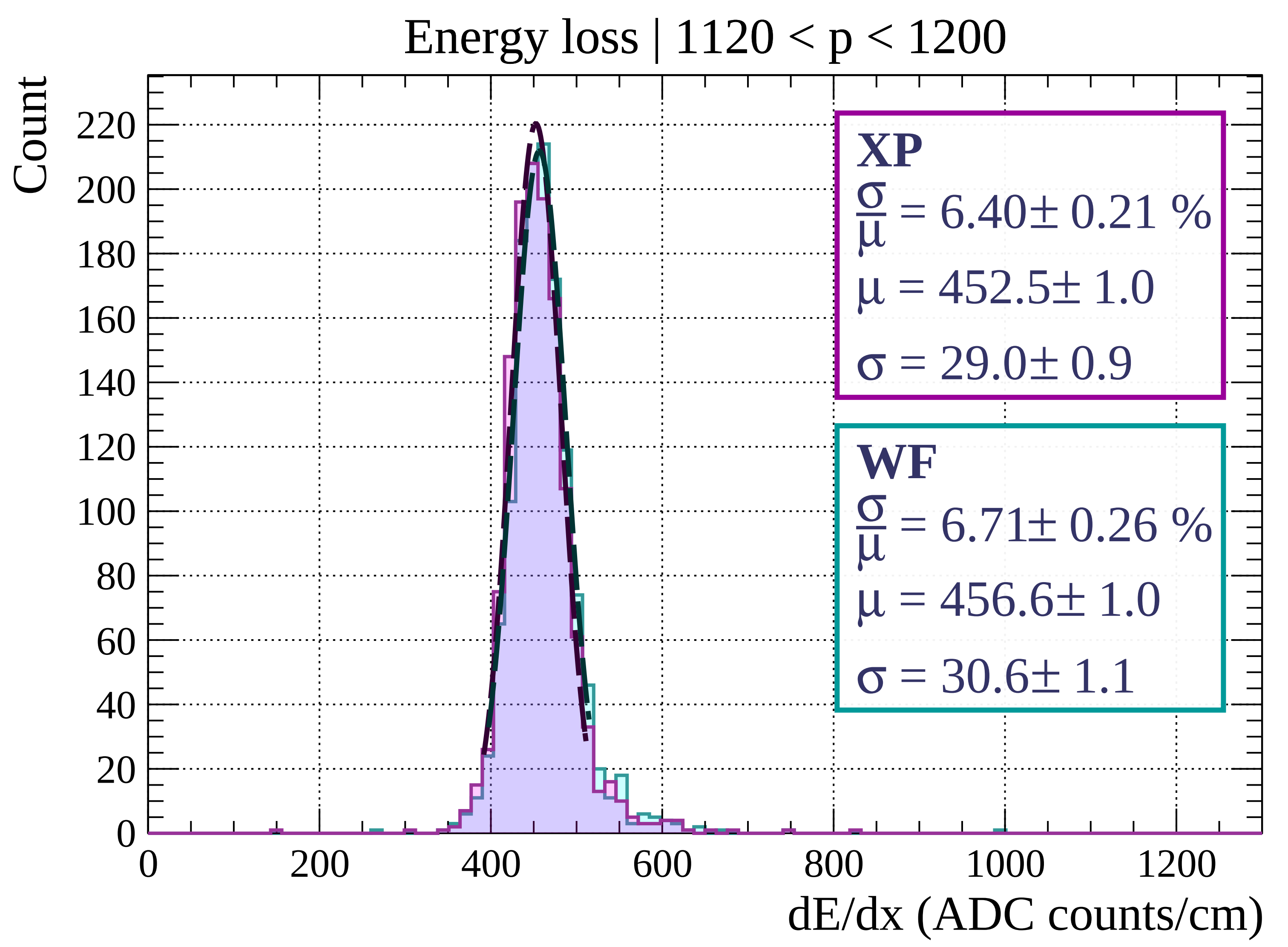
Energy loss | 800 < p < 880
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.32± 0.19 %
μ = 434.6± 0.8
σ = 27.4± 0.8
WF
σ/μ = 6.23± 0.19 %
μ = 438.7± 0.8
σ = 27.3± 0.8


Energy loss | 880 < p < 960
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 6.31± 0.18 %
μ = 440.5± 0.8
σ = 27.8± 0.8
WF
σ/μ = 6.58± 0.22 %
μ = 445.2± 0.9
σ = 29.3± 0.9

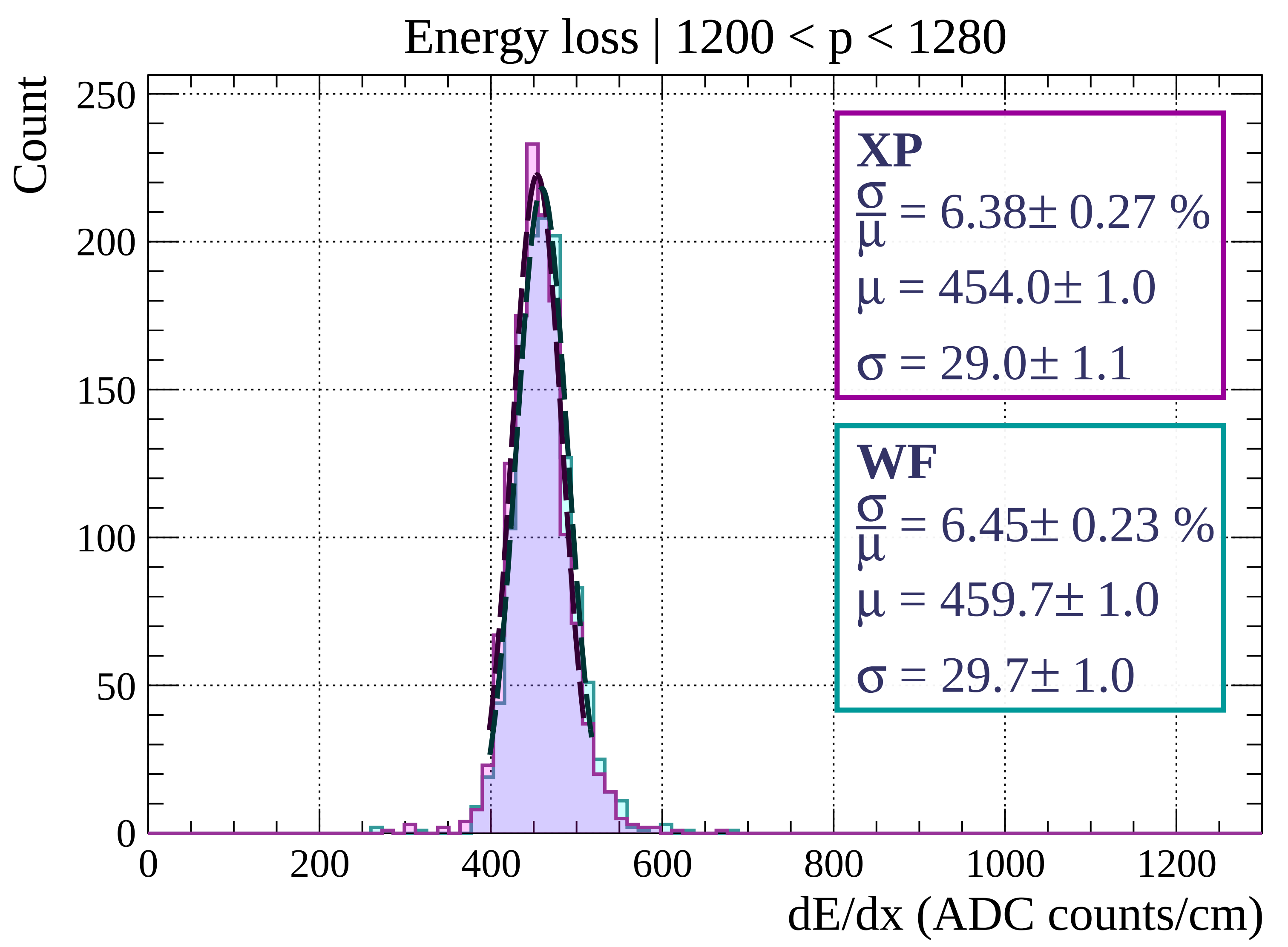

Energy loss | 960 < p < 1040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 6.01± 0.21 %
μ = 444.0± 0.9
σ = 26.7± 0.9
WF
σ/μ = 6.14± 0.18 %
μ = 448.6± 0.8
σ = 27.6± 0.8



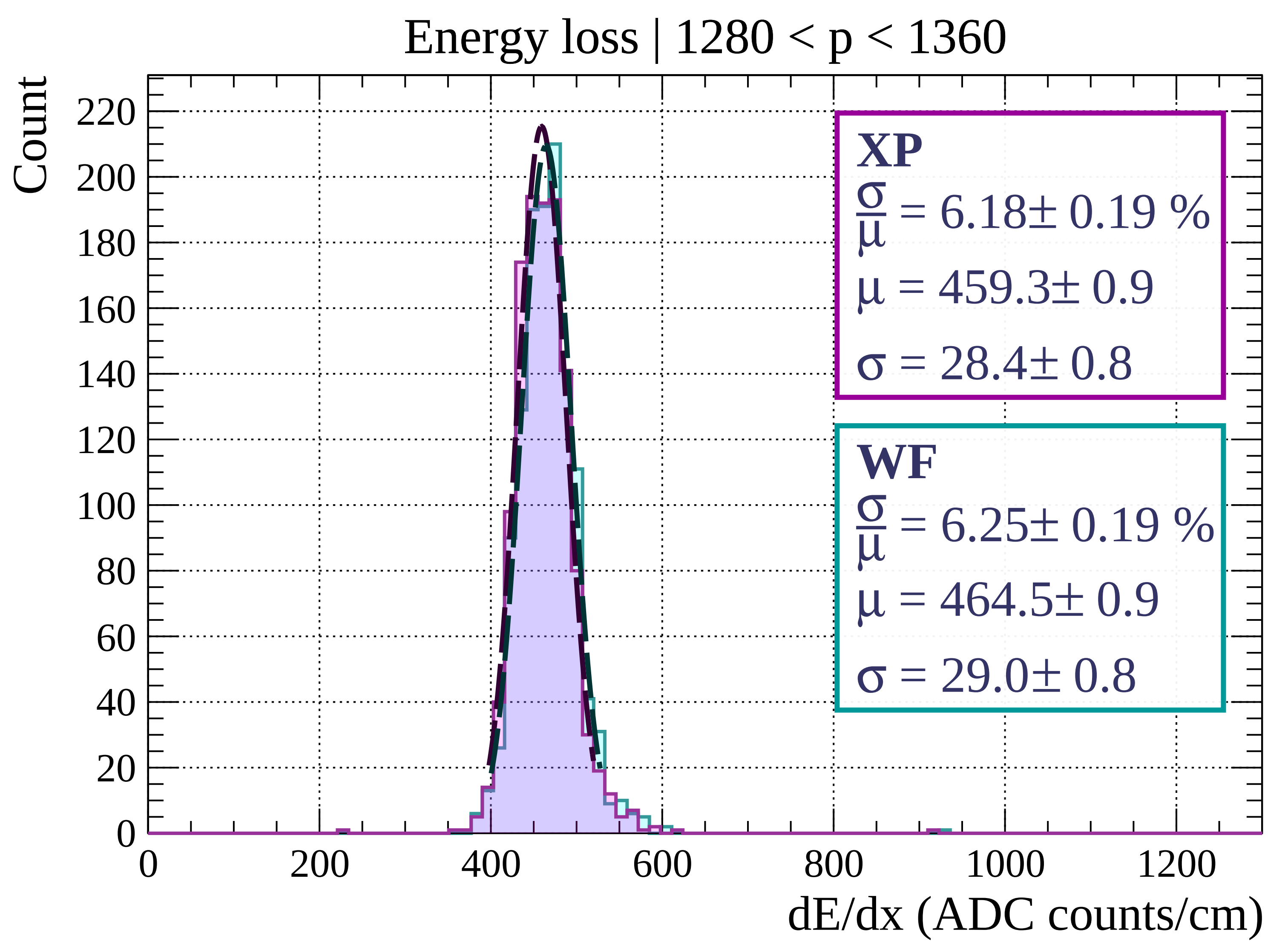
Energy loss | 1040 < p < 1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 6.20± 0.20 %
μ = 449.5± 0.9
σ = 27.8± 0.8
WF
σ/μ = 6.47± 0.23 %
μ = 453.5± 1.0
σ = 29.3± 1.0

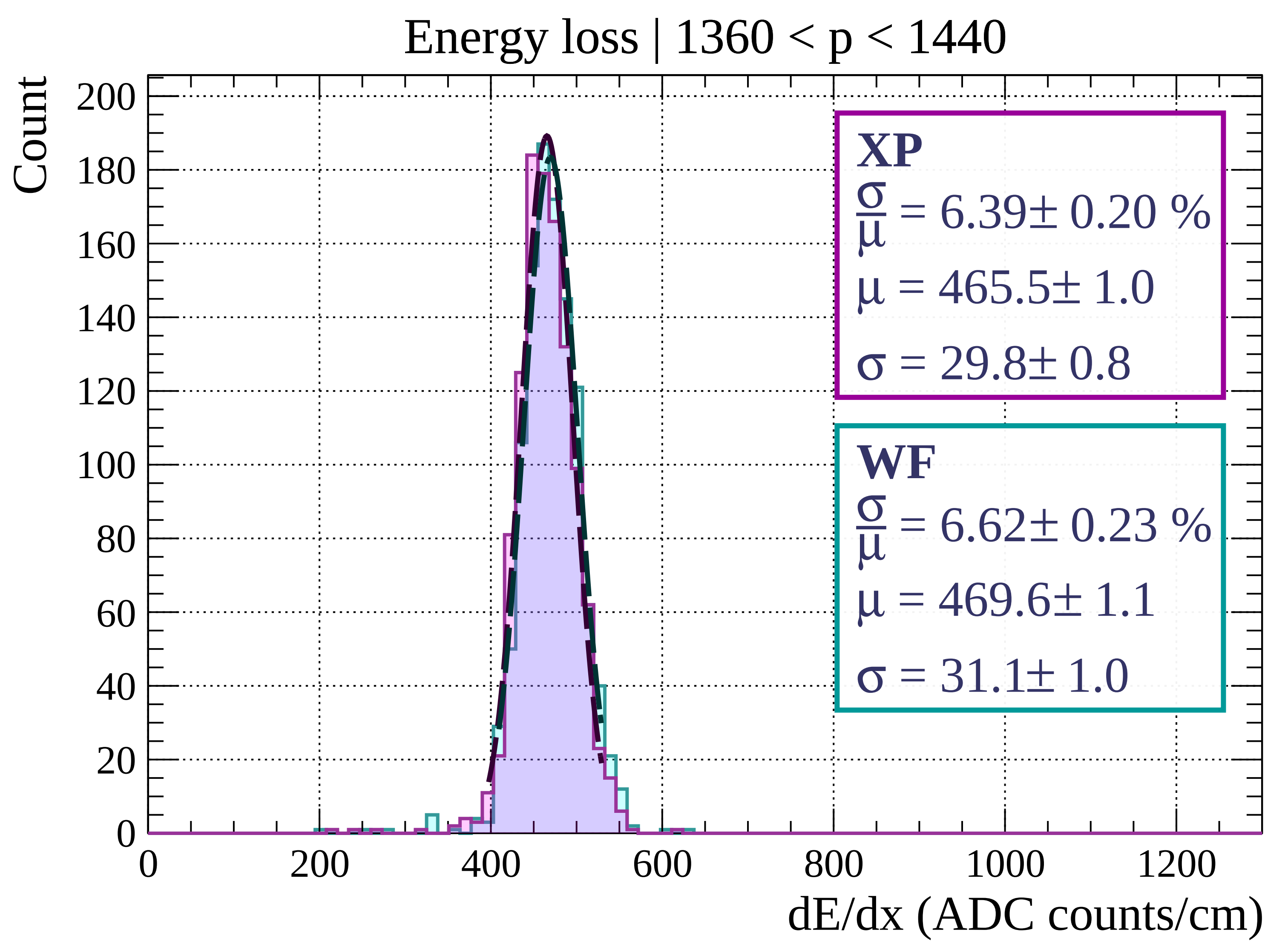


Energy loss | 1120 < p < 1200
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.40± 0.21 %
μ = 452.5± 1.0
σ = 29.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.71± 0.26 %
μ = 456.6± 1.0
σ = 30.6± 1.1

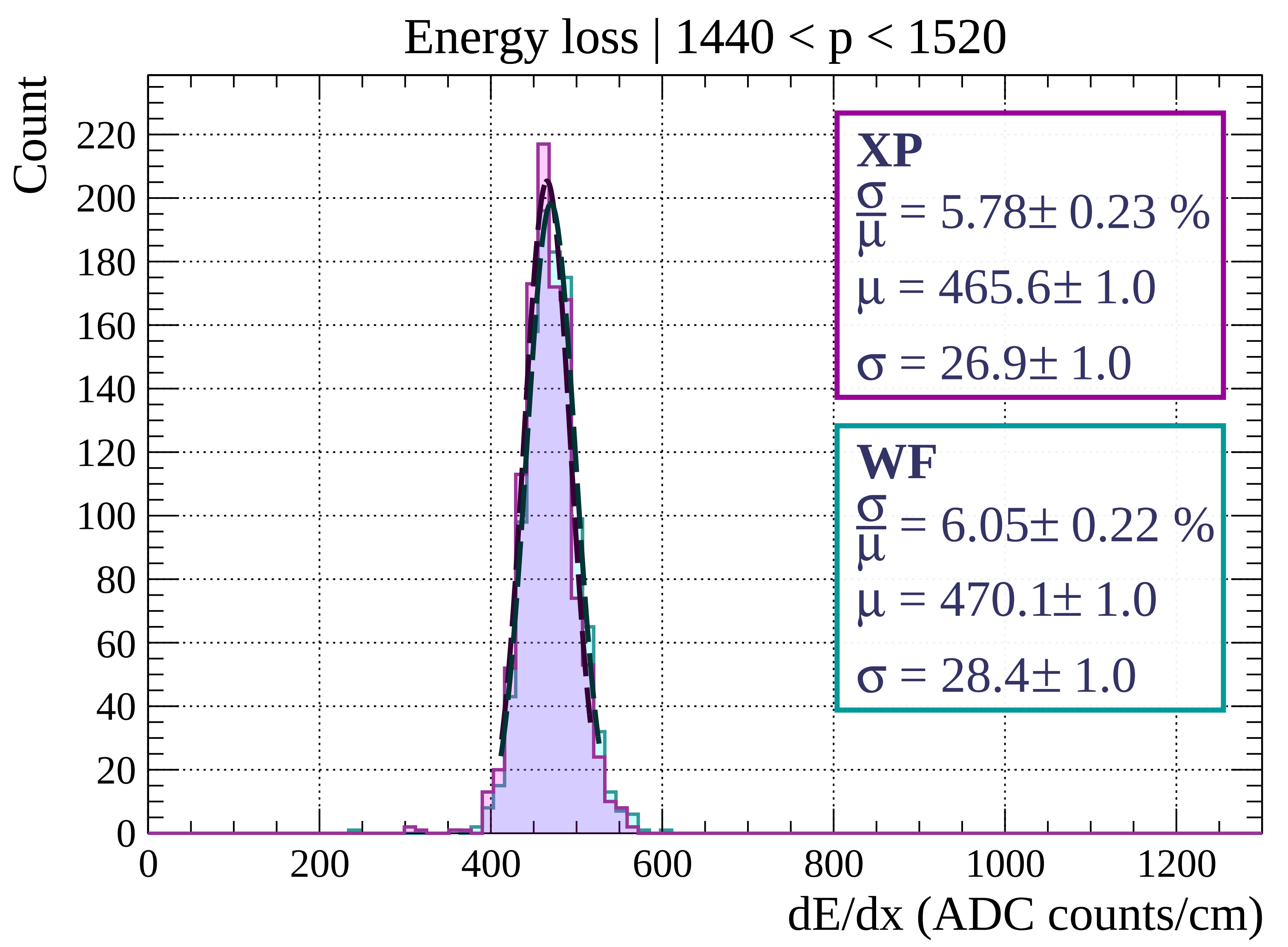


Energy loss | 1200 < p < 1280
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.38± 0.27 %
μ = 454.0± 1.0
σ = 29.0± 1.1
WF
$\frac{\sigma}{\mu}$ = 6.45± 0.23 %
μ = 459.7± 1.0
σ = 29.7± 1.0

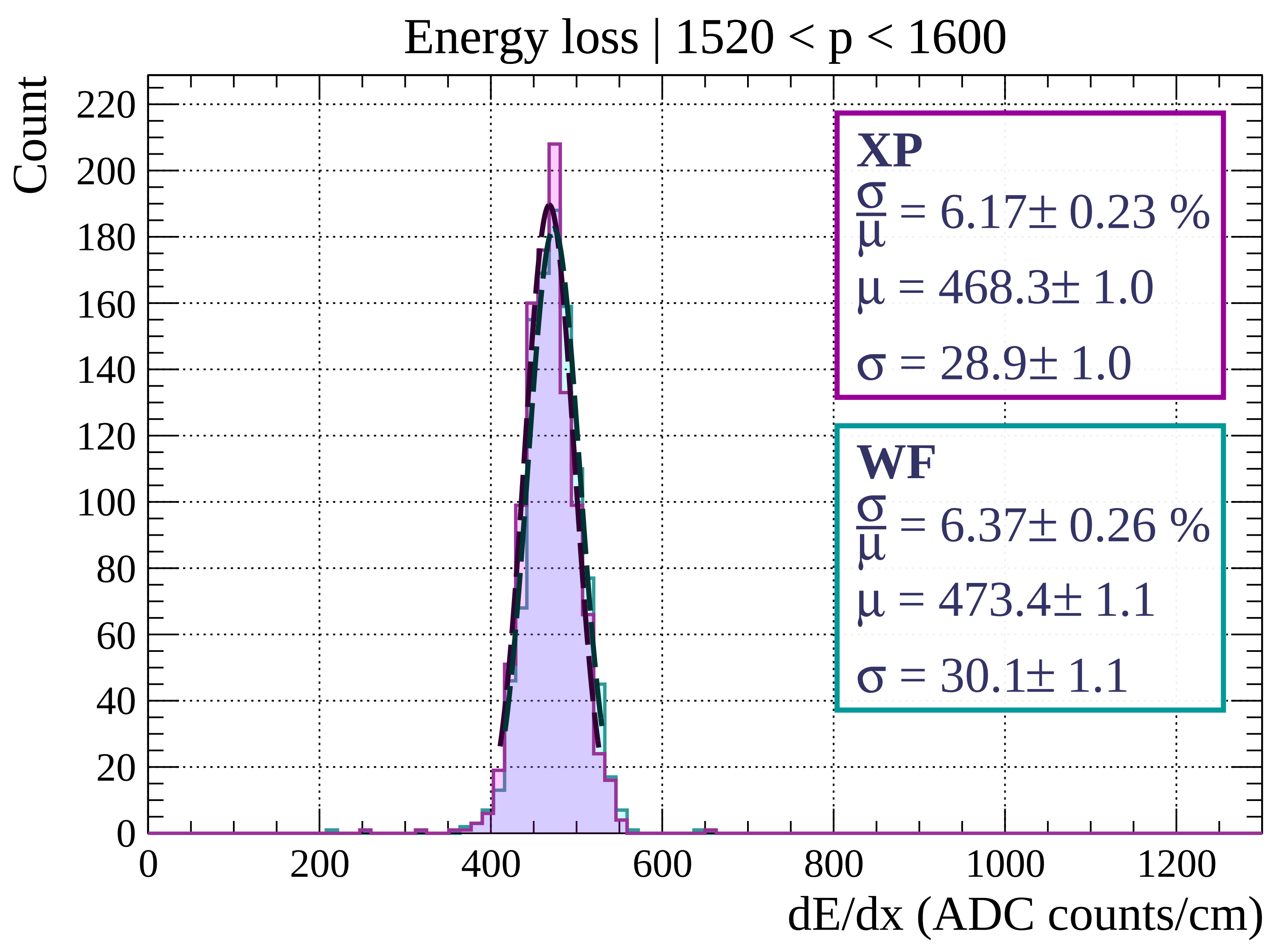

Energy loss | 1280 < p < 1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu}$ = 6.18± 0.19 %
μ = 459.3± 0.9
σ = 28.4± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.25± 0.19 %
μ = 464.5± 0.9
σ = 29.0± 0.8

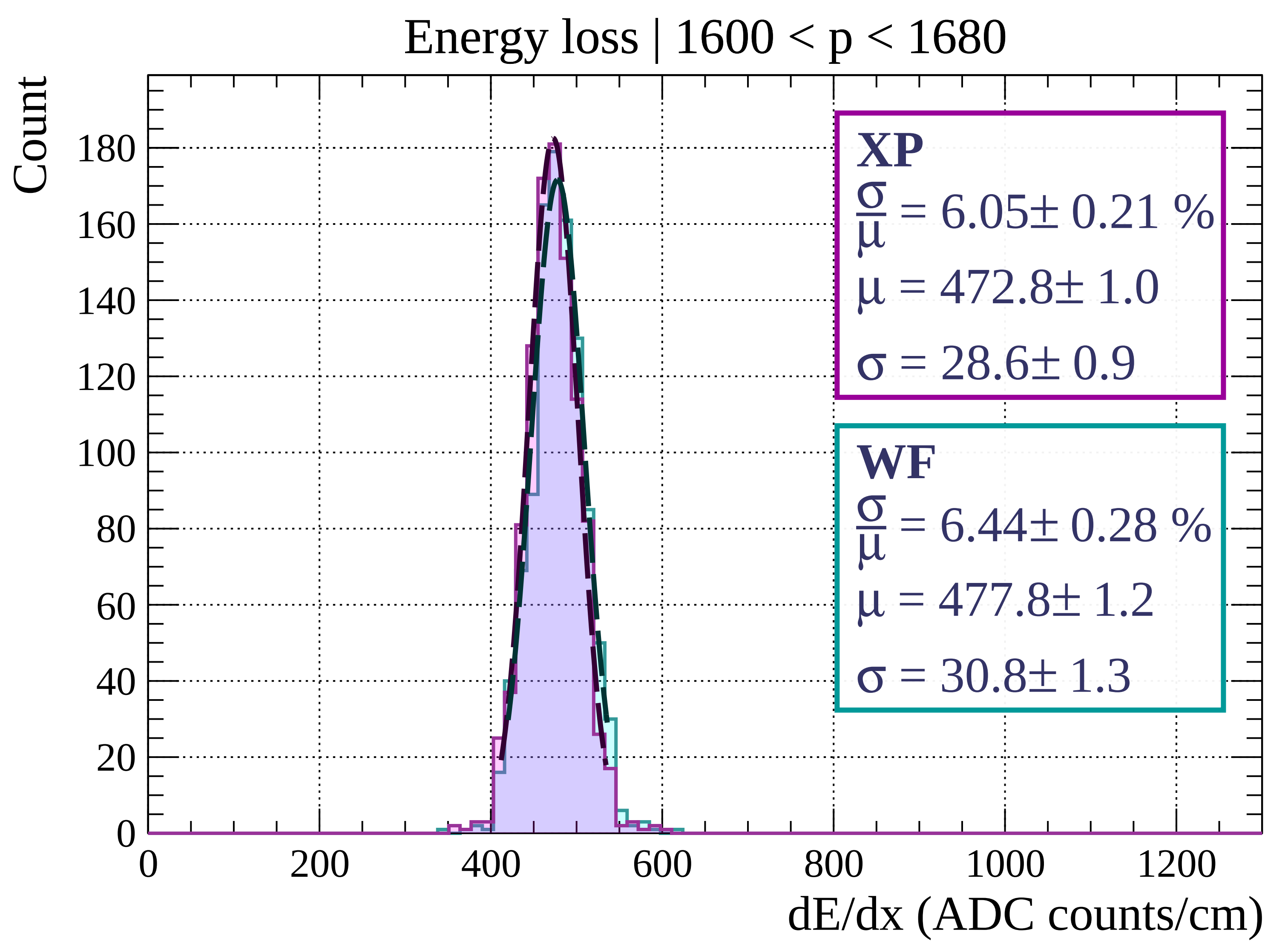


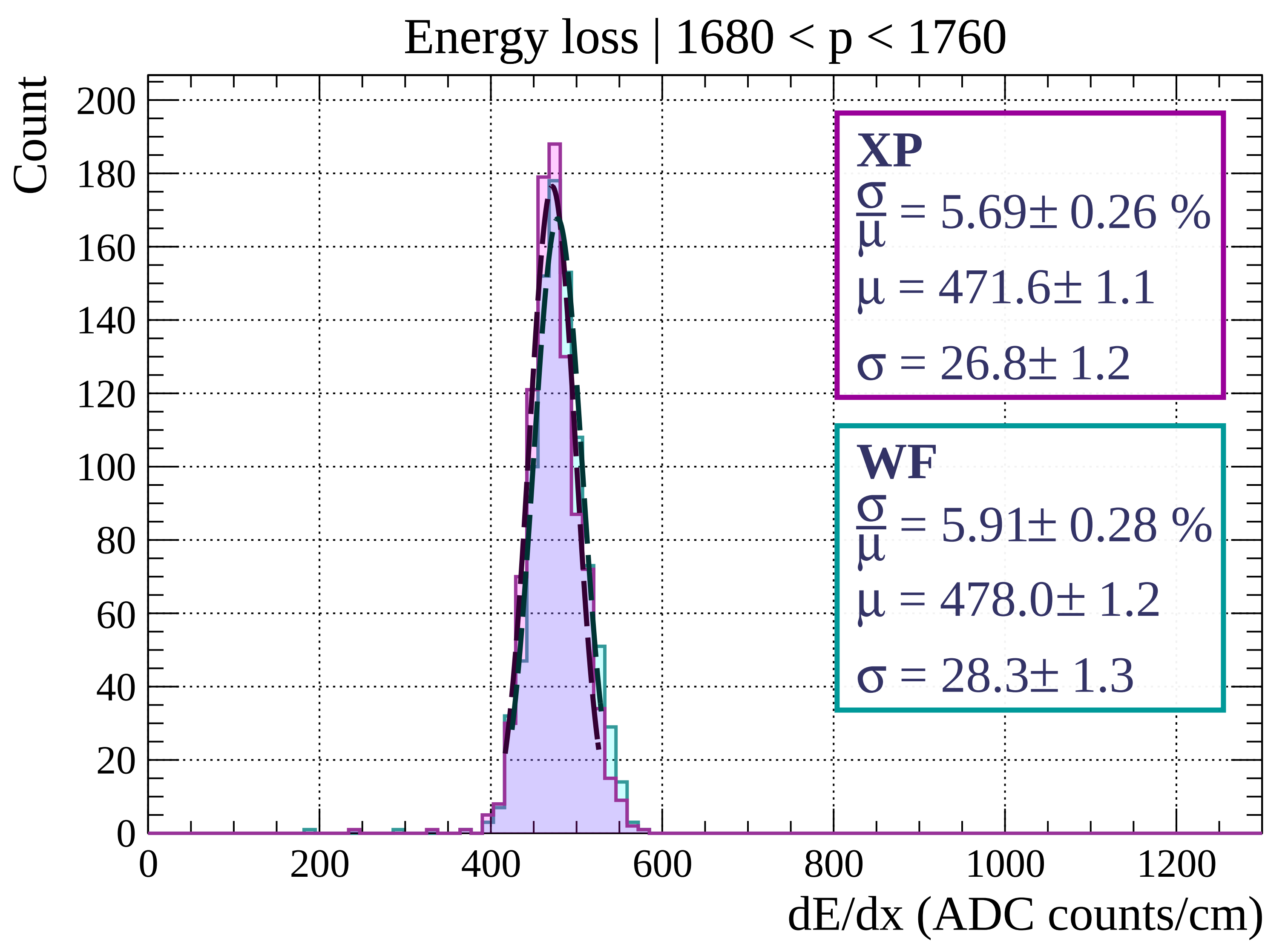
Energy loss | 1360 < p < 1440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.39± 0.20 %
μ = 465.5± 1.0
σ = 29.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 6.62± 0.23 %
μ = 469.6± 1.1
σ = 31.1± 1.0



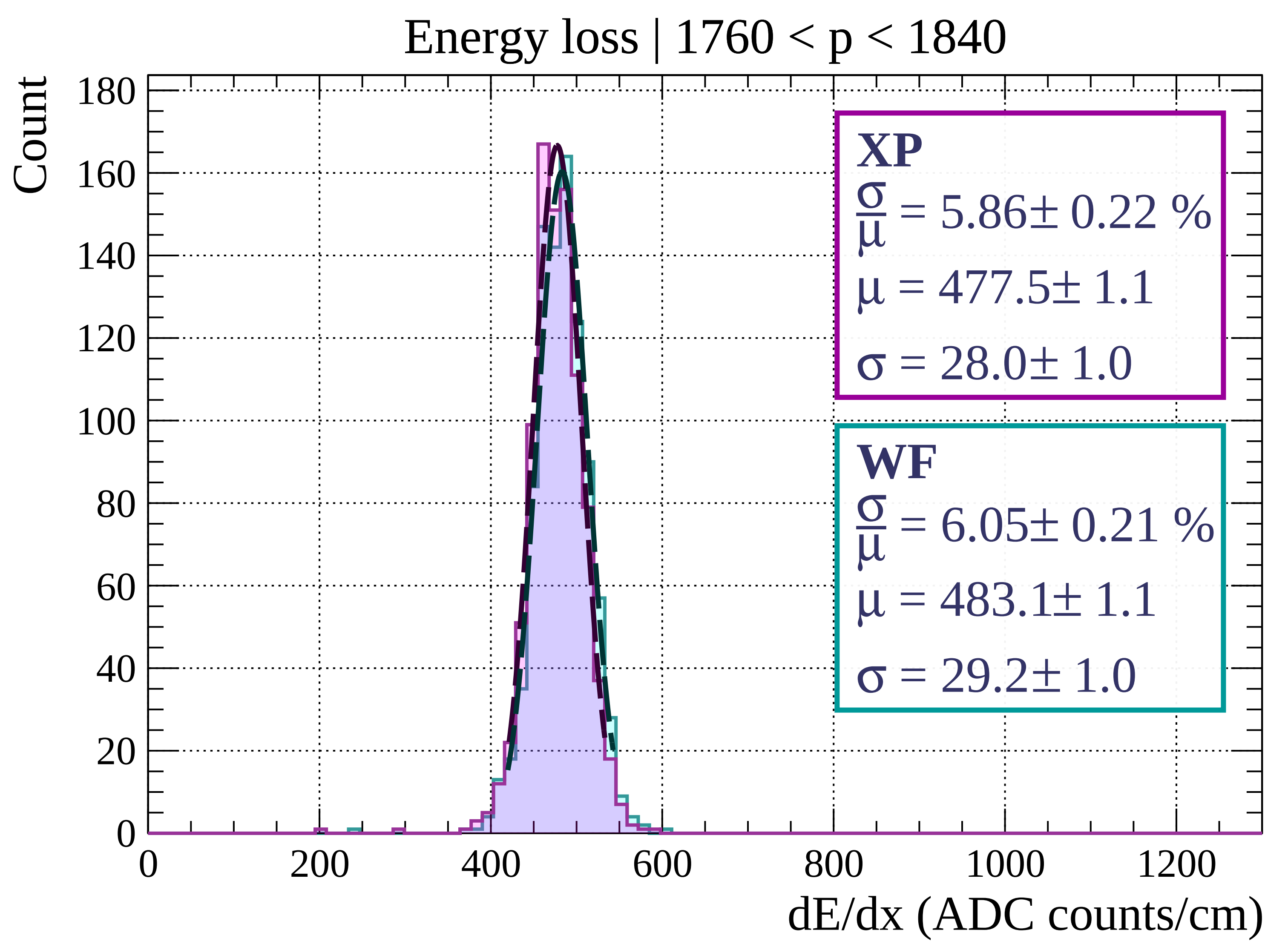
Energy loss | 1440 < p < 1520
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 5.78± 0.23 %
μ = 465.6± 1.0
σ = 26.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.05± 0.22 %
μ = 470.1± 1.0
σ = 28.4± 1.0


Energy loss | 1520 < p < 1600
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
220
XP
$\frac{\sigma}{\mu}$ = 6.17± 0.23 %
μ = 468.3± 1.0
σ = 28.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.37± 0.26 %
μ = 473.4± 1.1
σ = 30.1± 1.1

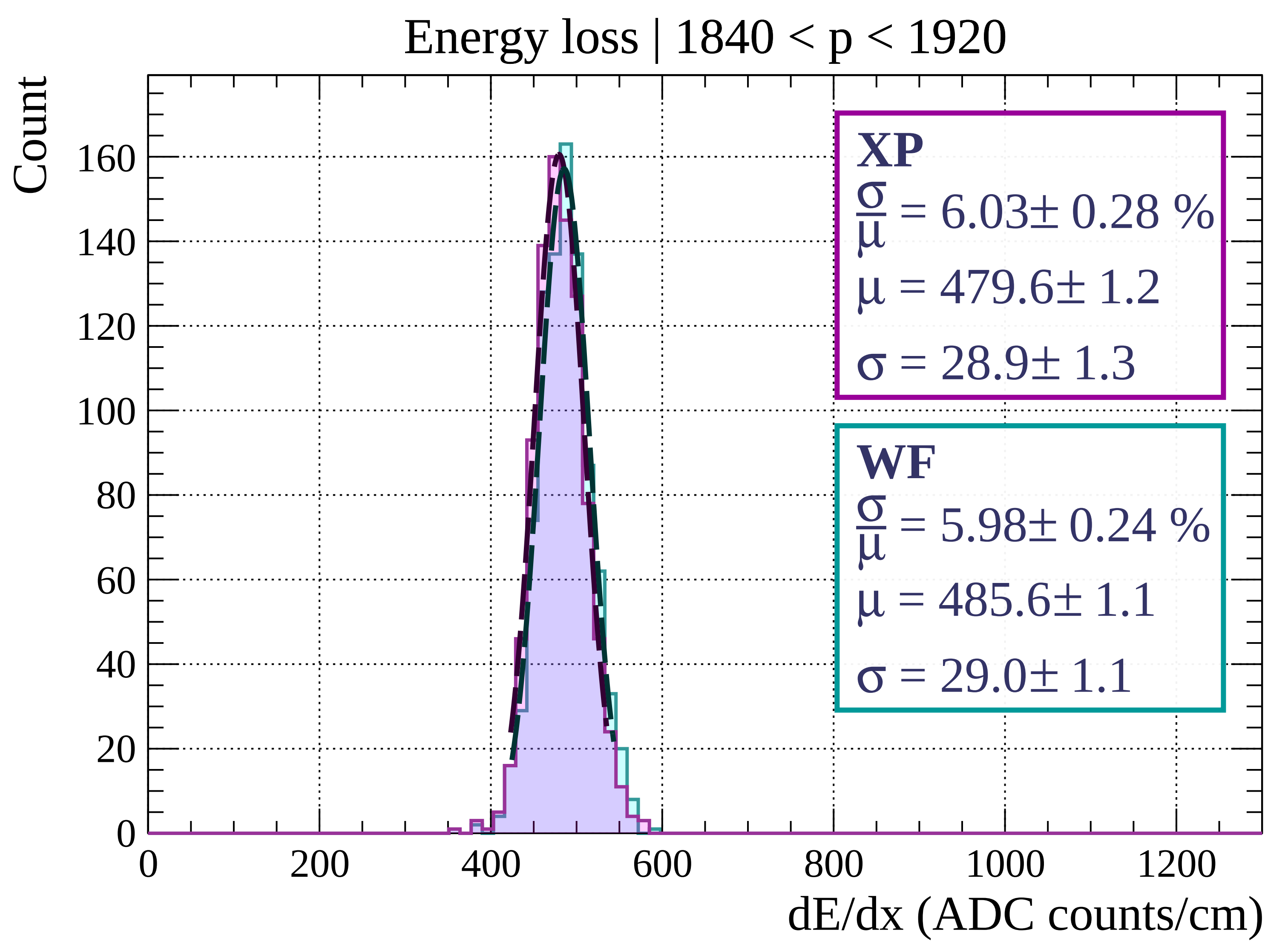

Energy loss | 1600 < p < 1680
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.05± 0.21 %
μ = 472.8± 1.0
σ = 28.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.44± 0.28 %
μ = 477.8± 1.2
σ = 30.8± 1.3

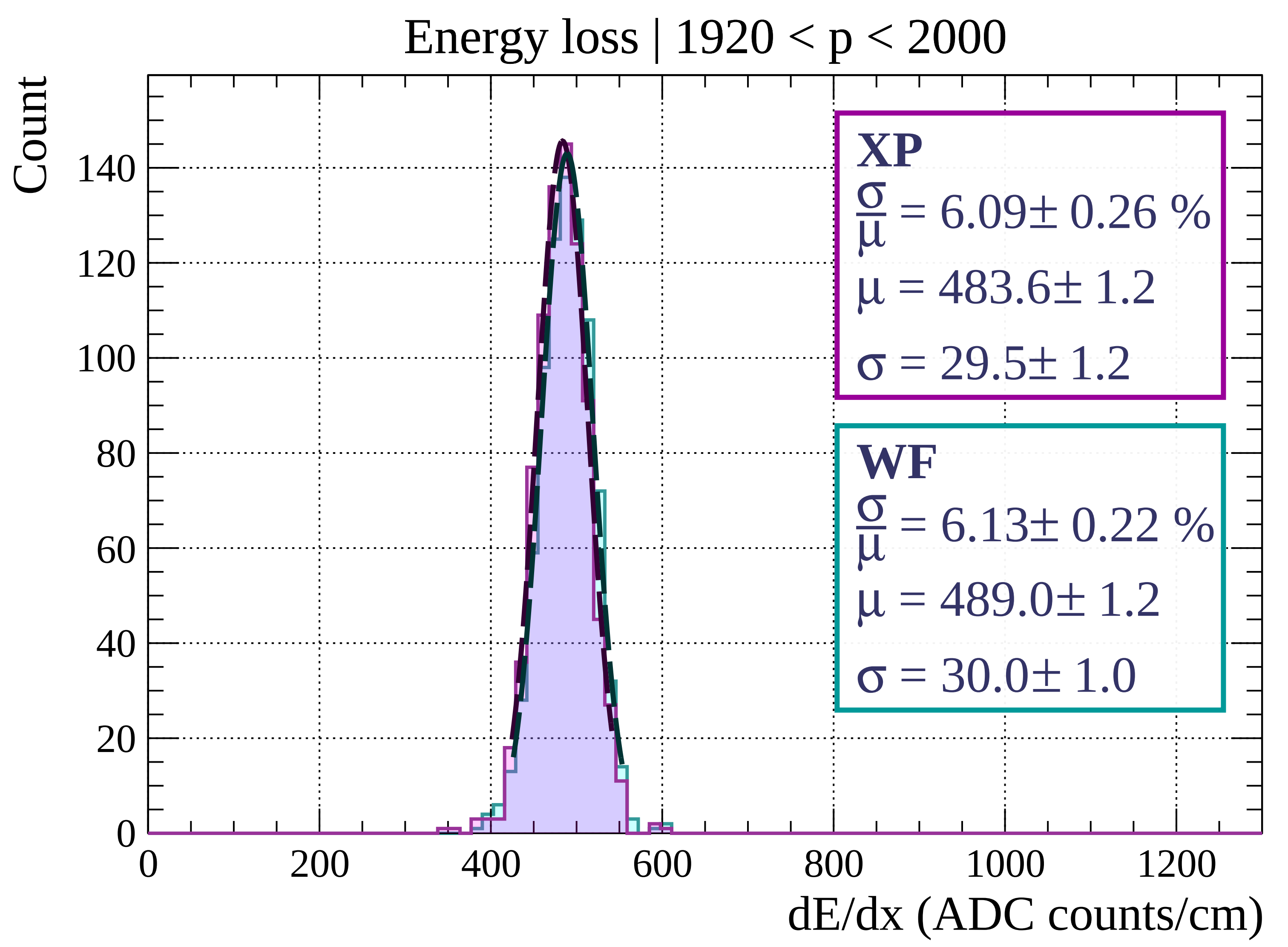


Energy loss | 1680 < p < 1760
Count
0
20
40
60
80
100
120
140
160
180
200
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 5.69± 0.26 %
μ = 471.6± 1.1
σ = 26.8± 1.2
WF
σ/μ = 5.91± 0.28 %
μ = 478.0± 1.2
σ = 28.3± 1.3

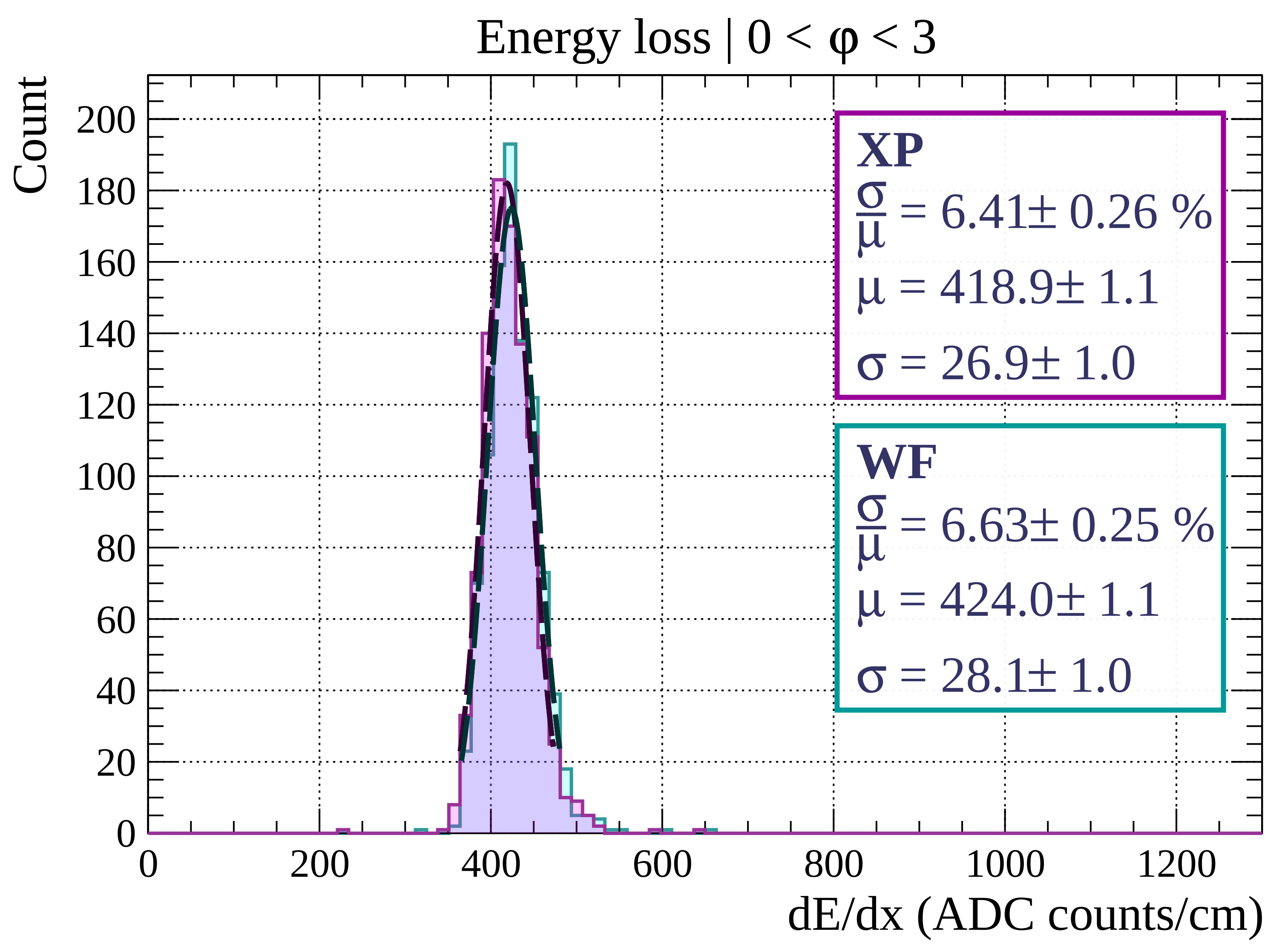


Energy loss | 1760 < p < 1840
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 5.86± 0.22 %
μ = 477.5± 1.1
σ = 28.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.05± 0.21 %
μ = 483.1± 1.1
σ = 29.2± 1.0

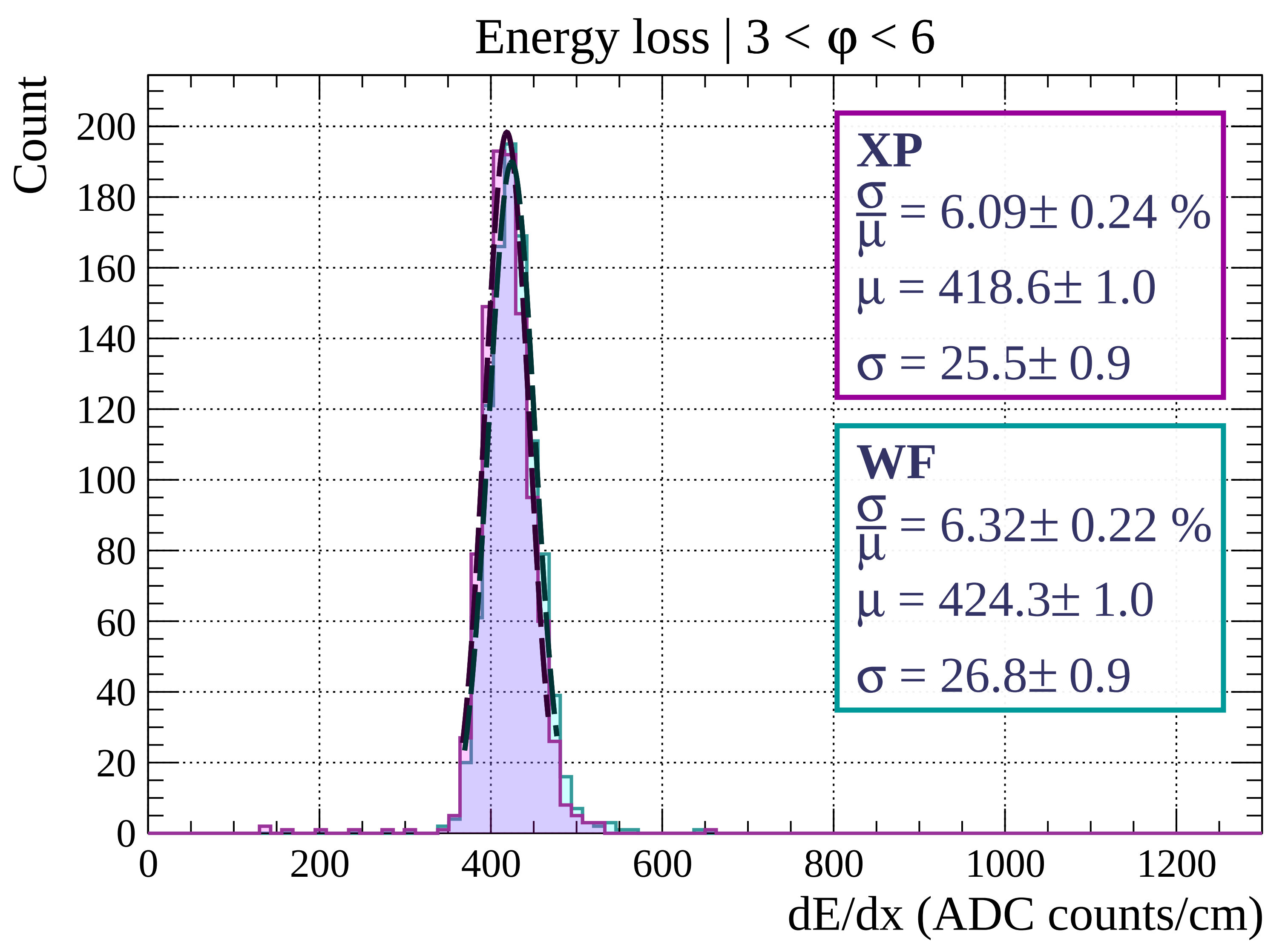

Energy loss | 1840 < p < 1920
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.03± 0.28 %
μ = 479.6± 1.2
σ = 28.9± 1.3
WF
$\frac{\sigma}{\mu}$ = 5.98± 0.24 %
μ = 485.6± 1.1
σ = 29.0± 1.1
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



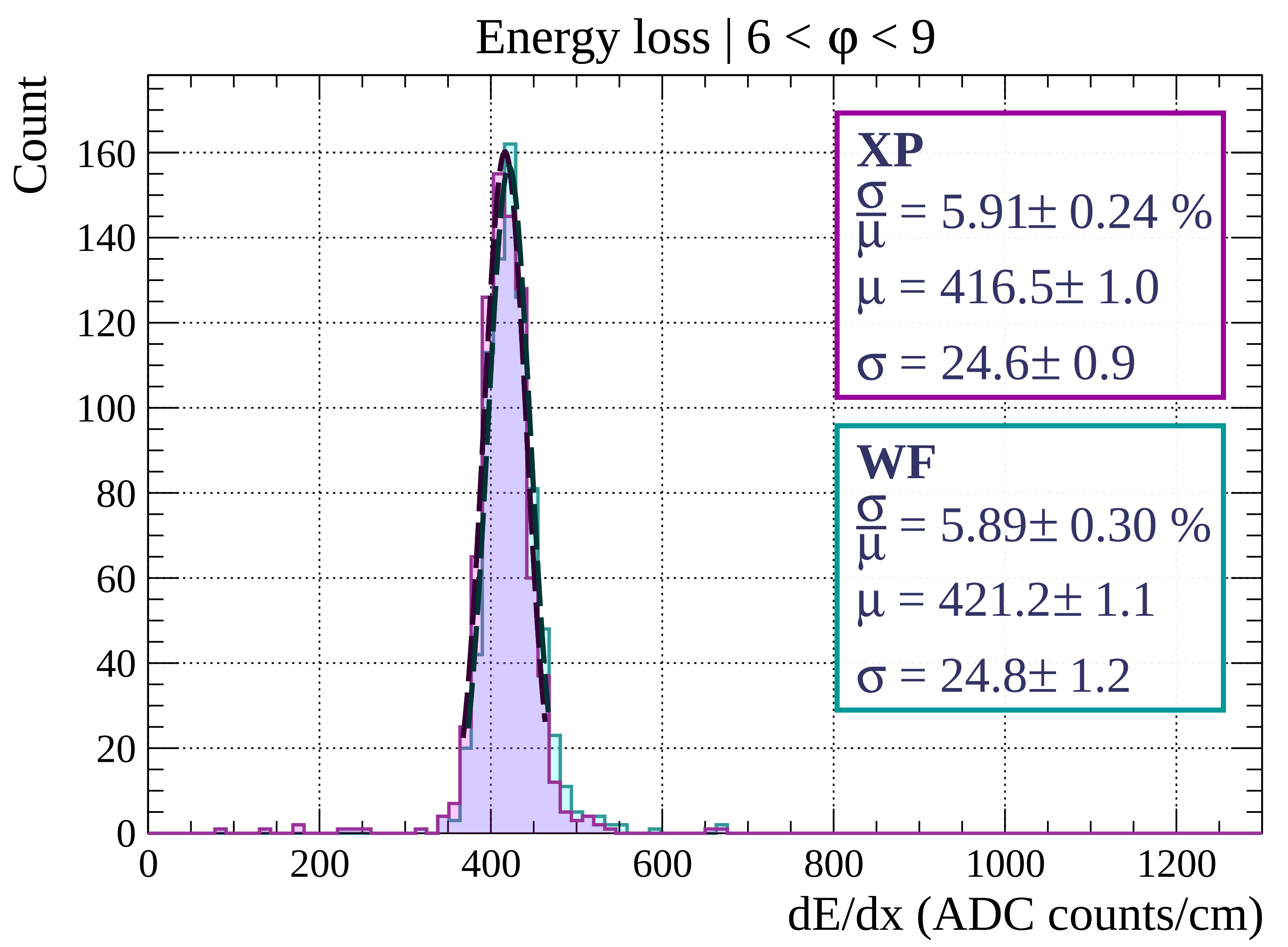
Energy loss | 1920 < p < 2000
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
XP
σ/μ = 6.09± 0.26 %
μ = 483.6± 1.2
σ = 29.5± 1.2
WF
σ/μ = 6.13± 0.22 %
μ = 489.0± 1.2
σ = 30.0± 1.0

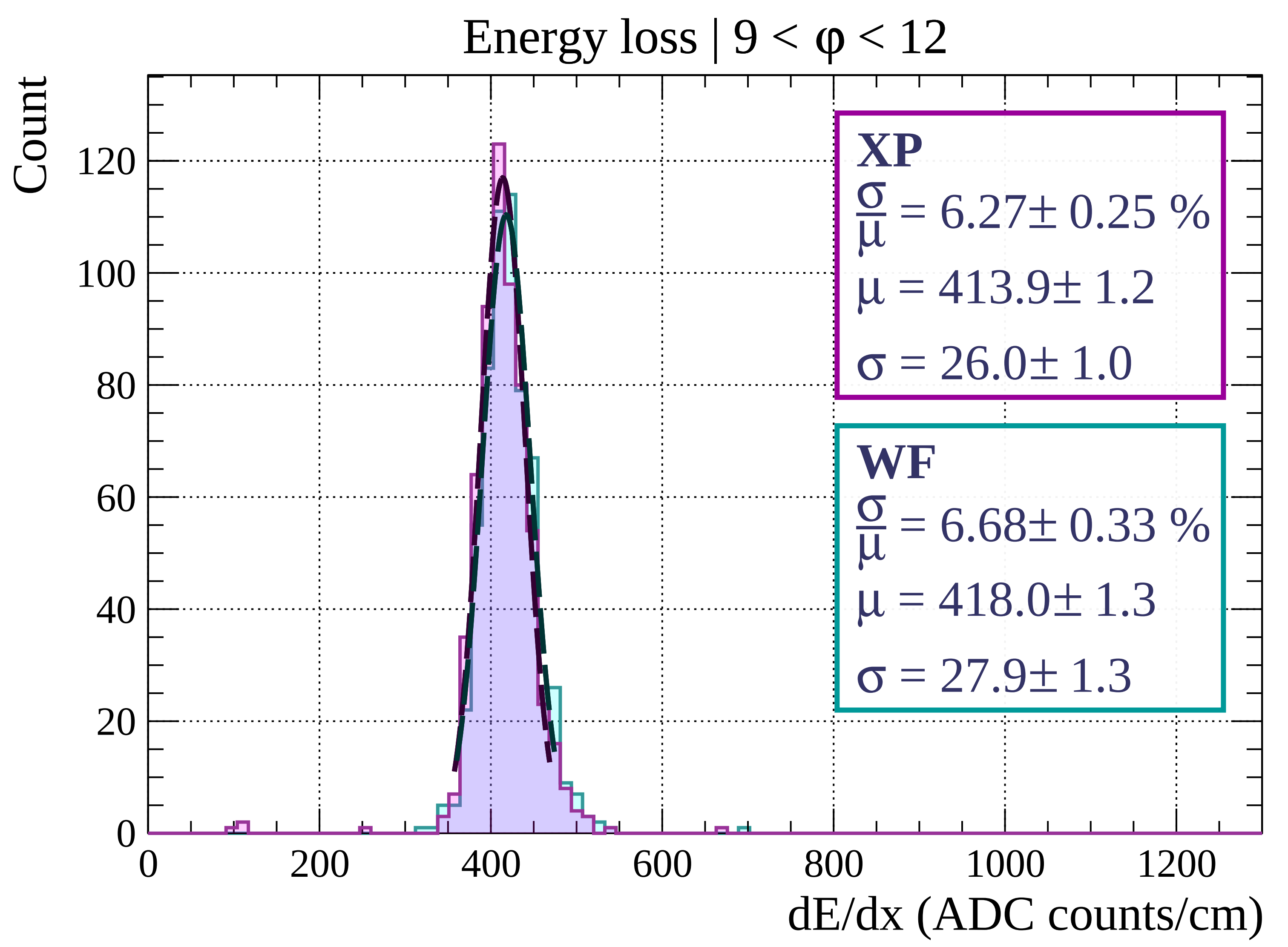


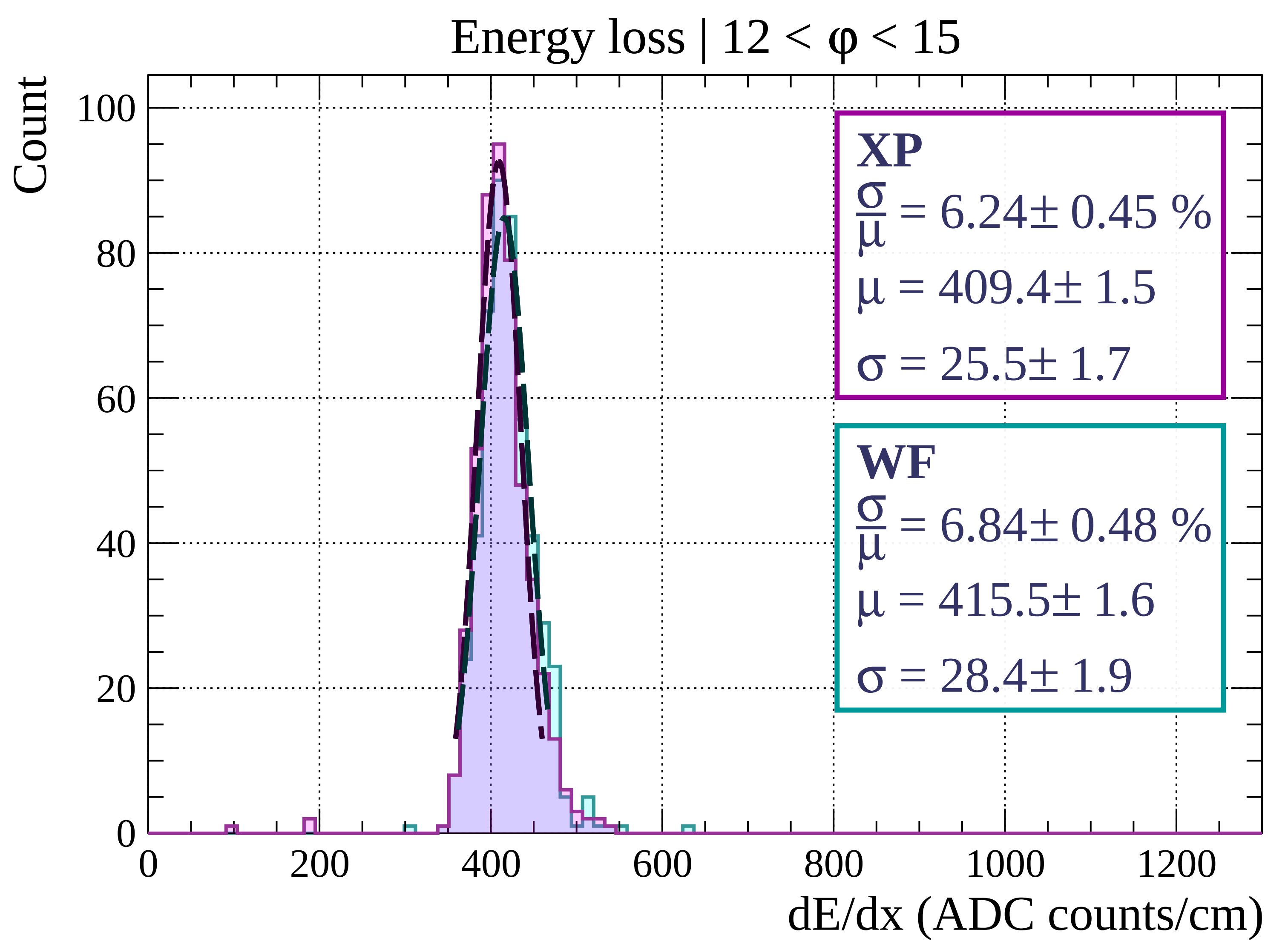
Energy loss | 0 < φ < 3
Count
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.41± 0.26 %
μ = 418.9± 1.1
σ = 26.9± 1.0
WF
σ/μ = 6.63± 0.25 %
μ = 424.0± 1.1
σ = 28.1± 1.0



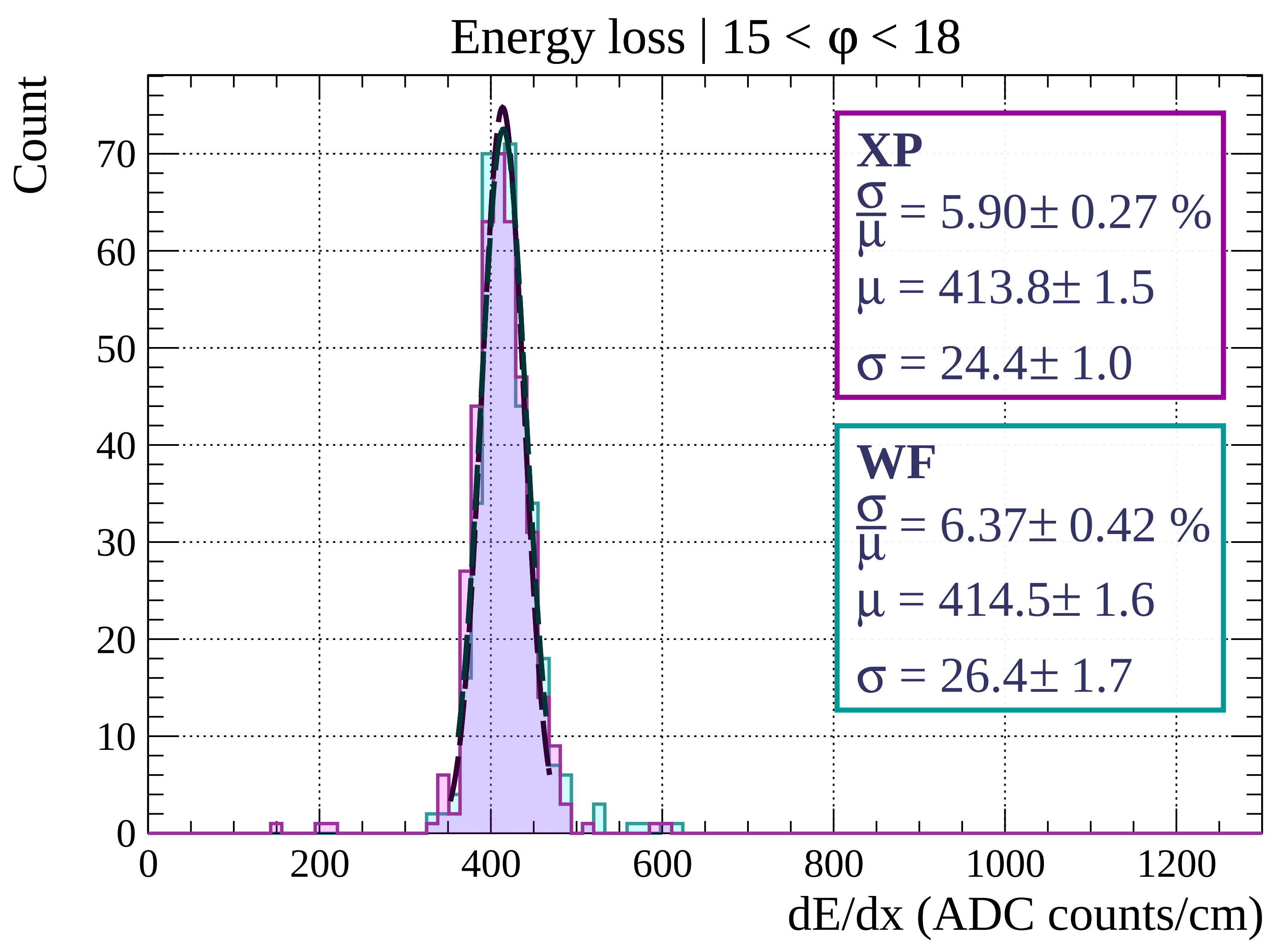
Energy loss | 3 < φ < 6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
σ/μ = 6.09± 0.24 %
μ = 418.6± 1.0
σ = 25.5± 0.9
WF
σ/μ = 6.32± 0.22 %
μ = 424.3± 1.0
σ = 26.8± 0.9


Energy loss | 6 < φ < 9
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
XP
σ/μ = 5.91± 0.24 %
μ = 416.5± 1.0
σ = 24.6± 0.9
WF
σ/μ = 5.89± 0.30 %
μ = 421.2± 1.1
σ = 24.8± 1.2

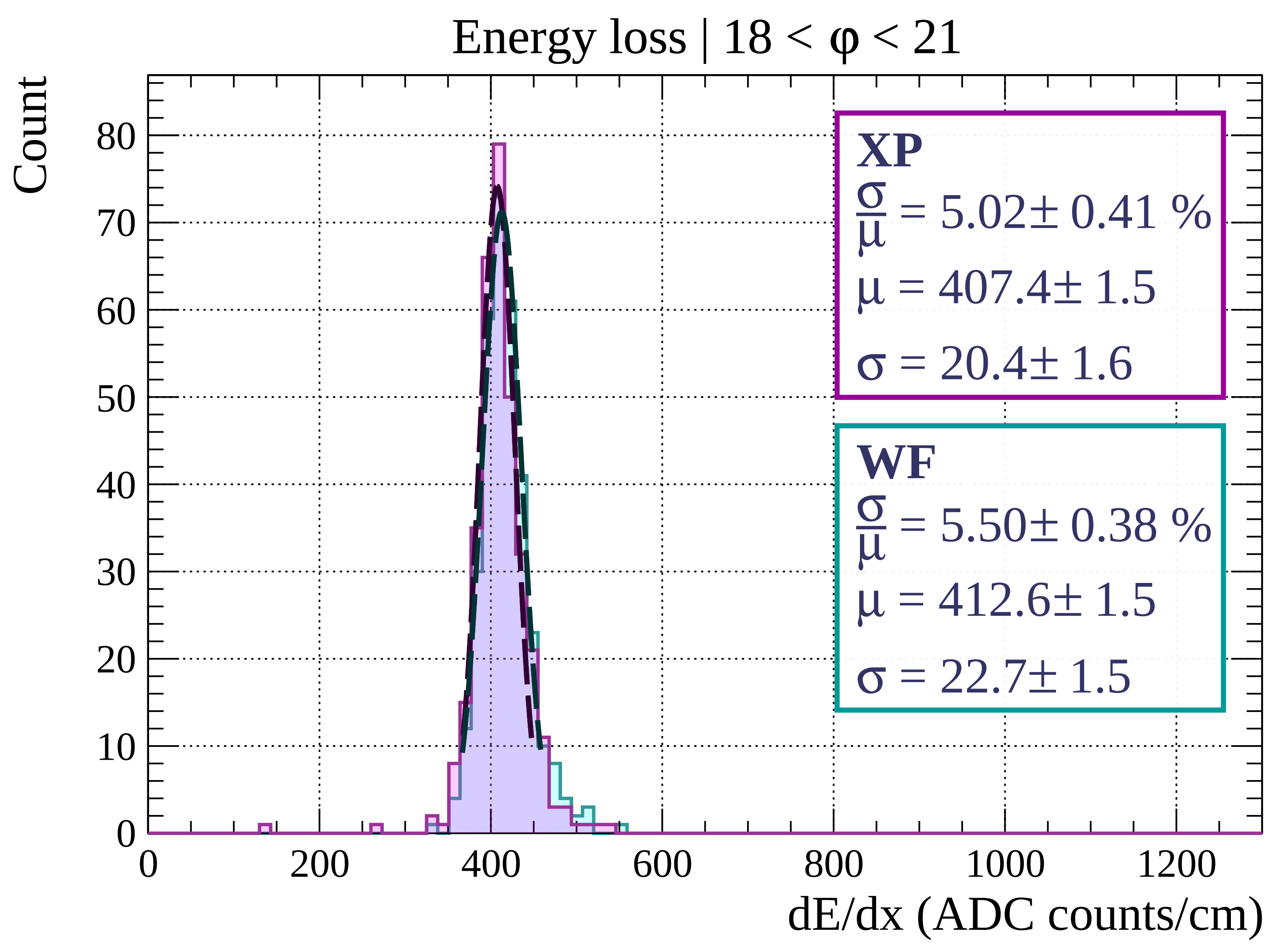

Energy loss | 9 < φ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.27± 0.25 %
μ = 413.9± 1.2
σ = 26.0± 1.0
WF
σ/μ = 6.68± 0.33 %
μ = 418.0± 1.3
σ = 27.9± 1.3
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200

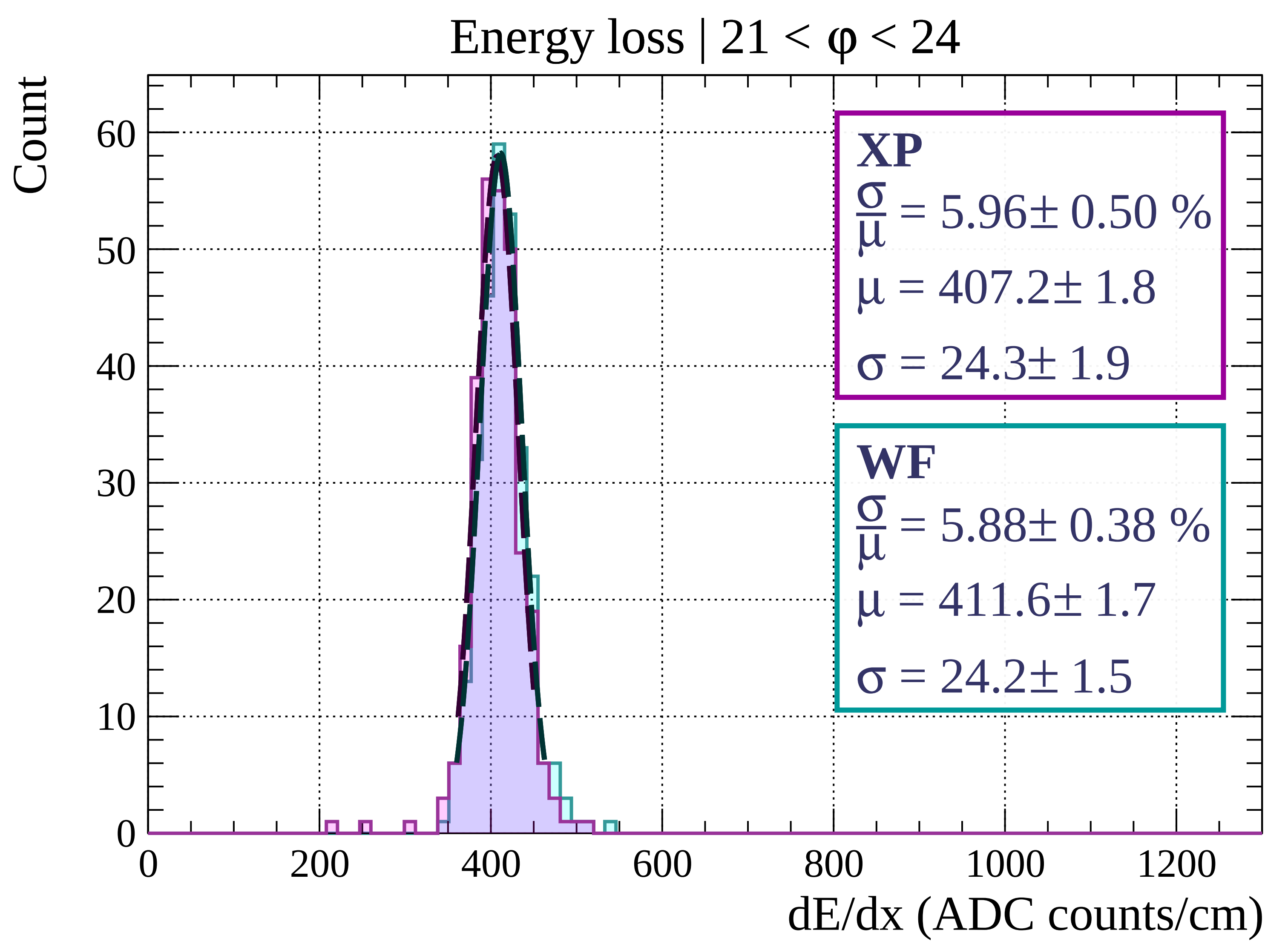


Energy loss | 12 < φ < 15
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.24± 0.45 %
μ = 409.4± 1.5
σ = 25.5± 1.7
WF
$\frac{\sigma}{\mu}$ = 6.84± 0.48 %
μ = 415.5± 1.6
σ = 28.4± 1.9

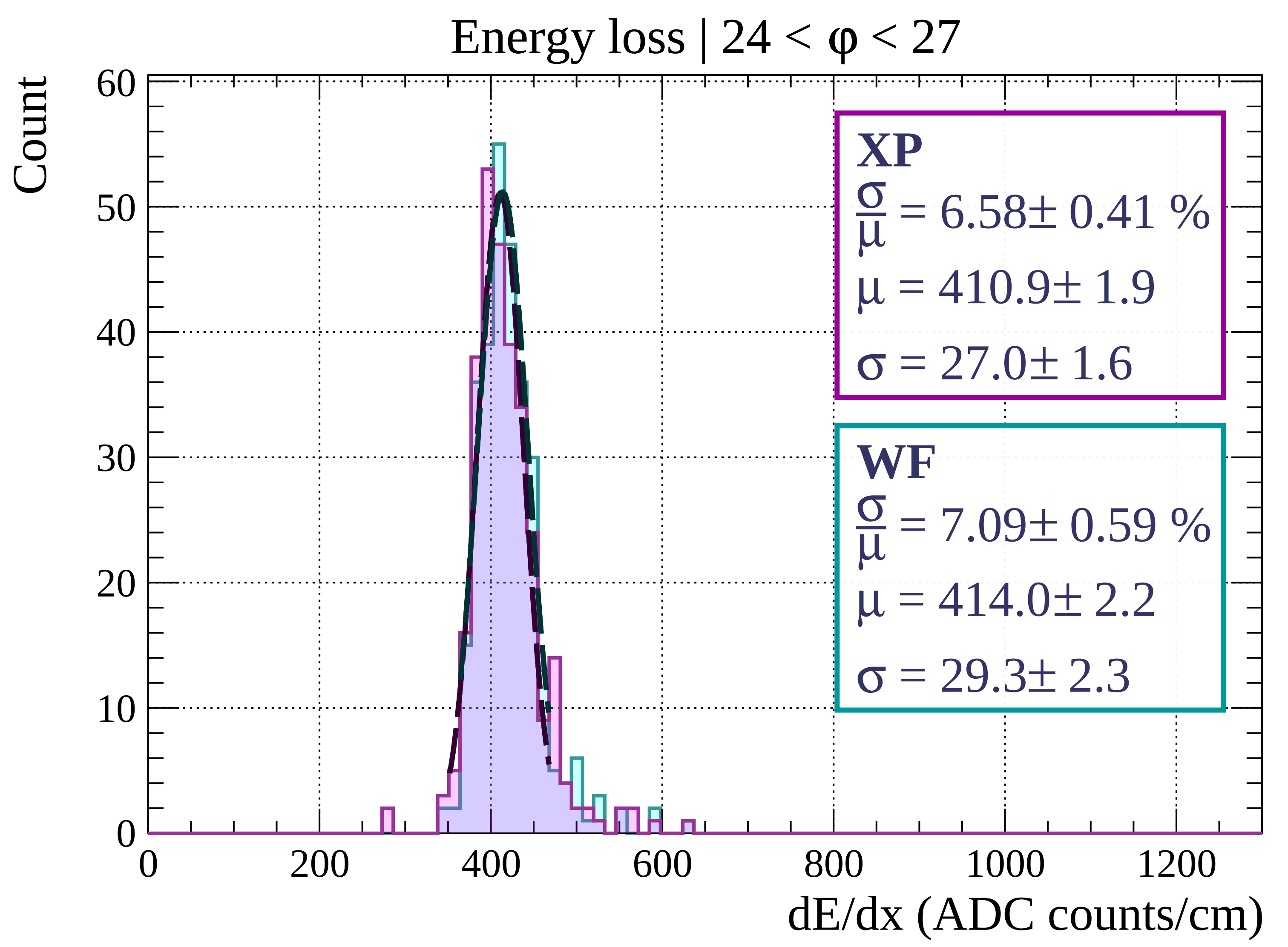


Energy loss | 15 < φ < 18
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
XP
$\frac{\sigma}{\mu}$ = 5.90± 0.27 %
μ = 413.8± 1.5
σ = 24.4± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.37± 0.42 %
μ = 414.5± 1.6
σ = 26.4± 1.7

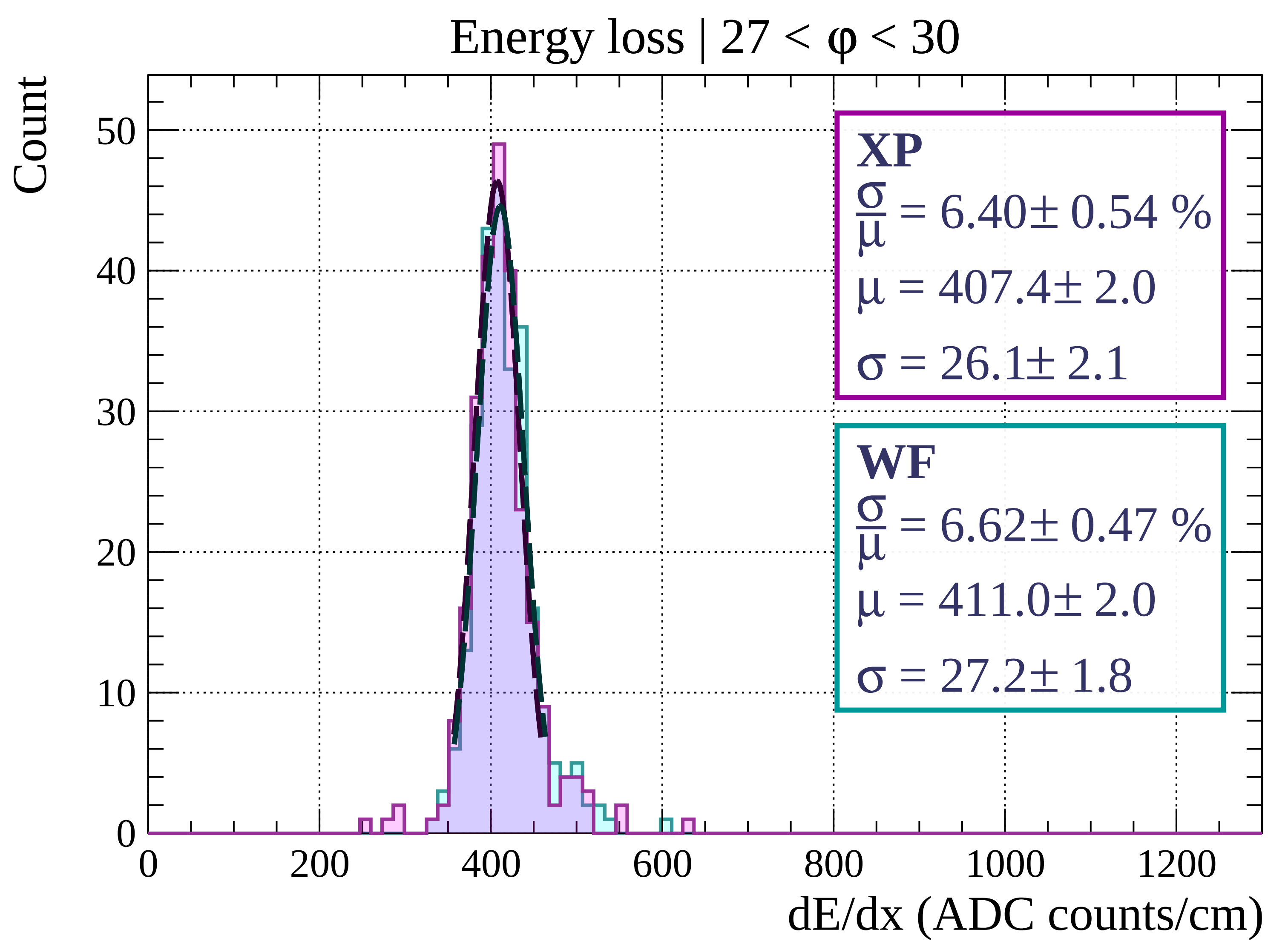

Energy loss | 18 < φ < 21
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
70
80
XP
σ/μ = 5.02± 0.41 %
μ = 407.4± 1.5
σ = 20.4± 1.6
WF
σ/μ = 5.50± 0.38 %
μ = 412.6± 1.5
σ = 22.7± 1.5



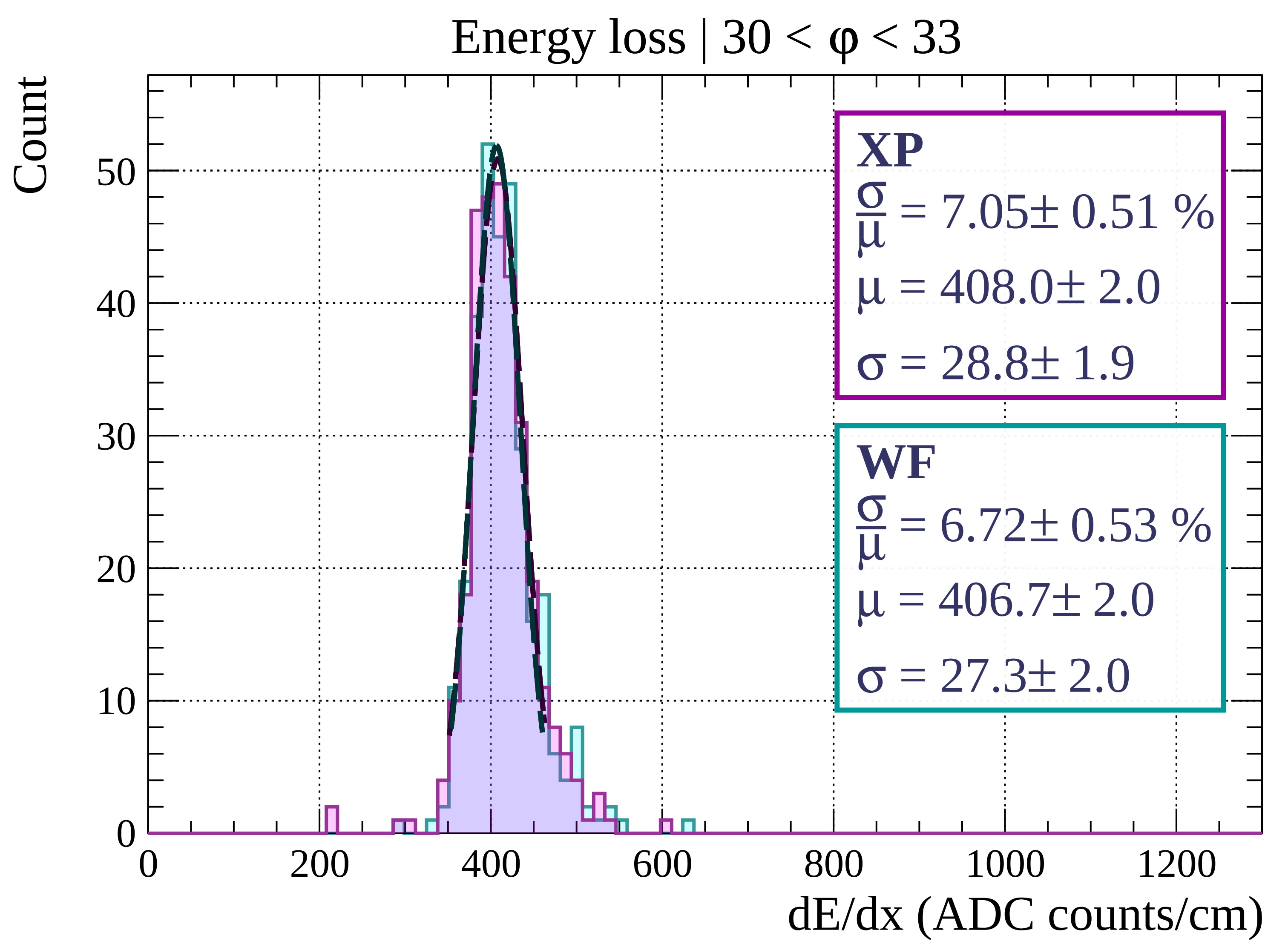
Energy loss | 21 < φ < 24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
60
XP
σ/μ = 5.96± 0.50 %
μ = 407.2± 1.8
σ = 24.3± 1.9
WF
σ/μ = 5.88± 0.38 %
μ = 411.6± 1.7
σ = 24.2± 1.5



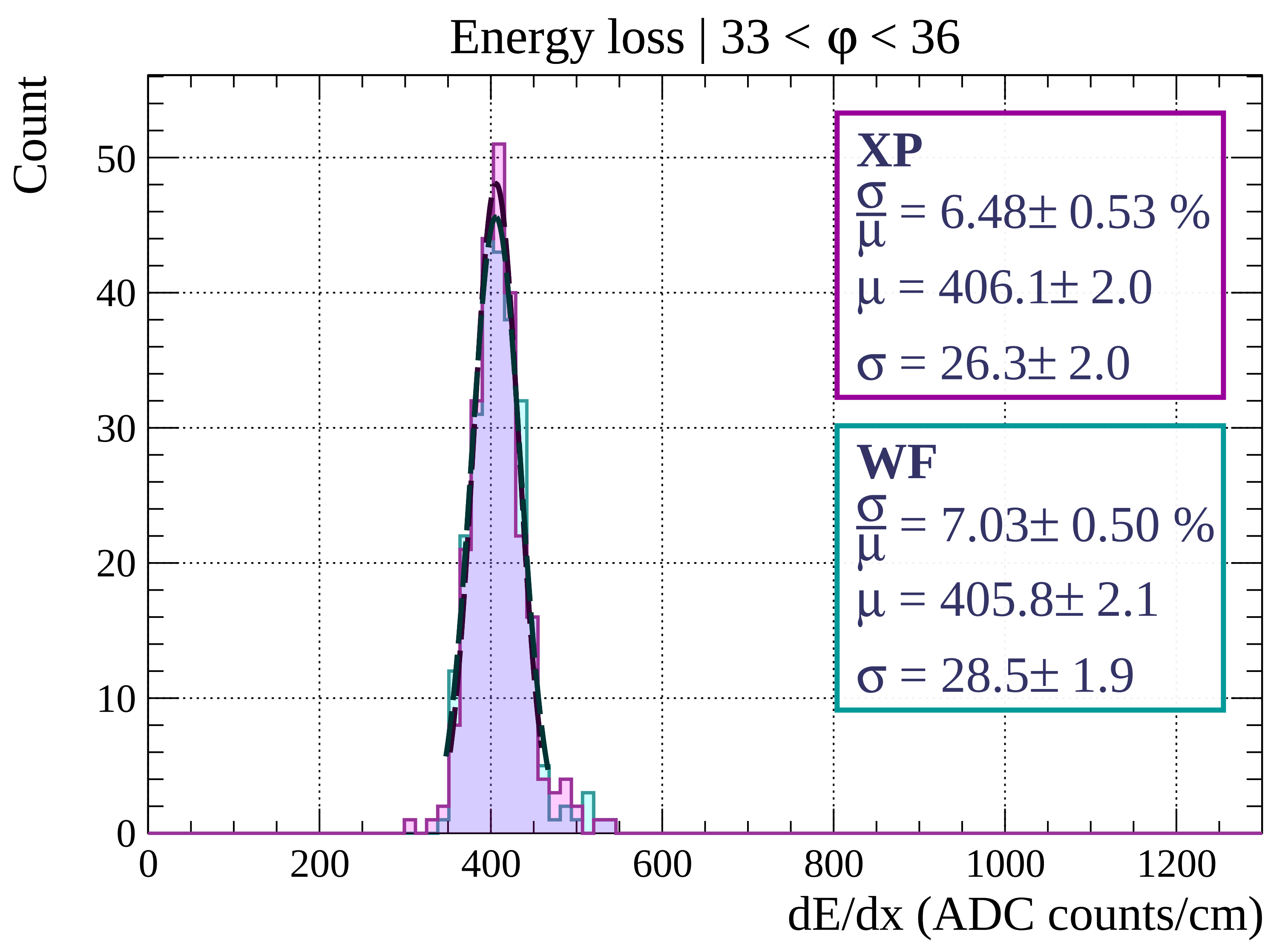
Energy loss | 24 < φ < 27
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 6.58± 0.41 %
μ = 410.9± 1.9
σ = 27.0± 1.6
WF
σ/μ = 7.09± 0.59 %
μ = 414.0± 2.2
σ = 29.3± 2.3



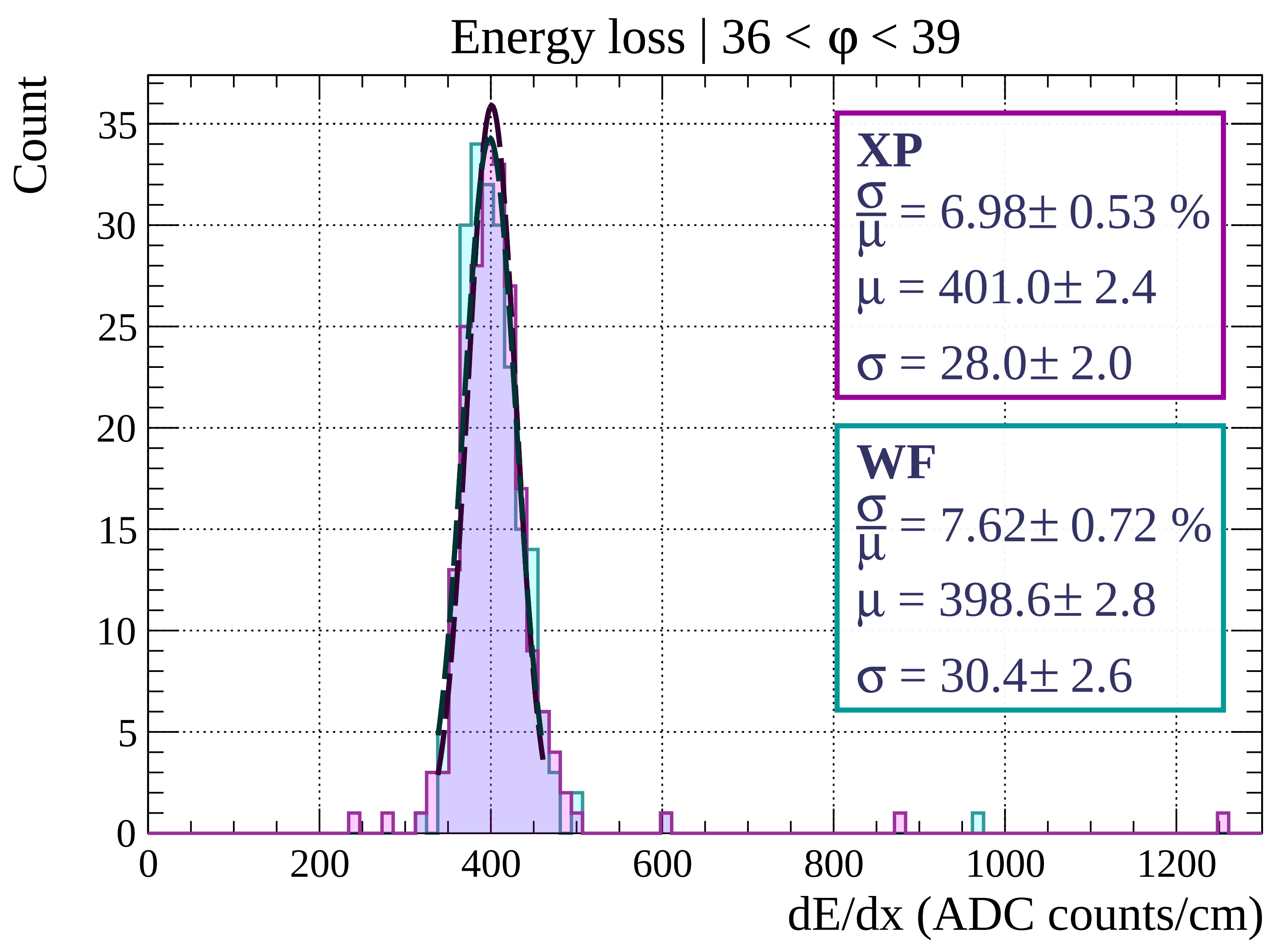
Energy loss | 27 < φ < 30
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.40± 0.54 %
μ = 407.4± 2.0
σ = 26.1± 2.1
WF
$\frac{\sigma}{\mu}$ = 6.62± 0.47 %
μ = 411.0± 2.0
σ = 27.2± 1.8



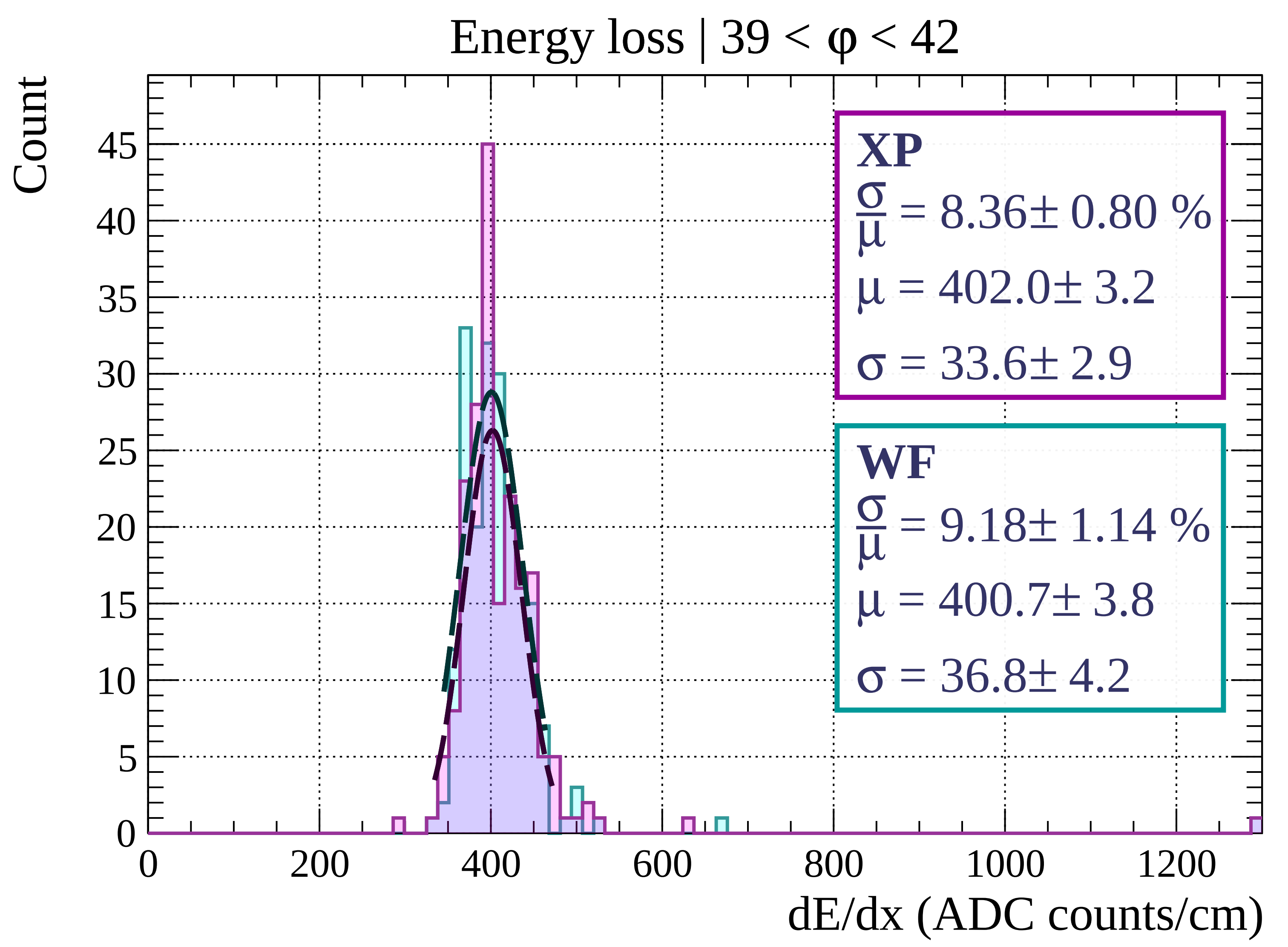
Energy loss | 30 < φ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
10
20
30
40
50
XP
σ/μ = 7.05± 0.51 %
μ = 408.0± 2.0
σ = 28.8± 1.9
WF
σ/μ = 6.72± 0.53 %
μ = 406.7± 2.0
σ = 27.3± 2.0

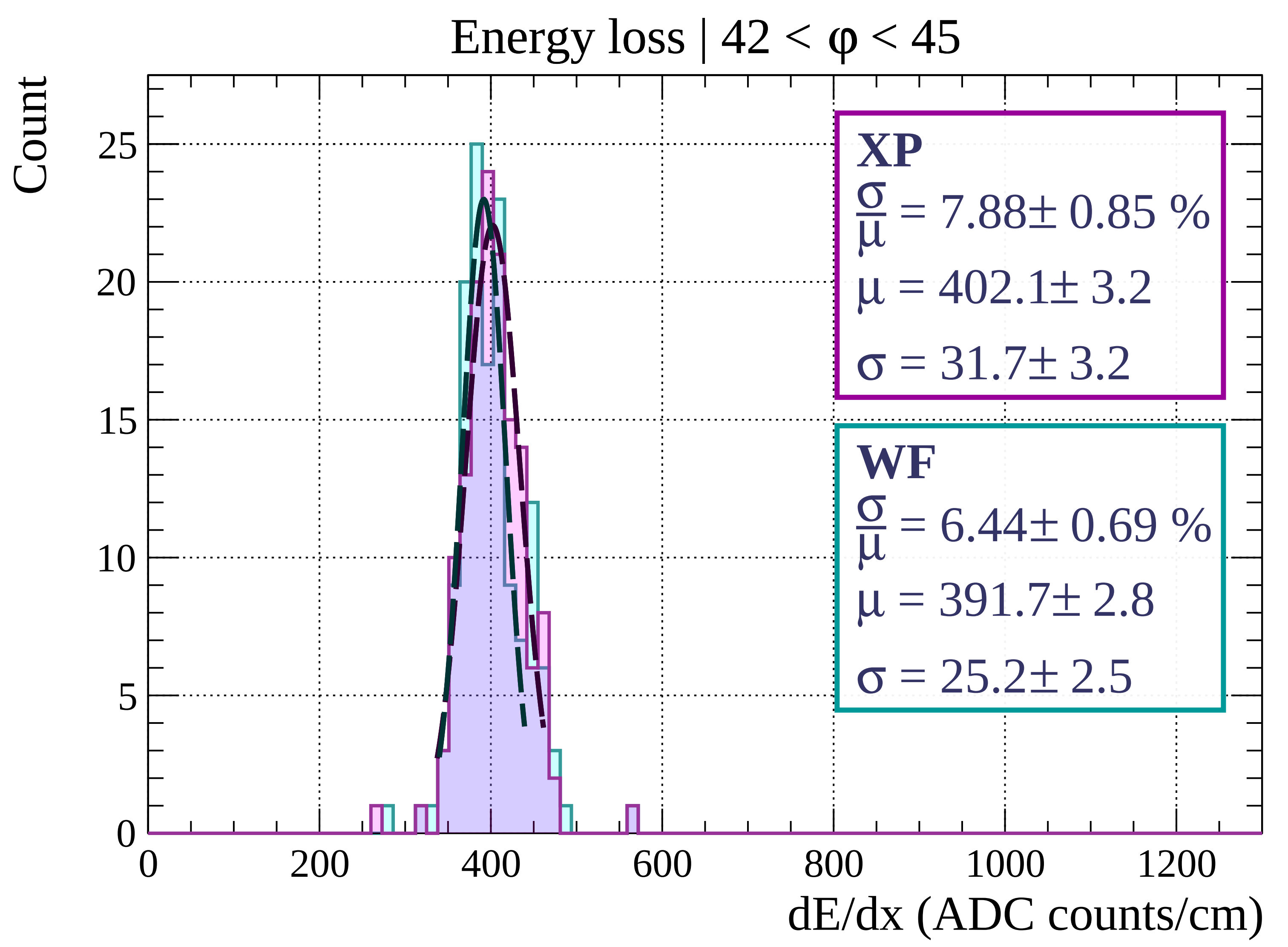


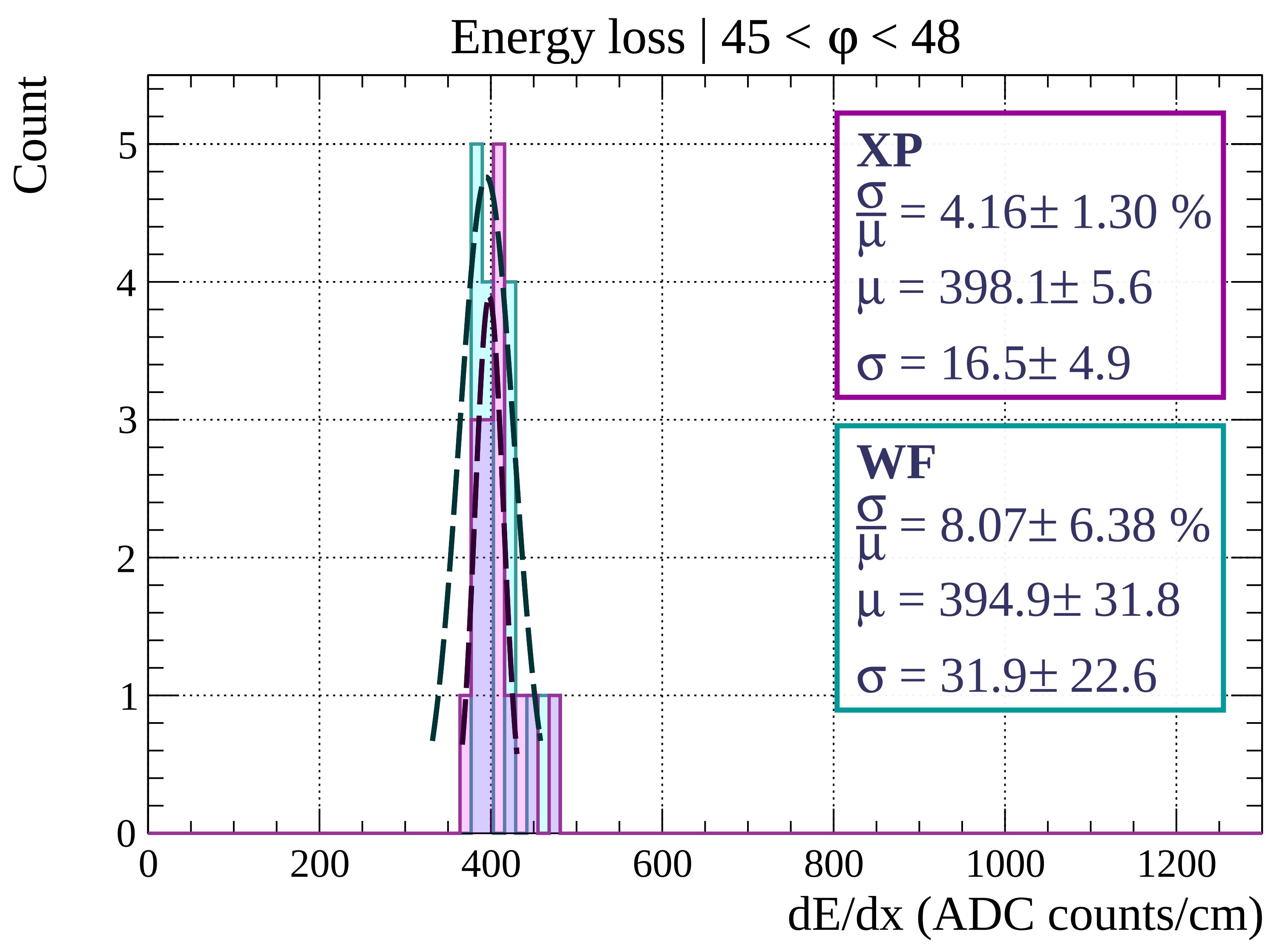
Energy loss | 33 < φ < 36
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
10
20
30
40
50
XP
σ/μ = 6.48± 0.53 %
μ = 406.1± 2.0
σ = 26.3± 2.0
WF
σ/μ = 7.03± 0.50 %
μ = 405.8± 2.1
σ = 28.5± 1.9

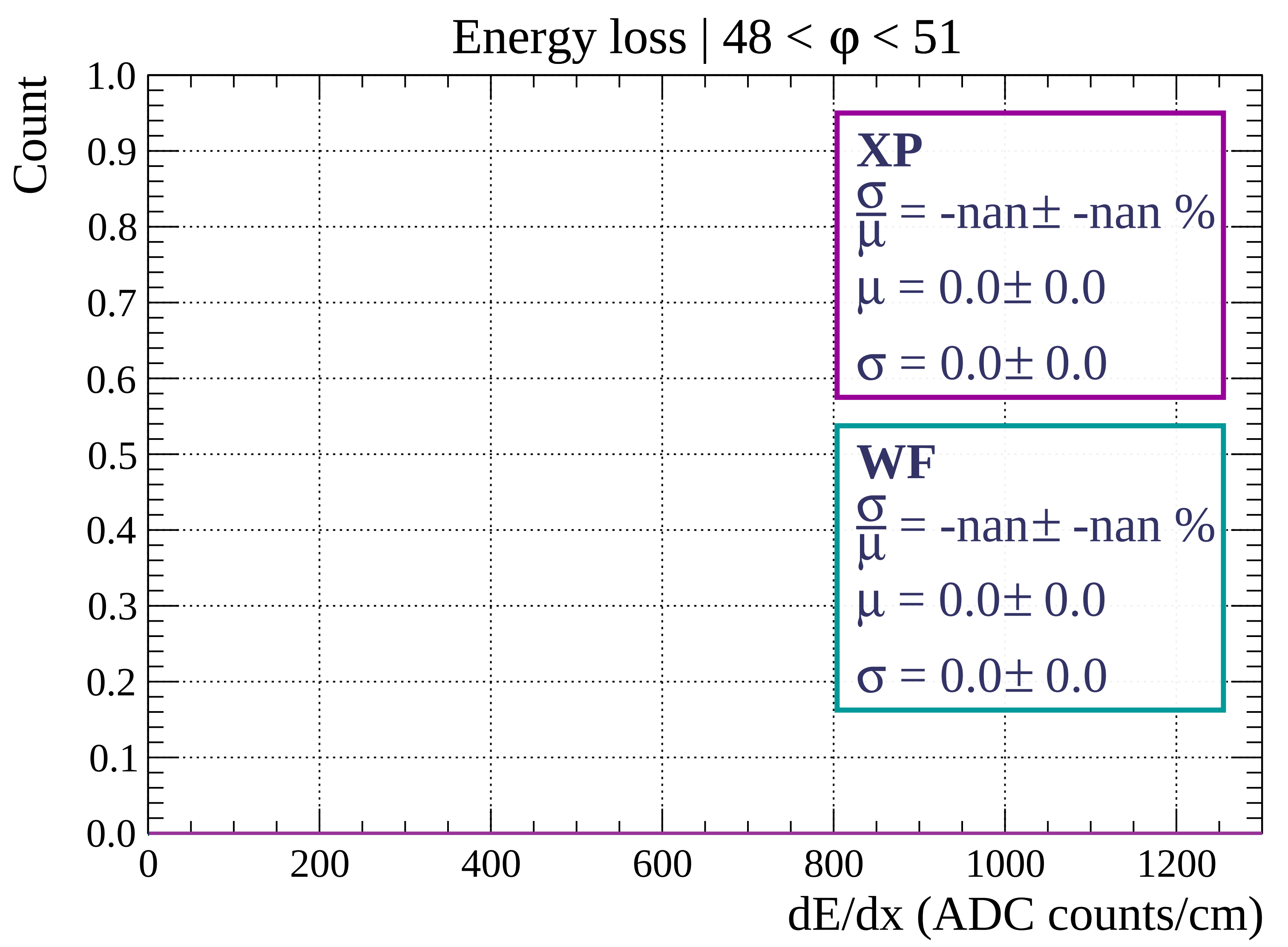


Energy loss | 36 < φ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
XP
σ/μ = 6.98± 0.53 %
μ = 401.0± 2.4
σ = 28.0± 2.0
WF
σ/μ = 7.62± 0.72 %
μ = 398.6± 2.8
σ = 30.4± 2.6

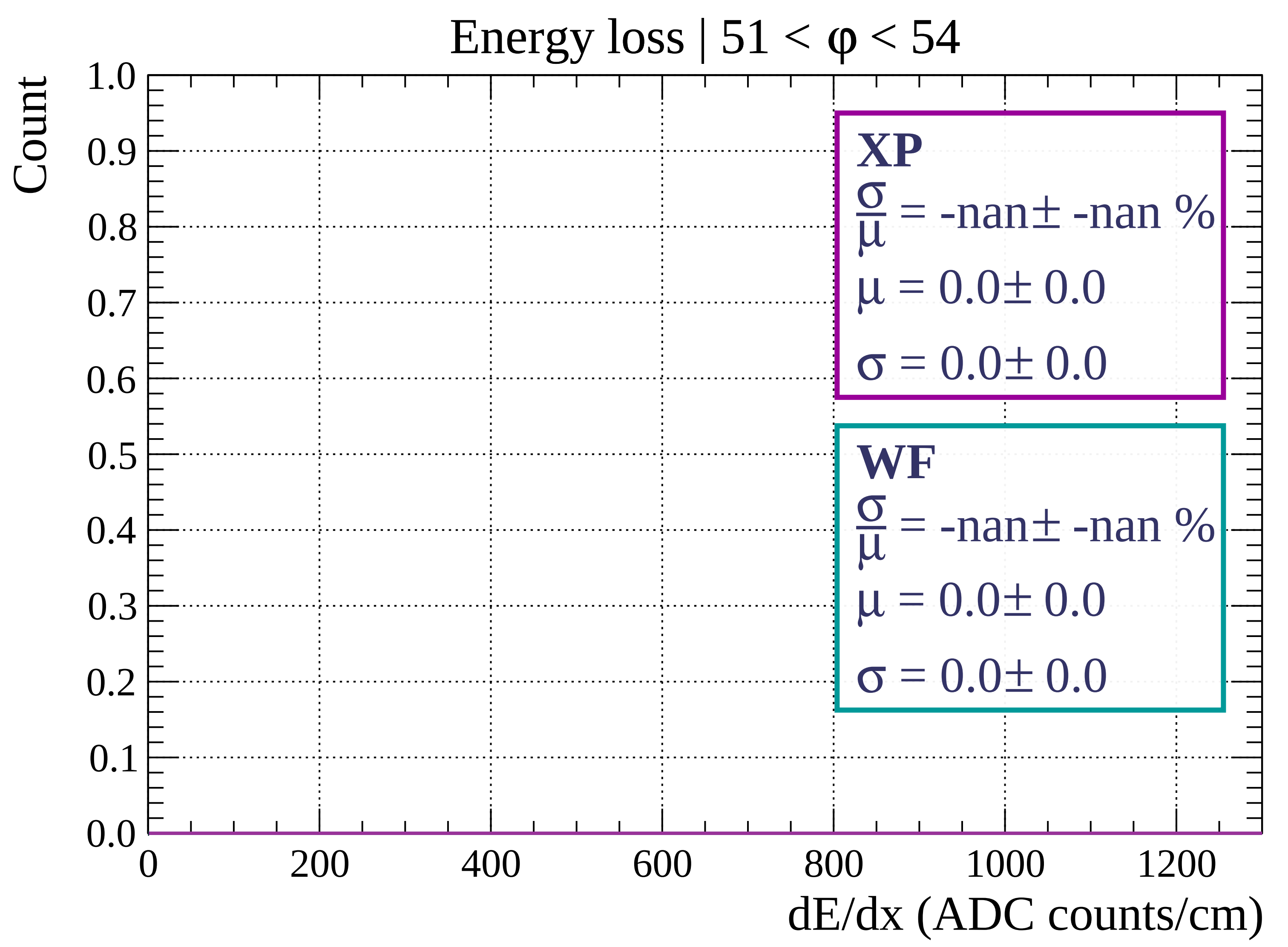

Energy loss | 39 < φ < 42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.36± 0.80 %
μ = 402.0± 3.2
σ = 33.6± 2.9
WF
σ/μ = 9.18± 1.14 %
μ = 400.7± 3.8
σ = 36.8± 4.2
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200


Energy loss | 42 < φ < 45
Count
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 7.88± 0.85 %
μ = 402.1± 3.2
σ = 31.7± 3.2
WF
σ/μ = 6.44± 0.69 %
μ = 391.7± 2.8
σ = 25.2± 2.5

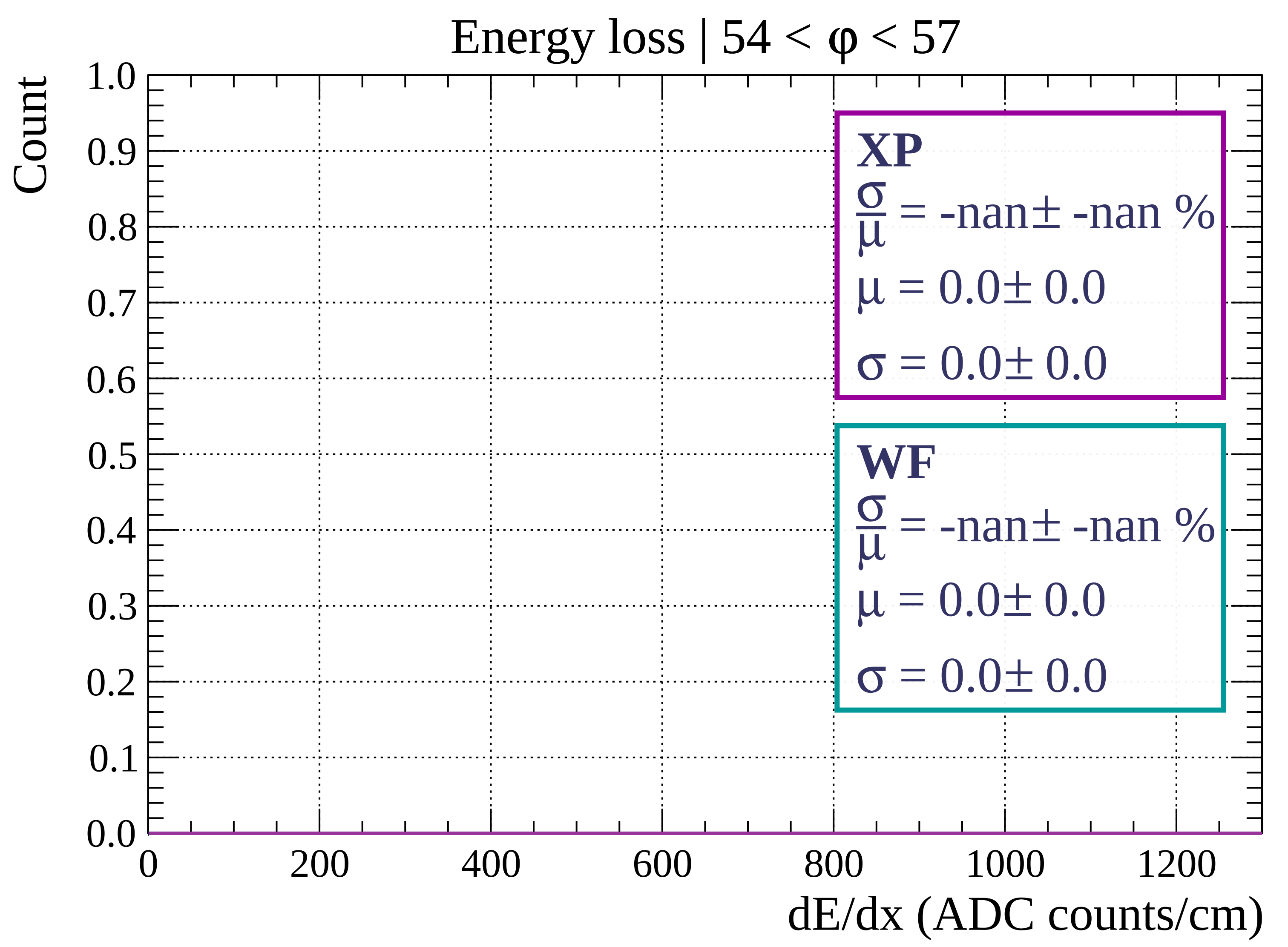

Energy loss | 45 < φ < 48
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
$\frac{\sigma}{\mu}$ = 4.16± 1.30 %
μ = 398.1± 5.6
σ = 16.5± 4.9
WF
$\frac{\sigma}{\mu}$ = 8.07± 6.38 %
μ = 394.9± 31.8
σ = 31.9± 22.6

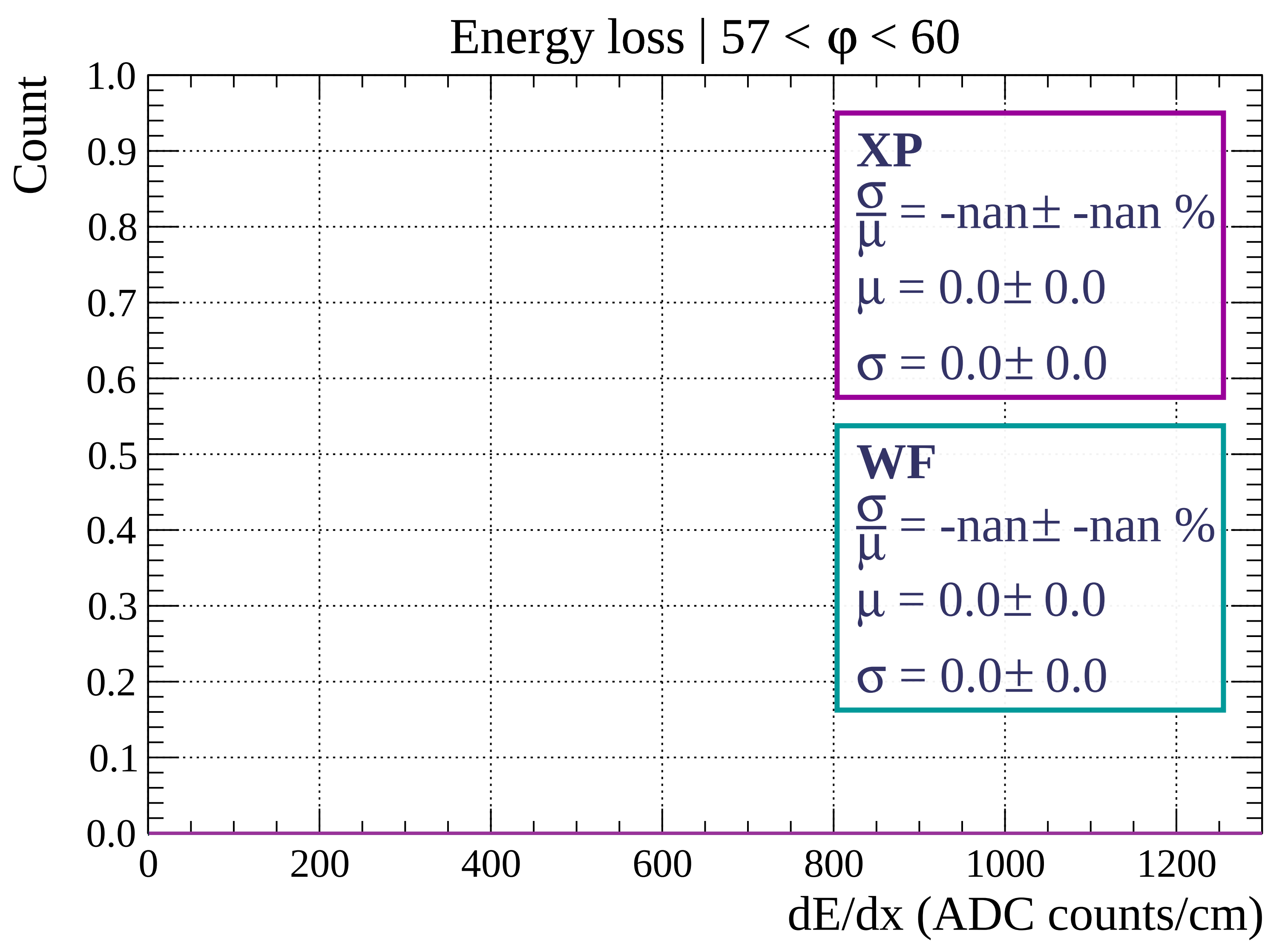


Energy loss | 48 < φ < 51
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

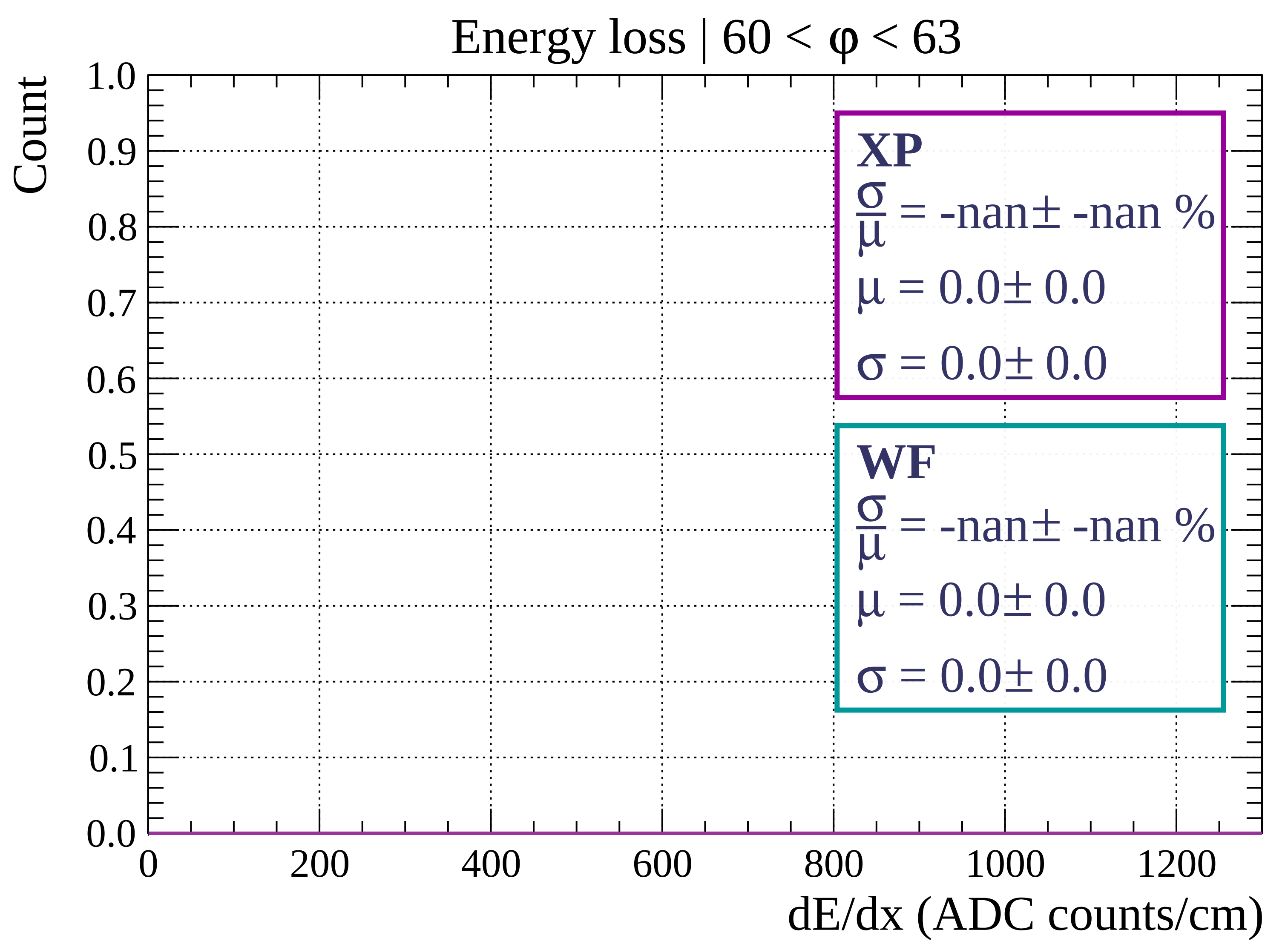


Energy loss | 51 < φ < 54
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

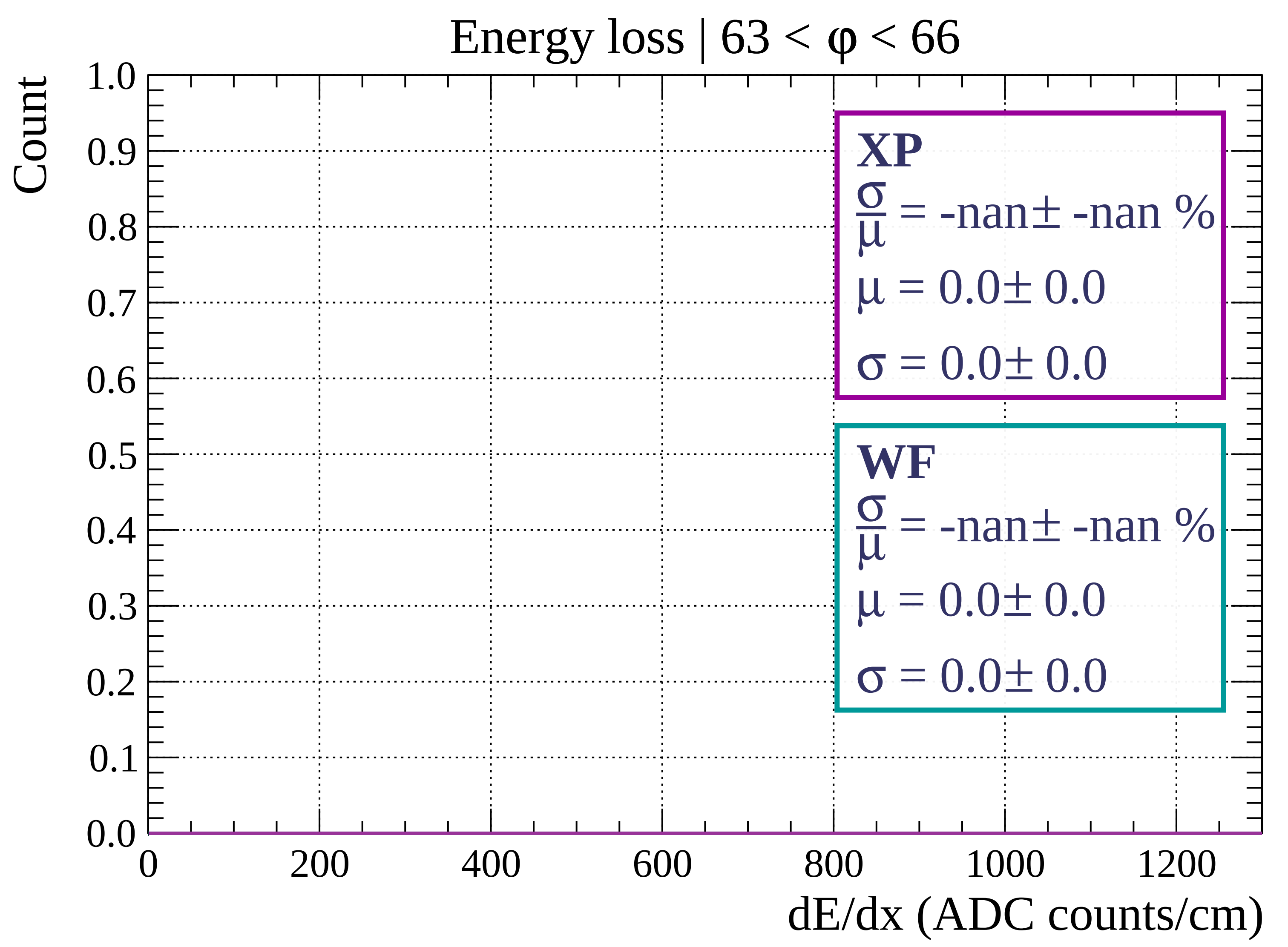

Energy loss | 54 < φ < 57
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

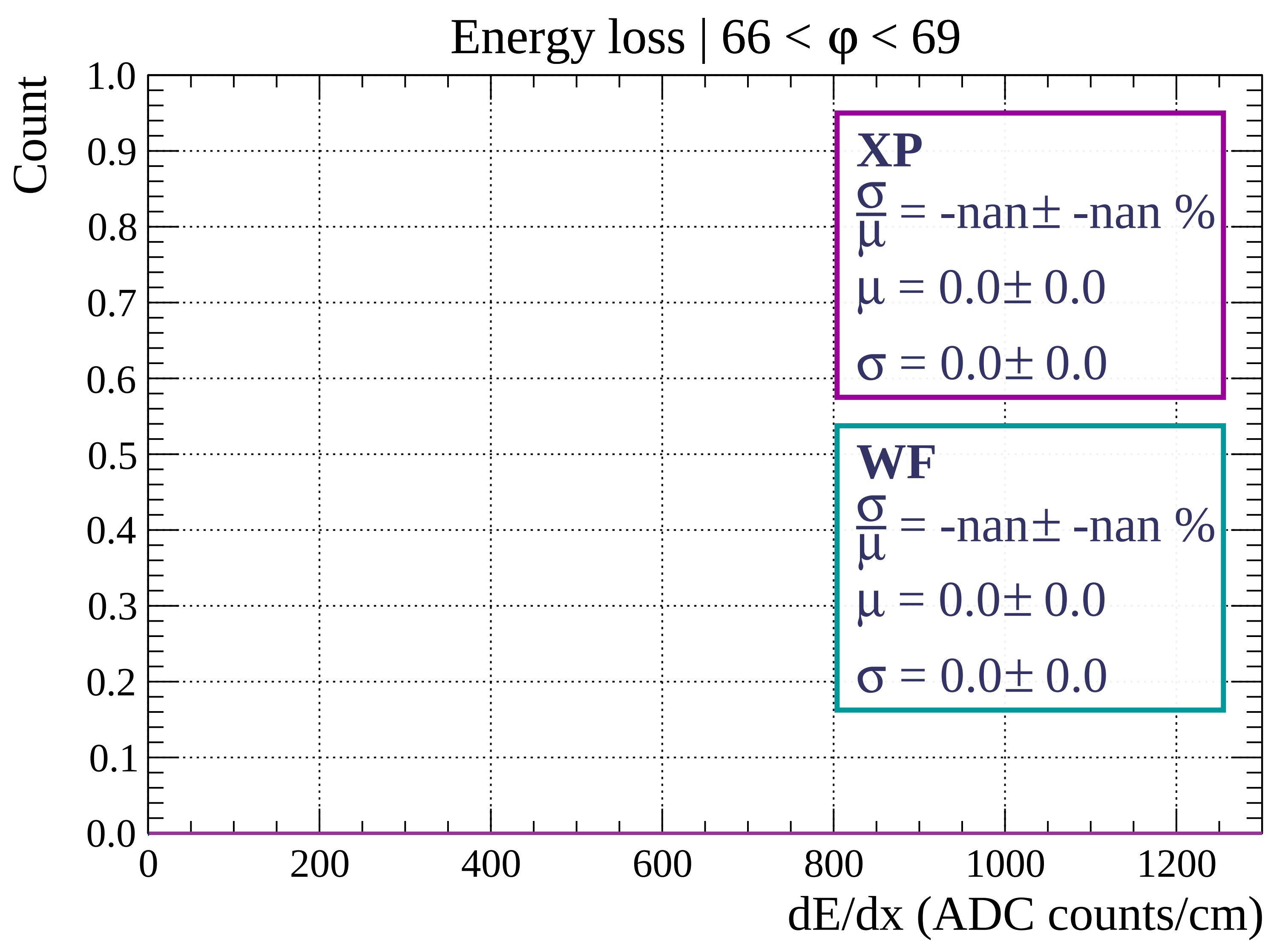


Energy loss | 57 < φ < 60
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200

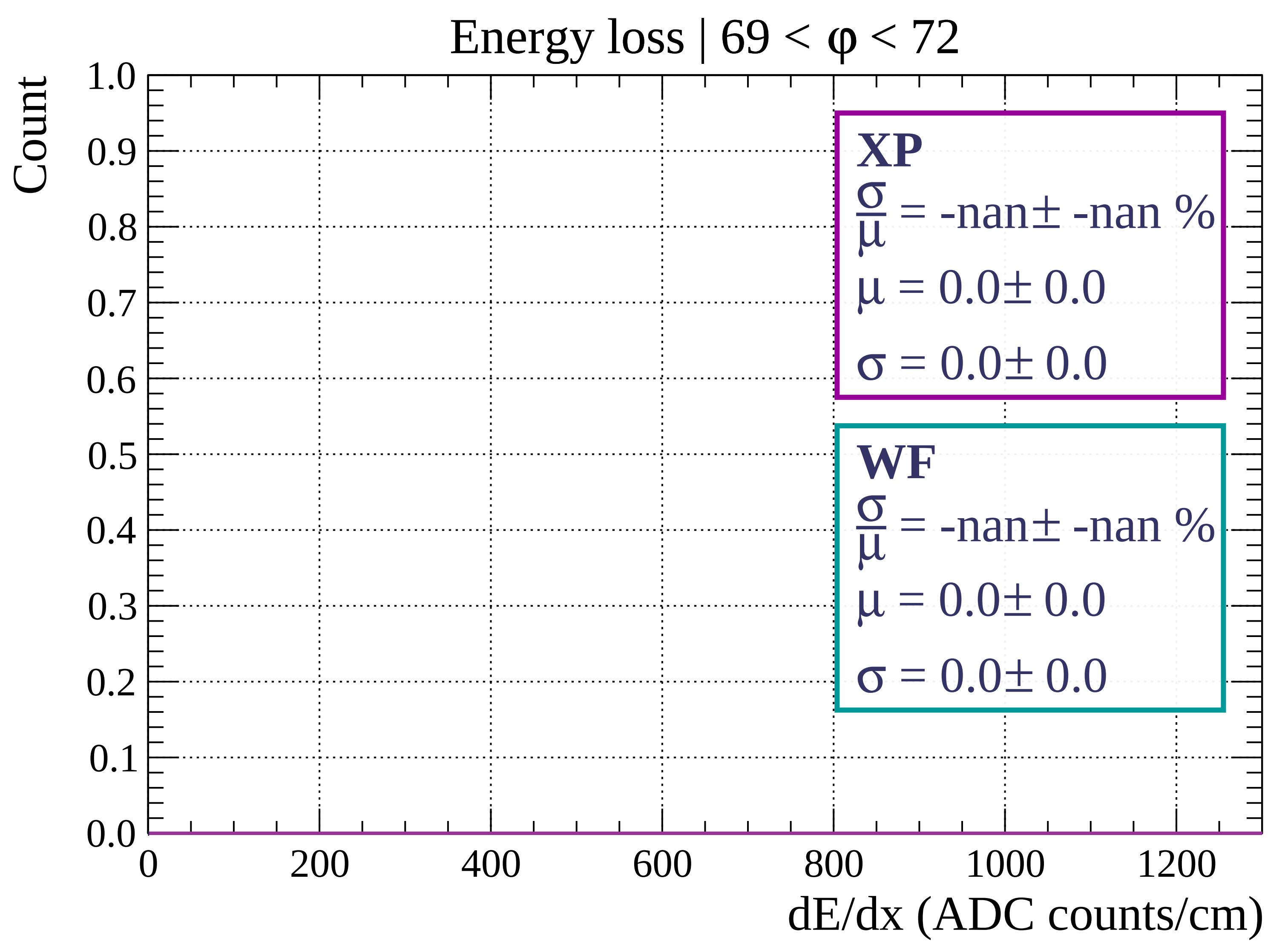


Energy loss | 60 < φ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

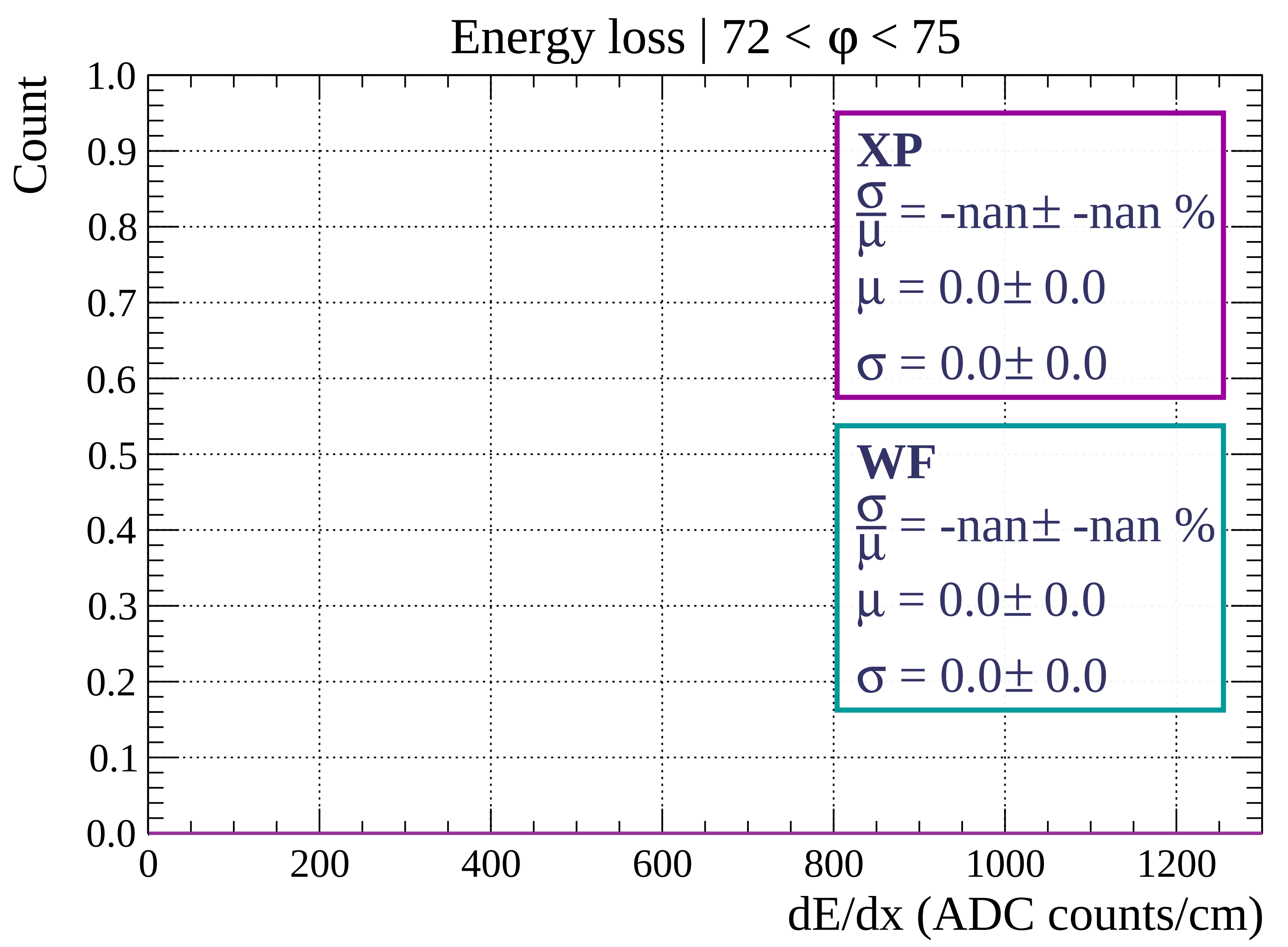

Energy loss | 63 < φ < 66
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



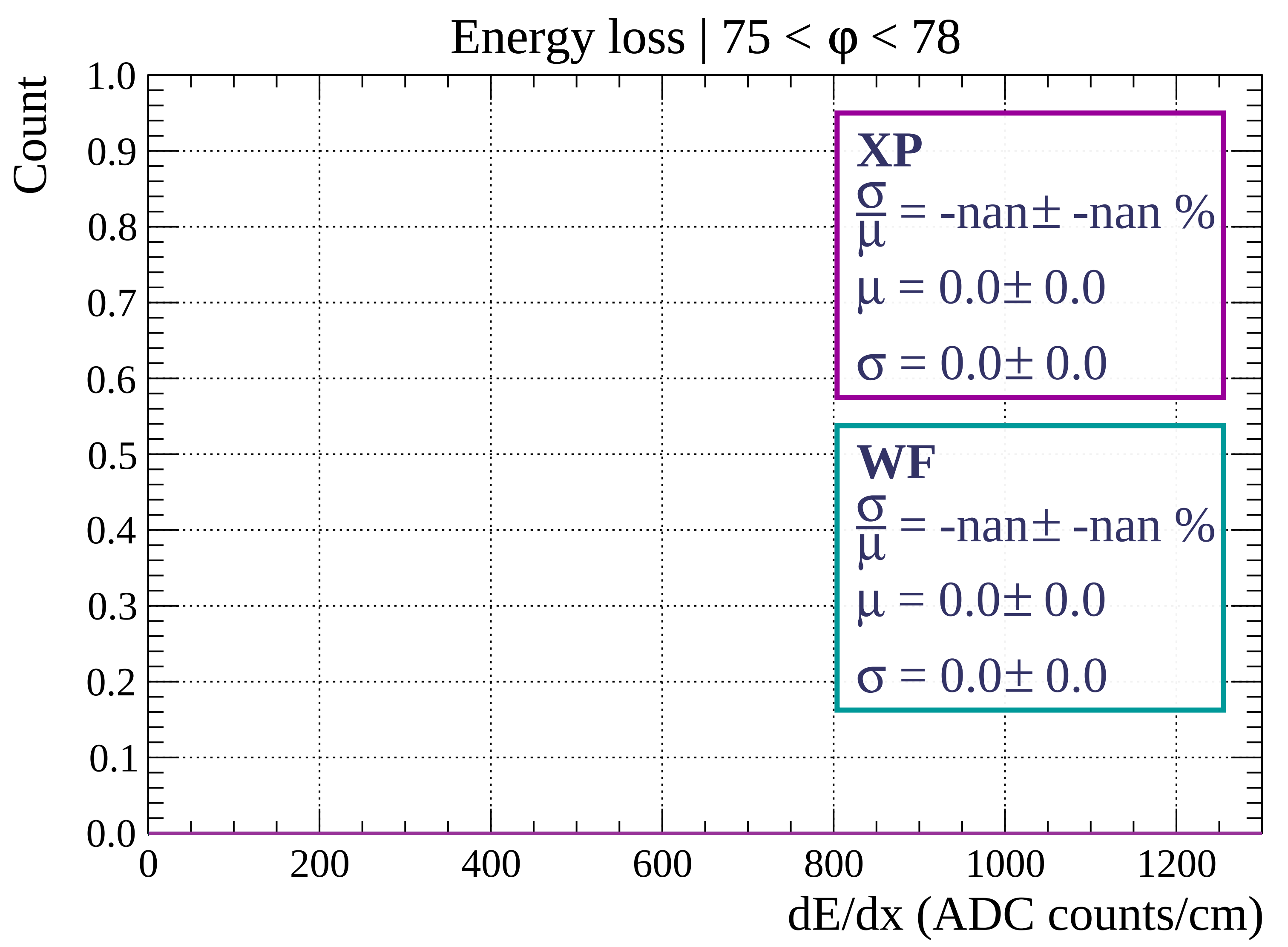
Energy loss | 66 < φ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



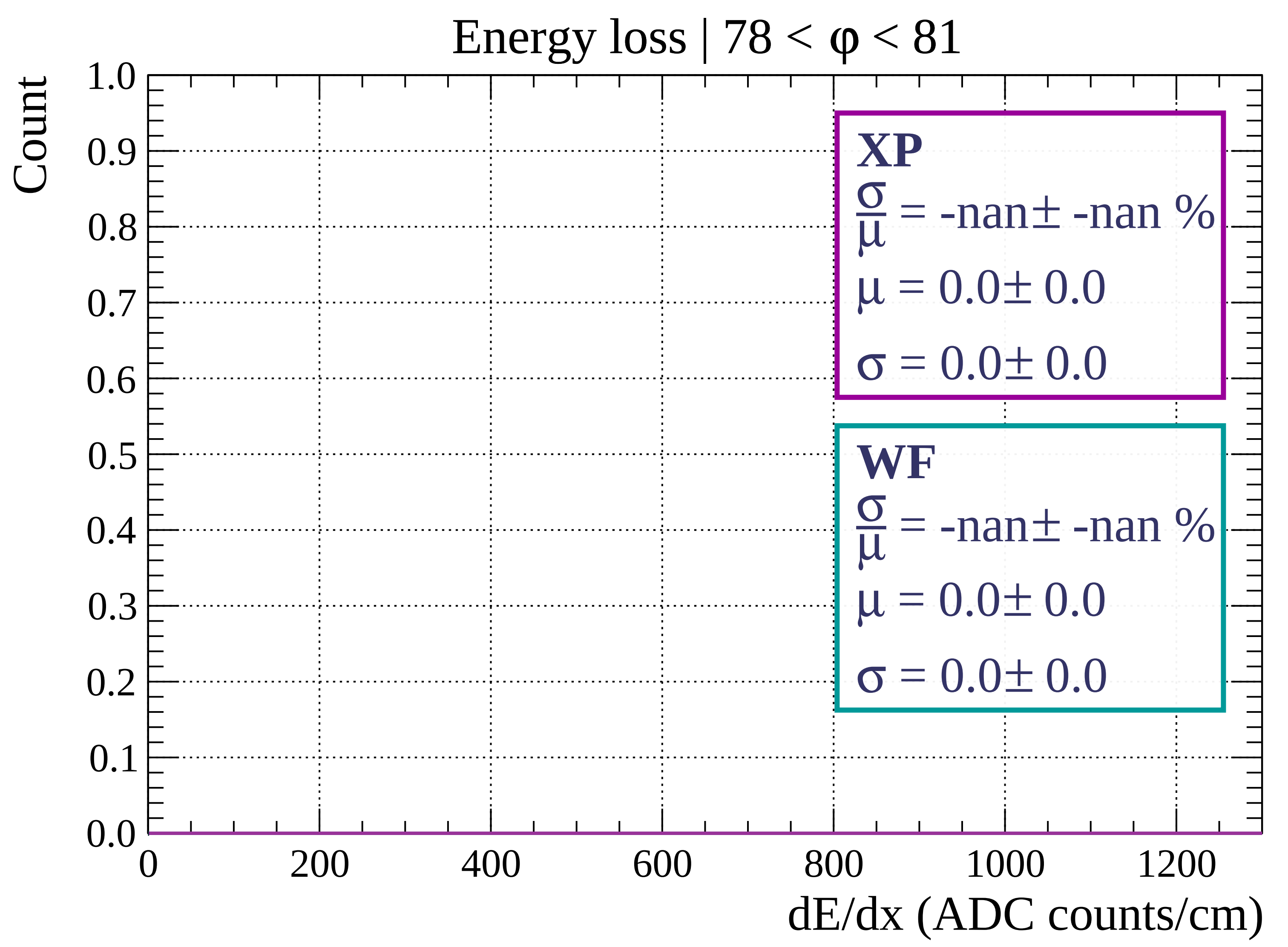
Energy loss | 69 < φ < 72
Count
dE/dx (ADC counts/cm)
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

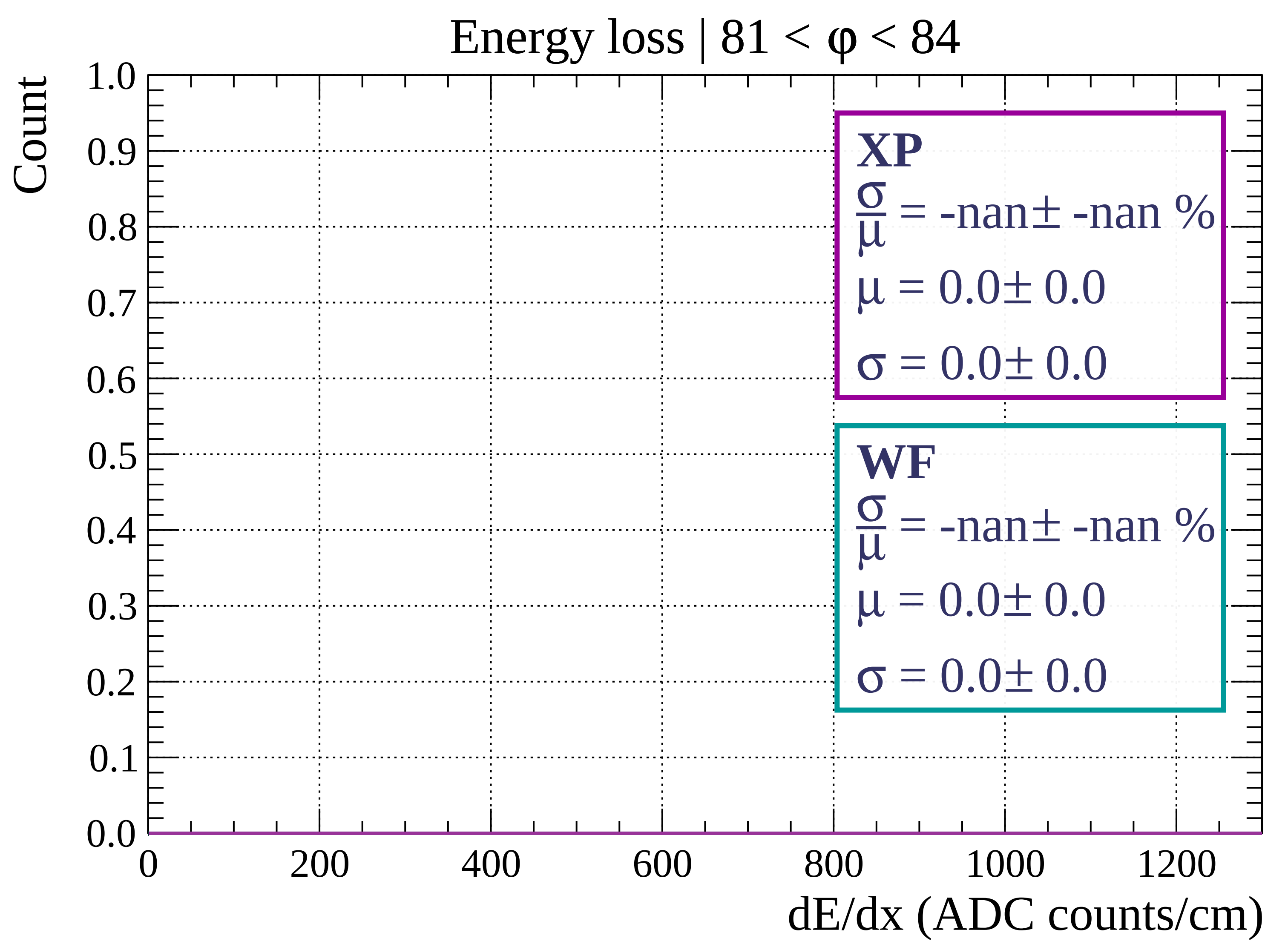


Energy loss | 72 < φ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

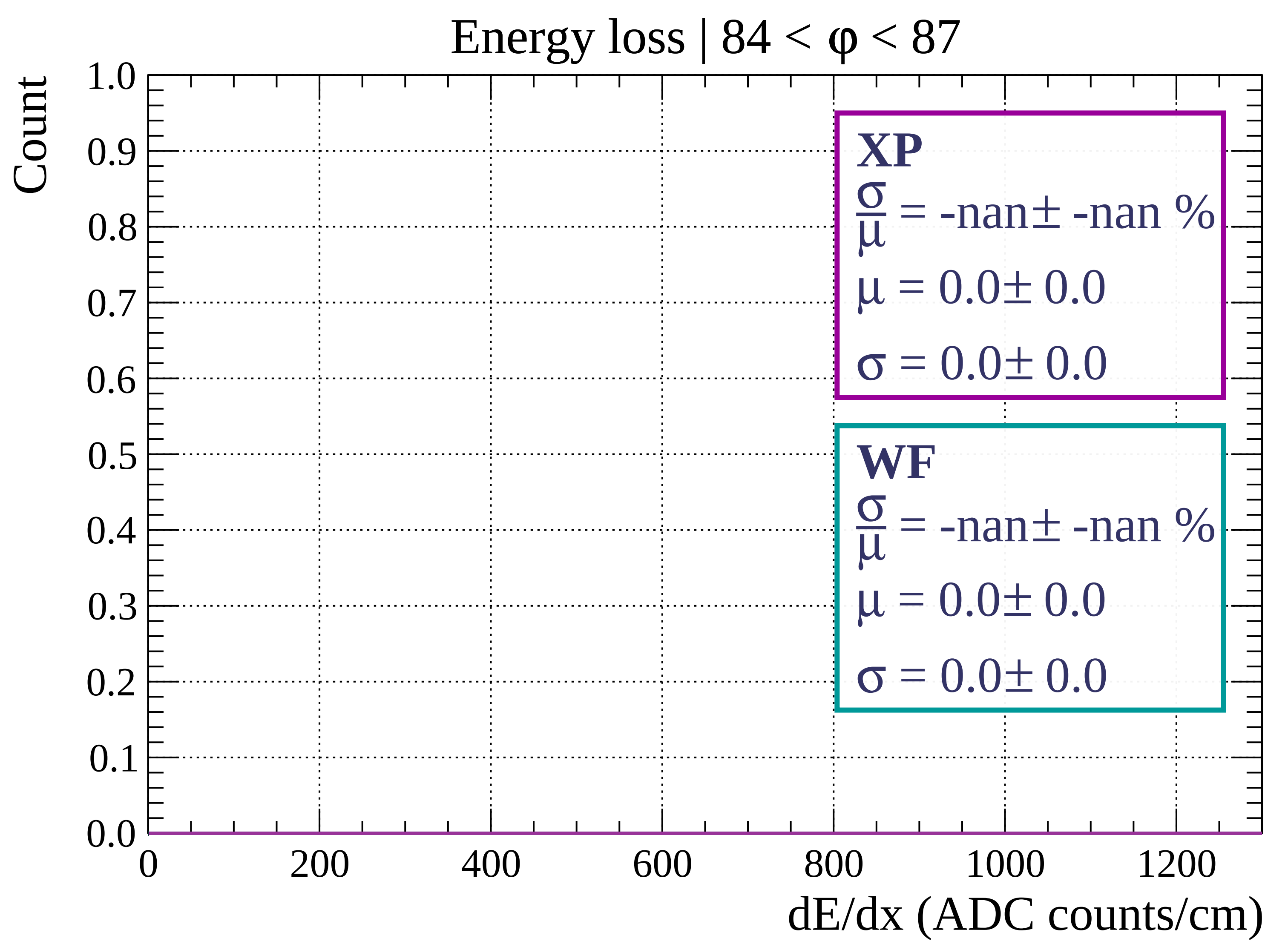


Energy loss | 75 < φ < 78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

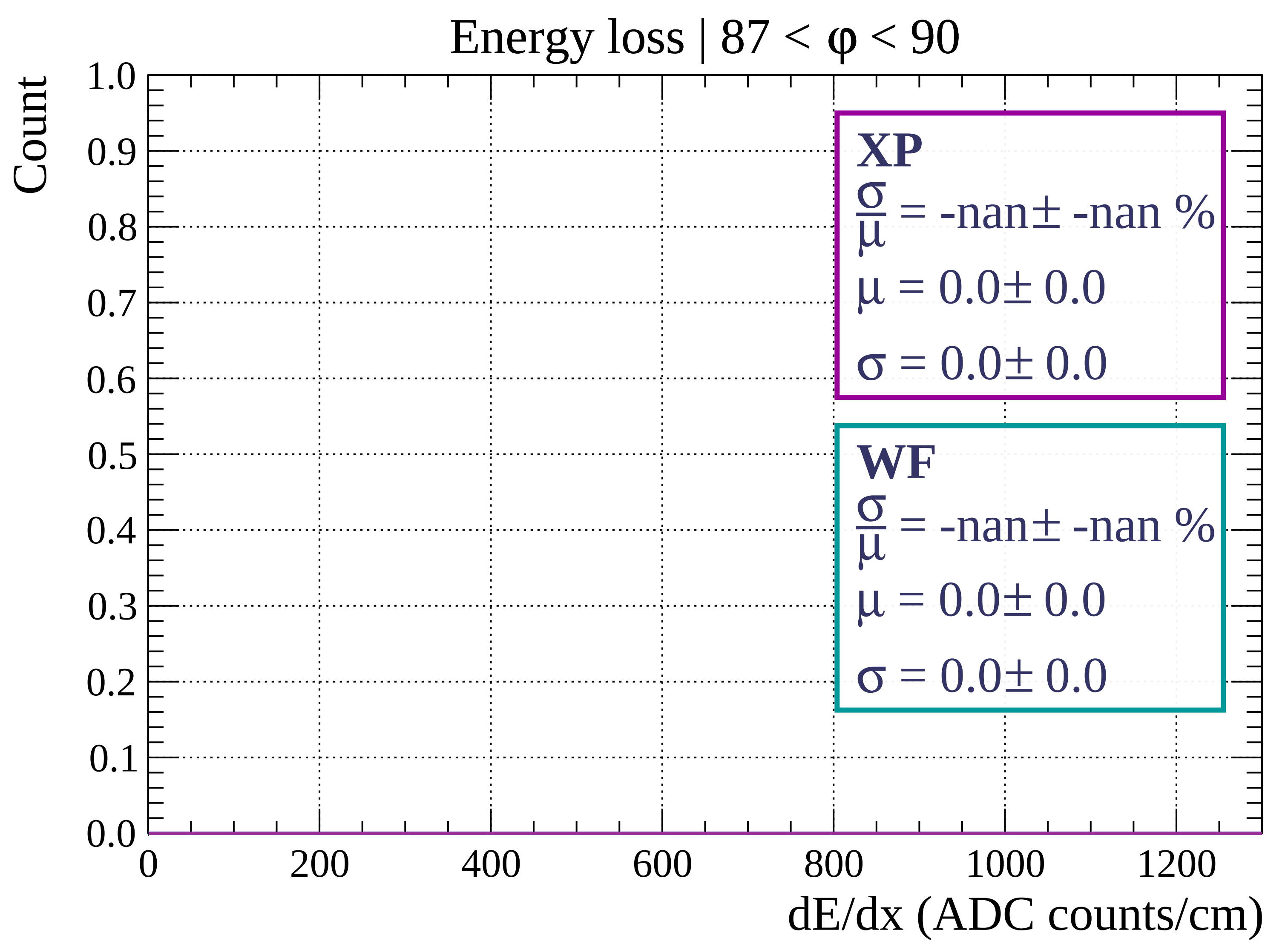

Energy loss | 78 < φ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



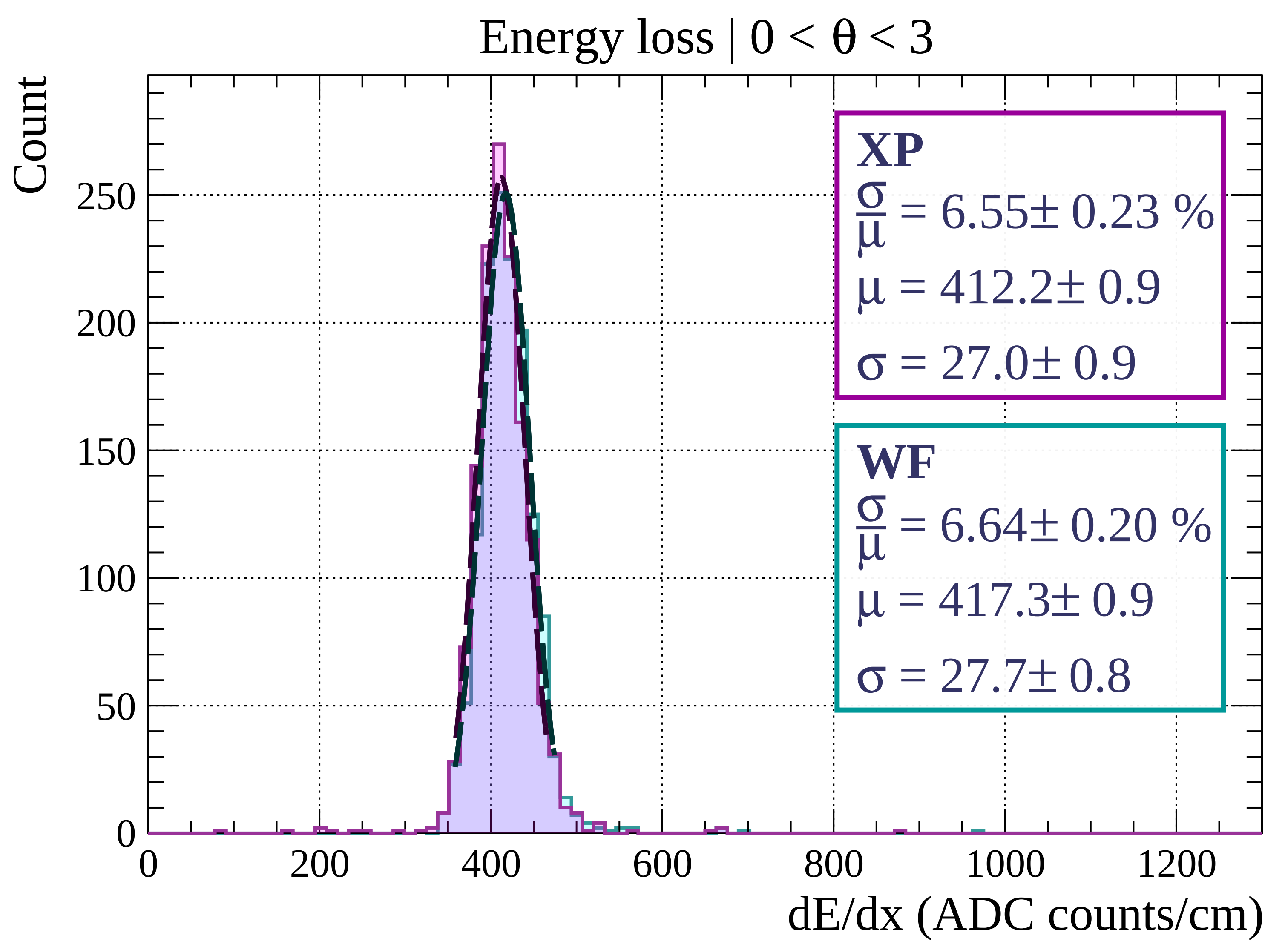
Energy loss | 81 < φ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



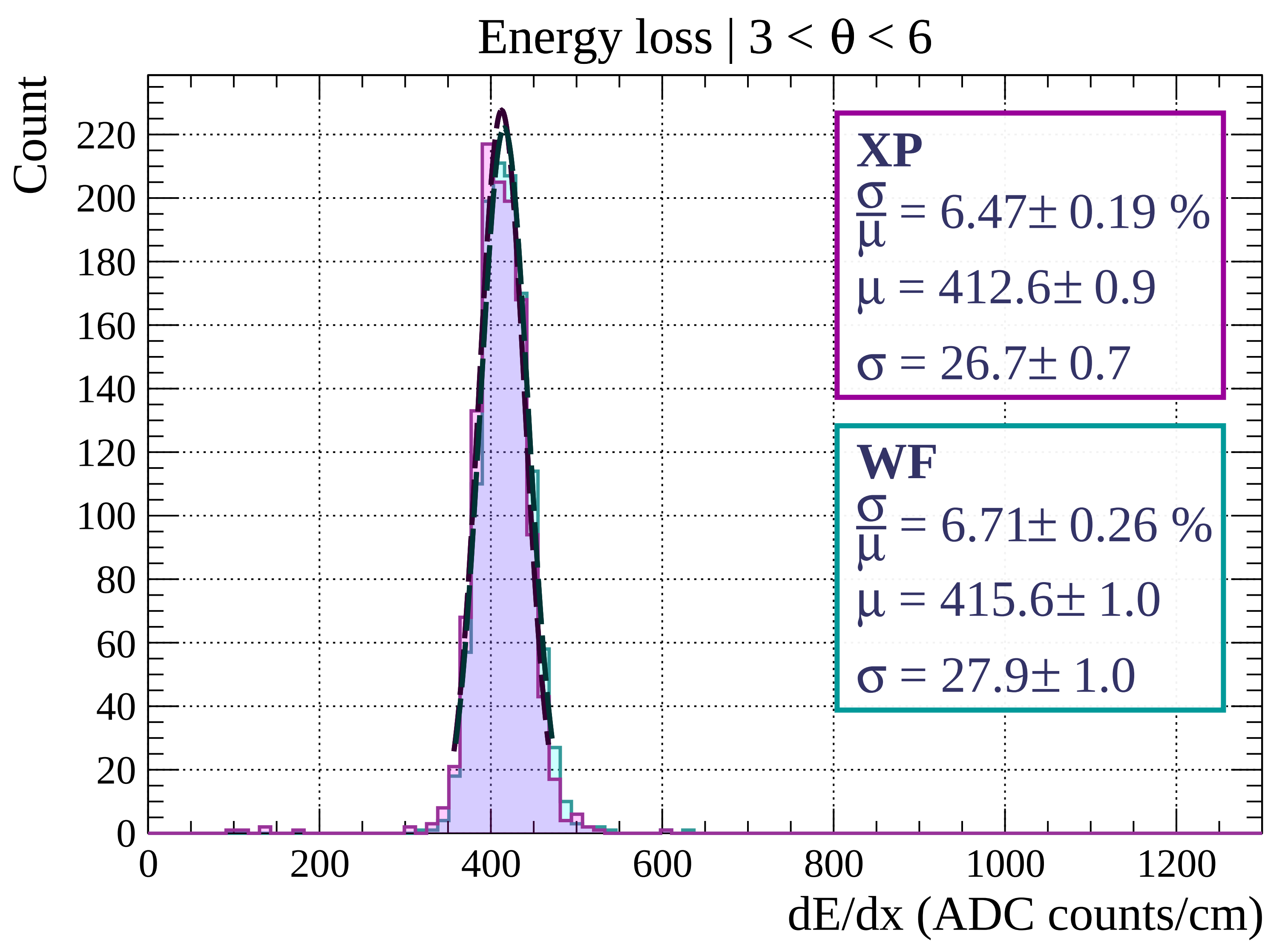
Energy loss | 84 < φ < 87
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

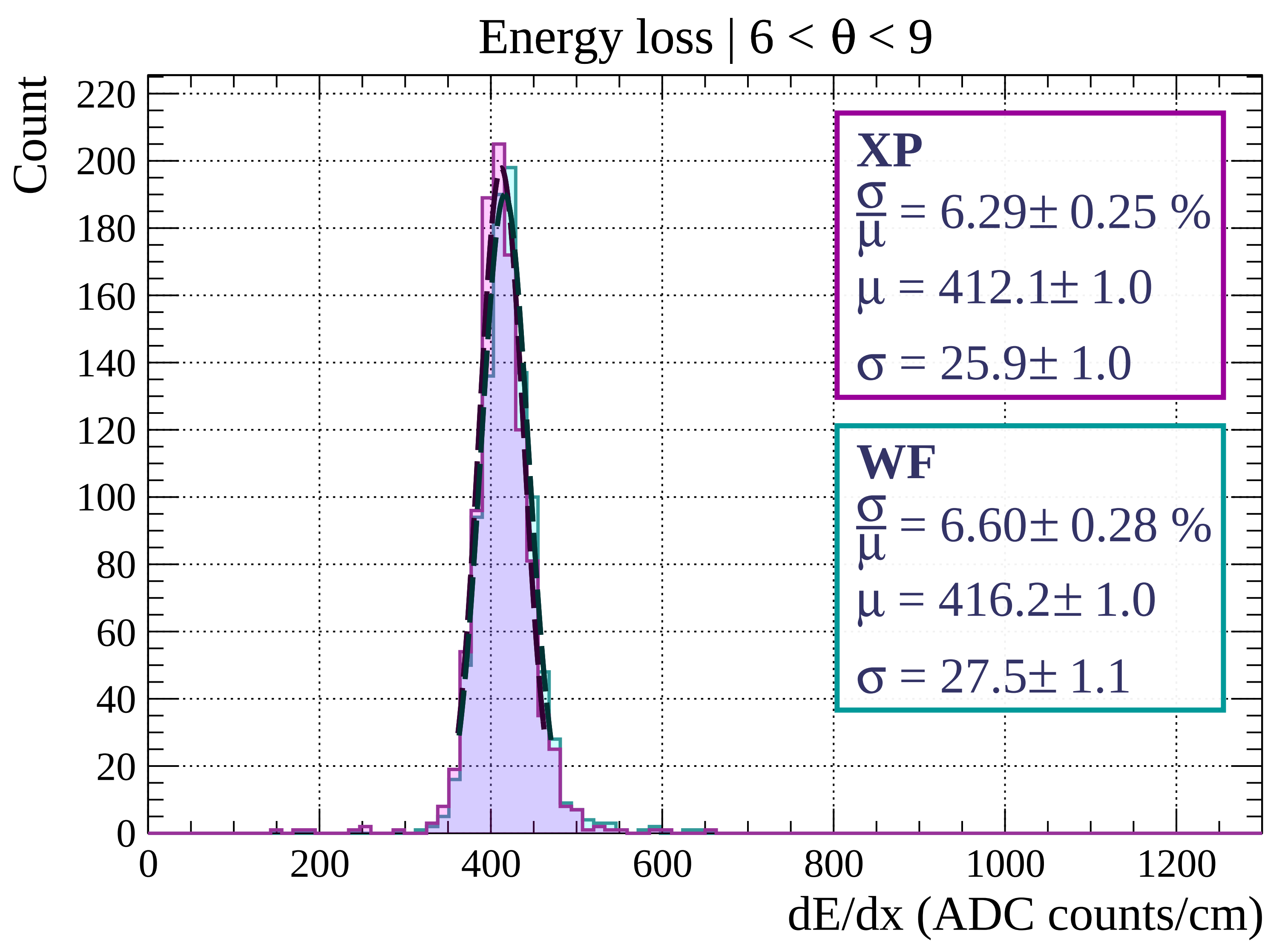


Energy loss | 87 < φ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

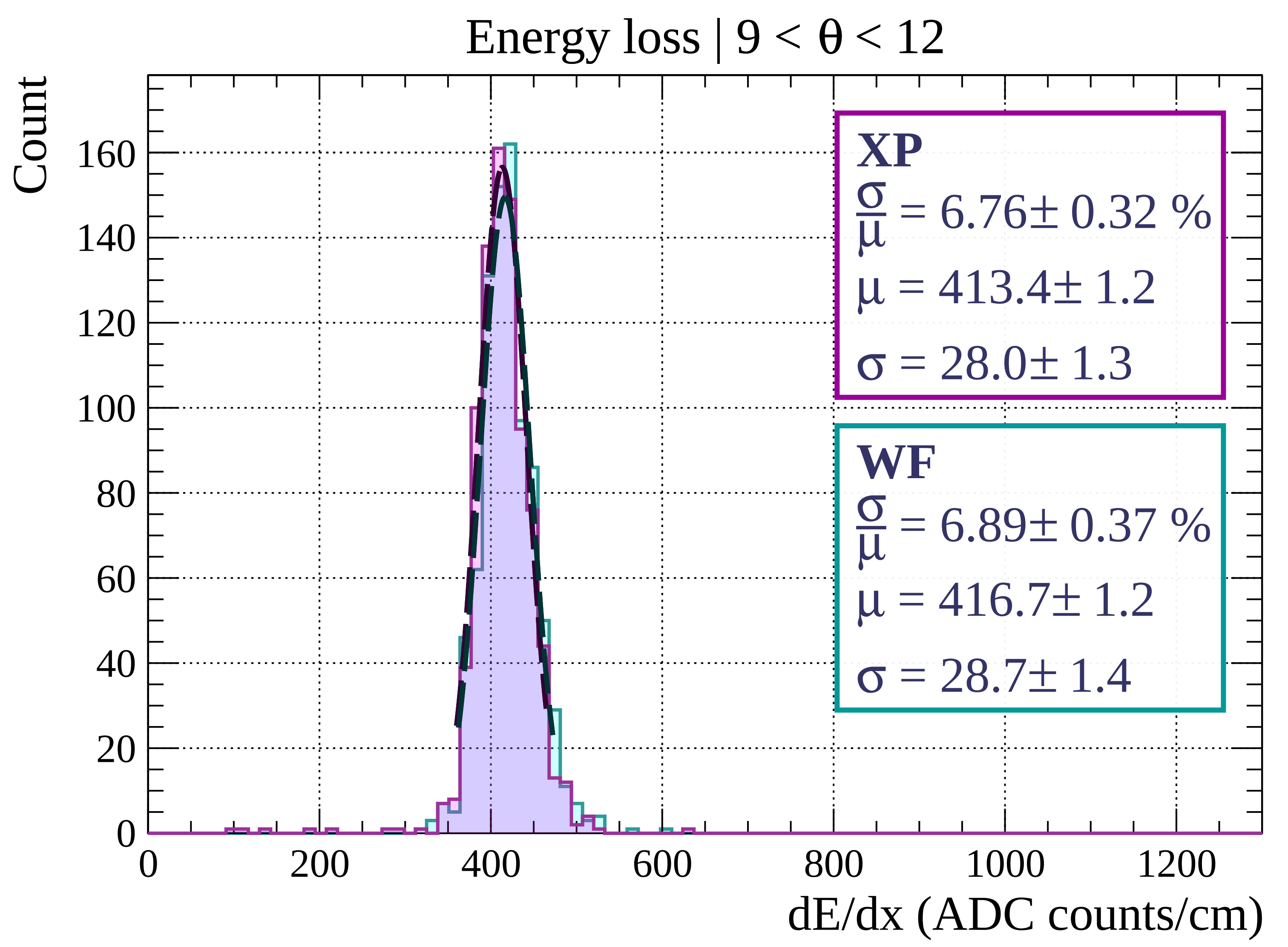


Energy loss | 0 < θ < 3
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.55± 0.23 %
μ = 412.2± 0.9
σ = 27.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 6.64± 0.20 %
μ = 417.3± 0.9
σ = 27.7± 0.8

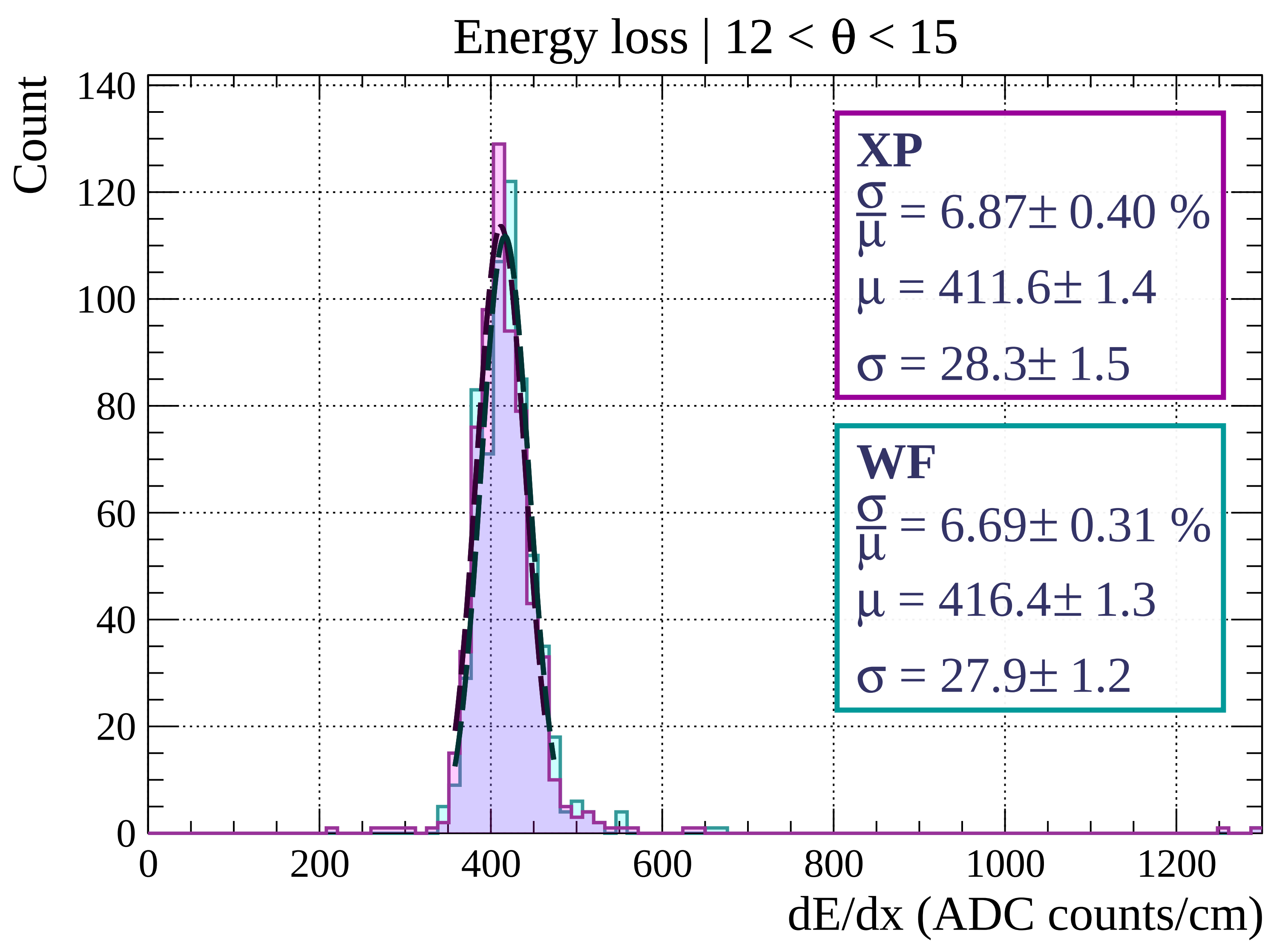

Energy loss | 3 < θ < 6
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 6.47± 0.19 %
μ = 412.6± 0.9
σ = 26.7± 0.7
WF
σ/μ = 6.71± 0.26 %
μ = 415.6± 1.0
σ = 27.9± 1.0

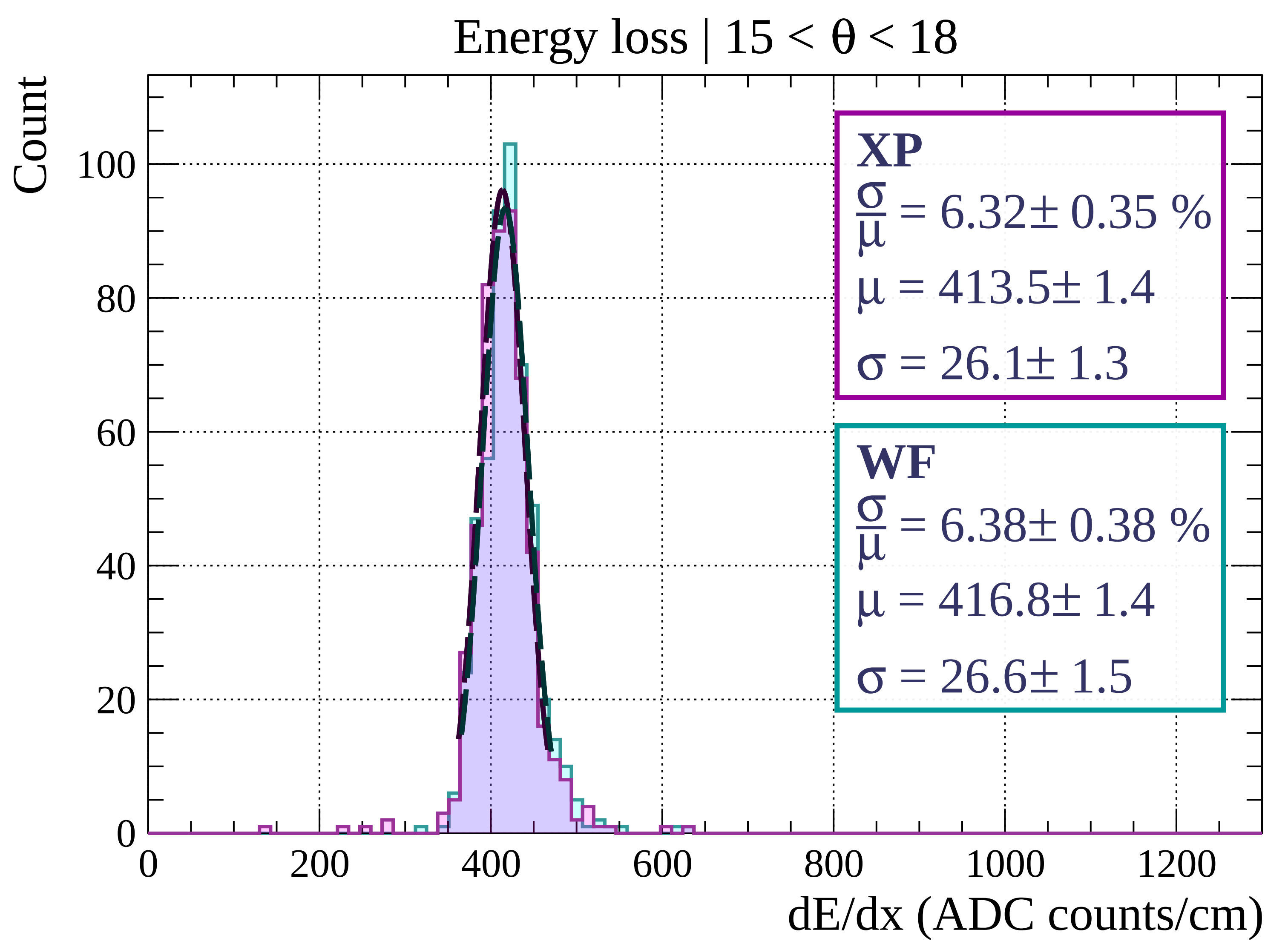


Energy loss | 6 < θ < 9
Count
XP
$\frac{\sigma}{\mu}$ = 6.29± 0.25 %
μ = 412.1± 1.0
σ = 25.9± 1.0
WF
$\frac{\sigma}{\mu}$ = 6.60± 0.28 %
μ = 416.2± 1.0
σ = 27.5± 1.1
dE/dx (ADC counts/cm)

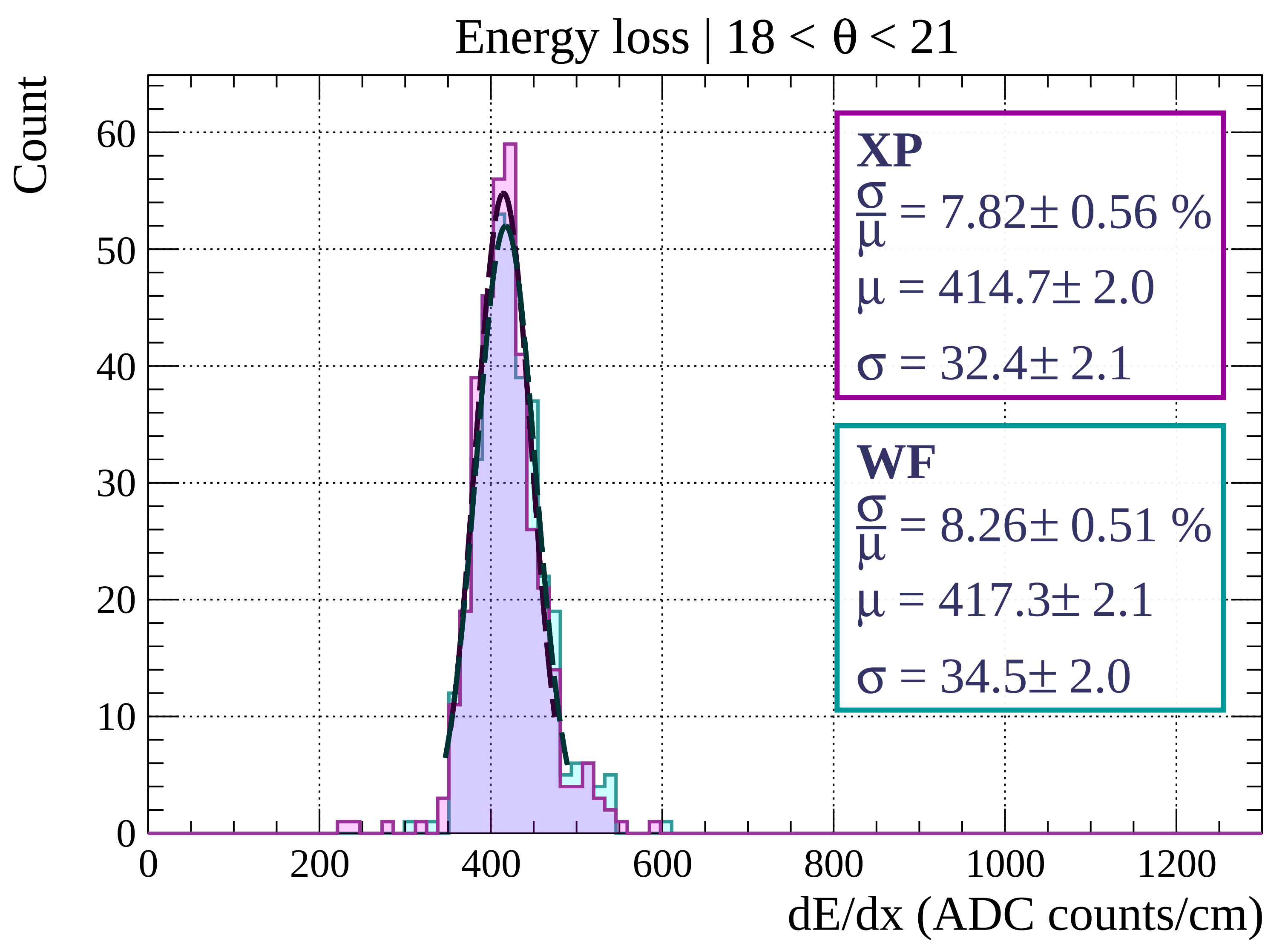


Energy loss | 9 < θ < 12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 6.76± 0.32 %
μ = 413.4± 1.2
σ = 28.0± 1.3
WF
σ/μ = 6.89± 0.37 %
μ = 416.7± 1.2
σ = 28.7± 1.4

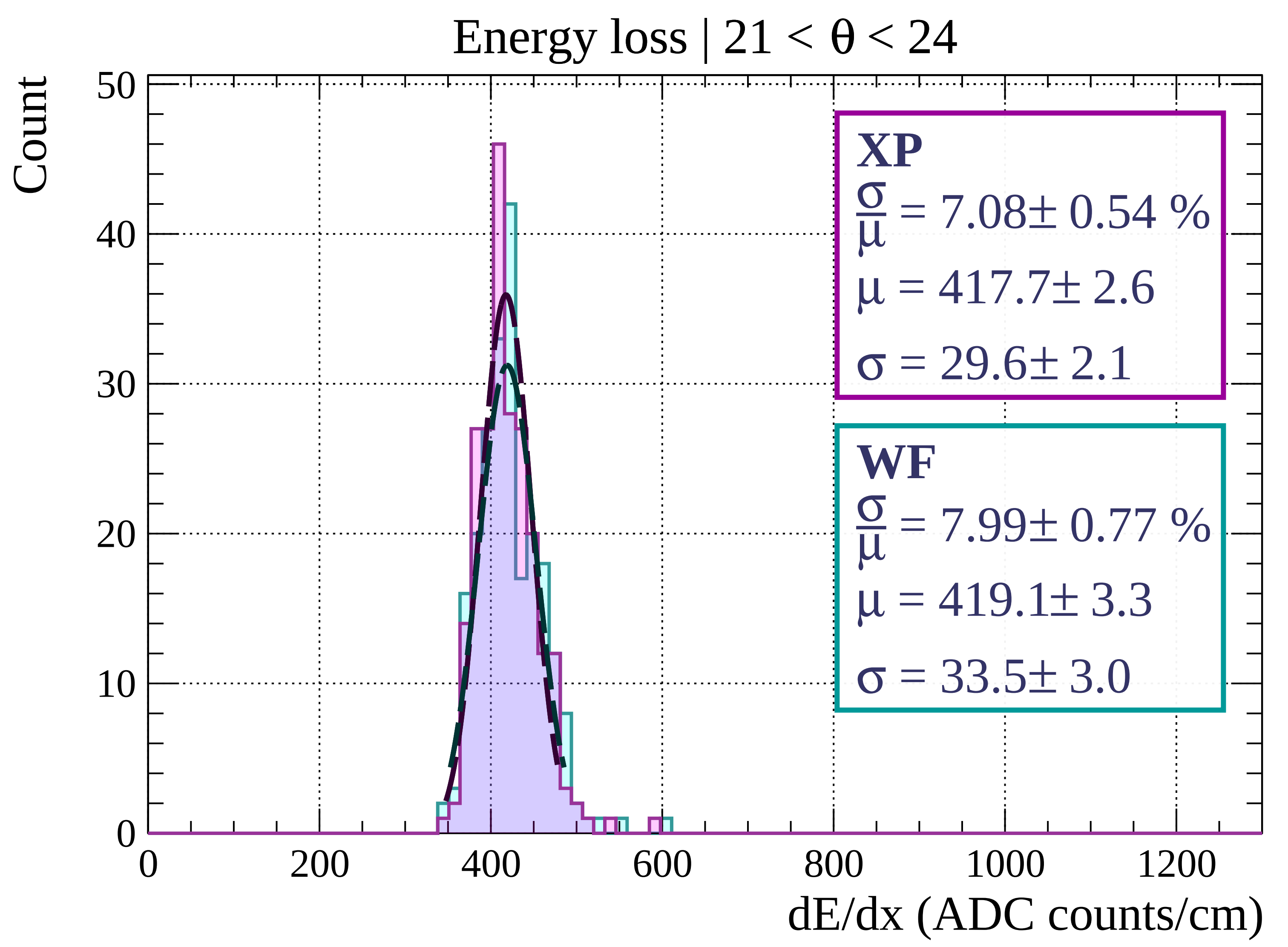

Energy loss | 12 < θ < 15
Count
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.87± 0.40 %
μ = 411.6± 1.4
σ = 28.3± 1.5
WF
$\frac{\sigma}{\mu}$ = 6.69± 0.31 %
μ = 416.4± 1.3
σ = 27.9± 1.2



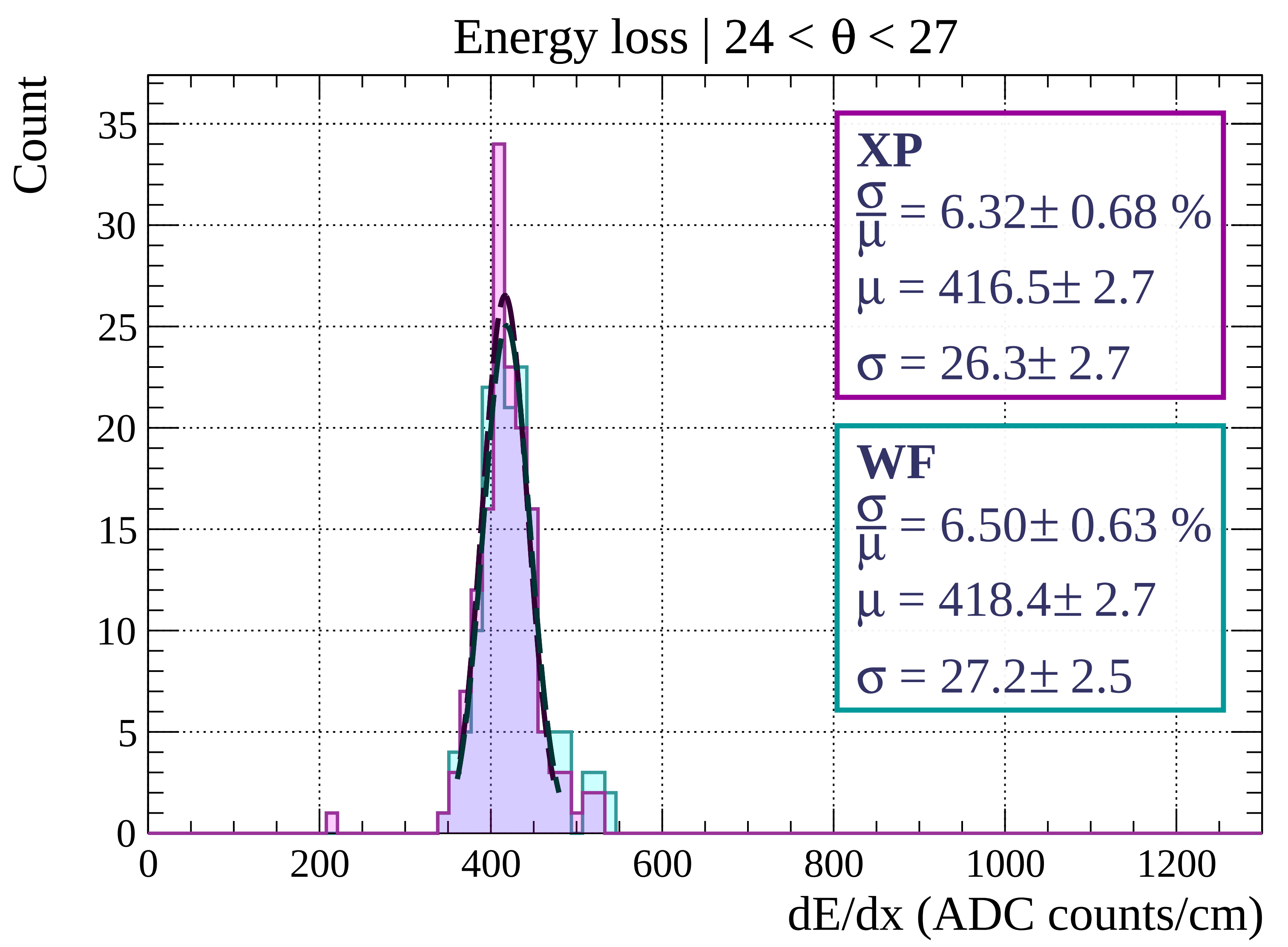
Energy loss | 15 < θ < 18
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 6.32± 0.35 %
μ = 413.5± 1.4
σ = 26.1± 1.3
WF
$\frac{\sigma}{\mu}$ = 6.38± 0.38 %
μ = 416.8± 1.4
σ = 26.6± 1.5



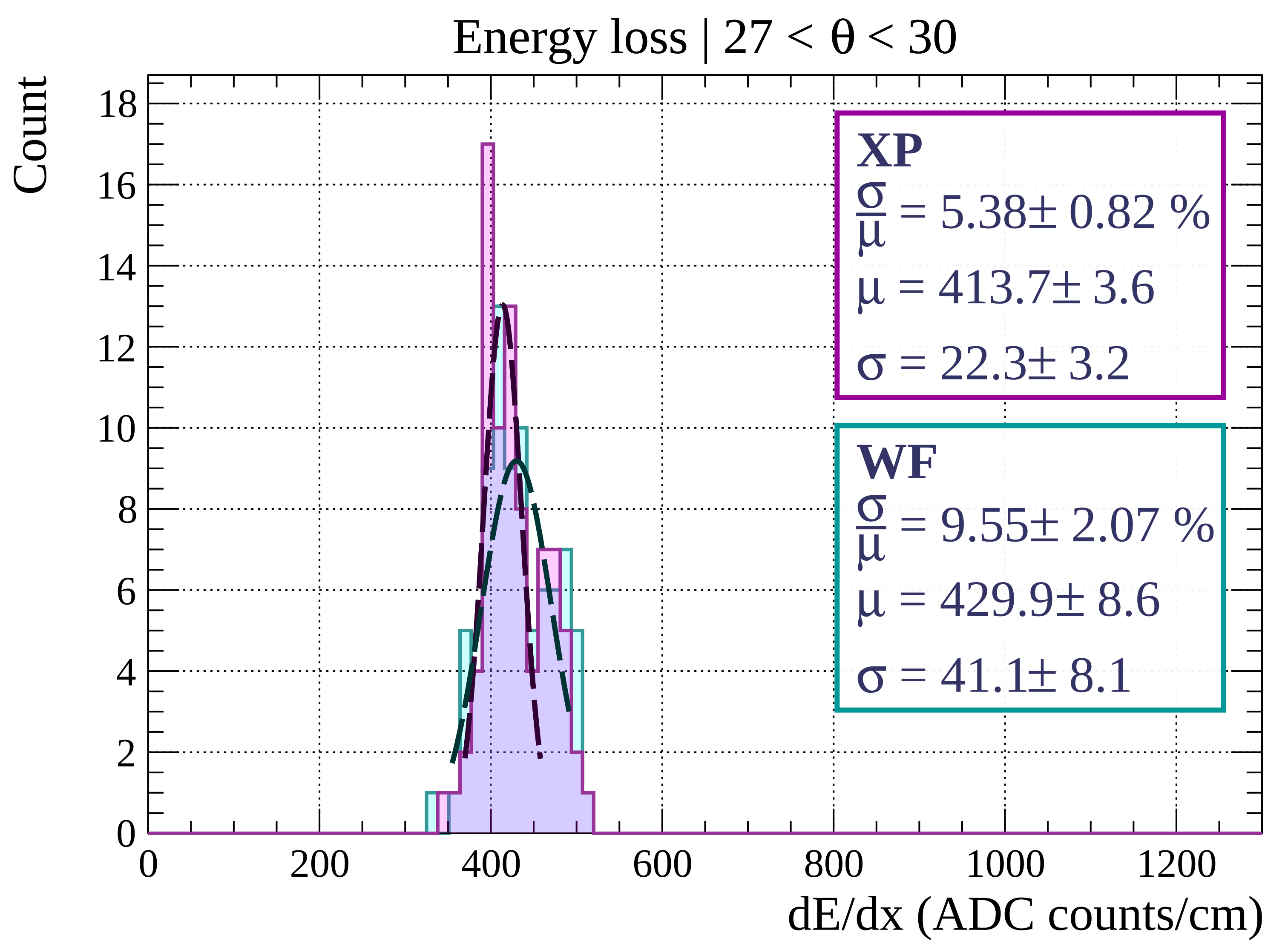
Energy loss | 18 < θ < 21
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200
XP
σ/μ = 7.82± 0.56 %
μ = 414.7± 2.0
σ = 32.4± 2.1
WF
σ/μ = 8.26± 0.51 %
μ = 417.3± 2.1
σ = 34.5± 2.0



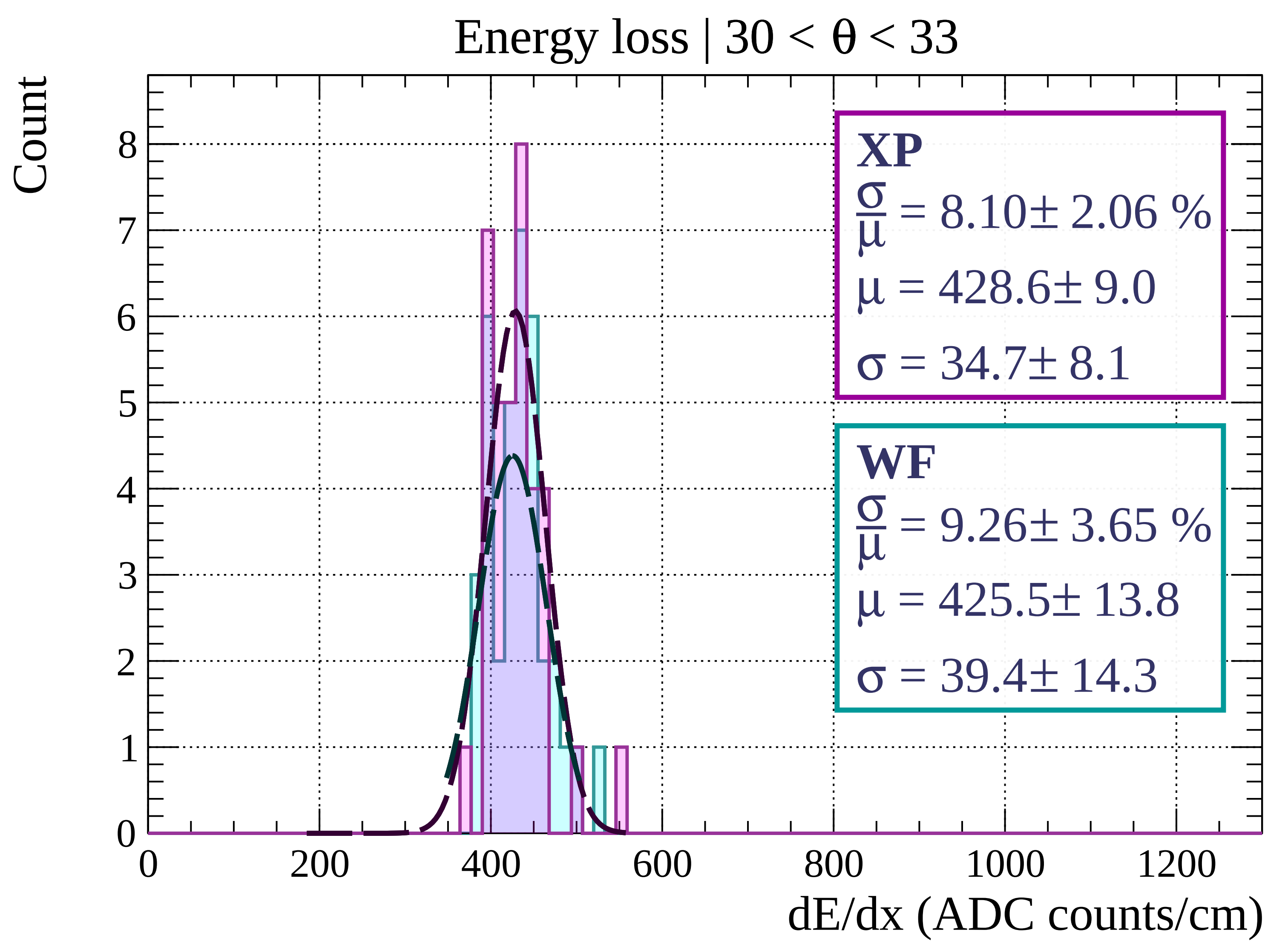
Energy loss | 21 < θ < 24
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
σ/μ = 7.08± 0.54 %
μ = 417.7± 2.6
σ = 29.6± 2.1
WF
σ/μ = 7.99± 0.77 %
μ = 419.1± 3.3
σ = 33.5± 3.0

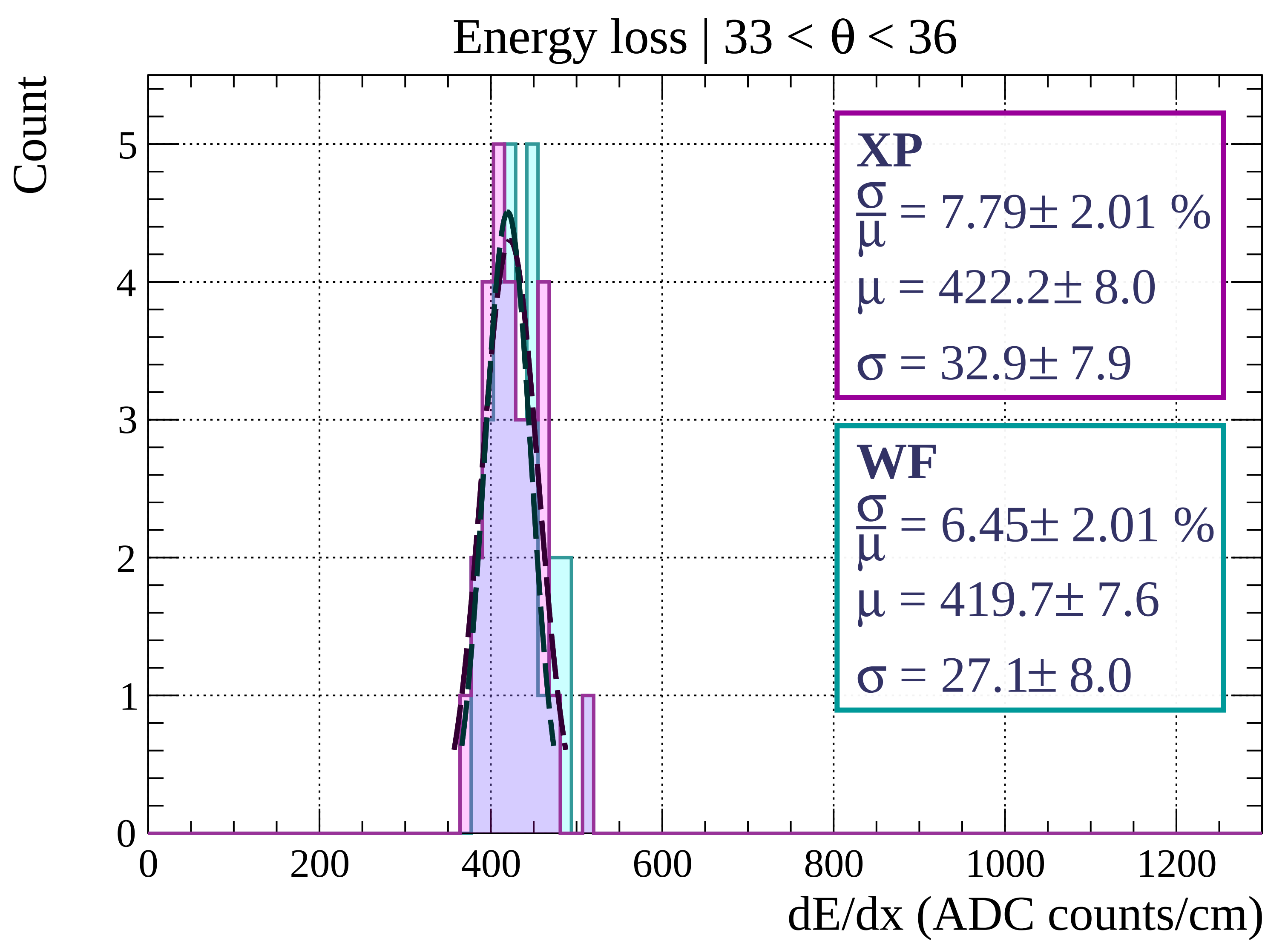


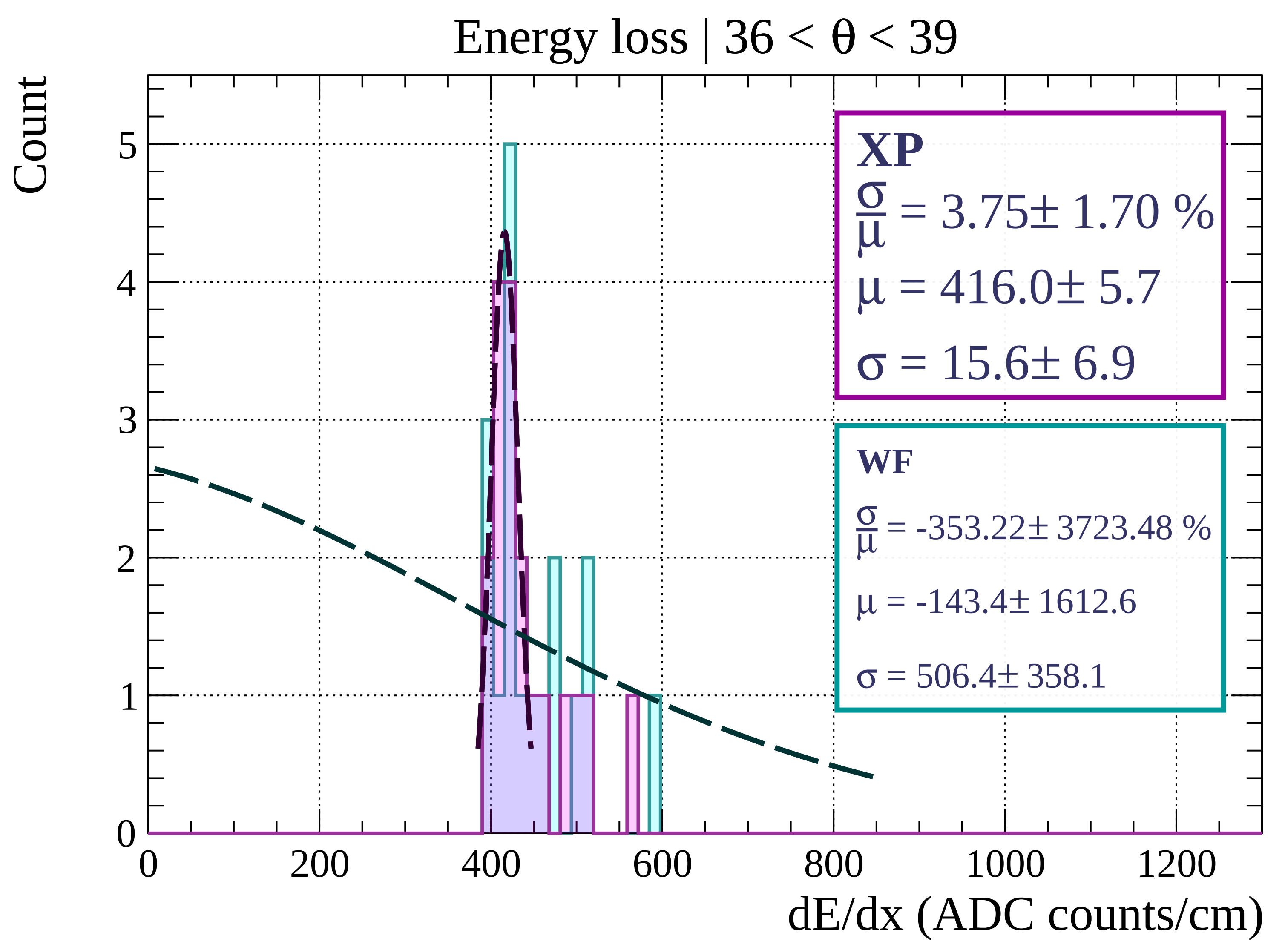
Energy loss | 24 < θ < 27
Count
35
30
25
20
15
10
5
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 6.32± 0.68 %
μ = 416.5± 2.7
σ = 26.3± 2.7
WF
$\frac{\sigma}{\mu}$ = 6.50± 0.63 %
μ = 418.4± 2.7
σ = 27.2± 2.5



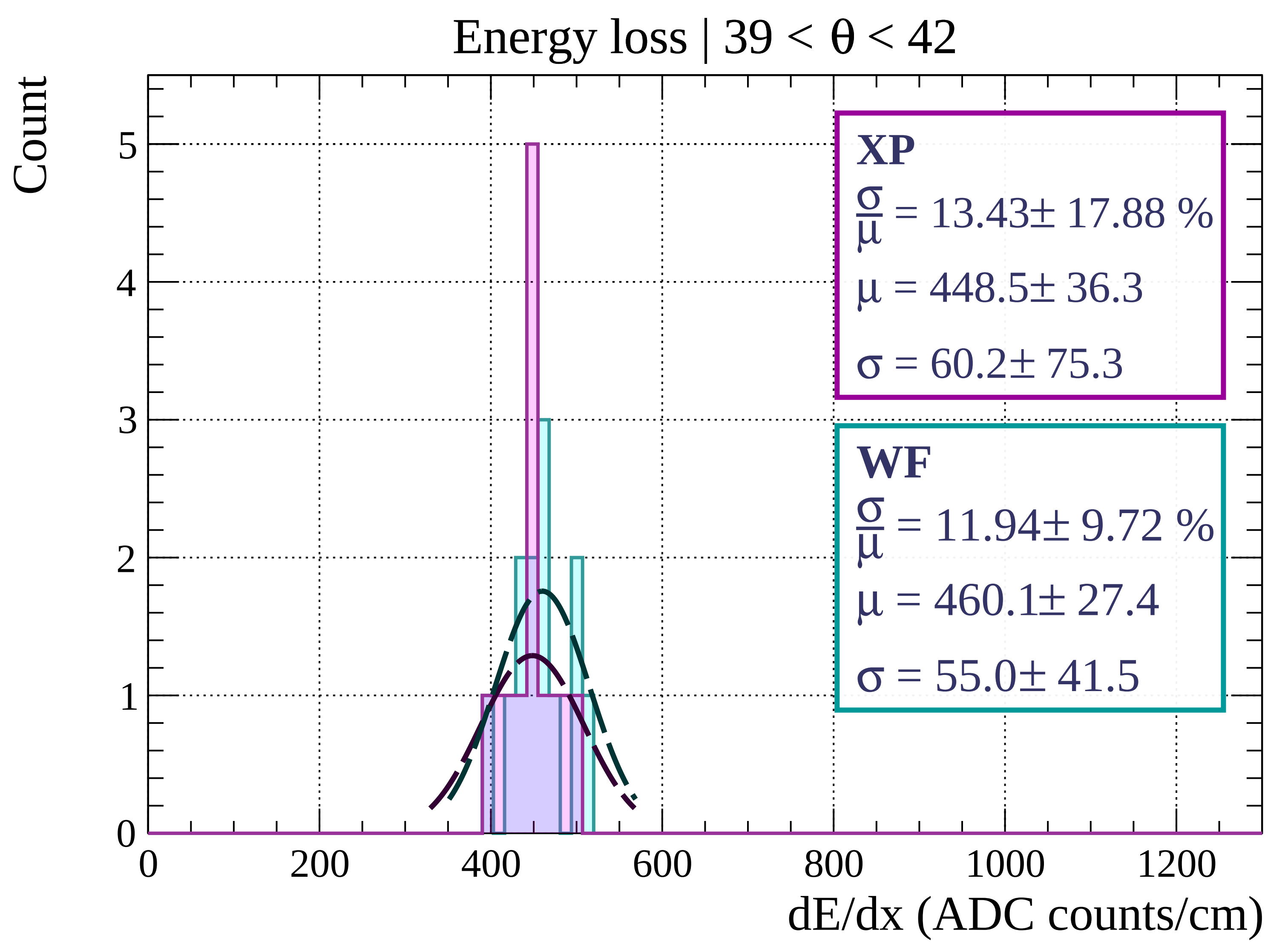
Energy loss | 27 < θ < 30
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
0
200
400
600
800
1000
1200
XP
σ/μ = 5.38± 0.82 %
μ = 413.7± 3.6
σ = 22.3± 3.2
WF
σ/μ = 9.55± 2.07 %
μ = 429.9± 8.6
σ = 41.1± 8.1


Energy loss | 30 < θ < 33
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
6
7
8
XP
$\frac{\sigma}{\mu}$ = 8.10± 2.06 %
μ = 428.6± 9.0
σ = 34.7± 8.1
WF
$\frac{\sigma}{\mu}$ = 9.26± 3.65 %
μ = 425.5± 13.8
σ = 39.4± 14.3

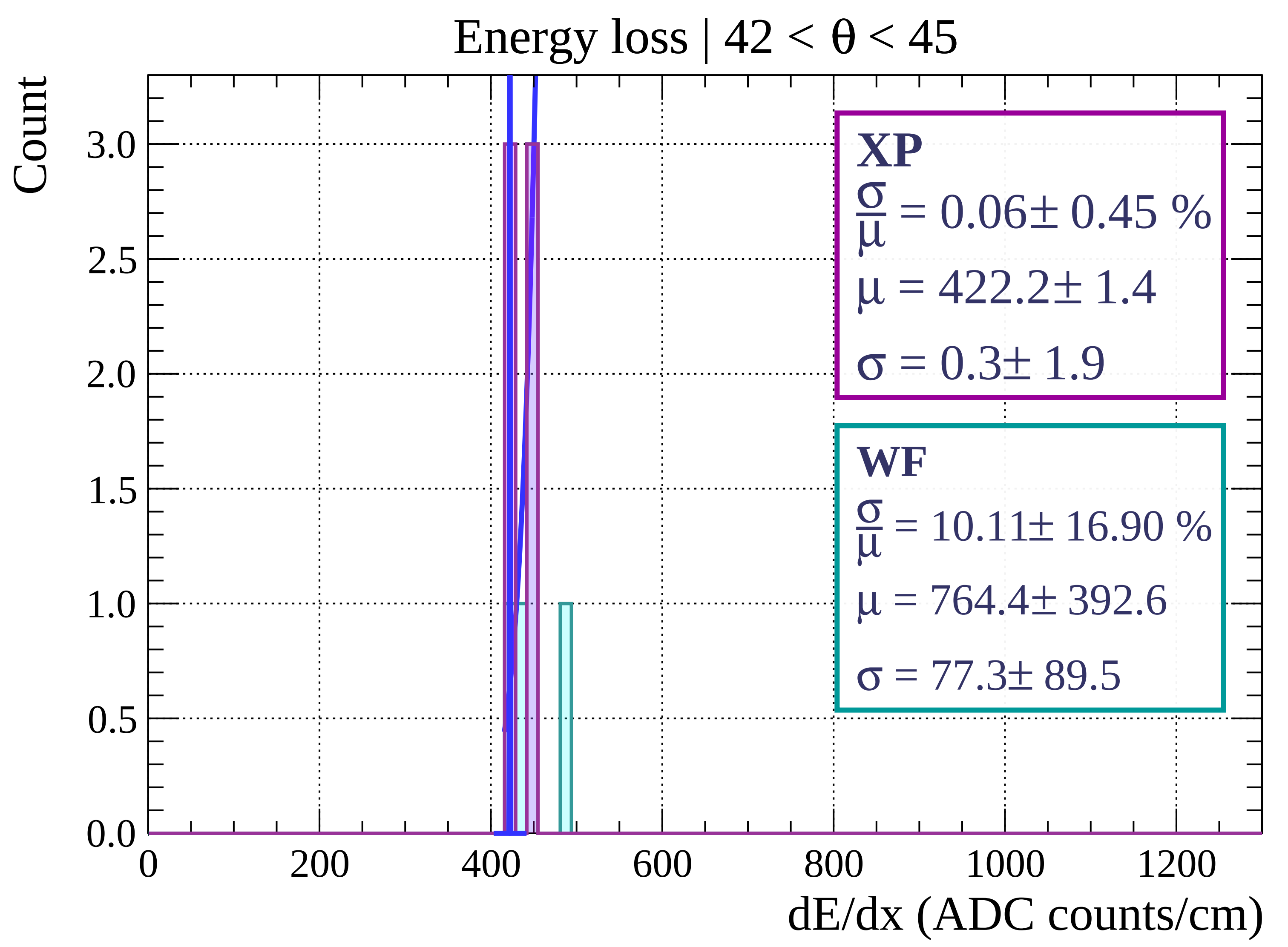

Energy loss | 33 < θ < 36
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 7.79± 2.01 %
μ = 422.2± 8.0
σ = 32.9± 7.9
WF
σ/μ = 6.45± 2.01 %
μ = 419.7± 7.6
σ = 27.1± 8.0



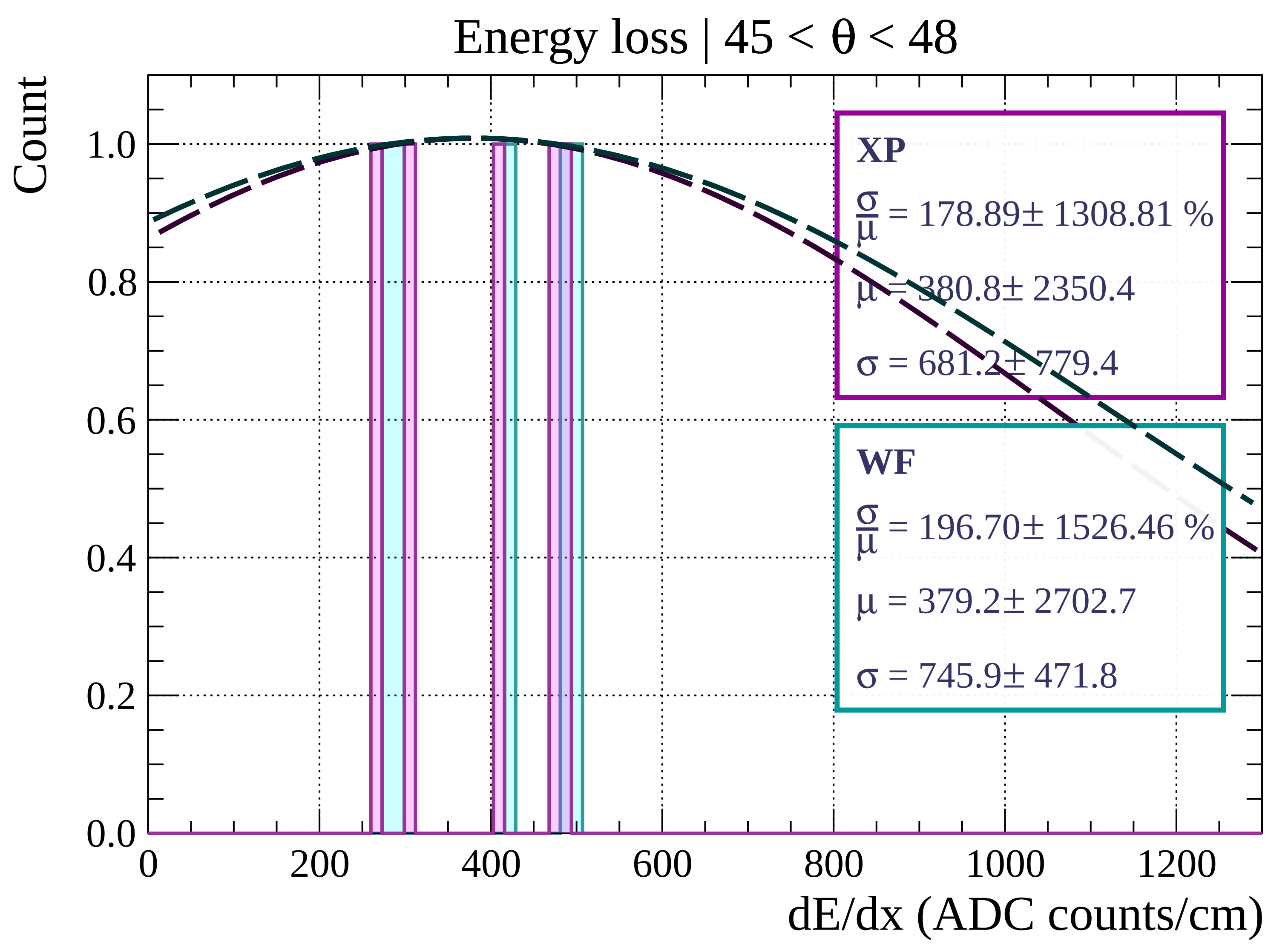
Energy loss | 36 < θ < 39
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 3.75± 1.70 %
μ = 416.0± 5.7
σ = 15.6± 6.9
WF
σ/μ = -353.22± 3723.48 %
μ = -143.4± 1612.6
σ = 506.4± 358.1

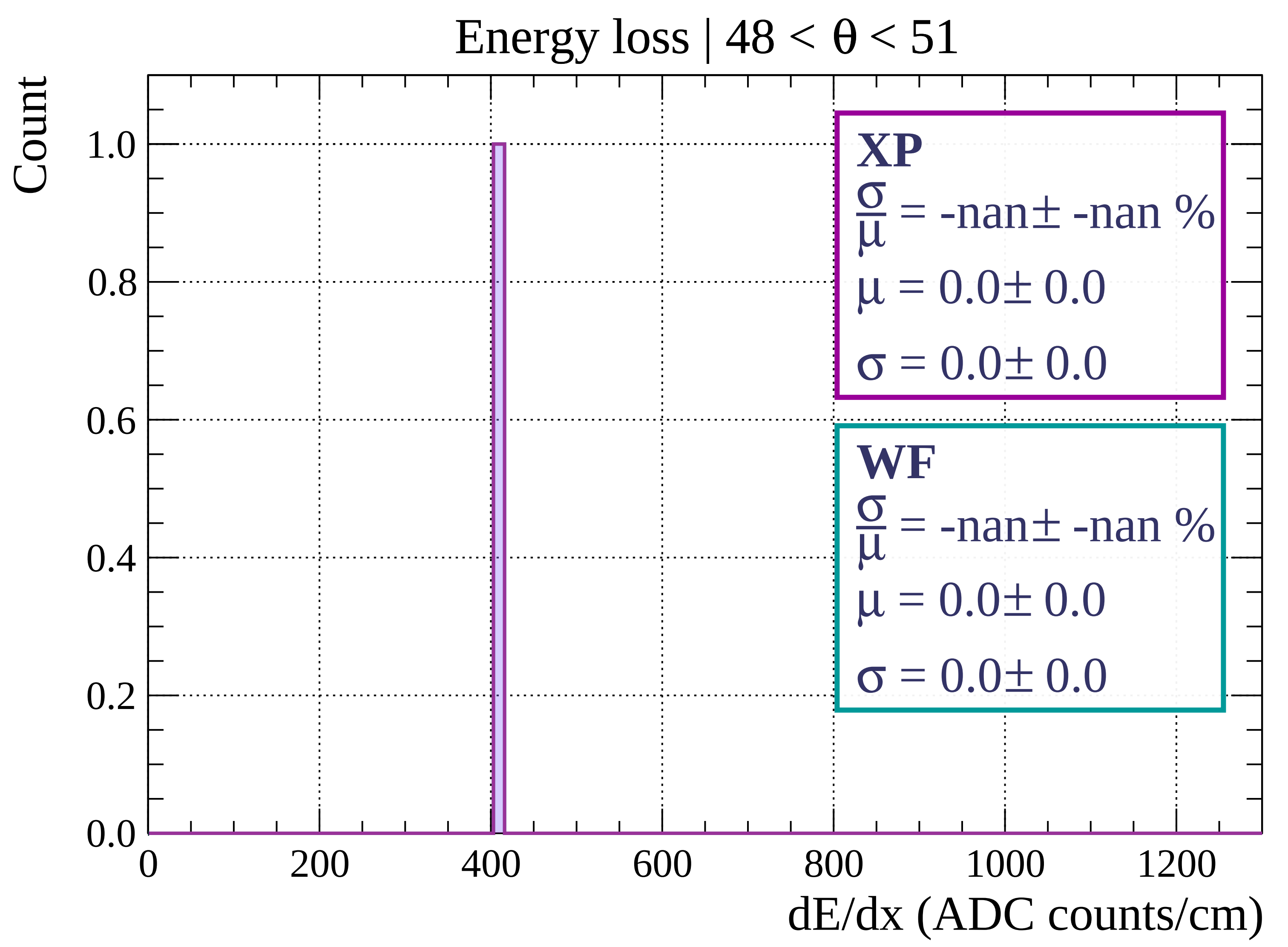


Energy loss | 39 < θ < 42
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
1
2
3
4
5
XP
σ/μ = 13.43± 17.88 %
μ = 448.5± 36.3
σ = 60.2± 75.3
WF
σ/μ = 11.94± 9.72 %
μ = 460.1± 27.4
σ = 55.0± 41.5

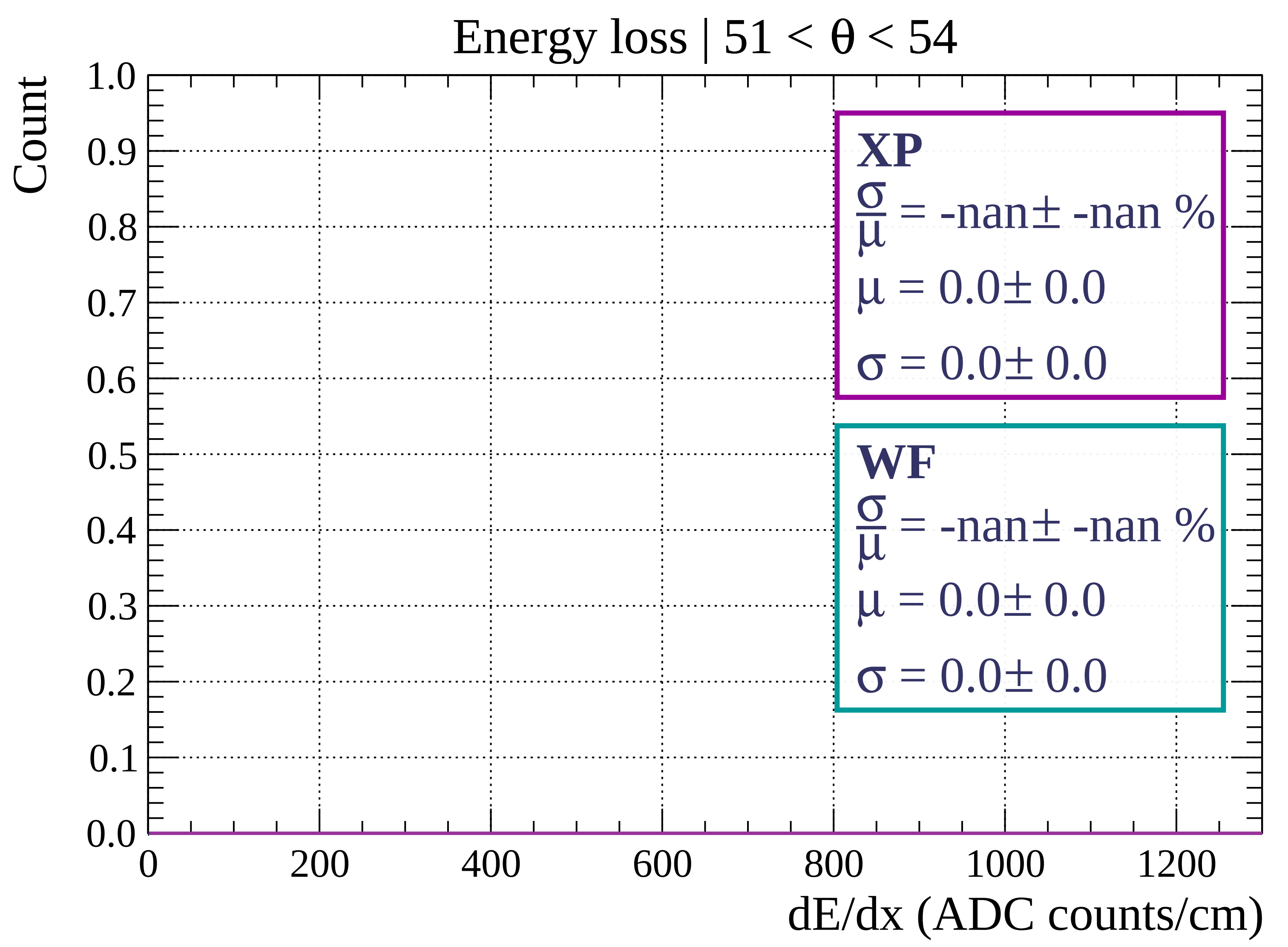


Energy loss | 42 < θ < 45
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.06± 0.45 %
μ = 422.2± 1.4
σ = 0.3± 1.9
WF
σ/μ = 10.11± 16.90 %
μ = 764.4± 392.6
σ = 77.3± 89.5

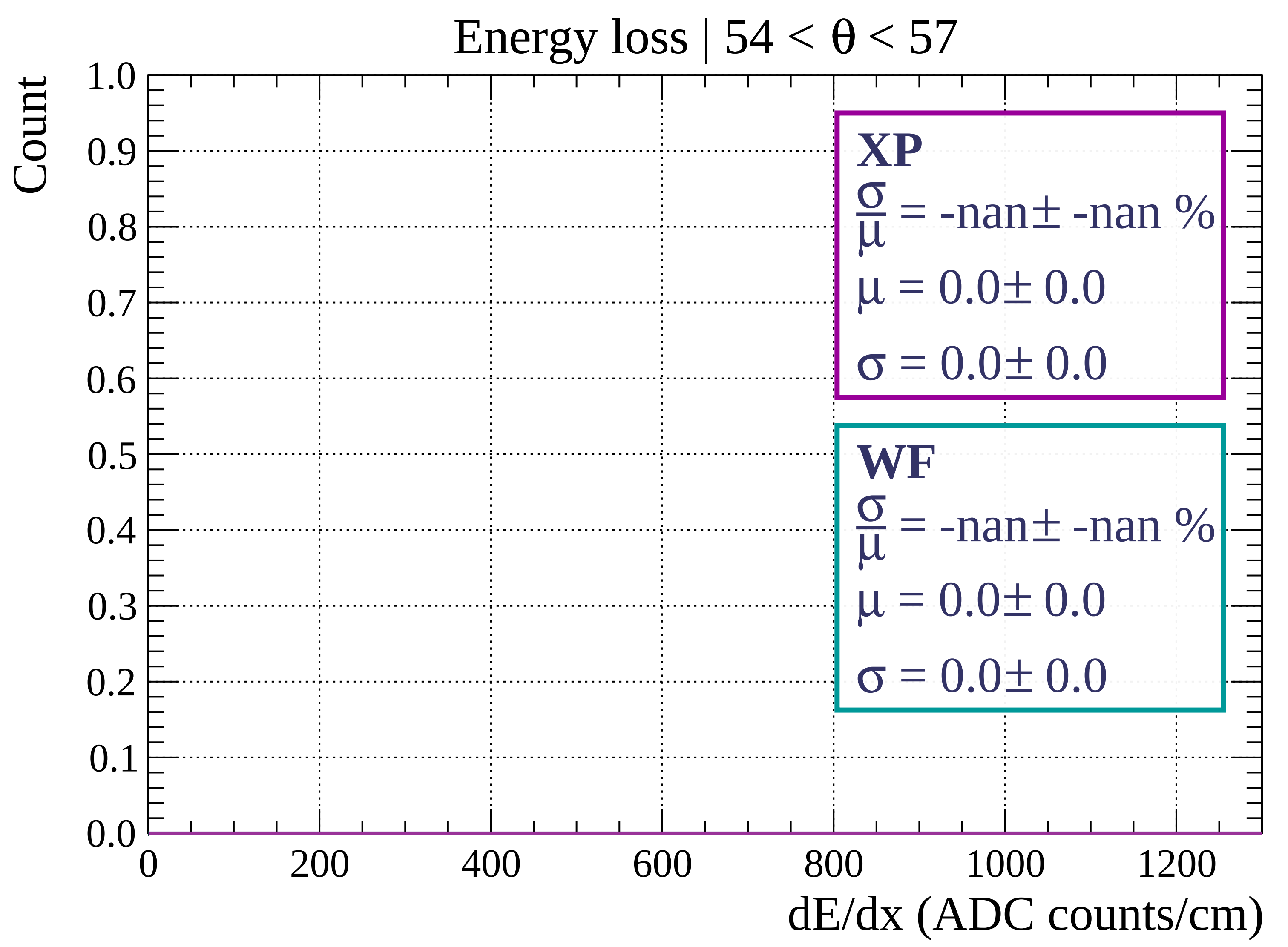

Energy loss | 45 < θ < 48
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 178.89± 1308.81 %
μ = 380.8± 2350.4
σ = 681.2± 779.4
WF
σ/μ = 196.70± 1526.46 %
μ = 379.2± 2702.7
σ = 745.9± 471.8



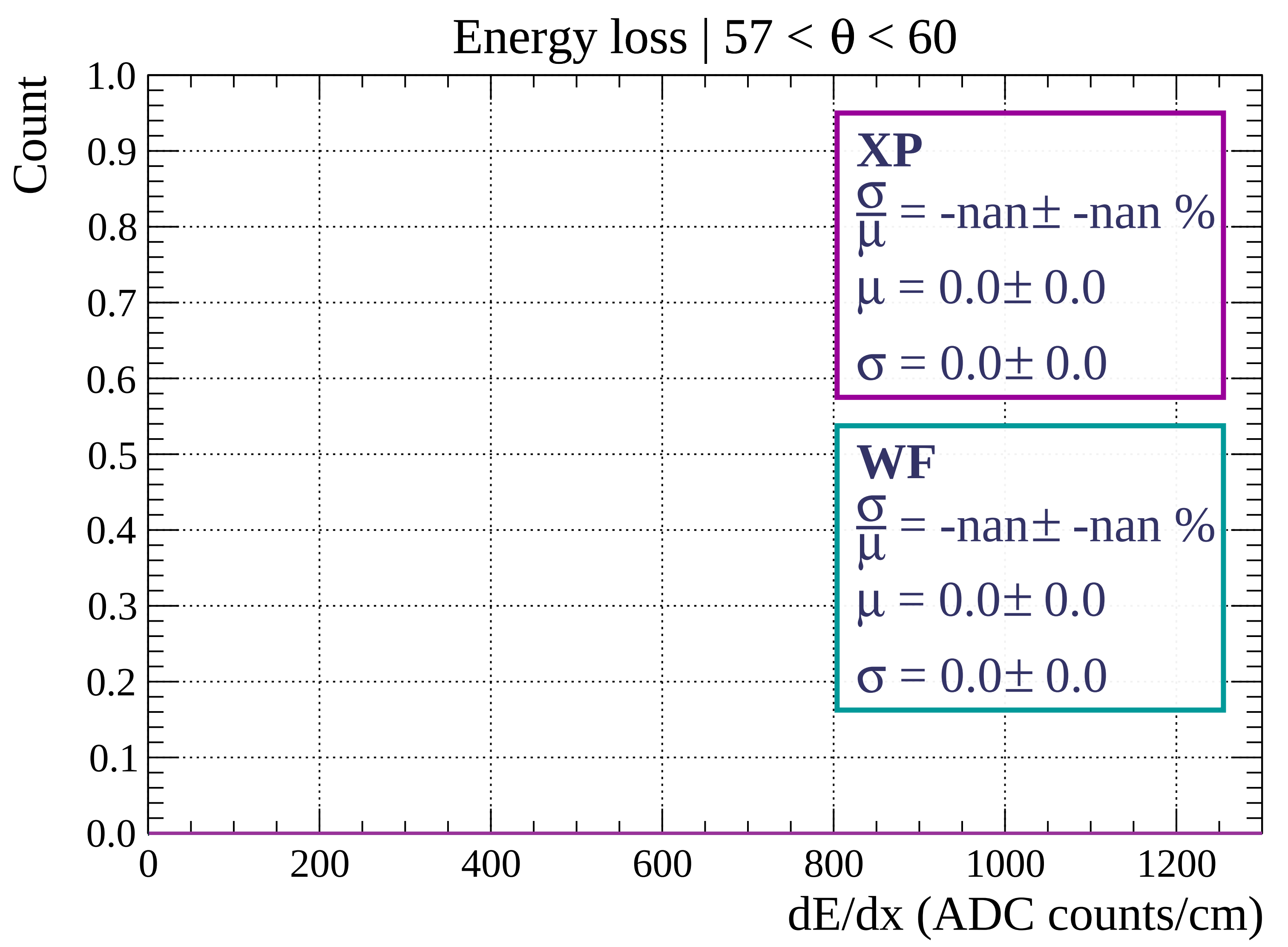
Energy loss | 48 < θ < 51
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

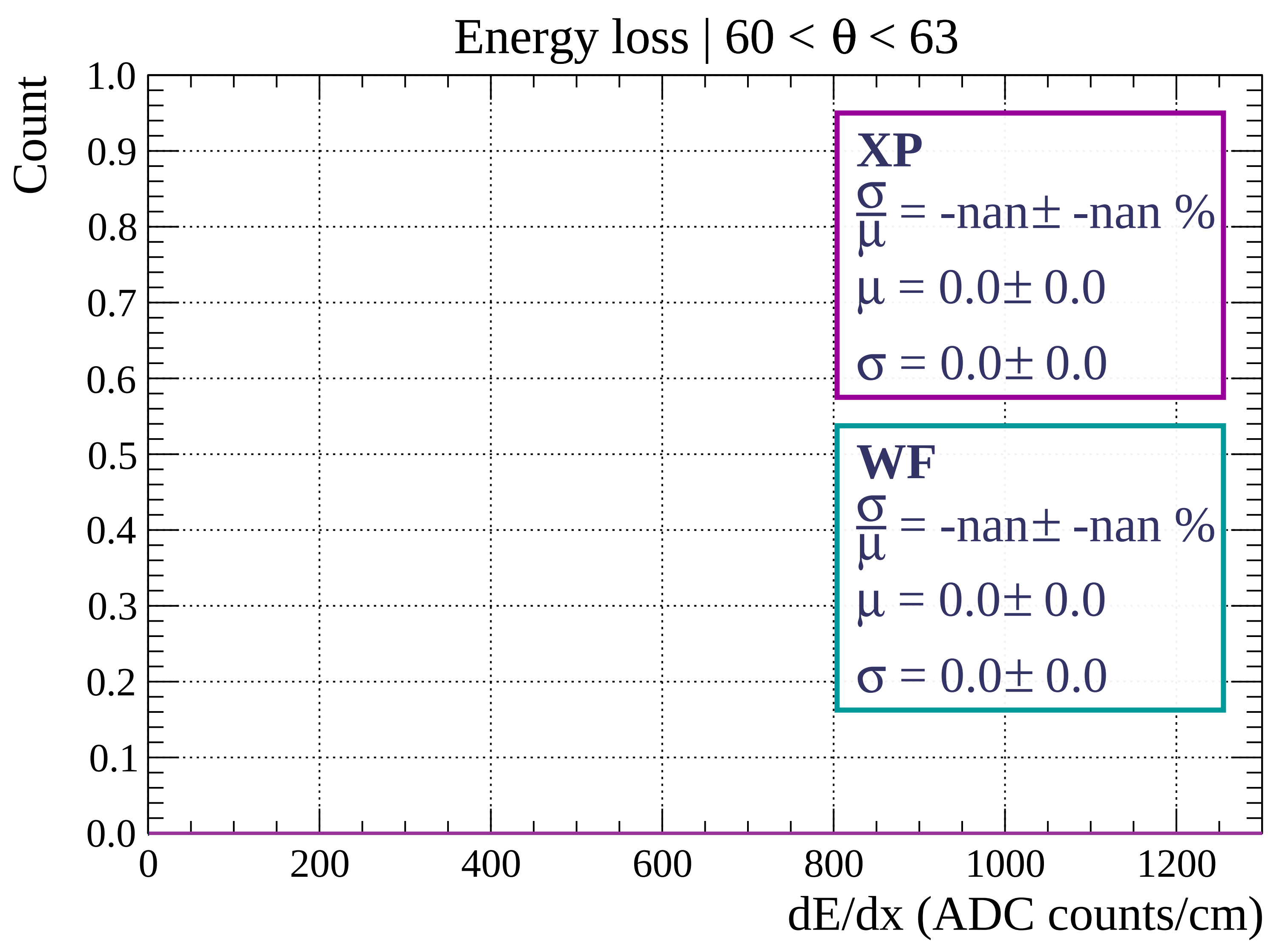


Energy loss | 51 < θ < 54
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

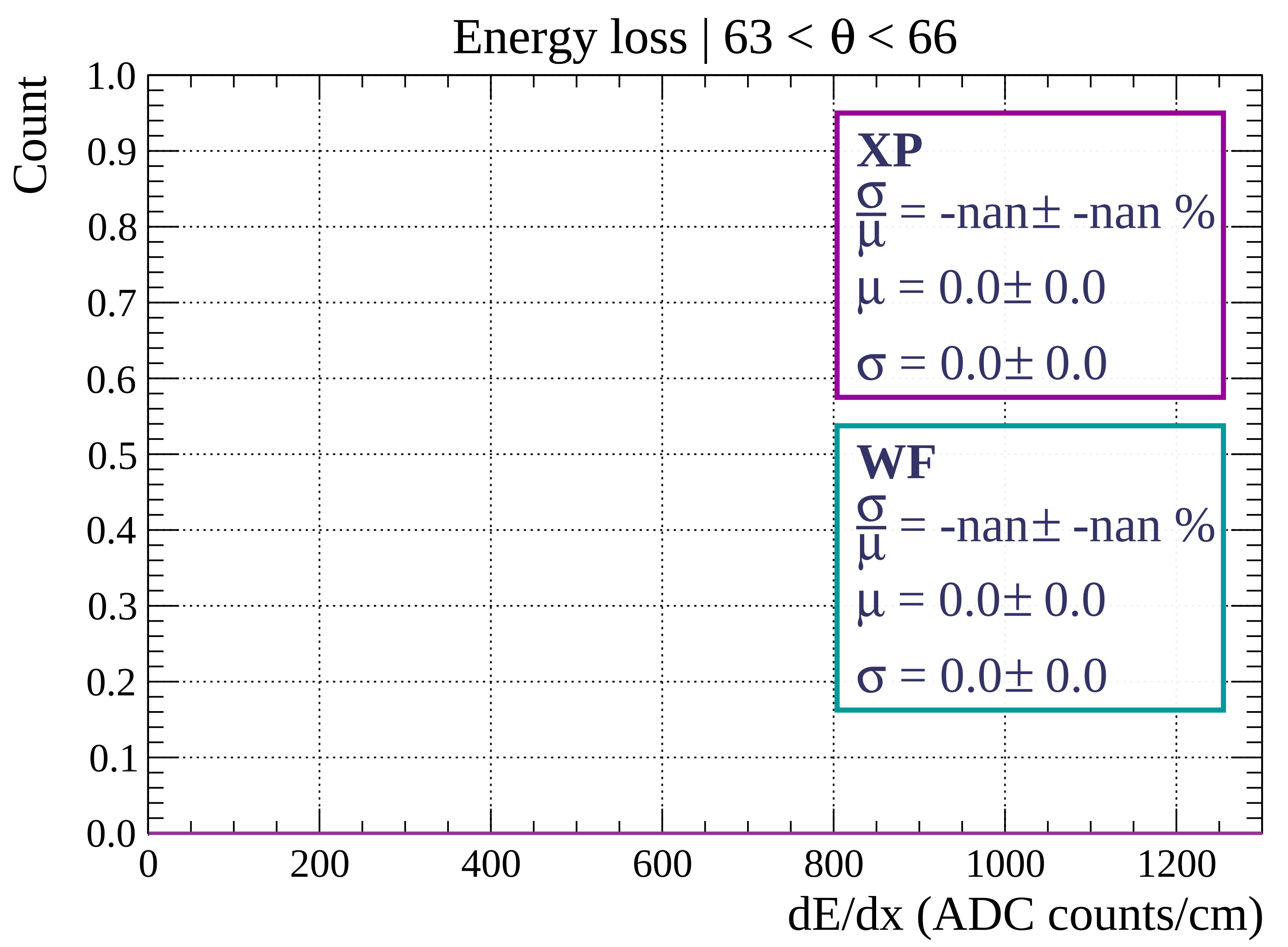


Energy loss | 54 < θ < 57
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

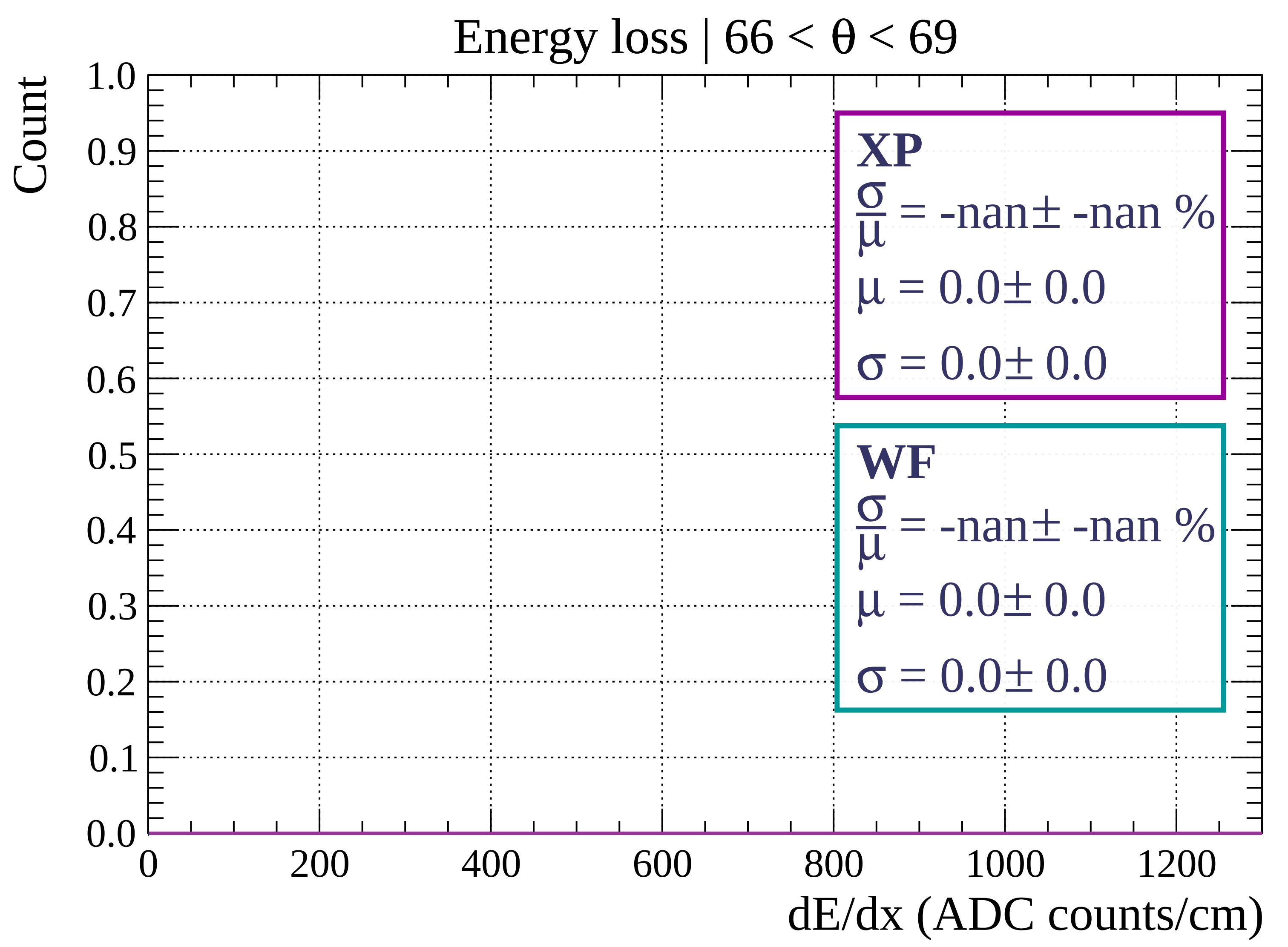

Energy loss | 57 < θ < 60
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



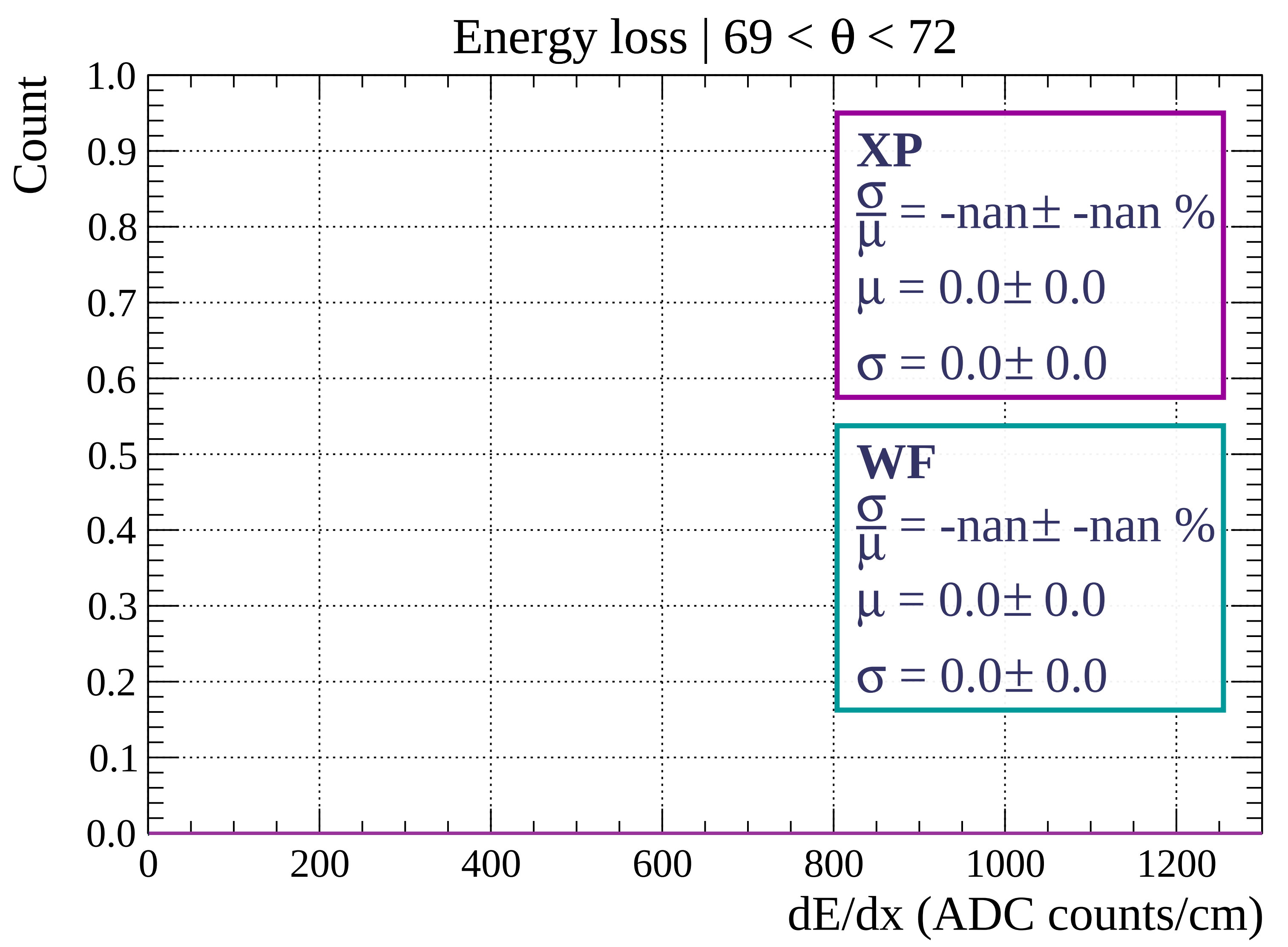
Energy loss | 60 < θ < 63
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

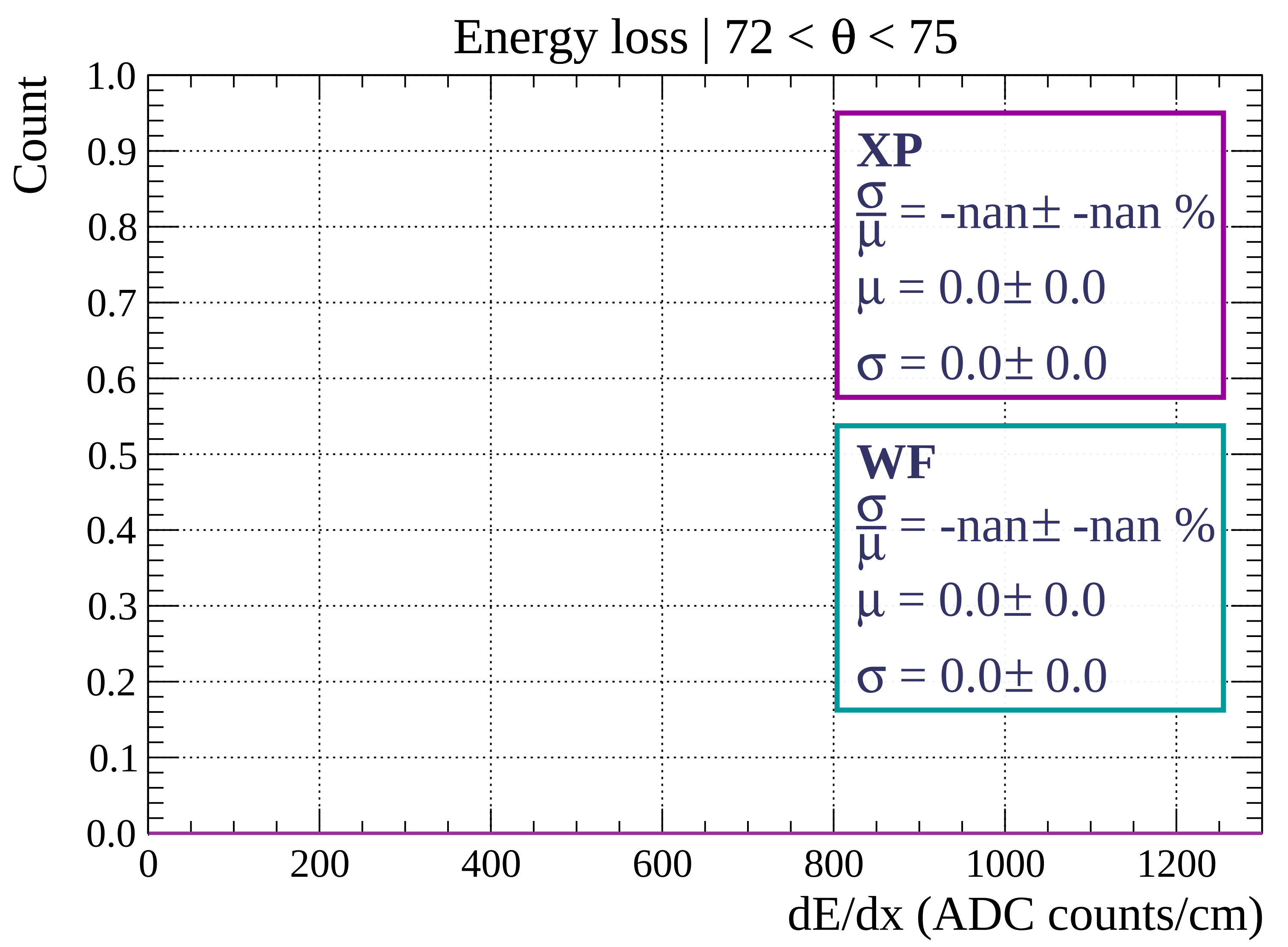


Energy loss | 63 < θ < 66
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

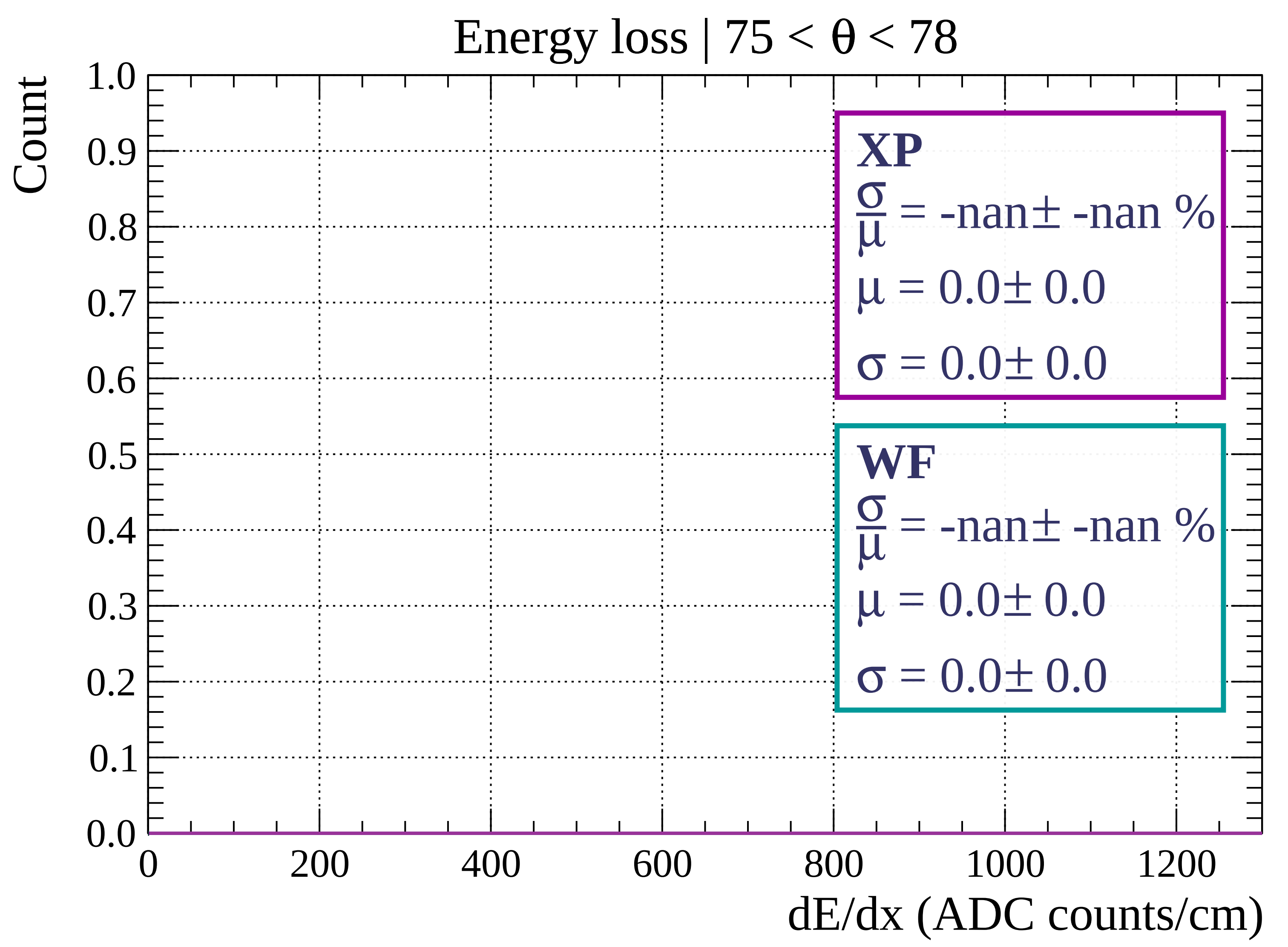


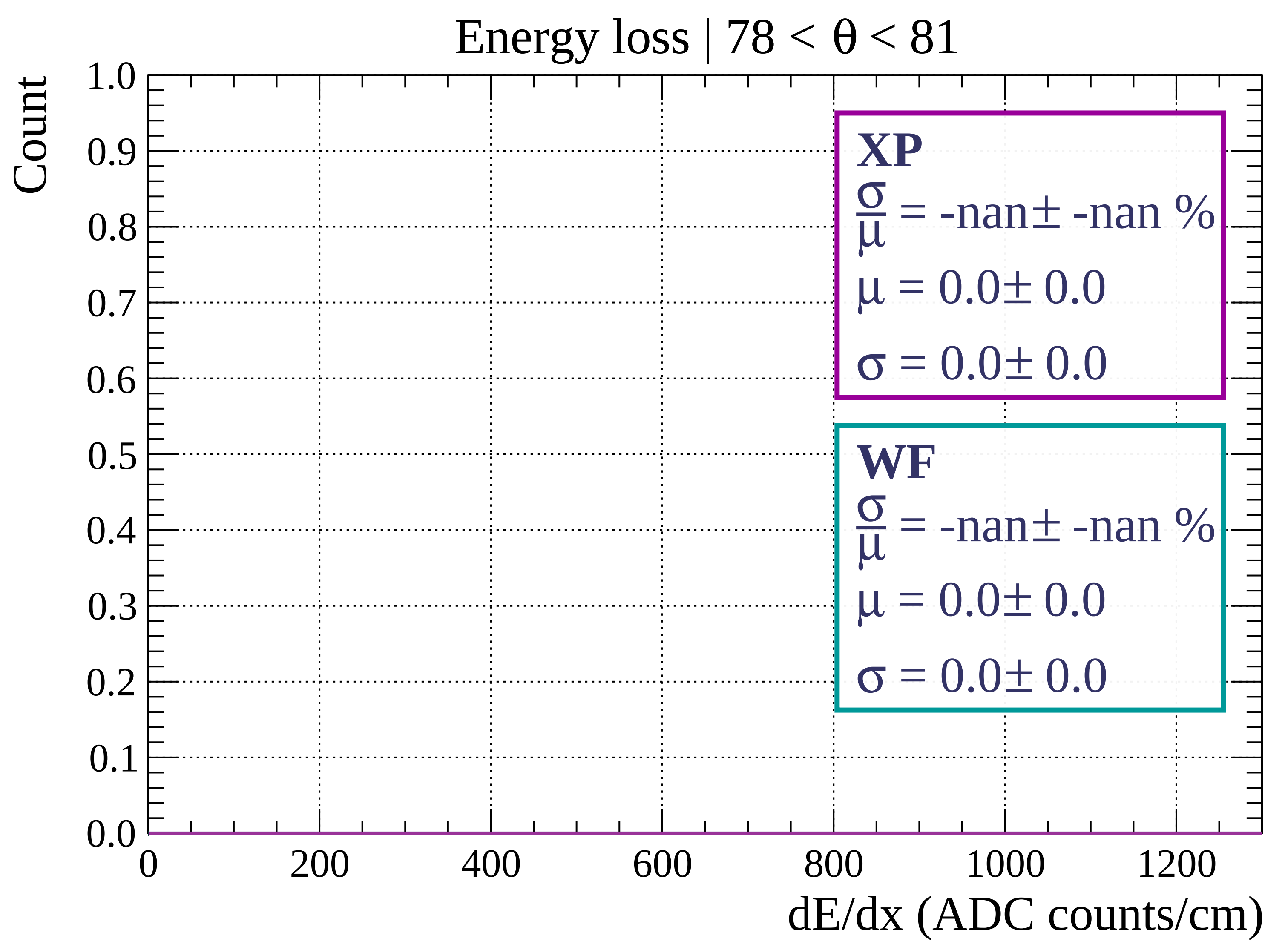
Energy loss | 66 < θ < 69
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 69 < θ < 72
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

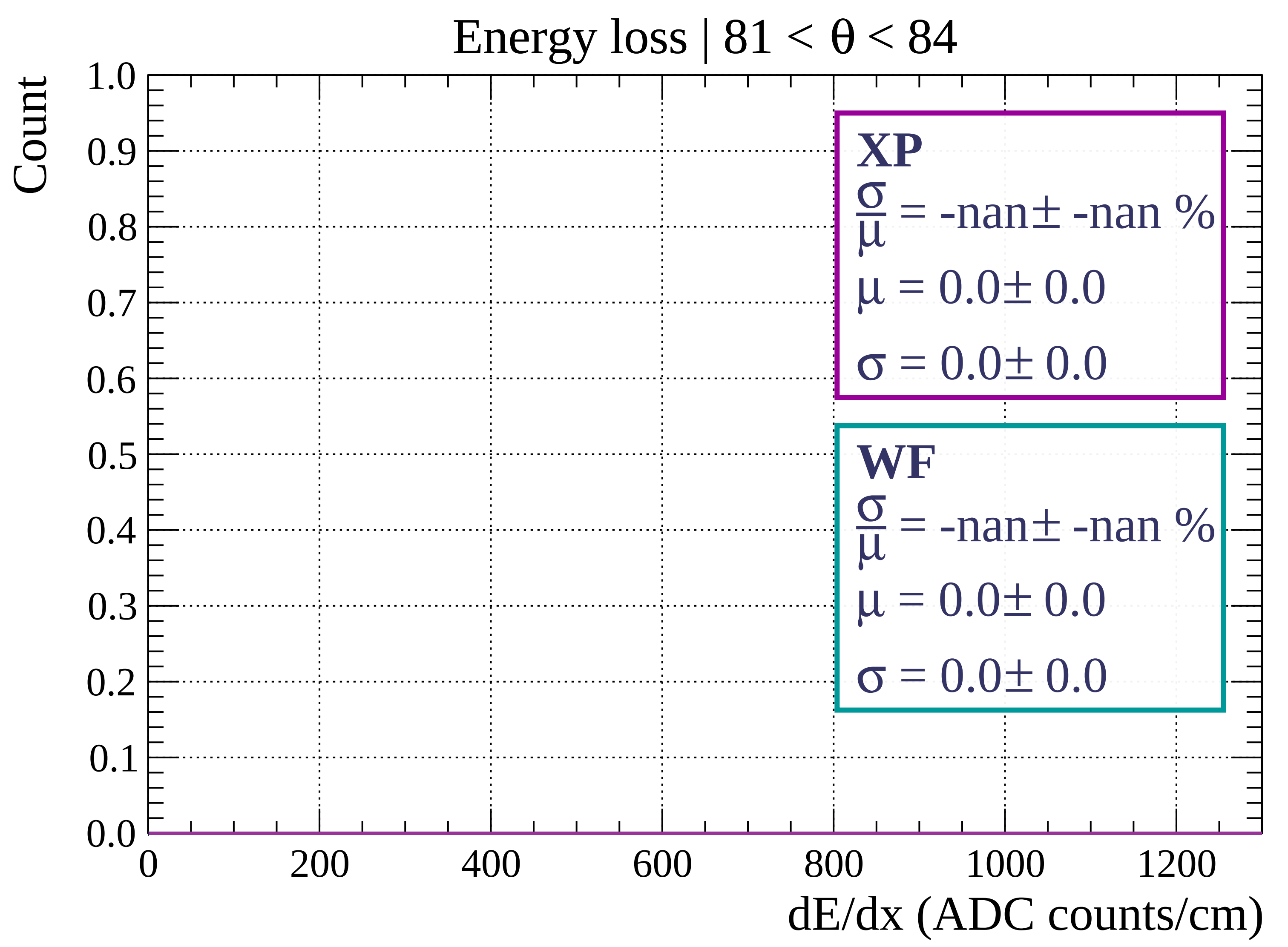


Energy loss | 72 < θ < 75
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

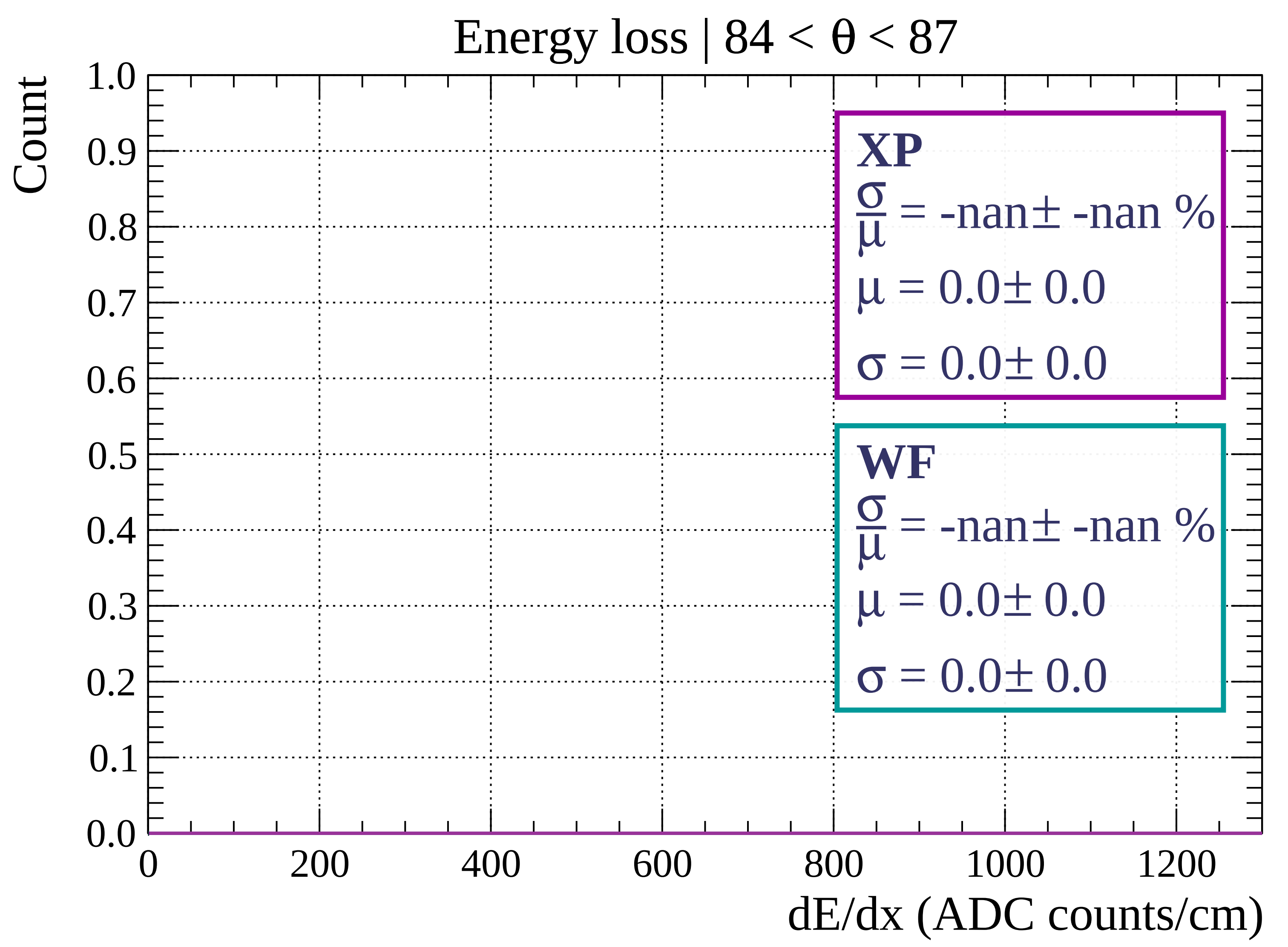

Energy loss | 75 < θ < 78
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



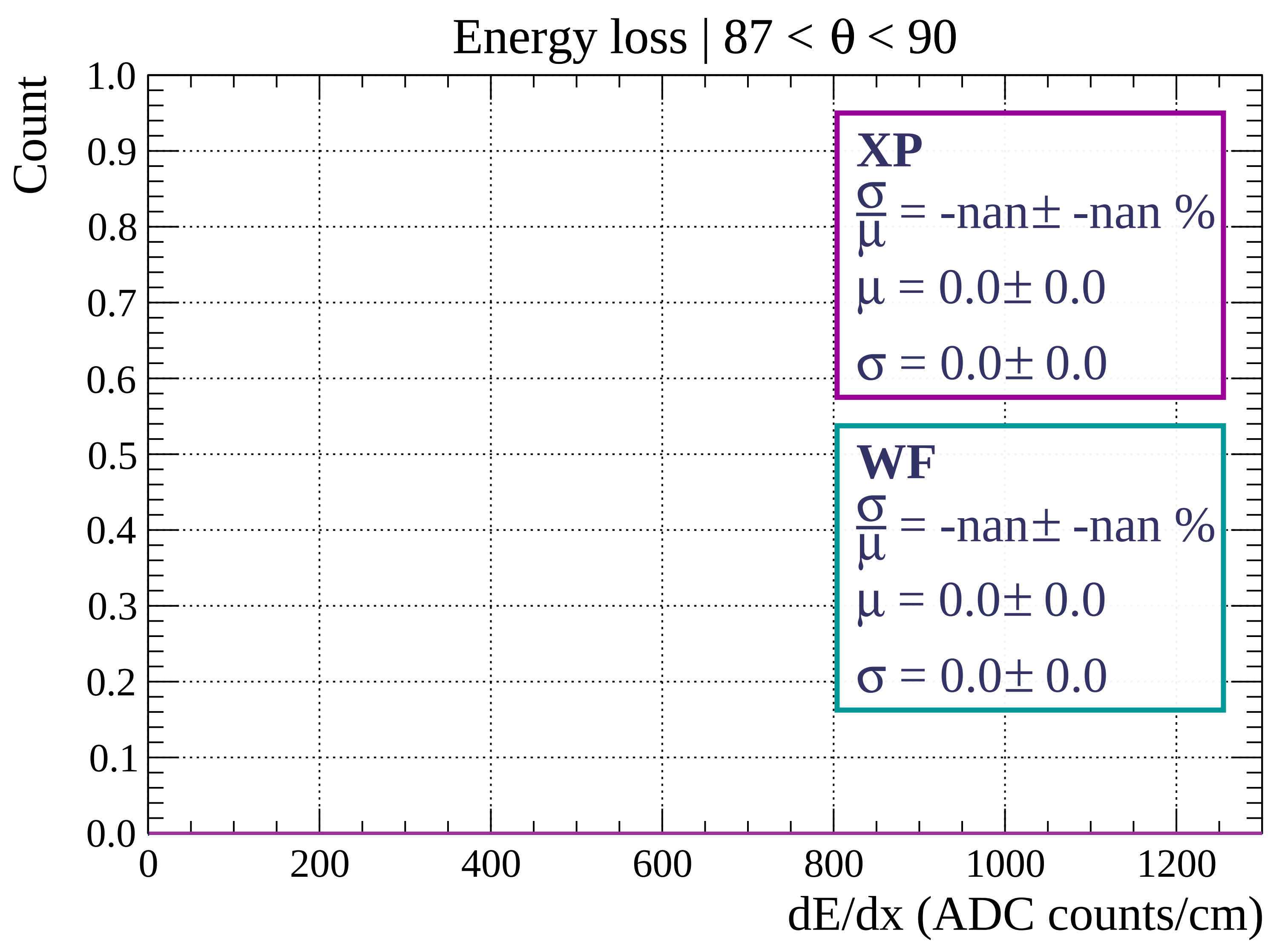
Energy loss | 78 < θ < 81
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

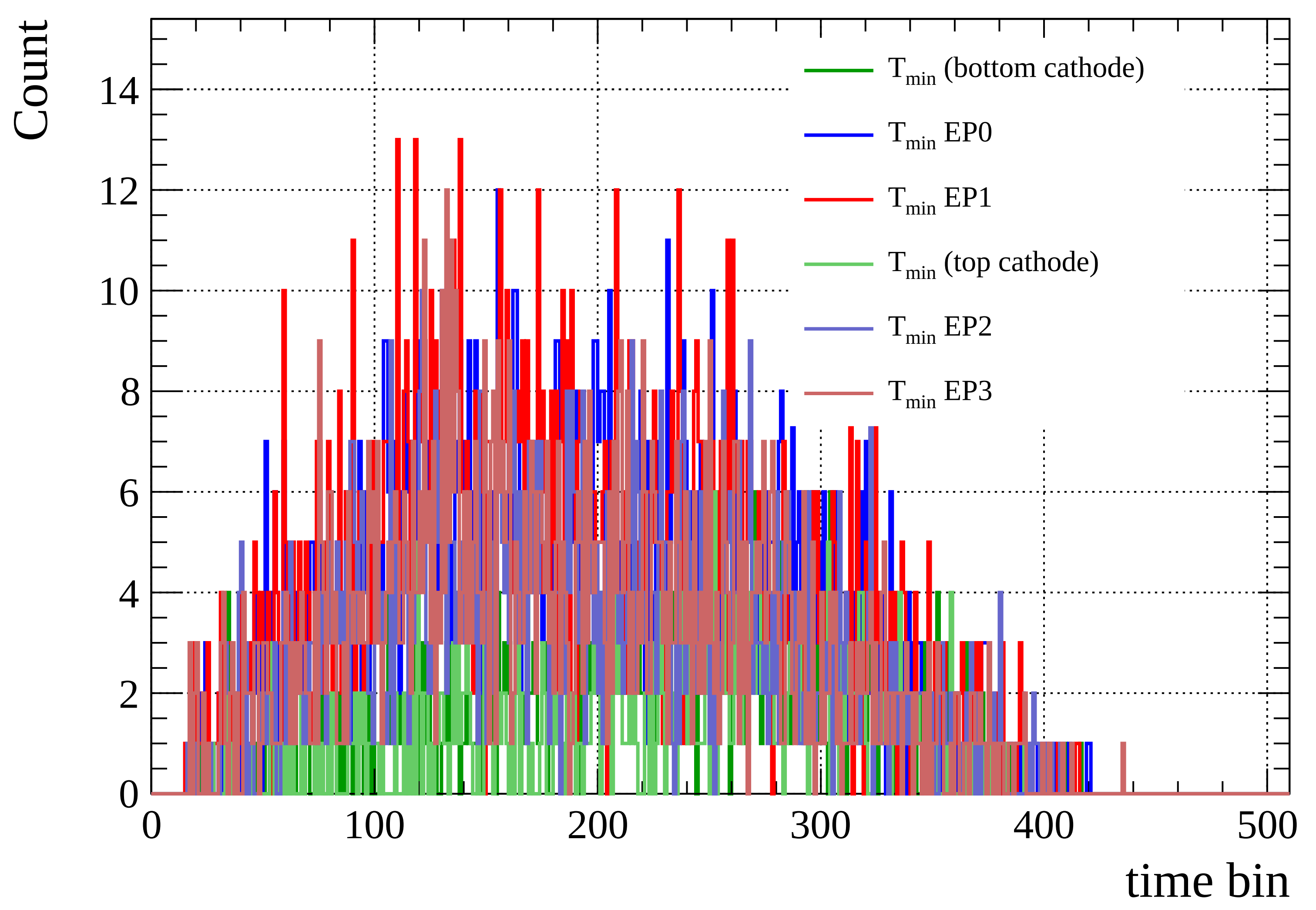


Energy loss | 81 < θ < 84
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

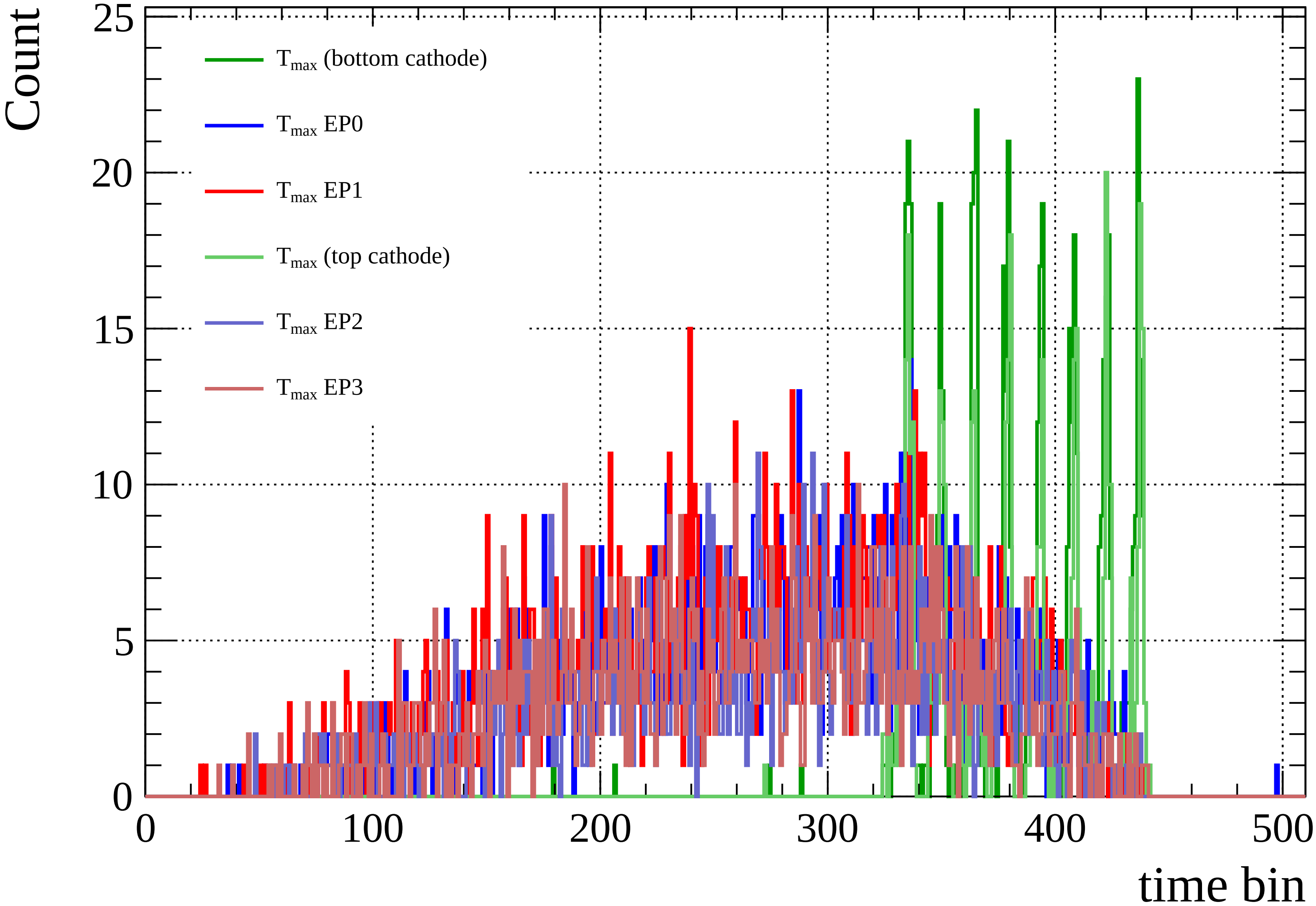


Energy loss | 84 < θ < 87
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200


Energy loss | 87 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
0
200
400
600
800
1000
1200
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0



Count
time bin
0
100
200
300
400
500
0
2
4
6
8
10
12
14
T_{min} (bottom cathode)
T_{min} EP0
T_{min} EP1
T_{min} (top cathode)
T_{min} EP2
T_{min} EP3



Count
25
20
15
10
5
0
0
100
200
300
400
500
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3